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COMPREHENSIVE MID-FIDELITY SIMULATION ENVIRONMENT FOR AEROELASTIC STABILITY ANALYSIS OF TILTROTORS WITH PILOT-IN-THE-LOOP

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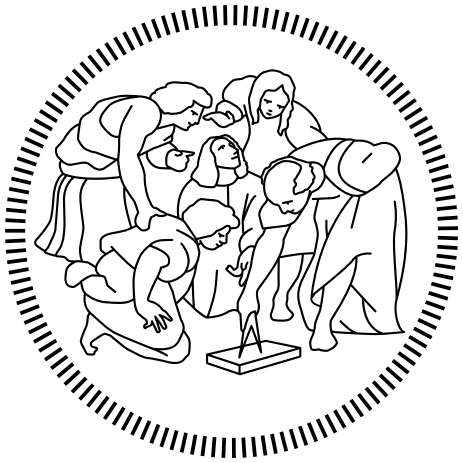

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Dare to dream impossible thing,
then wake up and start working on them.
— Paolo Nespoli

Dedicated to my family that supports me in my studies

ABSTRACT

Tiltrotor aircraft are challenging machines to design considering the various operating conditions and multipurpose missions that are expected to be performed by this complex type of aircraft. This machine combines the characteristic of a conventional turboprop aircraft with the vertical take-off and landing (VTOL) capability of a helicopter. In the past, some catastrophic accidents involved tiltrotors, which were caused by a combination of factors: complex three-dimensional unsteady aerodynamic flow field, structural loads, aeroelastic instabilities, and pilot control inputs. Therefore, the development of a comprehensive tool capable of capturing the combination of all these phenomena is of paramount importance for designing faster, lighter, and safer tiltrotors.

The main objective of this work is to develop, validate, and evaluate the limits of aeroelastic and aeromechanical stability of a tiltrotor aircraft using a comprehensive approach in mid-fidelity. Thanks to a large amount of freely available data, the Bell XV-15 tiltrotor equipped with Advanced Technology Blades (ATB) is chosen as a representative model; however, due to the modularity and parameterization of the model, other innovative rotary-wing configurations could be studied with the proposed approach.

The model is built using a co-simulation approach based on the coupling between a mid-fidelity aerodynamic solver based on the vortex particle method for wake modeling, DUST, and the multibody dynamics code, MBDyn. This coupling is managed through the partitioned multiphysics coupling library preCICE. In the first part of the work, the coupled tool is presented and validated by solving simple fixed-wing and rotary-wing problems from the open literature. In the second part, the multibody and aerodynamic modeling of each subcomponent, using the open-source software MBDyn and DUST, is illustrated and validated considering experimental and numerical results.

Each subcomponent is extensively validated against the available results and, starting from these, they are extended to enable the validation with other analytical tools. Globally, a good correlation is reached in the dynamics analyses for the whole system. In particular, the coupled model shows great advantages in terms of computational time and accuracy concerning high-fidelity tools when used to estimate hover performance.

A time-marching trim procedure is also proposed, and optimal longitudinal control of the tiltrotor is presented and validated to maintain altitude.

A detailed multibody biomechanical model of the upper body of a generic pilot is presented. It is made up of connecting a previously developed detailed model of the arms to a similarly detailed model of the spine. The entire model can be adapted to a specific subject identified by age, sex, weight, and height. The spine model and the scaling procedure are validated using experimental results for seat-to-head transmissibility. The coupled spine-arms model is used to evaluate the biodynamic response in terms of involuntary motion induced in the control inceptors, including related non-linearities.

Finally, the tiltrotor and pilot models are combined into a single comprehensive model in which the whirl flutter aeroelastic stability is investigated over the entire flight envelope, and the responses to discrete gusts were used to evaluate the effect of the pilot biomechanics and interactional aerodynamics. The results show that including an unsteady aerodynamic model is of paramount importance, especially when dealing with flight dynamics modes.

This model enables a wide spectrum of different analyses: for example, this model will be used to study passenger comfort, and performance in transient maneuvers such as conversion and pull-up, aeroelastic stability, pilot-induced-oscillation, and pilot-augmented-oscillation phenomena. The coupled multibody-mid-fidelity tool can be used for performing the large number of aeroelastic analyses required in the preliminary design phase of innovative rotary-wing configurations, not only in the rotorcraft field but also in other domains, from wind energy to micro-aerial vehicles.

Keywords: aeroelasticity; fluid-structure interaction; rotary-wing aerodynamics; multibody dynamics; tiltrotor; computational fluid dynamics; vortex particle method; pilot-biomechanis; pilot-augmented-oscillation; whirl flutter; gust response;

SOMMARIO

I convertiplani sono macchine impegnative da progettare considerando le varie condizioni operative e le missioni polivalenti che dovrebbero essere eseguite da questo complesso tipo di velivolo. Questa macchina combina le caratteristiche di un velivolo a turboelica convenzionale con la capacità di decollo e atterraggio verticale di un elicottero. In passato, alcuni incidenti catastrofici hanno coinvolto i convertiplani, causati da una combinazione di fattori: campo di flusso aerodinamico instabile tridimensionale, carichi strutturali, instabilità aeroelastiche e input di controllo del pilota. Pertanto, lo sviluppo di uno strumento completo in grado di catturare la combinazione di tutti questi fenomeni è di fondamentale importanza per progettare convertiplani più veloci, leggeri e sicuri.

L'obiettivo principale di questo lavoro è quello di sviluppare, convalidare e valutare i limiti di stabilità aeroelastica e aeromeccanica di un velivolo convertiplano utilizzando un approccio globale a media fedeltà. Grazie alla grande quantità di dati liberamente disponibili, il convertiplano Bell XV-15 dotato di Advanced Technology Blades (ATB) viene scelto come modello rappresentativo; tuttavia, data la modularità e la parametrizzazione del modello, potrebbero essere studiate altre configurazioni innovative ad ala rotante con l'approccio proposto.

Il modello è costruito utilizzando un approccio di co-simulazione basato sull'accoppiamento tra un risolutore aerodinamico di media fedeltà basato sul metodo delle particelle vorticose per la modellazione della scia, DUST e la dinamica multicorpo codice MBDyn. Questo accoppiamento è gestito attraverso la libreria di accoppiamento multifisico preCICE. Nella prima parte del lavoro, lo strumento accoppiato viene presentato e validato risolvendo semplici problemi ad ala fissa e ad ala rotante presenti in letteratura. Nella seconda parte, viene illustrata e validata la modellazione multicorpo e aerodinamica di ciascun sottocomponente, utilizzando il software open-source MBDyn e DUST, considerando i risultati sperimentali e numerici. Ogni sottocomponente è ampiamente validato rispetto ai risultati disponibili e, a partire da questi, vengono estesi per consentire la validazione con altri strumenti analitici. A livello globale, si raggiunge una buona correlazione nelle analisi dinamiche per l'intero sistema. In particolare, il modello accoppiato mostra grandi vantaggi in termini di tempo computazionale e accuratezza rispetto agli strumenti ad alta fedeltà quando utilizzato per stimare le prestazioni di volo stazionario.

Viene inoltre proposta una procedura per l'ottenimento dell'assetto mediante una simulazione integrata nel tempo e viene presentato e

convalidato il controllo longitudinale ottimale del convertiplano per mantenere l'altitudine.

Da ultimo, viene presentato un modello biomeccanico multicorpo dettagliato della parte superiore del corpo di un pilota generico. Consiste nel collegare un modello dettagliato delle braccia sviluppato in precedenza a un modello altrettanto dettagliato della colonna vertebrale. L'intero modello può essere adattato a un soggetto specifico individuato per età, sesso, peso e altezza. Il modello della colonna vertebrale e la procedura di ridimensionamento sono convalidati utilizzando risultati sperimentali per la trasmissibilità sedile-testa. Il modello accoppiato colonna vertebrale-braccia viene utilizzato per valutare la risposta biodinamica in termini di movimento involontario indotto negli incettori di controllo, comprese le relative non linearità.

Infine, i modelli del convertiplano e del pilota sono combinati in un unico modello completo in cui la stabilità aeroelastica a whirl-flutter viene studiata sull'intero inviluppo di volo e le risposte a raffiche discrete sono state utilizzate per valutare l'effetto della biomeccanica del pilota e dell'aerodinamica interazionale. I risultati mostrano che l'inclusione di un modello aerodinamico instabile è di fondamentale importanza, specialmente quando si ha a che fare con le modalità dinamiche di volo.

Questo modello consente un ampio spettro di analisi diverse: ad esempio, questo modello verrà utilizzato per studiare il comfort dei passeggeri e le prestazioni in manovre transitorie come la conversione e il pull-up, la stabilità aeroelastica, l'oscillazione indotta dal pilota e l'aumento del pilota fenomeni di oscillazione. Lo strumento accoppiato multibody-mid-fidelity può essere utilizzato per eseguire il gran numero di analisi aeroelastiche richieste nella fase di progettazione preliminare di configurazioni innovative ad ala rotante, non solo nel campo degli elicotteri ma anche in altri domini, dall'energia eolica alla micro-veicoli aerei.

Parole chiave: aeroelasticità; interazione fluido-struttura; aerodinamica ad ala rotante; dinamica multicorpo; convertiplano; fluidodinamica computazionale; metodo delle particelle vorticose; biomeccanica del pilota; oscillazione pilota aumentata; whirl-flutter; risposta a raffica;

PUBLICATIONS

During the my PhD, I co-authored the following publications:

[1] Gianni Cassoni, Alessandro Cocco, Aykut Tamer, Andrea Zanoni, Pierangelo Masarati, et al. "Tiltrotor Whirl-Flutter Stability Investigation Using Lyapunov Characteristic Exponents and Multibody Dynamics." In: *48th European Rotorcraft Forum (ERF 2022)*. 2022, pp. 1–6.

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ACRONYMS AND NOMENCLATURE

The following nomenclature and acronyms are used in this manuscript:

ATB	Advanced Technology Blade
VPM	Vortex Particle Method
API	Application Programming Interface
MBD	Multibody Dynamics
FEM	Finite Element Method
LPM	Lumped Parameter Model
BMI	Body Mass Index
FRF	Frequency Response Function
DOF	Degrees Of Freedom
AM	Apparent Mass
STHT	Seat To Head Transmissibility
DPMI	Driving Point Mechanical Impedance
CSD	Cross Spectral Density
PSD	Power Spectral Density
GA	Genetic Algorithm
MASST	Modern Aeroservoelastic State Space Tools
CMS	Component Mode Synthesis
RMS	Root Mean Square
RPC	Rotorcraft Pilot Coupling
BDFT	Biodynamic Feedthrough
PIO	Pilot Induced Oscillation
PAO	Pilot Augmented Oscillation
VTOL	Vertical Take-Off and Landing
MPE	Matrix Pencil Estimation Method
LTI	Linear Time Invariant
LTP	Linear Time Periodic
FMM	Fast Multiple Method
DAE	Differential Algebraic Equation
AR	Aspect Ratio
EA	Elastic Axis
CG	Center of Gravity
AoA	Angle of Attack

f	Frequency [Hz]
ξ	Damping factor
VL	Vortex lattice
SP	Surface panel
LL	Lifting line
$NL-VL$	Non-linear vortex lattice
U_∞	Free-stream speed
Ω	Rotor speed [RPM]
ρ	Air density [kg/m^3]
c	Aerodynamic chord [m]
R	Rotor radius [m]
N	Number of blades
σ	Rotor solidity $= (Nc)/(\pi R)$
N/rev	Number of rotor revolutions
T	Rotor thrust [N]
Q	Rotor torque [Nm]
C_T	Rotor thrust coefficient $= T/(\rho\pi R^4\Omega^2)$
C_Q	Rotor torque coefficient $= Q/(\rho\pi R^5\Omega^2)$
FM	Rotor figure of merit = $C_T\sqrt{C_T/2}/C_Q$
p	Static pressure [Pa]
p_∞	Free-stream static pressure [Pa]
C_P	Pressure coefficient $= (p-p_\infty)/(0.5\rho_\infty U_\infty^2)$
C_L	Lift coefficient = $L/(0.5\rho_\infty U_\infty^2 S)$
L	Lift [N]
ψ	Blade azimuthal angle [deg]
δ_a	Aileron deflection [deg]
I_{ij}	Moment of inertia about axis ij [kgm^2]

Part I

SOFTWARE TOOLS

1 INTRODUCTION AND THESIS OUTLINE

1.1 TILTROTOR HISTORY AND WIND TUNNEL TEST CAMPAIGNS

Tilting rotor aircraft, or tiltrotors, with their outstanding ability to take off and land vertically like helicopters and, at the same time, to achieve high speeds up to twice those of helicopters in forward flight when operating like conventional turboprop aircraft, represents one of the few successful examples of advanced vertical lift configurations.

After a long development phase that encompassed few experimental aircraft that successfully made it to flight noticeably the Transcendental Model 1-G (a tilting rotor concept) shown in Fig. 1a, which flew about 100 hours without ever completing a full conversion [109] and the Bell XV-3 (a tilting rotor configuration), shown in Fig. 1b, in the 1950s [51], and the Bell XV-15 (a tilting rotor/nacelle configuration) in the 1980s [109] shown in Fig. 2a, unlike other less fortunate technology demonstrators, the concept finally proved its soundness with the Bell-Boeing V-22, shown in Fig. 2b (a tilting rotor/nacelle configuration) [168], a military aircraft that after an almost 25 years long development phase, has been operated by the US Marine Corps with an increasing reward for the last 15 years, and is entering service for carrier onboard delivery in the US Navy. Remaining in the military arena, the Bell V-280 Valor, shown in Fig. 3a, (a tilting rotor configuration), after a test campaign started in 2017, is currently being evaluated by the US Army for its Future Long-Range Assault Aircraft (FLRAA) program [58]. However, with then Bell-Agusta and now Leonardo AW609,shown in Fig. 3b, (a tilting rotor/nacelle configuration) [144] about to become operational after a long and thorough development [63], the tiltrotor design appears to be mature enough to also enter the civil air transport market [1].

Nevertheless, tiltrotor design remains a rather challenging engineering task, considering the various operating conditions and multipurpose missions that are expected to be accomplished by this complex type of aircraft. In particular, the problem of assessing whirl flutter stability limits is at the same time fundamental and challenging. Whirl flutter is an aeroelastic stability phenomenon that affects both turboprop and tiltrotor aircraft [34]. When a rotor mounted on a flexible structure rotates, the normal vibration modes associated with the supporting structure may interact with the precession motion of the rotor.

(a) Transcendental Model 1-G

(b) Bell XV-3

Figure 1: Transcendental Model 1-G and Bell XV-3

(a) Bell XV-15

(b) Bell-Boeing V-22

Figure 2: Bell XV-15 and Bell-Boeing V-22

(a) Bell V-280 Valor

(b) Leonardo AW609

Figure 3: Bell V-280 Valor and Leonardo AW609

When its motion is perturbed, each point on the rotor axis of rotation draws paths about its reference position. This motion changes the way each rotor blade is affected by the incoming airspeed, correspondingly altering the overall aerodynamic loads. At the verge of whirl flutter, when this phenomenon is triggered, perturbations result in a periodic orbit. The resulting forces can cause divergence in the response of the system, which represents a true aeroelastic instability [165].

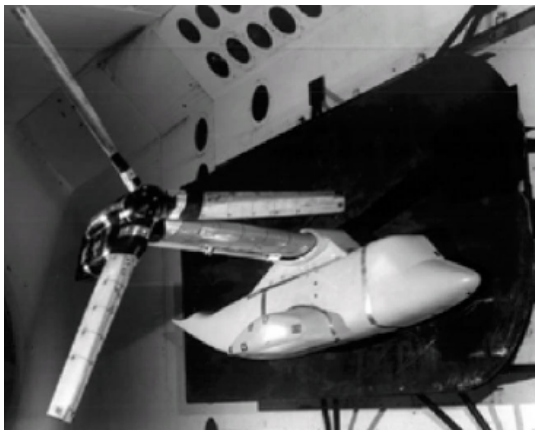
Today, although key aspects of the phenomenon in tiltrotor aircraft are understood, the ability to predict with high computational efficiency it is still limited, despite the availability of sophisticated aeroelastic analysis tools. The difficulty lies in its dependence on several factors, including geometric design, structural properties, actuator dynamics, and aerodynamic interaction between different components, which can all contribute to the whirl flutter characteristics in ways that are not always intuitive. The problem may be exacerbated when unconventional configurations are considered, as is the case of propeller- and rotor-driven advanced air mobility concepts [95].

Aeroelastic testing, especially when it comes to stability and flutter, can be extremely dangerous. For this reason, wind tunnel testing is needed to understand phenomena and increase the level of trust in predictions in support of their mandatory verification in flight. As a consequence, the possibility to validate numerical predictions using experimental data is of paramount importance from an industry point of view.

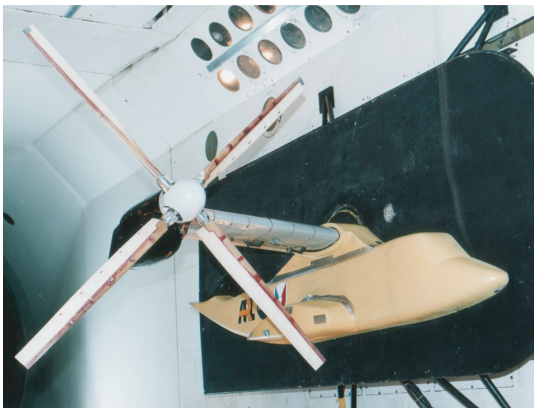
Wind-tunnel testing of tiltrotor configurations, and its use for validation and correlation with numerical predictions,have been widely used for tiltrotor development. Due to the need to aeroelastically scale also the wing portion of the model, of extreme miniaturization of the rotor consider, for example, the need to reproduce in detail rather complex blade root attachments and pitch control mechanisms, aeroelastic wind-tunnel models of tiltrotor aircraft can be even more complex than helicopter ones.

Several wind-tunnel models have been developed to explore the concept of whirl flutter in prop- and tiltrotor configurations. However, specific wind tunnel tests of configurations that were intended to be tested in flight date only back to the early development of the V-22 when Bell designed and manufactured a 1/5-scale semi-span model and used it for wind tunnel tests and correlation with the simulation capabilities available at that time [158]. Eventually, this model evolved in the Wing and Rotor Aeroelastic Test System (WRATS) 4, which has been in use for several years to explore various aspects of tiltrotor aeroelasticity, including aeroelastic tailoring of the wing [45, 139], several parametric studies (e.g., in [155]), active control for flutter suppression [70, 101], an interesting four-blade, articulated soft-inplane configuration for hub load reduction [140, 141], and several nu-

merical studies of stiff- and soft-inplane configurations successfully correlated with experiments [114, 121, 173, 174].



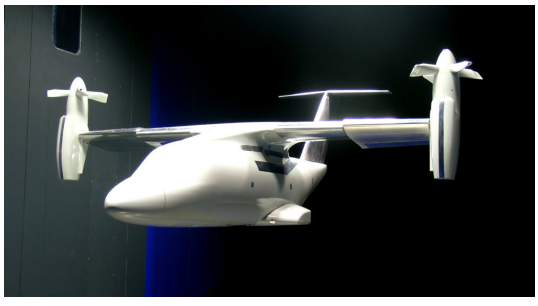
(a) Three-blade, stiff-in-plane



(b) Four-blade, articulated soft-in-plane

Figure 4: WRATS models

Another effort towards wind tunnel testing of the aeroelastic stability of tiltrotors took place in Europe, within the EUROFAR program [82]. Subsequent efforts were aimed at developing an original European tiltrotor concept, ERICA [137]. This concept is characterized by the tilt of the outer portions of the wing, affected by the downwash of the rotor. These efforts resulted, among others, in the ADYN project [22, 98], aimed at studying the aeroelasticity and noise characteristics of the concept.



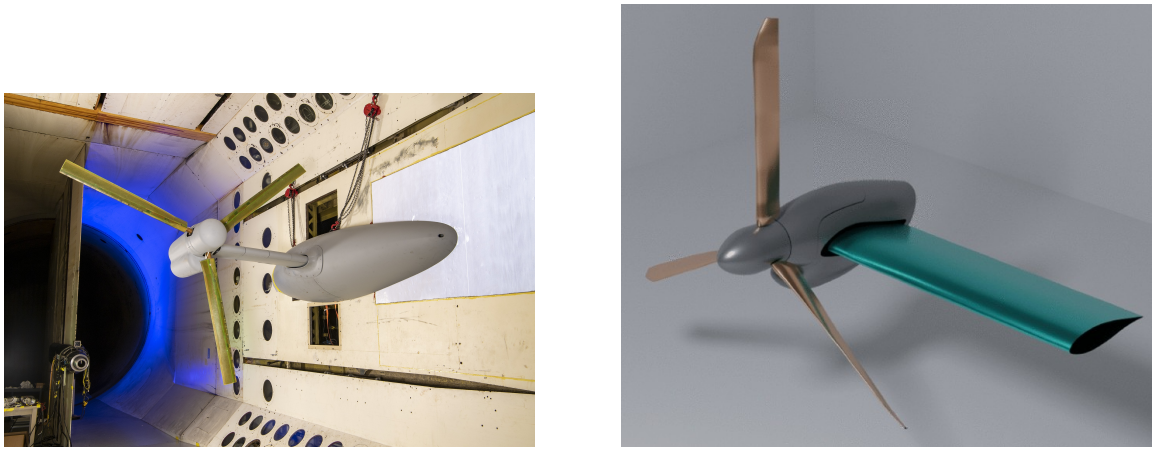
(a) ERICA

(b) ADYN

Figure 5: ERICA and ADYN

Currently, in the USA, a US Army-coordinated program called TRAST, shown in Fig. 6a, is ongoing to develop a testbed for research into tiltrotor aeroelasticity [15, 96, 203, 205, 214]. Another effort is underway at the University of Maryland, where a reconfigurable wind tunnel tiltrotor model is being developed [48, 183, 193] and recently underwent preliminary whirl flutter stability tests [74, 184, 204].

At about the same time, another effort is ongoing in Europe, within the CleanSky2 initiative, within the ATTILA project (a rendering is shown in Fig. 6b), to design, manufacture, and test a tiltrotor wind tunnel model in support of the Next-Generation Civil Tilt Rotor-Technology Demonstrator (NGCTR-TD), a fixed-engine tilting rotor configuration. The preliminary modeling efforts of the wind tunnel testbed were initially presented in [40, 42, 73] and a detailed comparison between the numerical models can be found in [41].



(a) TRAST wind tunnel model

(b) ATTILA multibody model [41]

Figure 6: TRAST and ATTILA models

1.2 INTRODUCTION TO WHIRL FLUTTER PHENOMENON

In airplane mode, under high flow conditions and when there are small variations in flow, considerable differences in load distribution cause the nacelle and the wing structure to bend or twist. Motion-dependent forces can then result in greater wing motion and pylon and hub motion, initiating a repetitive cycle of aeroelastic oscillations, which can, under certain circumstances, ultimately lead to structural failure.

1.2.1 *2-DOF model*

Whirl flutter is a very complex type of instability. Although more and more comprehensive models have been developed over the years, a very simplified representation can still provide a basic understanding of the phenomenon.

The first description proposed consists of a truly rigid propeller on a pylon with pitch and yaw degrees of freedom. A detailed discussion is presented in [164], and the representation of the model is shown in Figure 7.

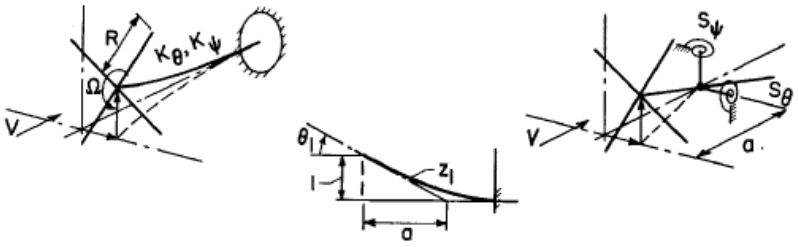


Figure 7: 2-dof Proprotor Model [164]

The governing equations of motion (Equation 1) include contributions from various forces and moments acting on the system. In addition to conventional inertia, damping, and elastic forces, the rotating propeller introduces gyroscopic and aerodynamic forces.

$$\begin{cases} I\ddot{\theta} + c_\theta\dot{\theta} + S_\theta\theta - I_x\Omega\dot{\psi} = aL_z + M_y \\ I\ddot{\psi} + c_\psi\dot{\psi} + S_\psi\psi + I_x\Omega\dot{\theta} = aL_y + M_z \end{cases} \tag{1}$$

Aerodynamic terms, on the right-hand side of Eq. 1, include the net effect of the vertical force L_z and the pitching moment M_y and the horizontal force L_y and the yawing moment M_z. The rotation of the propeller also affects the natural vibration modes, as shown in Figure 8.

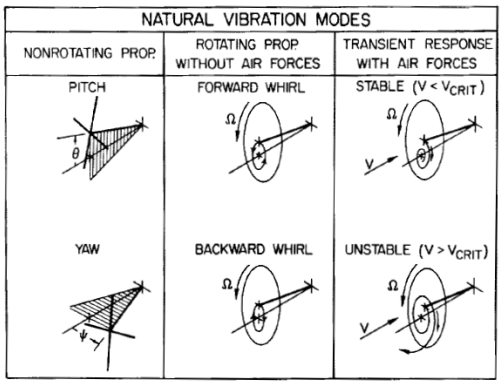


Figure 8: Proprotor Natural Vibration Modes [164]

In particular, while in the non-rotating case the pitch and yaw modes can be excited independently, this is not true for the rotating propeller, for which the gyroscopic action introduces a coupling effect. The modes then become characterized by a whirl or precession mode, with reference to how the hub is with respect to the static thrust axis. In this case, even for the symmetric stiffness properties of the system, the expected frequencies for the whirl modes are different. The forward whirl, defined as occurring in the same direction of propeller rotation, manifests itself at higher frequency, whereas the

backward whirl, defined as opposed to propeller rotation, occurs at lower frequency.

When these whirl modes produce a change in the angle of attack of the blade element, the aerodynamic forces generated are those that could cause the instability mechanism. At airstream velocities $V < V_{crit}$ the mechanism is stable, so a disturbance causes the hub to trace a spiral path that converges to the initial equilibrium position, in cases where $V > V_{crit}$ a divergent condition occurs. For a rigid propeller configuration, the instability occurs in the backward-whirl mode.

When analyzing the separate contributions in terms of propeller aerodynamics, a significant impact is given by the yaw moment due to pitch $M_z(\theta)$ or the pitch moment due to yaw $M_y(\psi)$, which generates a cross-stiffness (destabilizing) moment. In the backward whirl mode, this driving moment is resisted by the aerodynamic and structural damping moments, so that oscillations may become neutrally stable when the contributions of driving and resisting terms are exactly counterbalanced.

Several studies and experimental test campaigns conducted on this type of configuration highlighted that the most important parameters found to influence the critical velocity at which whirl flutter occurs are stiffness, damping, pivot location, and propeller advance ratio. The main effect and role of each term can be summarized as shown in Fig. 9.

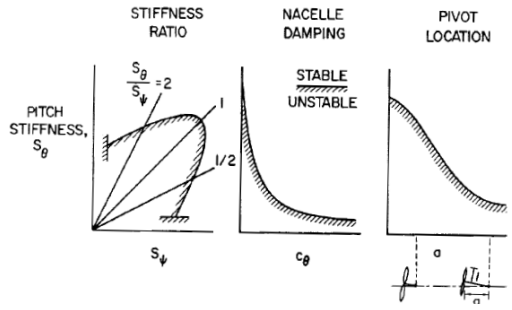


Figure 9: Effects of Parameters on 2-dof Whirl Flutter Model [164]

In particular, in relation to parameter variations, it is interesting to note that:

- In terms of stiffness, a system having equal pitch and yaw stiffness is significantly more subject to flutter when compared to a case where one stiffness is greatly reduced
- In terms of damping, an increased damping is significantly stabilizing
- In terms of pivot location, moving the pivot aft is significantly stabilizing

The 2-dof model presented, often referred to as the classical whirl flutter model, is mainly included for historical reasons and due to the

fact that the derivation of the exact expression for the stability boundaries is possible. However, it is commonly acknowledged that it is at least necessary to consider a 4-dof model (flapping rotor on a pylon with pitch and yaw degree of freedom) to obtain a more appropriate representation, especially when considering a tiltrotor proprotor. A schematic view of this configuration is shown in Fig. 10.

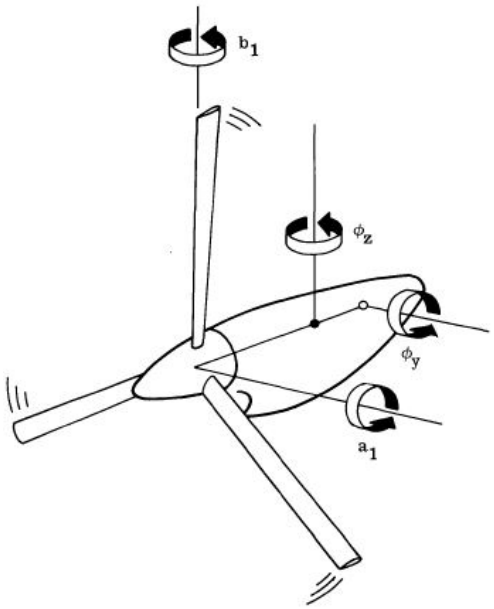


Figure 10: 4-dof Proprotor Model [100]

Also, for this case, which at least overcomes the limitations given by rigid propeller modeling, it may be required to add at least a few more degrees of freedom for both rotor and support in order to obtain meaningful results.

Johnson [87] later proposed a theoretical model that provided the ability to investigate the dynamics of the proprotor on a cantilever wing, as shown in Fig. 11.

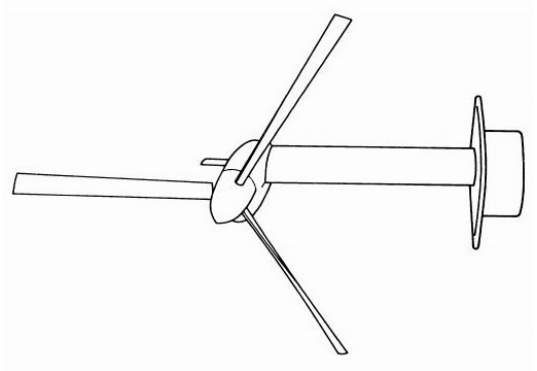


Figure 11: Proprotor - Cantilever Wing Model [86]

The discussed model was applicable to the case of axial flight and included a modal representation for the wing, considering a total of nine degrees of freedom (6 for the three-bladed rotor system and 3 fundamental wing modes). The analytical results obtained correlated well with the full-scale test results and the model was then extended to include elastic blade characteristics and helicopter and conversion modes of operation, effectively forming the basic tiltrotor model theory for the complete CAMRAD rotorcraft code [32, 88].

Although this review of theoretical models is by no means meant to be exhaustive, it was of interest to discuss some of the modeling choices and typical procedures involved. In general, also for more sophisticated models, as discussed in [164], it is very convenient to adopt a two-phase approach, where the main components are analyzed separately and then the complete aircraft behavior is examined:

1. In the first phase, the major components should be studied individually, for example, by considering a flexible propeller-nacelle system mounted on a rigid wing structure or a flexible wing with the assumption that the contribution of the nacelle-powerplant can be treated as rigidly attached to the wing
2. In the second phase, the equations of motion of each component should be coupled together.

1.3 IMPACT OF TECHNOLOGY

In this Section, a brief overview of the contributions of the major tiltrotor components is presented. The considerations and findings of the work by Nixon [139], [140] and Acree [9], [6], [7], [8] are outlined on the impact of technology and design options on aeroelastic stability.

1.3.1 *Rotor*

1.3.1.1 *Hub Concept*

Possible solutions to the design of the rotor hub include the use of:

- gimballed or hinged configurations
- hingeless configurations

Two different hub concepts, mounted on the WRATS model for airplane mode testing, are shown in Fig. 4.

The gimballed solution is currently the preferred approach and common practice and has been adopted for the XV-15, V-22, V-280, and AW609. Due to the kinematic constraints of the configuration, gimballed rotors do not scale well to a number of blades greater than three.

The semi-articulated solution, for example, that proposed by Bell Helicopters for the Army VGART program, having no gimbal but employing a standard blade flap hinge and elastomeric bearing for lag motion, was specifically conceived for a better scalability and growth potential by allowing the use of more than three blades.

The hingeless solution, having no blade flapping, lead-lag hinges, or rotor-flapping gimbal, contains fewer parts and would allow for higher helicopter mode pitch and yaw control moments, but is cause of higher loads for blade, hub, and main transmission, providing considerable penalties from a weight perspective and also regarding the life of the structural components. The hingeless hub has not received as much attention as the gimbal counterpart, but it is acknowledged that it seems to be less susceptible to whirl flutter.

In relation to the type of hub concept adopted, it is common to perform a distinction between soft-in-plane and stiff-in-plane configurations, in cases where the ratio between the in-plane natural frequency and main rotor harmonic is respectively below or above 1.

Stiff-in-plane rotors are not affected by concerns for ground resonance in hover and generally provide higher whirl flutter boundaries and adequate damping margins at subcritical speeds. However, particularly during maneuvers, very high in-plane dynamic loads may develop.

However, several studies including [140] have highlighted that soft-in-plane systems are beneficial from a load perspective, as they produce significantly lower dynamic loads, but provide reduced damping margins and stability boundaries compared to their stiff-in-plane counterparts under the same operating conditions.

1.3.1.2 *Rotor Couplings*

PITCH-FLAP COUPLING The kinematic coupling between blade flapping and pitch is called δ_3. This pitch-flap coupling creates an aerodynamic spring force that moves the flap frequency away from the rotation frequency (1P) and thus away from a resonant condition. Following common convention, a positive δ_3 implies a nose-down pitching for upward flapping. Even if counter-intuitive, this type of coupling is destabilizing for some modes, typically lag modes, as it creates an aerodynamic-based increase in flapping stiffness, leading to an eventual mode coalescence as the collective is increased with airspeed. For this reason, a negative value for δ_3 is adopted in the conventional tiltrotor design. The typical scheme for pitch-flap coupling is shown in Fig. 12 for a gimballed configuration for which an offset pitch horn is used.

It should also be noted that a negative δ_3 is as effective as a positive δ_3 in reducing the maximum transient flapping response and that the potential range of values actually implementable from a practical point of view is limited by the blade spacing.

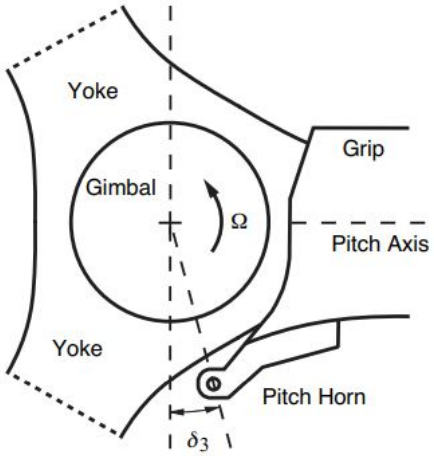


Figure 12: Pitch-Flap Coupling [6]

PITCH-LAG COUPLING Pitch-Lag coupling is strongly influenced by rotor precone and control system stiffness.

In relation to precone, the value of β_p, as shown in Figure 13, is chosen to improve blade loads in helicopter mode, when the disk loading is high and large blade bending moments arise. In this condition, the precone angle allows a component of centrifugal force that counteracts the contribution of the lift bending moment, so the net moment is reduced compared to the case without precone.

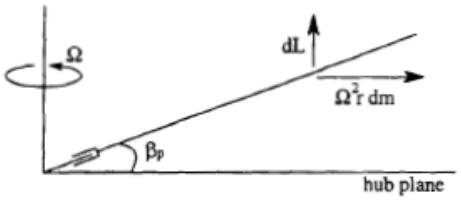


Figure 13: Precone Effect [139]

However, an excessive precone leads to a destabilizing effect in airplane mode, due to the role of a generated pitch-lag coupling. In cruise condition, disk loading is considerably reduced with respect to hover, and the centrifugal force component is reduced as well, owing to the lower rotor speed. The significant imbalance that is then created causes a torsion moment about the blade inboard section proportional to the lag bending deflection, as shown in Fig. 14, generating a destabilizing coupling effect.

The mentioned pitch-lag coupling is also affected to some extent by the stiffness of the control system. Studies conducted in relation to the stability of a rigid hub have shown that decreasing the stiffness of the control system had a destabilizing effect, while analyses of a flexed hub configuration demonstrated that this negative effect is less severe or has little impact on stability considerations [155]. The flexed

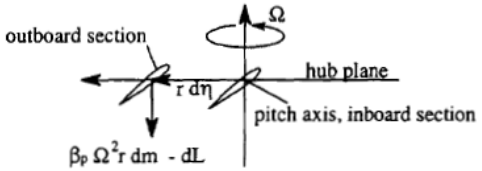


Figure 14: Torsion about Inboard Sections due to Lag Motion [139]

hub is designed specifically to allow the rotor to flatten out in cruise flight, reducing the design precone angle β_p by the angle β_h and consequently reducing the *total* pitch-lag coupling.

1.3.1.3 *Blade Design*

The blade aerodynamic design is what primarily determines the twist of the blade and the definition of the planform. The twist and chord distributions are typically based on a compromise between the figure of merit in hover and propeller efficiency in airplane mode, with chord selection subject to low-speed maneuvering capability requirements.

The blade has an intermediate twist between that of a conventional rotor and that of a propeller and is characterized by an airfoil selection which depends on the radial location. The adopted profiles are thicker at the root for structure build-up, are selected to provide a maximum lift-to-drag ratio in the middle, and are thin outboard. Blade taper improves hover efficiency, but reduces the propeller one.

Regarding blade design for improved stability margins, several studies have discussed the impact of variations in the center of gravity of the blade and the aerodynamic center along the blade. In particular, Acree [9] has shown that small forward offsets of the aerodynamic center with respect to the elastic and pitch axis, obtained by moving the airfoil quarter-chord, considerably increased stability boundaries, while small forward offsets of the center of gravity had a similar but less pronounced effect. The blade parametric variations were distributed from root to tip along four radial stations, and the offset contributions were analyzed separately, which also showed that the effect was largest in the outboard blade segments. The findings of the work were interesting because they proved that an increased chordwise distance between the center of gravity of the blade and the aerodynamic center, the classical approach adopted to increase stability for an isolated airfoil, was also applicable to the entire rotor-wing dynamic system, given the highly coupled nature of the whirl-mode instability. An important aspect that emerged from the investigation was related to the role of the proposed variations in the mean and oscillatory rotor loads. While the potential to define variations that increase stability should be considered, it should always be verified

that such modifications do not have a severe adverse effect on loads in both helicopter and airplane modes.

Acree [6] later extended the focus to other design modifications, with an approach that clearly grew from previous efforts. Specifically, in relation to blade design, both the sweep of the outboard sections and the moving of the balance weights at the tip ahead of the elastic axis were evaluated. Figure 15 shows the case for the considered swept blade modification derived from the baseline V-22 rotor.

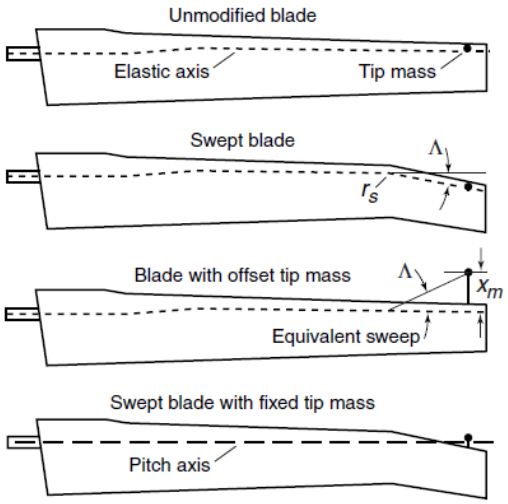


Figure 15: V-22 Blade Modifications - Sweep and Tip Mass Offset [6]

The research revealed how the most favorable combinations greatly increased the damping of the critical modes in terms of stability, fully stabilizing the rotor.

In a more general analysis of advanced geometry blades on whirl flutter stability, the tip anhedral has also been shown to increase the speed of flutter, while the planform taper, the tip dihedral and the forward sweep have proven detrimental on the stability margins [178].

1.3.1.4 *Operational Speed*

An important driver to consider in the structural design phase, especially in relation to the strong impact it has on blade dynamics, is the operational speed of the rotor. The value adopted is generally defined according to different considerations. In airplane mode, it is about 80% the value in helicopter mode. In this wide range of operating conditions, it is of paramount importance to guarantee adequate separation of the fundamental flap, lag, and torsion frequencies of the blade from harmonics associated with the operating speeds of the rotor during hover and cruise and to avoid any destabilizing frequency crossings.

While it is true that a reduction in rotor speed reduces the forces that drive whirl flutter, the issues related to frequency placement and modal coupling are potentially exacerbated.

Ultimately, it is clear that rotor system design must always consider the important trade-offs that exist between the necessary required stability margins and blade loads that arise when adopting one solution rather than the other.

1.3.2 *Wing*

The current practice in tiltrotor design is to employ a high-mounted wing with a tilting engine nacelle configuration. Some studies have also been performed considering low-wing configurations, for example in relation to the NASA Large Civil Tiltrotor (LCTR) program. Low-mounted wing designs have associated requirements that include fixed engines with tilting shafts, longer rotor shafts or extreme dihedral for fuselage clearance in hover, and hingeless rotors for adequate pitch control in hover. As the hingeless rotor tends to be less susceptible to whirl flutter, this wing configuration would potentially allow for a less strict torsional stiffness requirement. However, a wing with a tilting shaft and fixed engines will have different maximum torsional design loads with respect to the high-wing tilting engine nacelle counterpart because of the different offset between the rotor thrust vector and the wing center of gravity [7].

Conventional high-wing configurations are very stiff and thick (with a thickness-to-chord ratio around 0.23) and are characterized by a limited aspect ratio. Although a reduced wing thickness would be beneficial from a high-speed performance point of view, by allowing for a great reduction in profile drag, it would drastically limit the possibility of meeting the aeroelastic stability requirements. For this reason, the authors, including Nixon [139], have proposed solutions based on aeroelastic wing modifications to introduce beneficial couplings without incurring excessive weight penalties. Composite tailoring, in particular, would provide the opportunity to increase stability margins when considering a thin-wing configuration by modifying mode shapes and frequency placement of the main wing modes. However, in this case, the internal structure of the wings must still be able to accommodate the transmission cross shaft and the fuel tanks, so a thickness reduction driven by performance considerations alone may not always be possible.

In addition to the dynamics of the wings itself, the stability margins of the whirl flutter are greatly affected by the characteristics of the associated wing components, such as the stiffness of the transmission adapter, mast and pylon, conversion spindle, and downstop [139].

Among these, the downstop plays a particularly important role, which is the mechanism that allows locking the pylon to the wing

for a tiltrotor in airplane mode configuration. When not engaged, the stiffness of the attachment is governed by that of the conversion actuator only and is greatly reduced compared to the downstop-on configuration. The contribution of the drive system should also be considered for a proper modeling approach, as it has a significant impact on both the pylon/wing system and on the proprotor aeroelastic behavior. In relation to the effects of the drivetrain, Acree [6] considered the impact of eliminating the drivetrain entirely or eliminating only the flexibility of the drivetrain (thus considering a rigid drivetrain model). In both cases, it appeared that the presence of the drive train caused a significant loss of predicted damping on symmetric and antisymmetric wing beamwise-bending modes. Figure 16 shows the comparison between the results considering the symmetric wing bending mode (SWB).

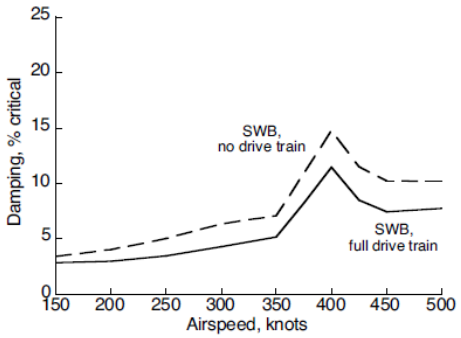


Figure 16: Drive-train Effect [6]

As neglecting the presence of the drive system leads to non-conservative results in terms of damping, this contribution should definitely be taken into account in both the design and the testing phase.

1.3.3 *Additional Considerations*

From the discussion presented, it is clear that the rotor and wing cannot be designed independently. Although it is true that performance goals for hover and cruise determine the initial rotor design and the set wing area and maximum thickness, loads associated with the sizing condition determine the structural design. The need to account for the potential onset of dynamic instabilities should be considered early in the design, as performance in airplane mode may be limited by whirl flutter rather than by aerodynamics or installed power. A posteriori corrections of the design may not always be feasible and are always cause of cost penalties and schedule delays, and should thus be avoided.

In general, there are several options for improving tiltrotor aeroelastic stability margins, and it is either possible to consider passive or active techniques, as well as a combination of both.

In relation to active techniques, a very promising solution consists of the adoption of active stability augmentation system [128]. However, flutter suppression systems are not accepted, at least up to date, from a regulatory point of view: the tiltrotor should be free (with margins) from any instability phenomena in the entire flight envelope.

In relation to passive techniques, a summary of the most important aspects and considerations is presented in view of the design options discussed in previous sections and the findings from [75]:

1. The rotor design parameters that have the greatest influence on whirl flutter stability are pitch-flap and pitch-lag couplings
2. The wing design parameters that have the greatest influence on whirl flutter stability are wing vertical bending and torsion stiffness, as well as vertical bending-torsion coupling
3. Modest changes in key design parameters for rotor and wing may provide a significant flutter speed increase
4. Generally, flutter speed has shown a greater sensitivity towards variations in rotor rather than wing design

Ultimately, having identified whirl flutter as a critical design requirement for the tiltrotor configuration, it is evident that the tools adopted in the design phase should be able to capture the behavior of parametric modifications on key design parameters and allow to easily analyze each component both separately and as part of the integrated vehicle.

From a modeling point of view, tools that provide the capability to use anisotropic beam theory are often the only viable option to reproduce the most important coupling effects in the design without leading to excessive computational times. This is especially true when the goal is to assess the impact of different design options at an early design stage, as excessive computational time for each analysis would drastically hinder the applicability of the approach.

1.4 TILTROTOR-ROTOR PILOT COUPLINGS

Rotorcraft-pilot-couplings (RPC) [149] are events that are caused by the pilot's erroneous perception of vehicle dynamics. RPC phenomena can be divided into two families:

- Pilot-Induced Oscillations (PIO): when the pilot inadvertently excites divergent vehicle oscillations by applying control inputs that are in the wrong direction or have phase lag with aircraft motion.
- Pilot-Assisted Oscillations (PAOs) are the result of involuntary control inputs by the pilot in the loop that may destabilize the

aircraft due to inadvertent man-machine couplings. Generally, for PAO to occur, an involuntary involvement of the pilot due to his biodynamic response to vibrations is required.

PAOs usually occur in the frequency range between $2-8$ Hz. In PAOs, airframe vibrations are transferred through the biodynamic feed-through (BDFT) to the control input, which in turn causes new structural vibrations. A block diagram of the pilot-aicraft close loop is shown in Fig. 17.

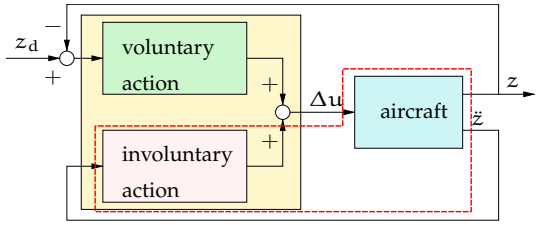


Figure 17: Aircraft-Pilot control loop

Although PAO phenomena have also been identified in fixed-wing aircraft, it is in rotorcraft that they showed a very high incidence [55, 147, 148]. Rotorcrafts are more susceptible to those high-frequency occurrences since their high-order dynamics play a prominent role, as shown in Fig. 18. The control inputs given by the pilots through the cockpit inceptors are transmitted through the swashplate to the blades, thus affecting the rotor flapping and control moments transferred to the airframe.

The rotating mechanism can generate, for any control at frequency ω, oscillations at $\omega \pm \Omega$, where Ω is the angular speed of the rotor. Thus, low-frequency pilot inputs generate high-frequency blade excitation. An interesting analysis of the effects of pilot BDFT on the V-22 was presented in [145], while a more comprehensive analysis of all PAO phenomena that can occur on a tiltrotor, without considering the interaction with FCS, can be found in [135].

Recently, in the work of Muscarello and Quaranta [136], the structural coupling problems for tiltrotors are investigated, considering not only the interaction of the flight control system with the flexible structure, but also the potentially adverse effects on aeroservoelastic stability. The investigation is focused on the analysis of side effects that could appear at high speed in the flight regime of aircrafts, where the whirl flutter boundaries may be significantly reduced. A detailed linearized tiltrotor model, representative of the Bell XV-15 and of a flight control system, has been built and joined with a linearized pilot biodynamic model acting on the power lever and on the center stick. The model has been developed in MASST [120] that connects the substructures using a Craig-Bampton component mode synthesis approach [47].

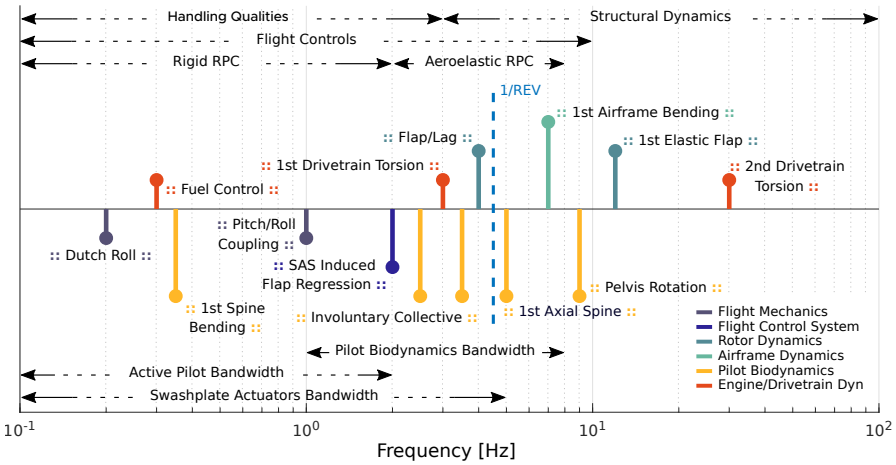


Figure 18: Frequency placement of the helicopter modes and the pilot biomechanical modes

A dramatic accident in which a PAO phenomenon led to the loss of the tiltrotor and the two pilots was the one that occurred in AW609 in 2015. The accident occurred during high-speed dive maneuvers. As stated in the final accident report[1]:

"The third dive, test R20, was commenced during a left 180° *turn. After rollout, a slight lateral-directional oscillations have begun to develop (roll at t=4 sec. and yaw at t=5 sec. and further on) such as to cause an "out-of-trim" condition by the FCS developed. At such oscillations, the crew did not initially react with inputs that were in contrast to them. Later, when the oscillations became greater, this phenomenon was initially contrasted with counterphase input roll maneuvers by the PF (roll tracking) and finally by pedal inputs, to contrast the yaw oscillations. However, these actions did not dampen the oscillations, which instead became divergent, bringing the sideslip angle to reach values above the maximum allowed in those speed conditions."*

Finally, the report stated that the causes of the accident were mainly three.

1. The development of latero-directional oscillations
2. to the FCS control laws unable to maintain conditions of controlled flight
3. to the project simulator (SIMRX), which did not foresee the event in any way:

In particular, the flight simulator was not representative at such high speeds of aircraft behavior, since the aerodynamic model behind was too weak: *"Anyway, the test on SIMRX would have not been representative, at high speed, of real aircraft behavior, as confermed also by ANSV verification on simulator, possibly due to unexpected aerodynamic characteristics*

1 https://www.ansv.it/cgi-bin/ita/FinalReportN609AG.pdf

in this extreme flight condition of speed, having extreme: AOA values, flaperon deflection angles and elevator deflection angles. This condition occurs, in particular, for cross-coupled sideslip and roll (example: left sideslip/right roll)."

1.5 THESIS OBJECTIVE AND OUTLINE

The main objective of this work is to develop, validate, and evaluate the aeroelastic and aeromechanical stability limits of a tiltrotor aircraft using a comprehensive approach.

The final goal is to provide a set of validated, open-source, and multipurpose tools to the scientific and industrial community that can simulate in a single environment all of the ingredients that participate in the aeroelastic loop: this objective is motivated by the lack of such framework in the current literature, which is a gap to be filled to enable considering whirl flutter and also phenomena such as PIO and PAO early in the design process of future tiltrotors while accounting for effects such as the detailed resolution of aerodynamic interactions, general flexible structures, and coupling with the pilot dynamics.

For what concerns the aerodynamic part, two models are adopted: a low-fidelity model based on pure 2D strip theory and a mid-fidelity model based on the vortex particle method developed in open source code DUST. The structural part and the control part are modeled using the multibody code MBDyn. The communication between the two software is handled by the open-source coupling library preCICE. Finally, the completely non-linear biomechanical model of the pilot is coupled with the aeroservoelastic tiltrotor model: this model is developed in MBDyn.

Thanks to the large amount of public data [4, 25, 31, 61, 62] the XV-15 tiltrotor aircraft equipped with Advanced Technology Blade (ATB) is chosen as the base model.

This work is organized as follows.

- *Chapter 2*: In this chapter, the used and developed software tools are presented from a theoretical perspective. A brief theoretical overview of the MBDyn code is given, and instead, a detailed theoretical overview of the DUST code is provided. Finally, the tight coupling between the two codes is presented, focusing on the implementation of the interpolation scheme that is needed to correctly deform the aerodynamic mesh and send the aerodynamic loads to the structure.

  Contribution: The non-linear vortex lattice elements are implemented, then the coupling between DUST and MBDyn is developed improving, in particular, the interpolation scheme and the wake shedding.

- *Chapter 3*: In this chapter, validation, and verification of the developed software tools is presented; in particular, the coupling between DUST and MBDyn is verified, evaluating the flutter speed of the Goland wing[71] against the results present in the literature. In the second part of the chapter, the new DUST nonlinear vortex lattice element is validated, starting from a fixed-wing case, then considering a wing-propeller example, and finally evaluating the aerodynamic performances of the XV-15 metallic blade rigid rotor against high-fidelity simulation and experimental data.

  Contribution: data collection, building the numerical models, running the analyses, and postprocessing the results.

- *Chapter 4*: In this chapter, the aeroelastic ATB blade rotor model is presented and extensively validated against the experimental results [60]. The two aerodynamic models are compared from an aeroelastic perspective in both hover and axial flow conditions.

  Contribution: Implementation of parametric tiltrotor generator TiPa, data collection, building the numerical models, running the analyses, and postprocessing the results.

- *Chapter 5*: In this chapter, the aeroelastic stick model of the XV-15 tiltrotor is presented and validated from a dynamic point of view. Then, a simple optimal control regulator needed to maintain altitude is presented and tested under a discrete gust external input.

  Contribution: data collection, building the numerical model, implementation of the trim procedure, running the analyses, and postprocessing the results.

- *Chapter 6*: In this chapter, the upper-body pilot model is presented: the model is build-up by joining a detailed torso multibody model with the upper-limb multibody model.

  Contribution: Implementation of the pilot's model parameterization technique, joining the spine model with the upper limb model, running the analyses, and postprocessing the results.

- *Chapter 7*: In this chapter, all the ingredients are combined into a single comprehensive tiltrotor model. First, the identification technique needed to evaluate the aeroelastic stability of the aircraft is presented, and then the whirl flutter boundaries of the XV-15 machine are investigated covering the complete flight envelope of the machine. Finally, the pilot is inserted into the loop and its effect is compared by looking at the response of the aircraft after a discrete gust. The effects of the two aerodynamic models are therefore compared.

Contribution: Coupling the tiltrotor model with the pilot model, running the analyses, and postprocessing the results.

- *Chapter 8*: In this chapter, the conclusions and further development are discussed.

2

SOFTWARE TOOLS USED/DEVELOPED

The design of complex rotary-wing vehicles, such as tiltrotors, represents a challenge for engineers and scientists. Being able to perform complete aeroelastic simulations considering the coupling of rotary-wing aerodynamics with the structural dynamics of the vehicle is essential for the development of novel aircraft configurations [17]. Sophisticated numerical tools have been developed to support the analysis of rather different operating flight conditions typical of VTOL vehicles with increasing level of detail. In fact, substantial effort has been made in recent years to develop sophisticated structural dynamics codes (CSD) that were used effectively for rotorcraft applications (see, e.g., references [20, 89, 207]). In detail, the structural dynamics of the rotary wings vehicles was typically investigated using the multibody approach [69, 160], which takes into account the non-linear dynamics of the interconnected rigid and flexible bodies that represent the components of the aircraft during transients. The multibody approach was also used to investigate aeroelastic phenomena, especially in flight in airplane mode, where whirl flutter instabilities can occur [127]. A particular effort in this research field was made at the Politecnico di Milano, where, starting in the 1990s, a free general purpose multibody software called MBDyn (`https://public.gitlab.polimi.it/DAER/mbdyn`) was developed, with the aim of gaining autonomous modeling capabilities of generic problems related to the dynamics of complex aeroelastic systems, specifically rotorcraft and tiltrotors [119]. A tiltrotor flight mission is characterized by vertical take-off and landing, typical of helicopters, and by a cruise flight condition typical of fixed-wing airplanes. The transition between the two flight conditions occurs through the so-called conversion maneuver, in which the proprotor-nacelle system is tilted. Therefore, an accurate model for the evaluation of unstable loads that occur during transients must be considered for a correct analysis of the complex dynamics characterizing these vehicles. However, multibody solvers, such as MBDyn, are typically equipped with simplified aerodynamics models based on Blade Element/Momentum Theory (BE/MT). This type of approach does not consider aerodynamic interference between the rotors and the actual geometry of the lifting surfaces, which can lead to misrepresentation of aerodynamic loads and loss of information related to periodic actions.

Accurate computation of unsteady aerodynamic loads for rotary wing vehicles was the main goal of several research groups that developed, in recent decades, different high fidelity Computational Fluid

Dynamics (CFD) codes employing the use of high-fidelity Navier-Stokes equations solvers. To name a few examples, ONERA, University of Glasgow, DLR, Airbus Helicopters Deutschland and Politecnico di Milano developed, respectively, elsA [33], HBM3 [27], FLOWer [97], and ROSITA [26], high-fidelity CFD codes based on the block-structured grid, finite volume, and Chimera approach for the simulation of rotating bodies. These tools were developed in Europe for studies of rotorcraft and tilt rotor applications [52, 56, 84]. Similarly, considerable research effort was dedicated in the USA to the numerical study of rotorcraft, particularly to tiltrotor aerodynamics, as shown for instance by the works of Meakin [129], Potsdam and Strawn [159] and Wissink et al. [201], where hover configurations were simulated using different implementations of the Navier-Stokes equations, and by the recent works of Lim Tran et al. [105, 190–192] that investigated the aerodynamic interaction between rotors and the wing of the XV-15 tiltrotor aircraft.

The combination of CSD codes with high-fidelity CFD solvers has been successfully investigated and implemented in the last two decades for aeroelastic simulation of rotorcraft applications [12, 49, 92, 176, 177, 206]. The coupled CSD/CFD numerical approach was successfully validated against experimental results, e.g., for the flutter calculations of a vertical tail model [65] and for the analysis of the structural loads of the rotor blade of a complete helicopter model tested in a transonic wind tunnel [126]. However, despite continuous advances in the field of high-performance computing, coupled simulations of CSD and time-accurate URANS simulations of complete rotorcraft configurations still require a robust computational effort, not suitable for the preliminary design stage of novel VTOL aircraft configurations such as tiltrotors, which requires a great number of simulations to reproduce the different flight conditions that characterize their mission.

In recent years, considerable effort was dedicated by several research groups to the development of mid-fidelity aerodynamic solvers based on the use of the vortex particle method (VPM) [46, 199] for wake modeling. This numerical methodology showed a fairly accurate representation of the aerodynamic interactions among several bodies typical of complex rotorcraft configurations and a limited computational time with respect to URANS CFD simulations. To cite few examples, Lu et al. [107] developed an optimization methodology for helicopter design based on a viscous VPM model combined with an unsteady panel hybrid method. Alvarez and Ning developed a VPM-based code [13] for the investigation of multirotor configurations. Tan et al. [187] used a vortex-based approach coupled with a viscous boundary model to study rotor-rotor interactional problems that occur during shipboard operations [187]. Recently, Politecnico di Milano developed a novel, flexible mid-fidelity computational tool,

called DUST (`https://public.gitlab.polimi.it/DAER/dust`) aimed at representing a fast and reliable asset for the simulation of the aerodynamics of complex rotorcraft configurations, such as the electrical Vertical Take Off and Landing (eVTOL) aircraft.

DUST is an open source code, released under MIT license,that integrates different aerodynamic models for solid bodies, as thick surface panels, thin vortex lattice elements, and lifting line elements. Moreover, a VPM method was implemented for wake modeling that provides a stable Lagrangian description of the free-vorticity flow field, which is suitable for numerical simulations of configurations characterized by strong aerodynamic interactions. DUST was thoroughly validated by comparison with results from high-fidelity CFD simulations and experimental data on complex rotorcraft configurations, such as eVTOLs [194] and tiltrotors [211]. Consequently, this novel open-source tool is reaching maturity for simulation of the aerodynamics of complex rotorcraft configurations, accurately reproducing the interaction between rotors and wings.

The combination of a multibody dynamics solver with a mid-fidelity aerodynamic tool aims at representing an ideal trade-off between speed of execution and accuracy of the solution, aimed at the preliminary design of novel rotary-wing aircraft configurations. A novel open-access aeroelastic tool was built by tightly coupling MBDyn to DUST. The coupling of the two codes is based on the partitioned multiphysics coupling library preCICE [30], a very useful and robust tool for managing communication between different solvers. After a brief description of the multibody dynamics and a detailed description of the theory behind the mid-fidelity aerodynamics solvers, updated in the latest release, the chapter describes the details of the methodology used for the coupling. An interesting novelty proposed by this tool is the ability to model the deflection of a control surface, which represents an essential aspect in the simulation of aircraft maneuvers. This chapter represents an extension of the work [171].

2.1 MBDYN

MBDyn is a general-purpose multibody solver, developed at the Department of Aerospace Science and Technology of Politecnico di Milano and distributed as free software.

MBDyn automatically writes and solves the equations of motion of a system of entities possessing degrees of freedom (nodes) connected through algebraic constraints and subjected to internal and external loads. Constraint equations are explicitly taken into account, follow-

ing a redundant coordinate set approach. Thus, the resulting system of Differential-Algebraic Equations (DAE) takes the form

$$\mathbf{M}(\mathbf{x}, t)\dot{\mathbf{x}} = \mathbf{p} \tag{2a}$$

$$\dot{\mathbf{p}} = \phi_{/\mathbf{x}}^{\mathsf{T}}\boldsymbol{\lambda} + \mathbf{f}_i(\dot{\mathbf{x}}, \mathbf{x}, t) + \mathbf{f}_e(\dot{\mathbf{x}}, \mathbf{x}, t) \tag{2b}$$

$$\mathbf{0} = \phi(\mathbf{x}) \tag{2c}$$

where $\mathbf{x}$ is the vector of the kinematic unknowns, $\mathbf{p}$ that of the momentum unknowns, $\boldsymbol{\lambda}$ that of the algebraic Lagrangian multipliers, $\mathbf{M}$ is a configuration- and time-dependent inertia matrix, $\mathbf{f}_i$, $\mathbf{f}_e$ are arbitrary internal and external forces, $\phi(\mathbf{x})$ is the vector of the (usually nonlinear) algebraic equations that express kinematic (holonomic) constraints, and $\phi_{/\mathbf{x}}$ is the Jacobian matrix of the constraints with respect to the kinematic unknowns.

Each node instantiates the corresponding balance equations 2b, while only nodes with associated inertia properties instantiate the definitions of the related momenta 2a. Additional states associated with scalar fields (namely, hydraulic pressure, temperature, electric potential) and thus the corresponding balance equations can be taken into account through dedicated sets of nodes.

The nodes that describe the kinematics of the structural problem can be connected either by elastic/viscoelastic internal forces (namely lumped structural components [113], beams [19, 68], shells [131], Component Mode Synthesis (CMS) elements [47]) expressed by $\mathbf{f}_i$, with a variety of viscoelastic constitutive laws, or by the kinematic constraints of Equation (2c). Simple aerodynamics can be modeled by built-in elements that exploit the 2D strip theory model by look-up tables of the aerodynamic coefficients and classical rotor inflow models based on momentum theory.

The DAE system can be integrated using several different A/L stable integration methods, among which is an original multistep method with tunable algorithmic dissipation, specifically designed for the class of problems usually solved with MBDyn.

Coupling with an external solver is performed through an external force element. This type of element communicates with the external solver by passing the kinematics of the model and receiving the corresponding loads. The coupling can be `loose` (that is, an explicit scheme) or `tight` (that is, an implicit scheme). For further details on the possible type of coupling with an external solver, the reader is referred to [115].

The external forces can be formulated directly in the absolute frame, or referred to a reference node. In the former case, operations are straightforward; in the latter, the kinematics are first reported in the reference frame of the reference node and then sent to the peer along with the motion of the reference node. The latter returns nodal forces and moments oriented according to the reference frame of the reference node.

STABILITY ANALYSIS The stability analysis of a nonlinear multibody model, formulated in an absolute reference frame, is critical, as the problem can be time-variant. In principle, since in the configuration under analysis the flow is essentially axial, this specific problem could be reduced to time-invariant through the so-called multiblade coordinates, with the possibility to determine a steady solution, when asymptotically stable, for its linearization and analysis using methods for linear, time-invariant (LTI) problems. However, due to the flexibility of the wing and the opportunity to trim the aircraft at non-zero angles of attack, at the trim point, the flow may not be perfectly axial; furthermore, aerodynamic interferences between the rotor and the wing may break the symmetry of the axial flow, introducing 1/rev periodicity in the motion of each blade and N_b/rev periodicity in the overall response of the system.

Thus, it is convenient to use methods specifically designed to address problems of this kind. A typical approach, based on Floquet's theory as introduced in the stability analysis of rotorcraft by Peters & Hohenemser in 1971 [152], evaluates the stability of linear, time-periodic (LTP) problems. It requires the integration of the state transition matrix of the problem over a revolution, which is called the monodromy matrix; this is often impractical when using multibody solvers, which formulate the problem as a large number of non-linear DAEs, as discussed in [21]. A generalization of Floquet's approach to non-linear, aperiodic problems, possibly subjected to random excitation, has been recently proposed in [185], based on Lyapunov's theory of characteristic exponents (LCE); however, the applicability of this method to systems of DAE is still debated [117], and hardly practical from a computational viewpoint for large systems. An interesting approach, based on proper orthogonal decomposition (POD) of snapshots of large sets of coordinates of the problem, taken during free response transients at each revolution of the rotor, has been proposed in [161] and has been used successfully with MBDyn. The equivalents of Floquet's characteristic coefficients are obtained from the eigenanalysis of a matrix that corresponds to Floquet's monodromy matrix, resulting in a least-squares fitting of the snapshots.

All the previously mentioned methods can, to a different extent, evaluate the contraction characteristic of the free response of the system (e.g., the real part of the eigenvalues, when defined, or the characteristic coefficients of Floquet's theory, or the characteristic exponents of Lyapunov's theory), but some are unable to determine its periodicity, when the free response is oscillatory ([21], [161], [185]), while others suffer from limitations in the interpretation of the imaginary part of the eigenvalues of the monodromy matrix ([152]), although attempts have been made to overcome them (see, for example, [153]).

2.2 DUST

DUST is an open-source software designed using object-oriented paradigms of the latest FORTRAN standards. Different classical potential-based elements are implemented in the code, such as lifting line [66, 156], surface panels [133], and linear vortex lattice elements [91] and non-linear vortex lattice element [39]. Even if incompressible potential flow is considered in the code assumptions, compressibility effects are considered in the computations. In particular, a Prandtl-Glauert correction [103] is considered for the calculation of steady aerodynamic loads with surface panels and vortex lattice elements. On the other hand, lifting line and non-linear vortex lattice elements intrinsically include compressibility and viscous effects introduced by Mach-depending tabulated sectional aerodynamic coefficients, obtained by experiments or by 2D RANS numerical simulations. Figure 19 summarizes the typical workflow needed to build and run a DUST model.

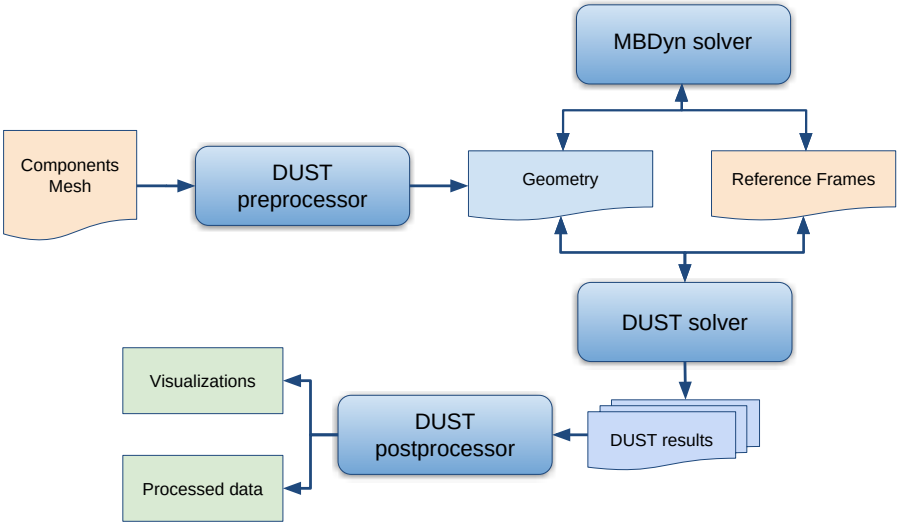


Figure 19: DUST user workflow

	Lifting Line (LL)	Vortex Lattice (VL)	Non-Linear VL (NL-VL)	Surface Panel (SP)
3D Unsteady	Yes	Yes	Yes	Yes
Thickness	Yes (Via database)	No	Yes (Via database)	Yes
Camber	Yes (Via database)	Yes	Yes	Yes
Viscous effects	Yes (Via database)	No	Yes (Via database)	No
Compressibility	Yes (Via database)	Yes (Via correction)	Yes (Via database)	Yes (Via correction)
Physical shape	No	No	No	Yes
Numerical scheme	Explicit	Implicit	Implicit	Implicit
Computational cost	Low	Low	Medium	High

Table 1: Summary of DUST available components and their characteristics

DUST simulations only require surface meshes, as the core of the mid-fidelity aerodynamic code is the vortex particle method (VPM)

[46, 199], a grid-free Lagrangian approach that is used to model the free vorticity of wakes and does not require a volume mesh of the flow surrounding the object of investigation. Vortex particles reproduction of the flow enabled one to obtain a robust representation of the interacting wakes caused by lifting surfaces and bodies, as typically occurs in rotary-wing vehicle applications. In particular, to reduce the computational cost associated with reproducing vortex particle interactions, a Cartesian fast multipole method (FMM) [106], which provides a hierarchical background decomposition of the domain into clusters of cells, is implemented. In the following paragraphs, we develop the theory behind DUST.

2.2.1 *Helmholtz's decomposition*

The mathematical formulation of the aerodynamic solver relies on a vorticity-velocity formulation of the aerodynamic problem, founded on the Helmholtz decomposition of the velocity field and a Lagrangian description of the vorticity field.

The Helmholtz decomposition of the velocity field states that the velocity field $\boldsymbol{u}(\boldsymbol{r}, t)$ can be written as the sum of an irrotational field, the potential velocity, $\boldsymbol{u}_\varphi = \boldsymbol{\nabla}\varphi$, and a soleinoidal field, the rotational velocity, $\boldsymbol{u}_\psi = \boldsymbol{\nabla} \times \boldsymbol{\psi}$,

$$\boldsymbol{u}(\boldsymbol{r}, t) = \boldsymbol{u}_\varphi(\boldsymbol{r}, t) + \boldsymbol{u}_\psi(\boldsymbol{r}, t) \tag{3}$$

Since the rotational velocity is a solenoidal vector field, the incompressibility constraint reduces to the Laplace equation for the kinetic potential $\boldsymbol{\varphi}(\boldsymbol{r}, t)$ in the fluid domain Ω,

$$0 = \boldsymbol{\nabla} \cdot \boldsymbol{u} = \boldsymbol{\nabla} \cdot \boldsymbol{u}_\varphi + \underbrace{\boldsymbol{\nabla} \cdot \boldsymbol{u}_\psi}_{\boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times\boldsymbol{\psi})\equiv 0} \quad \rightarrow \quad \Delta\boldsymbol{\varphi} = 0 \tag{4}$$

Since the potential velocity is an irrotational velocity field, the vorticity field $\boldsymbol{\omega}(\boldsymbol{r}, t)$ acts as a volume forcing of the Poisson equation for the vector potential $\boldsymbol{\psi}(\boldsymbol{r}, t)$,

$$\boldsymbol{\omega} = \underbrace{\boldsymbol{\nabla} \times \boldsymbol{u}_\Phi}_{\boldsymbol{\nabla}\times\boldsymbol{\nabla}\varphi\equiv 0} + \boldsymbol{\nabla} \times \boldsymbol{u}_\psi \quad \rightarrow \quad -\Delta\boldsymbol{\psi} = \boldsymbol{\omega} \tag{5}$$

given the vector identity $\Delta\boldsymbol{\psi} = \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \boldsymbol{\psi}) - \boldsymbol{\nabla} \times \boldsymbol{\nabla} \times \boldsymbol{\psi}$ and the gauge condition $\boldsymbol{\nabla} \cdot \boldsymbol{\psi} = 0$. Then, the Lagrangian description of the dynamical equation governing the vorticity field $\boldsymbol{\omega}(\boldsymbol{r}, t)$ of an incompressible flow reads:

$$\frac{D\boldsymbol{\omega}}{Dt} = (\boldsymbol{\omega} \cdot \boldsymbol{\nabla})\boldsymbol{u} + \nu\Delta\boldsymbol{\omega} \tag{6}$$

where ν is the flow kinematic viscosity and the material derivative of the vorticity field $D\boldsymbol{\omega}/Dt$ represents the material derivative of the vorticity associated with a material particle. Differential problems defined in the fluid domain Ω are manipulated using the Green function

method, to obtain a grid-free volume formulation of the aerodynamic problem.

2.2.2 *Potential velocity: Morino Formulation*

For completeness, in the following paragraphs the complete Morino's [132] formulation will be derived, starting from the Laplace equation written for the potential velocity and eventually highlighting the matrices that form the aerodynamic problem for the different elements.

2.2.2.1 *Mixed potential-velocity Morino formulation and panel method*

The idea of Morino's formulation is to solve the Laplace equation without making the hypothesis of simple solutions. The Green function method will then be applied to recast the differential problem governed by Laplace's equation for the perturbation kinetic potential, supplied with the appropriate boundary conditions, with an integral boundary problem.

The Green function method is applied to the potential aerodynamic problem, in order to recast a differential problem defined in the three-dimensional fluid domain as a boundary value problem, involving only the boundary of the potential flow domain Ω_φ. The idea is to solve Laplace's equation by some repetitive integrations.

Let us recall the Laplace problem and its Neumann- and Dirichlet boundary conditions:

$$\begin{cases} \Delta\boldsymbol{\varphi} = 0 & \text{in} \quad \Omega \\ \boldsymbol{\varphi} = b & \text{in} \quad \partial\Omega_d \\ \dfrac{\partial\boldsymbol{\varphi}}{\partial\hat{\boldsymbol{n}}} = a & \text{in} \quad \partial\Omega_n \end{cases} \tag{7}$$

The Laplace equation can be seen as a particular case of the Poisson equation: $\Delta\boldsymbol{\varphi} = f$.

Let $\mathcal{G}$ be the Green function; we can integrate in the Ω domain the Poisson equation pre-multiplied by the Green function:

$$\int_\Omega \mathcal{G}\Delta\boldsymbol{\varphi} = \int_\Omega \mathcal{G}f = \tag{8}$$

Integrating by parts the first term, we obtain:

$$\int_\Omega \mathcal{G}\nabla\cdot(\nabla\boldsymbol{\varphi}) = \int_\Omega \nabla\cdot(\mathcal{G}\nabla\boldsymbol{\varphi}) - \int_\Omega \nabla\mathcal{G}\cdot\nabla\boldsymbol{\varphi} \tag{9}$$

For the divergence theorem, the first right-hand side term can be rewritten as

$$\int_\Omega \nabla\cdot(\mathcal{G}\nabla\boldsymbol{\varphi}) = \int_{\partial\Omega} \mathcal{G}\cdot\frac{\partial\boldsymbol{\varphi}}{\partial\hat{\boldsymbol{n}}} = \int_{\partial\Omega} \mathcal{G}\cdot\hat{\boldsymbol{n}}\cdot\nabla\boldsymbol{\varphi} = \tag{10}$$

Then, we get rid of one derivative:

$$\int_\Omega \mathcal{G}\nabla\cdot(\nabla\boldsymbol{\varphi}) = \int_{\partial\Omega} \mathcal{G}\cdot\frac{\partial\boldsymbol{\varphi}}{\partial\hat{\boldsymbol{n}}} - \int_\Omega \nabla\mathcal{G}\cdot\nabla\boldsymbol{\varphi} \tag{11}$$

Using the identity:

$$\varphi\nabla\cdot\nabla\mathcal{G} = \nabla\cdot(\varphi\nabla\mathcal{G}) - \nabla\mathcal{G}\nabla\boldsymbol{\varphi} \tag{12}$$

We obtain the following.

$$\int_{\partial\Omega} \mathcal{G}\cdot\frac{\partial\boldsymbol{\varphi}}{\partial\hat{\boldsymbol{n}}} - \int_\Omega \nabla\mathcal{G}\cdot\nabla\boldsymbol{\varphi} = \int_{\partial\Omega} \mathcal{G}\cdot\frac{\partial\boldsymbol{\varphi}}{\partial\hat{\boldsymbol{n}}} + \int_\Omega \varphi\nabla\cdot\nabla\mathcal{G} - \int_\Omega \nabla\cdot(\varphi\nabla\mathcal{G}) \tag{13}$$

We can apply the divergence theorem to the last term obtaining the following.

$$\int_\Omega \nabla\cdot(\varphi\nabla\mathcal{G}) = \int_{\partial\Omega} \frac{\partial\mathcal{G}}{\partial\hat{\boldsymbol{n}}} \tag{14}$$

Then the original Poisson equation is transformed as:

$$\int_\Omega \mathcal{G}\nabla\cdot(\nabla\varphi) = \int_\Omega \varphi\Delta\mathcal{G} + \int_{\partial\Omega} \mathcal{G}\frac{\partial\boldsymbol{\varphi}}{\partial\hat{\boldsymbol{n}}} - \int_{\partial\Omega} \varphi\frac{\partial G}{\partial\hat{\boldsymbol{n}}} \tag{15}$$

$$\int_\Omega \varphi\Delta\mathcal{G} = \int_\Omega \mathcal{G}f + \int_{\partial\Omega} \mathcal{G}\frac{\partial\boldsymbol{\varphi}}{\partial\hat{\boldsymbol{n}}} - \int_{\partial\Omega} \varphi\frac{\partial G}{\partial\hat{\boldsymbol{n}}} \tag{16}$$

As a result, we find that in the domain Ω the potential derivative φ is no longer present; instead, its first derivative $\partial\varphi/\partial\hat{\boldsymbol{n}}$ is evaluated along the boundary $\partial\Omega$.

The Green function method exploits the property of the Dirac delta that its integral is equal to the unit:

$$\int_\Omega \delta(\boldsymbol{r}-\boldsymbol{r_0}) = 1 \tag{17}$$

Then, if we compute applying the Dirac delta to a generic function f, we obtain:

$$\int_\Omega f(\boldsymbol{r})\delta(\boldsymbol{r}-\boldsymbol{r_0}) = f(\boldsymbol{r_0}) \tag{18}$$

We can now define Green's function as $\mathcal{G} = \mathcal{G}(\boldsymbol{r}, \boldsymbol{r}_0)$, a solution of the following problem:

$$\begin{cases} \Delta\mathcal{G} = \delta(\boldsymbol{r}-\boldsymbol{r_0}) & \text{in} \quad \Omega \\ \mathcal{G} = 0 & \text{in} \quad \partial\Omega_d \\ \dfrac{\partial\mathcal{G}}{\partial n} = 0 & \text{in} \quad \partial\Omega_n \end{cases} \tag{19}$$

We can rewrite equation 16, knowing that the boundary is the sum of the Neumann and Dirichlet boundary $\partial\Omega = \partial\Omega_n + \partial\Omega_n$, as:

$$\int_\Omega \varphi \nabla \mathcal{G} = \int_\Omega \varphi \delta(\boldsymbol{r} - \boldsymbol{r}_0) = \varphi(\boldsymbol{r}_0) = \tag{20}$$

$$= \int_\Omega \mathcal{G}(\boldsymbol{r}, \boldsymbol{r}_0) f + \tag{21}$$

$$+ \int_{\partial\Omega_d} \cancel{\mathcal{G}} \frac{\partial \boldsymbol{\varphi}}{\partial \hat{\boldsymbol{n}}} + \int_{\partial\Omega_n} \mathcal{G} \frac{\partial \boldsymbol{\varphi}}{\partial \hat{\boldsymbol{n}}} - \tag{22}$$

$$- \int_{\partial\Omega_d} \varphi \frac{\partial \cancel{G}}{\partial \hat{\boldsymbol{n}}} - \int_{\partial\Omega_n} \varphi \frac{\partial G}{\partial \hat{\boldsymbol{n}}} \tag{23}$$

Let us solve the Laplace equation for the Green function defined in a $\mathbb{R}^\infty$ space:

$$\int_\Omega \Delta \mathcal{G}_\infty(\boldsymbol{r}, \boldsymbol{r}_0) = \int_\Omega \delta(\boldsymbol{r} - \boldsymbol{r}_0) = 1 \quad \text{if} \quad r_0 \in \Omega \tag{24}$$

But we can rewrite the integral using the divergence theorem:

$$\int_\Omega \Delta \mathcal{G}_\infty = \int_\Omega \nabla \left(\nabla \cdot \mathcal{G}_\infty \right) = \int_{\partial\Omega} \hat{\boldsymbol{n}} \cdot \nabla \mathcal{G} \cdot = \int_{\partial\Omega} \frac{\partial \mathcal{G}}{\partial \hat{\boldsymbol{n}}} \tag{25}$$

In an isotropic space, the solution does not depend on the orientation of the space, so Green's function depends only on the absolute value r of the vector $\boldsymbol{r} - \boldsymbol{r_0}$. We can solve the integral in a spherical domain:

$$\int_{\partial\Omega} \frac{\mathcal{G}(\boldsymbol{R})}{\partial n} = \int_0^{2\pi} \int_0^{\pi} \frac{\partial \mathcal{G}(\boldsymbol{R})}{\partial n} R^2 \sin\gamma \, d\gamma \, d\theta = \tag{26}$$

$$= 2\pi \frac{\partial \mathcal{G}(\boldsymbol{R})}{\partial n} R^2 \int_0^{\pi} \sin\gamma \, d\gamma = \tag{27}$$

$$= 2\pi \frac{\partial \mathcal{G}(\boldsymbol{R})}{\partial n} R^2 \cdot 2 = \tag{28}$$

$$= 4\pi \frac{\partial \mathcal{G}(\boldsymbol{R})}{\partial n} R^2 = 1 \tag{29}$$

Then $\mathcal{G}(\boldsymbol{R})$ is equal to:

$$\mathcal{G}(\boldsymbol{R}) = -\frac{1}{4\pi R} \quad \rightarrow \mathcal{G}(\boldsymbol{r}, \boldsymbol{r_o}) = -\frac{1}{4\pi |\boldsymbol{r} - \boldsymbol{r_o}|} \tag{30}$$

where the integral is performed on the variable $\boldsymbol{r_o}$ and $E(\boldsymbol{r})$ is the indicator function of the domain Ω,

$$E(\boldsymbol{r}) = \begin{cases} 1 & \boldsymbol{r} \in \\ \dfrac{\alpha}{4\pi} & \boldsymbol{r} \in \partial\Omega \\ 0 & \boldsymbol{r} \notin \bar{V} \end{cases} \tag{31}$$

where α is the solid angle of the domain Ω, seen from the point $\boldsymbol{r}$ on its boundary $\partial\Omega$. As an example, if the boundary of the domain is smooth in $\boldsymbol{r}$, the solid angle Ω is equal to 2π, and thus $E(\boldsymbol{r}) = 1/2$.

From a direct comparison of the Green function (31) and the kinetic potential induced by a point source (245), it is possible to identify the Green function as the opposite of the kinetic potential induced in $\boldsymbol{r}$ by a unitary point source located in $\boldsymbol{r_o}$ (and viceversa):

$$\varphi^s(\boldsymbol{r}, \boldsymbol{r_o}) = -\mathcal{G}(\boldsymbol{r}, \boldsymbol{r_o}) \tag{32}$$

The gradient of the Green function with respect to the first argument $\boldsymbol{r}$ reads[1]

$$\boldsymbol{\nabla}\mathcal{G}(\boldsymbol{r}, \boldsymbol{r_o}) = -\frac{1}{4\pi}\frac{\boldsymbol{r} - \boldsymbol{r_o}}{|\boldsymbol{r} - \boldsymbol{r_o}|^3} \tag{34}$$

and its scalar product with a unit vector $\hat{\boldsymbol{v}}$ is equal to the kinetic potential (247) induced in $\boldsymbol{r}$ by the point doublet with axis $\hat{\boldsymbol{v}}$ located in $\boldsymbol{r_o}$,

$$\varphi^d(\boldsymbol{r}, \boldsymbol{r_o}) = \hat{\boldsymbol{v}} \cdot \boldsymbol{\nabla}\mathcal{G}(\boldsymbol{r}, \boldsymbol{r_o}) \tag{35}$$

2.2.2.2 *Boundary conditions for the aerodynamic configuration*

The boundary $\partial\Omega$ of the potential flow domain is the union of the surfaces of solid bodies ($\partial\Omega_s$) and both sides of the zero-thickness surfaces involved in vortex lattice ($\partial\Omega_v$), lifting line ($\partial\Omega_l$), free panel wake models ($\partial\Omega_w$), and the far-field control surface ($\partial\Omega_\infty$):

$$\partial\Omega = \partial\Omega_s \cup \partial\Omega_v \cup \partial\Omega_l \cup \partial\Omega_w \cup \partial\Omega_\infty \tag{36}$$

where $S_i = S_i^+ \cup S_i^-$, for $i \in \{v, l, w\}$.

On the body surface $\partial\Omega_s$ non-penetration boundary conditions are applied:

$$\hat{\boldsymbol{n}} \cdot (\boldsymbol{u}_b - \boldsymbol{U}_\infty - \boldsymbol{u}_\psi)(\boldsymbol{r}_0 t) = \hat{\boldsymbol{n}} \cdot \boldsymbol{u}_\varphi(\boldsymbol{r}_0 t) = \hat{\boldsymbol{n}} \cdot \boldsymbol{\nabla}_{\boldsymbol{o}}\varphi(\boldsymbol{r}_0 t) =: \sigma(\boldsymbol{r}_0 t) \tag{37}$$

At the far-field $\partial\Omega_\infty$ the following condition on the velocity is applied:

$$\boldsymbol{U} \cdot \hat{\boldsymbol{n}} = \boldsymbol{U}_\infty \cdot \hat{\boldsymbol{n}} \tag{38}$$

The boundary conditions across the wake can be derived by applying the mass and momentum conservation over a control cube, as described in Fig. 20.

We can make the hypothesis of a thin wake contained in the $x - y$ plane, where $\hat{\boldsymbol{n}}$ is the normal of the wake, $\hat{\boldsymbol{n}}^+$ and $\hat{\boldsymbol{n}}^-$ are, respectively, the normal of the upper and lower surfaces of the wake, such that $\hat{\boldsymbol{n}} = \hat{\boldsymbol{n}}^+ = -\hat{\boldsymbol{n}}^-$, while $\boldsymbol{U}_s$ is the velocity of the wake, $\boldsymbol{U}^+$ and $\boldsymbol{U}^-$

1 The gradient of the Green function with respect to the second argument is the opposite,

$$\boldsymbol{\nabla}\mathcal{G}(\boldsymbol{r}, \boldsymbol{r_o}) = -\boldsymbol{\nabla}_{\boldsymbol{o}}\mathcal{G}(\boldsymbol{r}, \boldsymbol{r_o}) \tag{33}$$

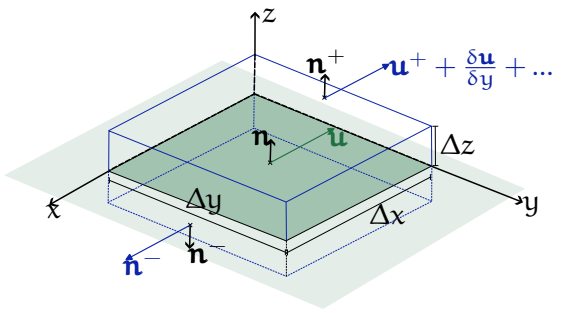


Figure 20: Control cube across the wake

are, respectively, the velocity perturbation of the upper and lower surface of the wake. We can write the mass balance:

$$\rho(\boldsymbol{U}^+ - \boldsymbol{U}_s) \cdot \hat{\boldsymbol{n}}^+ dA + \rho(\boldsymbol{U}^- - \boldsymbol{U}_s) \cdot \hat{\boldsymbol{n}}^- dA = 0 \tag{39}$$

$$\cancel{\rho}(\boldsymbol{U}^+ - \boldsymbol{U}_s) \cdot \hat{\boldsymbol{n}}^+ dA + \cancel{\rho}(\boldsymbol{U}^- - \boldsymbol{U}_s) \cdot \hat{\boldsymbol{n}}^- dA = 0 \tag{40}$$

$$(\boldsymbol{U}^+ - \cancel{\boldsymbol{U}_s}) \cdot \hat{\boldsymbol{n}}^+ dA + (\boldsymbol{U}^- - \cancel{\boldsymbol{U}_s}) \cdot \hat{\boldsymbol{n}}^- dA = 0 \tag{41}$$

$$\boldsymbol{U}^+ \cdot \hat{\boldsymbol{n}} - \boldsymbol{U}^- \hat{\boldsymbol{n}} = 0 \tag{42}$$

This means that the normal component of the velocity does not change across the wake.

The momentum conservation can be written, where p^+ and p^- are the pressure acting on the upper and lower surface.

$$\rho \boldsymbol{U}^+ (\boldsymbol{U}^+ - \boldsymbol{U}_s) \cdot \hat{\boldsymbol{n}}^+ + \rho \boldsymbol{U}^- (\boldsymbol{U}^- - \boldsymbol{U}_s) \cdot \hat{\boldsymbol{n}}^- = -p^+ \hat{\boldsymbol{n}}^+ - p^+ \hat{\boldsymbol{n}}^- \tag{43}$$

$$\tag{44}$$

Since this is a vectorial equation, we can treat the normal and tangential components separately:

$$\rho \boldsymbol{U}^+ \cdot \hat{\boldsymbol{n}}^+ (\boldsymbol{U}^+ - \boldsymbol{U}_s) \cdot \hat{\boldsymbol{n}} - \rho \boldsymbol{U}^- \cdot \hat{\boldsymbol{n}}^+ (\boldsymbol{U}^- - \boldsymbol{U}_s) \cdot \hat{\boldsymbol{n}} = -p^+ \hat{\boldsymbol{n}}^+ \cdot \hat{\boldsymbol{n}}_s - p^- \hat{\boldsymbol{n}}^- \cdot \hat{\boldsymbol{n}}$$

$$\tag{45}$$

$$\rho \boldsymbol{U}^+ \cdot \hat{\boldsymbol{n}}^+ (\boldsymbol{U}^+ - \boldsymbol{U}_s) - \rho \boldsymbol{U}^- \cdot \hat{\boldsymbol{n}}^+ (\boldsymbol{U}^- - \boldsymbol{U}_s) = -p^+ + p^- \tag{46}$$

$$\rho \boldsymbol{U}^+ \cdot \hat{\boldsymbol{n}}^+ \underbrace{\left[(\boldsymbol{U}^+ - \boldsymbol{U}_s) \hat{\boldsymbol{n}} - (\boldsymbol{U}^- - \boldsymbol{U}_s) \hat{\boldsymbol{n}} \right]}_{0, \text{for mass conservation}} = -p^+ + p^- = 0 \;\rightarrow\; p^+ = p^- \tag{47}$$

For the tangential part, we have the following:

$$\rho \boldsymbol{U}^+ \cdot \hat{\boldsymbol{\tau}} (\boldsymbol{U}^+ - \boldsymbol{U}_s) \cdot \hat{\boldsymbol{n}}_s - \rho \boldsymbol{U}^- \cdot \hat{\boldsymbol{\tau}} (\boldsymbol{U}^- - \boldsymbol{U}_s) \cdot \hat{\boldsymbol{n}}_s = 0 \tag{48}$$

The wake represents a discontinuity surface in the tangential component of the velocity, which means $\boldsymbol{U}^+ \cdot \hat{\boldsymbol{\tau}} \neq \boldsymbol{U}^- \cdot \hat{\boldsymbol{\tau}}$, so the only way to get the momentum conservation is:

$$(\boldsymbol{U}^+ - \boldsymbol{U}_s) \cdot \hat{\boldsymbol{n}}_s = (\boldsymbol{U}^- - \boldsymbol{U}_s) \cdot \hat{\boldsymbol{n}}_s = 0 \quad \rightarrow \quad \boldsymbol{U}^{+/-} \cdot \hat{\boldsymbol{n}}_s = \boldsymbol{U}_s \cdot \hat{\boldsymbol{n}}_s \tag{49}$$

Finally, we evaluate the jump of the potential across the wake through the unsteady Bernoulli theorem.

$$\begin{cases} \dfrac{\partial \varphi^+}{\partial t} + \dfrac{p^+}{\rho} + \dfrac{1}{2}\|\boldsymbol{U}^+\|^2 = C \\ \dfrac{\partial \varphi^-}{\partial t} + \dfrac{p^-}{\rho} + \dfrac{1}{2}\|\boldsymbol{U}^-\|^2 = C \end{cases} \tag{50}$$

Subtracting the first and second equations, we obtain the following.

$$\frac{\partial \varphi^+}{\partial t} - \frac{\partial \varphi^-}{\partial t} + \cancel{\frac{p^+ - p^-}{\rho}} + \frac{1}{2}\left(\|\boldsymbol{U}^+\|^2 - \|\boldsymbol{U}^-\|^2\right) = 0 \tag{51}$$

$$\frac{\partial \Delta\varphi}{\partial t} + \frac{1}{2}\left(\|\boldsymbol{U}^+\|^2 - \|\boldsymbol{U}^-\|^2\right) = 0 \tag{52}$$

$$\frac{\partial \Delta\varphi}{\partial t} + \underbrace{\frac{(\|\boldsymbol{U}^+\| + \|\boldsymbol{U}\|)}{2}}_{U_m}\left(\|\boldsymbol{U}^+\| - \|\boldsymbol{U}\|\right) = 0 \tag{53}$$

We know that $\boldsymbol{U}^+ = \boldsymbol{U}_\infty + \boldsymbol{u}^+$ and $\boldsymbol{U}^- = \boldsymbol{U}_\infty - \boldsymbol{u}^-$, where u is the velocity perturbation; that means $\boldsymbol{U}^+ - \boldsymbol{U}^- = \boldsymbol{u}^+ - \boldsymbol{u}^-$. We also know that the potential is defined starting from the velocity perturbation: $\boldsymbol{u} = \nabla\boldsymbol{\varphi}$, that is,

$$\boldsymbol{u}^+ - \boldsymbol{u}^- = \nabla\boldsymbol{\varphi}^+ - \nabla\boldsymbol{\varphi}^- = \nabla(\Delta\boldsymbol{\varphi}) \tag{54}$$

So, Bernoulli's theorem is written as the material derivative with respect to the mean velocity of the wake:

$$\frac{\partial \Delta\varphi}{\partial t} + (\boldsymbol{U}_m \cdot \boldsymbol{\nabla})\,\Delta\varphi = 0 \quad \rightarrow \quad \frac{D_{U_m}\Delta\varphi}{Dt} = 0 \tag{55}$$

This means that along the wake the potential jump remains constant on a particle that is moving at a mean velocity $\boldsymbol{U}_m$, in practice, the potential jump is transported with the average speed. Moreover, the particle must remain in the wake, so that the normal component of the mean velocity is equal to the normal component of the velocity: $\boldsymbol{U}_m \cdot \hat{\boldsymbol{n}} = \boldsymbol{U}_s \cdot \hat{\boldsymbol{n}}$.

In conclusion, the following conditions apply to the wake:

$$\begin{cases} \boldsymbol{U}^+ \cdot \hat{\boldsymbol{n}} = \boldsymbol{U}^- \cdot \hat{\boldsymbol{n}} \\ p^+ = p- \\ \boldsymbol{U}^{+/-} \cdot \hat{\boldsymbol{n}} = \boldsymbol{U}_s \cdot \hat{\boldsymbol{n}} \\ \dfrac{D_{U_m}\Delta\varphi}{Dt} = 0 \end{cases} \tag{56}$$

2.2.2.3 *Solution of the aerodynamic problem*

We can separate the integral into subdomains:

$$E\varphi(\boldsymbol{r}_0) = \int_{\partial\Omega}\left(\varphi\frac{\partial\mathcal{G}}{\partial\hat{\boldsymbol{n}}} - \mathcal{G}\frac{\partial\varphi}{\partial\hat{\boldsymbol{n}}}\right) = \tag{57}$$

$$= \int_{\partial\Omega_b}\left(\varphi\frac{\partial\mathcal{G}}{\partial\hat{\boldsymbol{n}}} - \mathcal{G}\frac{\partial\varphi}{\partial\hat{\boldsymbol{n}}}\right) + \int_{\partial\Omega_\infty}\left(\varphi\frac{\partial\mathcal{G}}{\partial\hat{\boldsymbol{n}}} - \mathcal{G}\frac{\partial\varphi}{\partial\hat{\boldsymbol{n}}}\right) + \tag{58}$$

$$+ \int_{\partial\Omega_{w^+}}\left(\varphi\frac{\partial\mathcal{G}}{\partial\hat{\boldsymbol{n}}} - \mathcal{G}\frac{\partial\varphi}{\partial\hat{\boldsymbol{n}}}\right) + \int_{\partial\Omega_{w^-}}\left(\varphi\frac{\partial\mathcal{G}}{\partial\hat{\boldsymbol{n}}} - \mathcal{G}\frac{\partial\varphi}{\partial\hat{\boldsymbol{n}}}\right) \tag{59}$$

where $\Omega_b = \partial\Omega_s \cup \partial\Omega_v \cup \partial\Omega_l$.

Let considered the far-field integral: we can represent the domain as a sphere of infinite radius:

$$\int_{\partial\Omega_\infty}\varphi\frac{\partial\mathcal{G}}{\partial\hat{\boldsymbol{n}}} = \lim_{R\to\infty}\int_0^{2\pi}\int_0^{\pi}\varphi\frac{\partial\mathcal{G}}{\partial R}R^2\sin\gamma\,d\gamma\,d\theta = \tag{60}$$

$$= \lim_{R\to\infty}\int_0^{2\pi}\int_0^{\pi}\frac{\varphi}{4\pi\cancel{R^2}}\cancel{R^2}\sin\gamma\,d\gamma\,d\theta = \tag{61}$$

$$= \lim_{R\to\infty}\int_0^{2\pi}\int_0^{\pi}\sum_{n=1}^{\infty}\frac{f_n(\gamma,\theta)}{R^n}\frac{1}{4\pi}\sin\gamma\,d\gamma\,d\theta \approx \tag{62}$$

$$\approx \lim_{R\to\infty}\int_0^{2\pi}\int_0^{\pi}\frac{f_n(\gamma,\theta)}{R}\frac{1}{4\pi}\sin\gamma\,d\gamma\,d\theta = 0 \tag{63}$$

and the other term is:

$$\int_{\partial\Omega_\infty}\mathcal{G}\frac{\partial\varphi}{\partial\hat{\boldsymbol{n}}} = \lim_{R\to\infty}\int_0^{2\pi}\int_0^{\pi}-\frac{1}{4\pi R}\left(-\frac{f(\gamma,\theta)}{\cancel{R^2}}\right)\cancel{R^2}\sin\gamma\,d\gamma\,d\theta = 0 \tag{64}$$

Consider the wake term. We know $\boldsymbol{u}^+\cdot\hat{\boldsymbol{n}} = \boldsymbol{u}^-\cdot\hat{\boldsymbol{n}}$ from the mass balance equation. We can expand the two terms of the condition as follows:

$$\begin{cases}\boldsymbol{u}^+\cdot\hat{\boldsymbol{n}}_s = \hat{\boldsymbol{n}}^+\cdot\nabla\varphi^+ = & \dfrac{\partial\varphi^+}{\partial\hat{\boldsymbol{n}}} \\ \boldsymbol{u}^-\cdot\hat{\boldsymbol{n}}_s = -\hat{\boldsymbol{n}}^+\cdot\nabla\varphi^+ = & \dfrac{\partial\varphi^-}{\partial\hat{\boldsymbol{n}}}\end{cases} \quad\rightarrow\quad \frac{\partial\varphi^+}{\partial\hat{\boldsymbol{n}}} = -\frac{\partial\varphi^-}{\partial\hat{\boldsymbol{n}}} \tag{65}$$

We can then simplify the wake term:

$$\int_{\partial\Omega_{w^+}}\left(\varphi\frac{\partial\mathcal{G}}{\partial\hat{\boldsymbol{n}}} - \mathcal{G}\frac{\partial\varphi}{\partial\hat{\boldsymbol{n}}}\right) + \int_{\partial\Omega_{w^-}}\left(\varphi\frac{\partial\mathcal{G}}{\partial\hat{\boldsymbol{n}}} - \mathcal{G}\frac{\partial\varphi}{\partial\hat{\boldsymbol{n}}}\right) \tag{66}$$

$$\int_{\partial\Omega_{w^+}}\varphi\frac{\partial\mathcal{G}}{\partial\hat{\boldsymbol{n}}} - \int_{\partial\Omega_{w^-}}\varphi\frac{\partial\mathcal{G}}{\partial\hat{\boldsymbol{n}}} - \int_{\partial\Omega_{w^+}}\cancel{\mathcal{G}\frac{\partial\varphi}{\partial\hat{\boldsymbol{n}}}} + \int_{\partial\Omega_{w^-}}\cancel{\mathcal{G}\frac{\partial\varphi}{\partial\hat{\boldsymbol{n}}}} \tag{67}$$

$$\int_{\partial\Omega_w}\varphi\frac{\partial\mathcal{G}}{\partial\hat{\boldsymbol{n}}} - \varphi^-\frac{\partial\mathcal{G}}{\partial\hat{\boldsymbol{n}}} = \int_{\partial\Omega_w}\Delta\varphi\frac{\partial\mathcal{G}}{\partial\hat{\boldsymbol{n}}} \tag{68}$$

In the end, the problem is transformed into an integral equation plus the non-penetration boundary condition, in which the unknown is the potential on $\partial\Omega_s$:

$$\begin{cases} E\varphi(\boldsymbol{r}_0) = \int_{\partial\Omega_s}\left(\varphi\dfrac{\partial\mathcal{G}}{\partial\hat{\boldsymbol{n}}} - \mathcal{G}\dfrac{\partial\varphi}{\partial\hat{\boldsymbol{n}}}\right) + \int_{\partial\Omega_w}\Delta\varphi\dfrac{\partial\mathcal{G}}{\partial\hat{\boldsymbol{n}}} \\ \dfrac{\partial\varphi}{\partial\hat{\boldsymbol{n}}} = \sigma(r_0, t) \end{cases} \tag{69}$$

KUTTA'S CONDITION Kutta condition represents the vorticity balance at the trailing edge and determines the intensity of the wake panel shed into the domain. Kutta condition is naturally satisfied by a vortex lattice and lifting line description of a lifting surface, whereas a row of implicit wake panels is introduced at the trailing edge of thick bodies modeled with surface panels, in order to satisfy the potential jump equivalent to the Kutta condition.

The intensity $\boldsymbol{\mu}_{w_{TE}}$ of the trailing edge doublets and the intensity $\tilde{\boldsymbol{\mu}}_w$ of the wake doublets released at the previous time steps are identified in the wake doublet vector, $\boldsymbol{\mu}_w = (\boldsymbol{\mu}_{w_{TE}}, \boldsymbol{\mu}_{\tilde{w}})$.

Kutta condition relates the intensity of the i_{TE}^{th} element of the first row of the wake panels with the doublet intensity of the surface panels at the trailing edge,

$$\mu_{w,\,i_{TE}} = \mu_{s^u,\,i_{TE}} - \mu_{s^l,\,i_{TE}} \tag{70}$$

In this way, we can incorporate the integral on the wake domain into the one on the body surface, since now the intensity of the wake is implicitly linked to the one of the surface panel.

$$E\varphi(\boldsymbol{r}_0) = \int_{\partial\Omega}\left(\varphi\frac{\partial\mathcal{G}}{\partial\hat{\boldsymbol{n}}} - \mathcal{G}\frac{\partial\varphi}{\partial\hat{\boldsymbol{n}}}\right) \tag{71}$$

2.2.2.4 *Solution of the discretized problem*

The problem is solved by the collocation method: the surface of the solid bodies $\partial\Omega_b$ and the potential part of the wake $\partial\Omega_w$ are subdivided into panels and the velocity is evaluated in the center of the panel for each element. We can divide the domain of the integral 71 into two subdomains: one for the surface panels $\partial\Omega_s$ and the remaining will include the flat elements (vortex lattice, lifting line and wake) $\partial\Omega\ \partial\Omega_s$:

$$E\varphi(\boldsymbol{r}_0) = \int_{\partial\Omega_s}\left(\varphi\frac{\partial\mathcal{G}}{\partial\hat{\boldsymbol{n}}} - \mathcal{G}\frac{\partial\varphi}{\partial\hat{\boldsymbol{n}}}\right) + \int_{\partial\Omega\ \partial\Omega_s}\left(\varphi\frac{\partial\mathcal{G}}{\partial\hat{\boldsymbol{n}}} - \mathcal{G}\cancel{\frac{\partial\varphi}{\partial\hat{\boldsymbol{n}}}}\right) \tag{72}$$

The last term cancels out for the non-penetration boundary condition on the flat body. Using a uniform-intensity panel discretisation and defining the intensity of the doublets, $\mu_{i_s} = -\varphi_{i_s}$, and the sources,

$\sigma_{i_s} = \hat{\boldsymbol{n}}_{i_s} \cdot (\boldsymbol{u}_b - \boldsymbol{U}_\infty - \boldsymbol{u}_\psi)_{i_s}$, the discrete counterpart of the problem reads

$$\sum_{k_s=1}^{N_s} A_{i_s k_s} \mu_{k_s} + \sum_{k_v=1}^{N_v} A_{i_s k_v} \mu_{k_v} + \sum_{k_l=1}^{N_l} A_{i_s k_l} \mu_{k_l} + \tag{73}$$

$$\sum_{k_s=1}^{N_s} B_{i_s k_s} \sigma_{k_s} + \sum_{k_w=1}^{N_w} A_{i_s k_w} \mu_{k_w} = 0 \quad \forall i_s = 1 : N_s \tag{74}$$

where A_{ik} and B_{ik} represents the induced potential of the k^{th} doublet and the source panel at the point $\boldsymbol{r}_i$,

$$\begin{aligned} A_{ik} &= -\int_{S_k} \frac{\partial \mathcal{G}(\boldsymbol{r}_o, \boldsymbol{r}_i)}{\partial \hat{\boldsymbol{n}}_{k_s}} \\ B_{ik} &= -\int_{S_k} \mathcal{G}(\boldsymbol{r}_o, \boldsymbol{r}_i) \end{aligned} \tag{75}$$

except for the self-induction potential Aerodynamic Influence Coefficient (AIC) of a doublet,

$$A_{i_s i_s} = E(\boldsymbol{r}_{i_s}) - \int_{S_{i_s}} \frac{\partial \mathcal{G}(\boldsymbol{r}_o, \boldsymbol{r}_{i_s})}{\partial \hat{\boldsymbol{n}}_{i_s}} = \frac{1}{2} \tag{76}$$

At each time step, the intensity σ_{k_s} of the sources is provided by the non-penetration boundary condition as a function of the velocity of the body surface $\boldsymbol{u}_b$ and the rotational velocity $\boldsymbol{u}_\psi$ at the collocation points $\boldsymbol{r}_0$, while the intensity μ_{k_w} of the wake doublet elements is known from previous time steps. The boundary element problem (73) for the solid bodies modeled with the surface panels is a linear system of N_s -equation in the $N = N_s + N_v + N_l$ unknown doublet intensities μ_k, associated with all surface panels, the vortex lattice, and the lifting line elements. This linear system is underdetermined if vortex lattice or lifting line elements are used to build the model, along with surface panels.

LOADS EVALUATION Using the Eulerian description of the system, the unsteady Bernoulli theorem for an irrotational flow of an incompressible fluid with constant density ρ, without any volume force states:

$$\frac{\partial \phi}{\partial t} + \frac{|\boldsymbol{U}|^2}{2} + \frac{P}{\rho} = C(t) \tag{77}$$

where ϕ is the kinetic potential, $\boldsymbol{U} = \boldsymbol{\nabla}\phi$ is the velocity field, and P the pressure field. The kinetic potential and velocity fields can be decomposed as the sum of the free-stream and perturbation contributions as

$$\phi = \phi_\infty + \varphi \quad \rightarrow \quad \boldsymbol{\nabla}\phi = \boldsymbol{\nabla}\phi_\infty + \boldsymbol{\nabla}\varphi \tag{78}$$

so that $\phi_\infty = \boldsymbol{U}_\infty \cdot \boldsymbol{r}$, being $\boldsymbol{r}$ the space variable.

In a panel method, it may be useful to write equations with an ALE (Arbitrary Lagrangian Eulerian) description to compute loads on a moving body by means of the Bernoulli equation. This approach has the advantage of using the spatial derivatives computed with the surface grid of the body and the temporal derivatives for given material points of the solid surface, as was done to compute the wake boundary condition. The time derivative of a physical quantity for a material point r_b (of the solid component) moving with velocity $u_b = \frac{\partial x}{\partial t}\Big|_{r_b}$ is expressed as follows.

$$\frac{\partial}{\partial t}\Big|_{r_b} = \frac{\partial}{\partial t} + u_b \cdot \nabla \tag{79}$$

Thus, the ALE description of the Bernoulli theorem (77) reads as follows:

$$\frac{\partial \phi}{\partial t}\Big|_{r_b} - u_b \cdot \nabla \phi + \frac{|U|^2}{2} + \frac{P}{\rho} = C(t) \tag{80}$$

By equating the decomposition (78) and the free-stream condition taken as reference, the Bernoulli equation (80) is simply as follows:

$$\frac{\partial(\cancel{\phi_\infty} + \varphi)}{\partial t}\Big|_{r_b} - u_b \cdot \nabla(\cancel{\phi_\infty} + \varphi) + \frac{|U|^2}{2} + \frac{P}{\rho} = \tag{81}$$

$$= \frac{\partial \cancel{\phi_\infty}}{\partial t}\Big|_{r_b} - u_b \cdot \nabla \cancel{\phi_\infty} + \frac{|U_\infty|^2}{2} + \frac{P_\infty}{\rho} \tag{82}$$

Thus, the pressure for a single panel can be calculated as follows.

$$P_i = P_\infty + \rho\frac{|U_\infty|^2}{2} - \left(\rho\frac{\partial \varphi_i}{\partial t}\Big|_{r_b} - u_b \cdot \nabla\varphi + \rho\frac{|U|^2}{2}\right) \tag{83}$$

$$P_i^{P-G} = \frac{P_i}{\sqrt{1 - Ma^2}} \tag{84}$$

The force for a single panel will then be:

$$F_i = -\left(P_\infty - P_i^{P-G}\right) dA_i \cdot \hat{n}_i \tag{85}$$

2.2.3 *Vortex Lattice*

Thin lifting bodies can be modeled as a zero-thickness two-dimensional vortex sheet $\partial\Omega_\nu$, where the non-penetration boundary condition must hold:

$$\hat{n} \cdot u_\varphi = \hat{n} \cdot (u_b - U_\infty - u_\psi) =: \sigma \quad \text{on} \quad \partial\Omega_\nu \tag{86}$$

The vortex-lattice method provides aerodynamic elements for the discrete representation of the mean surface of thin lifting bodies, modeled as a sheet of vortex rings of intensity Γ_{i_ν}, equivalent to a piecewise uniform surface doublet distribution with the same intensity

$\mu_{i_v} = \Gamma_{i_v}$, $i_v = 1 : N_v$. The boundary condition at the collocation point $\boldsymbol{r}_{i_v}$ reads:

$$\sum_{k_s=1}^{N_s} C_{i_v k_s} \mu_{k_s} + \sum_{k_v=1}^{N_v} C_{i_v k_v} \mu_{k_v} + \sum_{k_l=1}^{N_l} C_{i_v k_l} \mu_{k_l} + \tag{87}$$

$$+ \sum_{k_s=1}^{N_s} D_{i_v k_s} \sigma_{k_s} + \sum_{k_w=1}^{N_w} C_{i_v k_w} \mu_{k_w} = \sigma_{i_v} \quad \forall i_v = 1 : N_v \tag{88}$$

where C_{ik} and D_{ik} represents the induced velocity of the k^{th} doublet and the source panel at the point $\boldsymbol{r}_i$,

$$\begin{aligned} C_{ik} &= -\boldsymbol{\nabla} \int_{S_k} \frac{\partial \mathcal{G}(\boldsymbol{r}_o, \boldsymbol{r}_i)}{\partial \hat{\boldsymbol{n}}_{k_s}} \\ D_{ik} &= -\boldsymbol{\nabla} \int_{S_k} \mathcal{G}(\boldsymbol{r}_o, \boldsymbol{r}_i) \end{aligned} \tag{89}$$

Given the value of the surface variables σ_k, from the movement of the body and the rotational part of the velocity, and the intensity of the wake doublet elements μ_{k_w}, from previous time steps, the boundary conditions (87) are N_v linear equations in the N unknown doublet intensities μ_k.

LOADS EVALUATION For vortex lattice elements, aerodynamic loads are calculated using the unsteady formulation of the Kutta-Joukowsky theorem with a Prandtl-Glauert compressibility correction. For a stripe that contains n_{str} panels from $i_{str} = 1 : n_{str}$, starting from the leading edge, we see that the force acting on the i-th panels is:

$$\begin{cases} \boldsymbol{F}_i = \rho_\infty \dfrac{(\boldsymbol{V}_i \times \boldsymbol{d\ell}_i)}{\sqrt{(1 - Ma^2)}} \cdot (\Gamma_i - \Gamma_{i-1}) - \rho dA_i \dfrac{d\Delta\Gamma}{dt} \cdot \hat{\boldsymbol{n}}_i & \text{if } i \neq 1 \\ \boldsymbol{F}_i = \rho_\infty \dfrac{(\boldsymbol{V}_i \times \boldsymbol{d\ell}_i)}{\sqrt{(1 - Ma^2)}} \cdot (\Gamma_i) - \rho dA_i \dfrac{d\Gamma}{dt} \cdot \hat{\boldsymbol{n}}_i & \text{if } i = 1 \end{cases} \tag{90}$$

Where ρ_∞ is the flow density, $\boldsymbol{V}_i$ is the induced velocity on the panel by all elements, $\boldsymbol{d\ell}_i$ is the spanwise vector, Ma is the local Mach, $\frac{d\Delta\Gamma}{dt}$ is the time derivative of the circulation calculated by an explicit Euler, dA_i and $\hat{\boldsymbol{n}}_i$ are respectively the panel area and normal.

2.2.4 *Lifting Lines*

A lifting line is a 1-D model of thin slender lifting bodies, whose sectional aerodynamic coefficients are provided as a function of the local angle of attack α, the local Reynolds number Re and the local Mach number M,

$$\{c_\ell, c_d, c_m\}(\alpha, Re, M; \boldsymbol{r}) \tag{91}$$

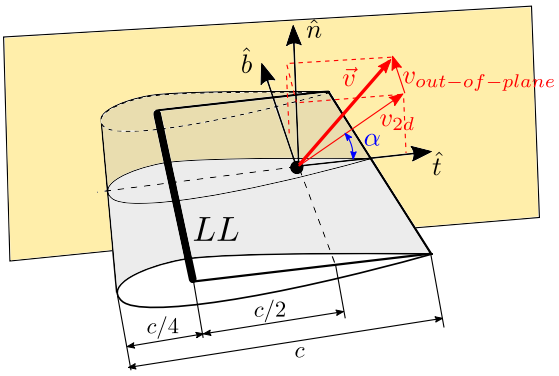


Figure 21: Lifting line angles

The circulation $\Gamma_l(\boldsymbol{r}, t)$ of the lifting line is determined as the solution of a non-linear problem, connecting $\Gamma_l(\boldsymbol{r}, t)$ with the tabulated aerodynamic coefficients of its lifting sections.

Both a loosely coupled α-method [156] and a Γ-method [66] solver are available; however, only the Γ-method can be used when the element is coupled with an external structural solver, such as MBDyn, although this method suffers from numerical instabilities for low aspect ratio wings when coupled with flexible elements.

The Γ-method consists in the equivalence of the semiempirical expression of the sectional lift and its analytical expression from the Kutta-Joukowski theorem.

$$\frac{1}{2}\rho\,|\boldsymbol{u}_{2D}(\boldsymbol{r}, t)|^2 c(\boldsymbol{r})\, c_\ell(\alpha(\boldsymbol{r}, t); \boldsymbol{r}) = -\rho\,|\boldsymbol{u}_{2D}(\boldsymbol{r}, t)|\,\Gamma(\boldsymbol{r}, t) \tag{92}$$

where $c(\boldsymbol{r})$ and $c_\ell(\alpha; \boldsymbol{r})$ represent the chord and the lift curve of the aerodynamic sections of the component, and the local angle of attack α and the relative velocity $\boldsymbol{u}_{2D}$ are functions of the circulation of the lifting line Γ_l through the induced velocity. The total velocity $\boldsymbol{v}_\ell(\boldsymbol{r}, t)$ acting on the control point at time t of the lifting line l can be calculated as follows:

$$\boldsymbol{v}_\ell(\boldsymbol{r}, t) = \boldsymbol{U}_\infty - \boldsymbol{u}_{\ell,b} + \boldsymbol{u}_{tot\setminus\ell}(\boldsymbol{r}, t) + \boldsymbol{u}_\ell(\boldsymbol{r}, t) \tag{93}$$

where $\boldsymbol{u}_\ell(\boldsymbol{r}, t)$ is the induced velocity of the lifting on the other lifting lines: this contribution is proportional to the circulation of the lifting calculated in the previous iteration $(k-1)$ times the induction of a point doublet in the control point: $\boldsymbol{u}_{tot\ \ell}(\boldsymbol{r}, t) = \Gamma_l^{(k-1)} \varphi^d(\boldsymbol{r}, \boldsymbol{r}_o)$. In fact, each lifting line element constitutes a vortex ring along with its trailing vortices, and the last line vortex released in the wake aligned with the span-wise direction. The intensity of this vortex ring is equivalent to a uniform surface doublet $\mu_l = \Gamma_l$. $\boldsymbol{u}_{tot\setminus\ell}(\boldsymbol{r}, t)$ is the velocity induced by the other elements on the lifting line and $\boldsymbol{u}_{\ell,b}$ is the body velocity of the lifting line. Then, the velocity is projected on the $\hat{\boldsymbol{n}} - \hat{\boldsymbol{t}}$ plane:

$$\boldsymbol{u}_{2D} = (\boldsymbol{v}_\ell(\boldsymbol{r}, t) \cdot \hat{\boldsymbol{n}}_l)\, \hat{\boldsymbol{n}}_l + (\boldsymbol{v}_\ell(\boldsymbol{r}, t) \cdot \hat{\boldsymbol{t}}_l)\, \hat{\boldsymbol{t}}_l \tag{94}$$

Angle of attack $\tilde{\alpha}_l$ is calculated as:

$$\tilde{\alpha}_l = \tan^{-1}\left(\frac{\boldsymbol{u}_{2D}\cdot\hat{\boldsymbol{n}}_l}{\boldsymbol{u}_{2D}\cdot\hat{\boldsymbol{t}}_l}\right) \tag{95}$$

This angle must be corrected by taking out the velocity induced by the lifing line itself, according to [156] this correction is equal to:

$$\alpha_{2D_l} = \frac{\Gamma_l^{(k-1)}}{\pi c_l |\boldsymbol{u}_{2D}(\boldsymbol{r},t)|} \tag{96}$$

Equation 96 is a simplification of the Kutta-Joukowski theorem for an airfoil with a lift slope coefficient of 2π. Eventually, the angle of attack used to enter in the look-up tables is computed as $\alpha = \tilde{\alpha} - \alpha_{2D}$, and the velocity used to calculate the Mach and Reynolds numbers is $\boldsymbol{u}_{2D}$.

The discrete version of equation 92 can be formally formulated as a fixed-point problem:

$$\mu_{i_l} = f_{i_l}(\mu_{k_s}, \mu_{k_v}, \mu_{k_l}, \sigma_{k_s}, \mu_{k_w}), \quad i_l = 1 : N_l \tag{97}$$

in which the convergence is reached when the difference between the singularity magnitudes between two consecutive iterations is below a certain threshold.

LOADS EVALUATION The loads for the lifting line are calculated in the same way as the vortex lattice (considering the case with only 1 in the chord) plus an additional term due to viscosity, calculated using strip theory:

$$\boldsymbol{F}_i^{visc} = \frac{1}{2}\rho_\infty |\boldsymbol{u}_{rel}(\boldsymbol{r},t)|^2 dA \cdot c_d \left(\sin(\alpha_i)\hat{\boldsymbol{n}} + \cos(\alpha_i)\hat{\boldsymbol{t}}\right) \tag{98}$$

Then, the total force acting on a single lifting line is:

$$\boldsymbol{F}_i = \rho_\infty(\boldsymbol{V}_i \times \boldsymbol{d\ell}_i)\cdot(\Gamma_i) - \rho dA_i \frac{d\Gamma}{dt}\cdot\hat{\boldsymbol{n}}_i + \boldsymbol{dF}_i^{visc} \tag{99}$$

2.2.5 *Non-linear Vortex Lattice*

To overcome both limitations of the lifting line (due to numerical scheme instabilities that occur when the lifting line is used to evaluate the lift of a short aspect ratio wing) and the vortex lattice (related to the capture of viscous effects), the two elements were joined together, providing a new non-linear vortex lattice element. In the first step, the potential solution of the linear vortex lattice is calculated by solving the linear system that imposes the non-penetration boundary condition on the vortex lattice panel as described in2.2.3. Then, the idea behind the method is to modify the right-hand side term of the complete aerodynamic system to obtain the match between the calculated lift coefficient and the tabulated one. The modification of the

right-hand side occurs only on the rows of the linear system that are linked to μ_{ν}^{nl}, which is the unknown intensity of the surface doublet for the $i-th$ non-linear vortex lattice panel. Any modification occurs on the left-hand side of the system, which means that the overall system is factorized only once for each time step.

After the solution of the system, it is possible to calculate the angle of attack of each stripe in the component as for the lifting line case. The total velocity $\boldsymbol{v}_{\nu_{nl}}(\boldsymbol{r}, t)$ acting on the control point (center) at time t of each stripe can be calculated as follows:

$$\boldsymbol{v}_{\nu_{nl}}(\boldsymbol{r}, t) = \boldsymbol{U}_{\infty} - \boldsymbol{u}_{\nu_{nl},b} + \boldsymbol{u}_{tot\setminus\nu_{nl}}(\boldsymbol{r}, t) + \boldsymbol{u}_{\nu_{nl}}(\boldsymbol{r}, t) \tag{100}$$

where $\boldsymbol{u}_{\nu_{nl}}(\boldsymbol{r}, t)$ is the induced velocity of the stripe on the other stripes. In this case, the total stripe intensity is calculated following the load calculation formula of the vortex lattice.

$$\Gamma_{\nu_{nl}} = d\Gamma_{\nu_{nl}^{1}} + \sum_{i=2}^{N} d\Gamma_{\nu_{nl}^{i}} - d\Gamma_{\nu_{nl}^{i-1}} \tag{101}$$

where $d\Gamma_{\nu_{nl}^{i}}$ is the intensity panel i -th of the stripe calculated in the previous iteration.

A fixed-point iterative problem is then solved, as for the lifting line Γ-method, by imposing a convergence between the lift coefficient derived from the aerodynamic look-up table and the one calculated from the stripe intensity. For each iteration, the vortex lattice linear system is solved by modifying only the right-hand side and taking as results the updated circulations. To numerically stabilize the method, two types of relaxation are available:

- Constant relaxation, where the updated right-hand side of the system is:

$$\mathbf{RHS}_{k} = \mathbf{RHS}_{k-1} + \alpha \cdot \mathbf{r} \tag{102}$$

  where α is the constant relaxation factor and $\mathbf{r}$ is the residual vector, defined as the difference between the lift coefficient solved from the linear system and the one obtained for the aerodynamic look-up table in iteration k.

- Aitken acceleration [99], where the updated right-hand side of the system is

$$\alpha_{k} = -\alpha_{k-1} \frac{\mathbf{r}_{k-1} \cdot \Delta\mathbf{r}_{k}}{\Delta\mathbf{r}_{k} \cdot \Delta\mathbf{r}_{k}} \tag{103}$$

  where α_{k} is the Aitken relaxation factor for iteration k, and $\mathbf{r}$ is the residual vector and $\Delta\mathbf{r}_{k}$ is the difference between the residuals in iteration k and $k-1$. Then **RHS** is updated as described in Eq. 102.

LOADS EVALUATION For non-linear vortex lattice elements, aerodynamic loads are calculated using the unsteady formulation of the Kutta-Joukowsky theorem, as for the linear case plus a viscous correction on the drag term:

$$\boldsymbol{dF}_{i\in s} = \boldsymbol{dF}^{sys}_{i\in s} + \frac{1}{2}\rho_\infty(\boldsymbol{u}(\boldsymbol{r},t))^2 dA_{i\in s} c^s_d \left(\cos(\alpha)\hat{\boldsymbol{t}} + \sin\alpha\hat{\boldsymbol{n}}\right) \quad (104)$$

where $i \in s$ stands for a panel i that belongs to the stripe s and $\boldsymbol{dF}^{sys}_{i\in s}$ is the force calculated from the system.

2.2.6 *Coupled governing equations*

The potential contribution of the velocity field is the result of the superposition of the free-stream velocity and the induced velocity from the doublet and source singularities, modeling the bound vorticity on the surface of the investigated bodies. Given the free stream velocity, the rotational component of the velocity field, the state of the wake, and the motion of the solid components of the model, the intensity of the surface doublets are the actual unknowns of the potential problem obtained combining Eqs. (73), (87) and (97) and the subdividing vortex lattice in the linear (ν_l) and non-linear case (ν_{nl}):

$$\begin{cases} \begin{bmatrix} A_{ss} & A_{s\nu_l} & A_{s\nu_{nl}} & A_{sl} \\ C_{\nu_l s} & C_{\nu_l\nu_l} & C_{\nu_l\nu_{nl}} & C_{\nu_l l} \\ C_{\nu_{nl}s} & C_{\nu_{nl}\nu_l} & C_{\nu_{nl}\nu_{nl}} & C_{\nu_{nl}l} \end{bmatrix} \boldsymbol{\mu} = \\ - \begin{bmatrix} B_{ss} & 0_{s\nu_l} & 0_{s\nu_{nl}} \\ D_{\nu_l s} & -I_{\nu_l\nu_l} & 0_{\nu_l\nu_{nl}} \\ D_{\nu_{nl}s} & 0_{\nu_{nl}\nu_l} & -I_{\nu_{nl}\nu_{nl}} \end{bmatrix} \boldsymbol{\sigma} - \begin{bmatrix} A_{sw} \\ C_{\nu_l w} \\ C_{\nu_{nl}w} \end{bmatrix} \boldsymbol{\mu}_w \\ \boldsymbol{\mu}_l = \boldsymbol{f}_l(\boldsymbol{\mu};\boldsymbol{\sigma},\boldsymbol{\mu}_w) \end{cases} \quad (105)$$

where the vector $\boldsymbol{\mu}_w$ collects the intensity of the wake doublets, the vector $\boldsymbol{\mu} = (\boldsymbol{\mu}_s, \boldsymbol{\mu}_{\nu_l}, \boldsymbol{\mu}_{\nu_{nl}}, \boldsymbol{\mu}_l)$ collects the intensity of the body doublets on the surface panel, the vortex lattice and the lifting line elements, the vector $\boldsymbol{\sigma} = (\boldsymbol{\sigma}_s, \boldsymbol{\sigma}_{\nu_l}, \boldsymbol{\sigma}_{\nu_{nl}})$ contains the unperturbed relative normal velocity at the collocation points of the surface panels and the vortex lattice elements, $0_{s\nu}$ is a null matrix $N_s \times N_\nu$, $I_{\nu\nu}$ is the identity matrix $N_\nu \times N_\nu$.

The aerodynamic problem (105) can be written in a compact form as

$$\begin{cases} Y\boldsymbol{\mu} = -Z\boldsymbol{\sigma} - Y_w\boldsymbol{\mu}_w \\ \boldsymbol{\mu}_l = \boldsymbol{f}_l(\boldsymbol{\mu};\boldsymbol{\sigma},\boldsymbol{\mu}_w) \end{cases} \quad (106)$$

2.2.7 *Wake shedding*

Lifting bodies release wakes from their trailing edges. Kutta condition represents the vorticity balance at the trailing edge and determines

the intensity of the wake panel shed into the domain. Kutta condition is naturally satisfied by a vortex lattice and lifting line description of a lifting surface, whereas a row of implicit wake panels is introduced at the trailing edge of thick bodies modeled with surface panels, in order to satisfy the potential jump equivalent to the Kutta condition.

The intensity $\boldsymbol{\mu}_{w_{TE}}$ of the trailing edge doublets and the intensity $\tilde{\boldsymbol{\mu}}_w$ of the wake doublets released at the previous time steps are identified in the wake doublet vector, $\boldsymbol{\mu}_w = (\boldsymbol{\mu}_{w_{TE}}, \boldsymbol{\mu}_{\tilde{w}})$.

Kutta condition relates the intensity of the i_{TE}^{th} element of the first row of the wake panels with the doublet intensity of the surface panels at the trailing edge,

$$\mu_{w,i_{TE}} = \mu_{s^u,i_{TE}} - \mu_{s^l,i_{TE}} \quad , \quad \boldsymbol{\mu}_{w_{TE}} = T\boldsymbol{\mu}_s \tag{107}$$

where s^u is the SP element with the unit normal approximately aligned with the unit normal vector of the wake surface, s_l is the SP on the opposite side of the trailing edge, and the connectivity matrix T provides the matrix form of the Kutta condition.

The implicit contribution of the trailing edge doublets in Eq. (106) are separated from the explicit wake terms,

$$Y_w\boldsymbol{\mu}_w = [Y_{w_{TE}}\ Y_{\tilde{w}}] \begin{bmatrix} \boldsymbol{\mu}_{w_{TE}} \\ \boldsymbol{\mu}_{\tilde{w}} \end{bmatrix} = Y_{w_{TE}} T\boldsymbol{\mu}_s + Y_{\tilde{w}}\boldsymbol{\mu}_{\tilde{w}} \tag{108}$$

so that the boundary value problem for the potential velocity becomes

$$\begin{cases} \tilde{Y}\boldsymbol{\mu} = -Z\boldsymbol{\sigma} - Y_{\tilde{w}}\boldsymbol{\mu}_{\tilde{w}} \\ \boldsymbol{\mu}_l = \boldsymbol{f}_l(\boldsymbol{\mu}; \boldsymbol{\sigma}, \boldsymbol{\mu}_{\tilde{w}}) \end{cases} \tag{109}$$

where the matrix $\tilde{Y}$ collects the AICs contributions of the body singularities along with the trailing edge wake doublets shed by SP elements,

$$\tilde{Y} = \begin{bmatrix} \tilde{Y}_s\ Y_v\ Y_l \end{bmatrix} \quad , \quad \tilde{Y}_s = Y_s + Y_{w_{TE}} T \, . \tag{110}$$

2.2.8 *Rotational Velocity*

The mathematical formulation of the solver implemented in DUST is based on the decomposition of the velocity field as the sum of the potential contribution, induced by the surface elements, and the rotational contribution described in this section.

The calculation of the rotational contribution $\boldsymbol{u}_\psi$ is based on a vortex particle method (VPM) to provide a grid-free Lagrangian description of the velocity-vorticity formulation of the governing equations of the fluid, as an alternative to classical Eulerian grid methods. This numerical method is particularly suited for these flows, where vorticity is confined only to a small fraction of the volume domain, such as in many aeronautical flows of interest. The vorticity field is represented as the sum of the elementary contributions associated with

the vortex particles, the material particles moving with the local fluid velocity. Thus, the evolution of the vorticity field is governed by the Lagrangian form of the vorticity equation and the kinematic equation of the material particles:

$$\begin{cases} \dfrac{D\boldsymbol{\omega}}{Dt} = (\boldsymbol{\omega}\cdot\boldsymbol{\nabla})\boldsymbol{u} - \nu\Delta\boldsymbol{\omega} \\ \dfrac{D\boldsymbol{r}(t)}{Dt} = \boldsymbol{u}(\boldsymbol{r}(t),\, t) \end{cases} \tag{111}$$

The evolution of the vorticity of a material particle is influenced by vortex stretching/tilting and viscous effects. In the VPM method, these terms are computed for each particle at each time step as the sum of the elementary interaction with all the other particles and potential elements. Since this is a $\mathcal{O}(N^2)$ operation and the number of vortex particles increases as the simulation advances in time, the computation of these interactions is accelerated by a Cartesian Fast Multiple Method (FMM), targeting the theoretical $\mathcal{O}(N)$ computational cost.

Following Helmholtz's decomposition of the velocity field, the rotational component of the velocity field is defined as the curl of the vector potential, $\boldsymbol{u}_\psi(\boldsymbol{r},t) = \boldsymbol{\nabla}\times\boldsymbol{\psi}(\boldsymbol{r},t)$, and the vorticity field corresponds to the curl of the rotational component of the velocity, $\boldsymbol{\omega}(\boldsymbol{r},t) = \boldsymbol{\nabla}\times\boldsymbol{u}_\psi(\boldsymbol{r},t)$. Using these two relations, we have a Poisson equation for the vector potential $\boldsymbol{\psi}(\boldsymbol{r},t)$ where the vorticity field $\boldsymbol{\omega}(\boldsymbol{r},t)$ acts as the volume forcing.

$$-\Delta\boldsymbol{\psi} = \boldsymbol{\omega} \tag{112}$$

is obtained, given the vector identity $\Delta\boldsymbol{\psi} = \boldsymbol{\nabla}(\boldsymbol{\nabla}\cdot\boldsymbol{\psi}) - \boldsymbol{\nabla}\times\boldsymbol{\nabla}\times\boldsymbol{\psi}$ and the divergence-free condition $\boldsymbol{\nabla}\cdot\varphi = 0$.

The potential vector field $\boldsymbol{\psi}(\boldsymbol{r},t)$ is computed as the solution of the Poisson equation using the Green function method, namely:

$$\boldsymbol{\psi}(\boldsymbol{r},t) = \int_{\Omega_0} \mathcal{G}(\boldsymbol{r},\boldsymbol{r}_0)\boldsymbol{\omega}(\boldsymbol{r}_0,t)\,d\Omega_0 = \frac{1}{4\pi}\int_{\Omega_0}\frac{\boldsymbol{\omega}(\boldsymbol{r}_0,t)}{|\boldsymbol{r}-\boldsymbol{r}_0|}d\Omega_0 \tag{113}$$

if the boundary contributions vanish, as in the case of unbounded aerodynamic flows.

The rotational part of the velocity field $\boldsymbol{u}_\psi$ is evaluated taking the curl of the vector potential field (113),

$$\begin{aligned} \boldsymbol{u}_\psi(\boldsymbol{r},t) = \boldsymbol{\nabla}\times\boldsymbol{\psi}(\boldsymbol{r},t) &= \frac{1}{4\pi}\boldsymbol{\nabla}\times\int_{\Omega_0}\frac{\boldsymbol{\omega}(\boldsymbol{r}_0,t)}{|\boldsymbol{r}-\boldsymbol{r}_0|}d\Omega_0 = \\ &= -\frac{1}{4\pi}\int_{\Omega_0}\frac{\boldsymbol{r}-\boldsymbol{r}_0}{|\boldsymbol{r}-\boldsymbol{r}_0|^3}\times\boldsymbol{\omega}(\boldsymbol{r}_0,t)\,d\Omega_0 = \end{aligned} \tag{114}$$

and can be interpreted as the convolution of the Biot-Savart kernel, $\boldsymbol{K}(\boldsymbol{x}) = \boldsymbol{\nabla}\widetilde{\mathcal{G}}(\boldsymbol{x}) = -\frac{1}{4\pi}\frac{\boldsymbol{x}}{|\boldsymbol{x}|^3}$, with the vorticity field.

2.2.8.1 *Vortex particles: singular model*

Vortex particle methods introduce a finite-dimensional approximation of the vorticity field $\boldsymbol{\omega}(\boldsymbol{r}, t)$ in the expression of the rotational part of the velocity: the vortex particle method assumes the following form of the vorticity field:

$$\boldsymbol{\omega}(\boldsymbol{r}, t) = \sum_{p=1}^{N_p} \boldsymbol{\alpha}^p(t)\delta\left(\boldsymbol{r} - \boldsymbol{r}_p(t)\right) \tag{115}$$

where $\boldsymbol{r}^p(t)$ denotes the coordinate of the p^{th} vortex particle, $\boldsymbol{\alpha}^p(t)$ is the intensity vector of the p^{th} vortex particle, and $\delta\left(\boldsymbol{r} - \boldsymbol{r}_p(t)\right)$ is the Dirac delta. Introducing the approximation (115) of the vorticity field in expression (113), the approximation of the vector potential field $\boldsymbol{\psi}(\boldsymbol{r}, t)$ becomes:

$$\begin{aligned}\boldsymbol{\psi}(\boldsymbol{r}, t) &= \frac{1}{4\pi}\sum_{p=1}^{N_p} \boldsymbol{\alpha}^p(t)\int_{\Omega_0} \frac{\delta(\boldsymbol{r}_0 - \boldsymbol{r}^p(t))}{|\boldsymbol{r} - \boldsymbol{r}_0|}\mathrm{d}\Omega_0 = \\ &= \frac{1}{4\pi}\sum_{p=1}^{N_p} \frac{\boldsymbol{\alpha}^p(t)}{|\boldsymbol{r} - \boldsymbol{r}_p(t)|}\end{aligned} \tag{116}$$

and the rotational part of the velocity fields follows,

$$\boldsymbol{u}_\varepsilon(\boldsymbol{r}, t) = -\frac{1}{4\pi}\sum_{p=1}^{N_p} \frac{\boldsymbol{r} - \boldsymbol{r}_p}{|\boldsymbol{r} - \boldsymbol{r}_p|^3} \times \boldsymbol{\alpha}^p(t) \tag{117}$$

Both the vector potential and the rotational velocity (and its space derivatives) are singular at the location of the vortex particles: the velocity intensity grows to infinity at that point. Regularized models are introduced to avoid infinite-intensity interactions between particles.

2.2.8.2 *Vortex particles: regular(ised) model*

In order to avoid singularities, the vortex particle method assumes the following form of the vorticity field.

$$\boldsymbol{\omega}_\varepsilon^h(\boldsymbol{r}, t) = \sum_{p=1}^{N_p} \boldsymbol{\alpha}^p \zeta_\varepsilon\left(\boldsymbol{r} - \boldsymbol{r}^p(t)\right) \tag{118}$$

where $\boldsymbol{r}^p(t)$ denotes the coordinates of the p^{th} vortex particle and the regular *cut-off* function $\zeta_\varepsilon(\boldsymbol{x})$ replaces the Dirac delta of the singular model. It can be proven that the regular expression (118) is equal to the convolution of the singular vorticity field (115) with the *cut-off* function $\zeta_\varepsilon(\boldsymbol{r})$. Several regularized Green functions are available in the literature with their respective regular Biot-Savart kernels. In

DUST, the Rosenhead kernel is implemented as a regularized version of the Biot-Savart kernel,

$$\boldsymbol{K}_{\varepsilon}(\boldsymbol{x}, \boldsymbol{y}) = -\frac{1}{4\pi} \frac{\boldsymbol{x} - \boldsymbol{y}}{(|\boldsymbol{x} - \boldsymbol{y}|^2 + \delta^2)^{3/2}} \tag{119}$$

defined as the gradient of the Plummer potential;

$$\mathcal{G}_{\varepsilon}(\boldsymbol{x}, \boldsymbol{y}) = \frac{1}{4\pi(|\boldsymbol{x} - \boldsymbol{y}|^2 + \delta^2)^{1/2}} \tag{120}$$

where the regularization parameter δ is proportional to the radius of the vortex particles. The vortex radius is automatically set with a radius equal to the radius of the circumscribed circle on the correspondent wake panel, which is converted into a particle. In this way, the overlap between the particles is always guaranteed.

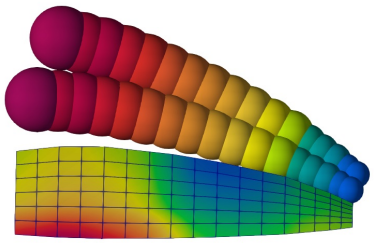


Figure 22: Vortex radius behavior for a rotating propeller

2.2.9 *Vortex particle method: dynamical equations*

A Lagrangian description of the rotational velocity and the vorticity fields is provided by the kinematic equation of the trajectory of the material particles and the vorticity equation:

$$\begin{cases} \dfrac{d\boldsymbol{r}_p(t)}{dt} = \boldsymbol{U}(\boldsymbol{r}_p(t), t) \\ \dfrac{D\boldsymbol{\omega}}{Dt} = (\boldsymbol{\omega} \cdot \boldsymbol{\nabla})\boldsymbol{U} + \nu\nabla^2\boldsymbol{\omega} \end{cases} \tag{121}$$

where $\boldsymbol{r}_p(t)$ is the position of the p^{th} material particles and D/Dt is the material derivative, which describes the time rate of change of a physical quantity of a material particle. These equations must be supplemented with appropriate initial conditions and are directly related to the wake-shedding criterion of solid bodies in the simulations of flows of aeronautical interest. The discrete counterpart of the dynamical equations (121) reads,

$$\begin{cases} \dfrac{d\boldsymbol{r}_p(t)}{dt} = \boldsymbol{u}(\boldsymbol{r}_p(t), t) \\ \dfrac{d\boldsymbol{\alpha}^p}{dt} = (\boldsymbol{\alpha}^p \cdot \boldsymbol{\nabla})\boldsymbol{u}(\boldsymbol{r}_p(t), t) + \nu\nabla^2\boldsymbol{\alpha}^p \end{cases} \tag{122}$$

where $\boldsymbol{u} = \boldsymbol{U}_\infty + \boldsymbol{u}_\varphi + \boldsymbol{u}_\psi$ is the complete velocity field and the operator $\nu\nabla^2\boldsymbol{\alpha}^p$ represents the viscous term that acts on the p -th particle:

$$\nu\nabla^2\boldsymbol{\alpha}^p = \nu\int_{V_p}\nabla^2\boldsymbol{\omega} \tag{123}$$

The irrotational velocity $\boldsymbol{u}_\varphi$ is the solution to the linear problem, the rotational velocity $\boldsymbol{u}_\psi$ is replaced by its regularized approximation of $\boldsymbol{u}_\varepsilon$. The diffusion term is approximated by means of Particle-Strength Exchange (PSE).

2.2.9.1 *Vortex stretching/tilting term*

For incompressible flows, three different equivalent expressions of the vortex stretching and tilting term are available;

$$\begin{aligned}
&(\boldsymbol{\omega}\cdot\boldsymbol{\nabla})\boldsymbol{u} && \text{(direct scheme)}\\
&(\boldsymbol{\omega}\cdot\boldsymbol{\nabla}^T)\boldsymbol{u} && \text{(adjoint scheme)}\\
&\frac{1}{2}\left[(\boldsymbol{\omega}\cdot\boldsymbol{\nabla})\boldsymbol{u} + (\boldsymbol{\omega}\cdot\boldsymbol{\nabla}^T)\boldsymbol{u}\right] && \text{(symmetric scheme)}
\end{aligned} \tag{124}$$

Although these schemes are equivalent in the continuous framework, their discrete counterparts are not. In the following, only the discrete version of the transpose scheme of the particle-to-particle direct interactions in the near field implemented in DUST is detailed. The transposed scheme is very special: it guarantees the exact conservation of the total vorticity and, moreover, leads to a weak solution of the vorticity equation[199]. The direct scheme can be easily derived by following the same procedure. The computation of panel-to-particle interactions requires the expression of the velocity gradient (or its transpose), while the far-field interactions are accelerated with the Cartesian FMM.

TRANSPOSE SCHEME The transpose scheme for inviscid flows is as follows.

$$\begin{cases}
\dfrac{d\boldsymbol{r}_p(t)}{dt} = \boldsymbol{u}(\boldsymbol{r}^p(t), t)\\
\dfrac{d\boldsymbol{\alpha}^p}{dt} = (\boldsymbol{\alpha}^p\cdot\boldsymbol{\nabla}^T)\boldsymbol{u}(\boldsymbol{r}^p(t), t)
\end{cases} \tag{125}$$

where the index expression of the vortex stretching/tilting term reads:

$$(\boldsymbol{\alpha}^p\cdot\boldsymbol{\nabla}^T)\boldsymbol{u}(\boldsymbol{r}^p(t), t)_i = \alpha_k^p\,\partial_i\,u_k(\boldsymbol{r}^p(t), t) \tag{126}$$

This term is evaluated using the Rosenhead kernel (119). The influence on the particle p of all particles $q = 1 : N$ is computed in vector form as:

$$\begin{aligned}
\frac{d\boldsymbol{\alpha}^p}{dt} &= -\frac{1}{4\pi}\sum_q\left[\frac{1}{R^3}\boldsymbol{\alpha}^q\times\boldsymbol{\alpha}^p - \frac{3}{R^5}\Delta\boldsymbol{r}^{pq}(\boldsymbol{\alpha}^q\cdot\boldsymbol{\alpha}^p\times\Delta\boldsymbol{r}^{pq})\right] =\\
&= -\frac{1}{4\pi}\sum_q\left[\frac{1}{R^3}\boldsymbol{\alpha}^q\times\boldsymbol{\alpha}^p - \frac{3}{R^5}\Delta\boldsymbol{r}^{pq}(\Delta\boldsymbol{r}^{pq}\cdot\boldsymbol{\alpha}^q\times\boldsymbol{\alpha}^p)\right]
\end{aligned} \tag{127}$$

where $R = (|\Delta \boldsymbol{r}^{pq}|^2 + \delta^2)^{1/2}$ and $\Delta \boldsymbol{r}^{pq} = \boldsymbol{r}^p - \boldsymbol{r}^q$.

2.2.9.2 *Viscous term: Particle Strength Exchange (PSE) method*

The viscous term in the vorticity equation is modeled with the PSE method, whose idea is to replace the differential Laplacian operator with an integral one. The motivation here is that integrals are much better suited for particle methods than second-order differential operators. It is possible to prove that the integral operator:

$$\Delta_\varepsilon \boldsymbol{\omega}(\boldsymbol{x}) = \varepsilon^{-2} \int_V [\boldsymbol{\omega}(\boldsymbol{y}) - \boldsymbol{\omega}(\boldsymbol{x})]\, \eta_\varepsilon(\boldsymbol{y} - \boldsymbol{x}) dV_{\boldsymbol{y}} \tag{128}$$

is a good approximation of the differential Laplacian operator. If the kernel function is defined as $\eta_\varepsilon(\boldsymbol{x}) = \frac{1}{\varepsilon^3}\eta\left(\frac{\boldsymbol{x}}{\varepsilon}\right)$ and $\eta(\boldsymbol{x})$ satisfies the following moment conditions:

$$\begin{aligned}
&\int_V x_i x_j \eta(\boldsymbol{x}) dV = 2\delta_{ij} \quad \text{for } i,j = 1,2,3 \\
&\int_V x_1^{i_1} x_2^{i_2} x_3^{i_3} \eta(\boldsymbol{x}) dV = 0 \quad \text{if } i_1 + i_2 + i_3 = 1 \text{ or } 3 \leqslant i_1 + i_2 + i_3 \leqslant r+1 \\
&\int_V |\boldsymbol{x}|^{r+2} |\eta(\boldsymbol{x})| < \infty
\end{aligned} \tag{129}$$

For vortex particle methods, the integral (128) can be approximated as the sum of all particles in the domain:

$$\Delta_\varepsilon \boldsymbol{\omega}(\boldsymbol{x}) \simeq \varepsilon^{-2} \sum_q V_q\, [\boldsymbol{\omega}_q - \boldsymbol{\omega}(\boldsymbol{x})]\, \eta_\varepsilon(\boldsymbol{r}_q - \boldsymbol{x}) \tag{130}$$

where V_q is the volume of the q^{th} vortex particle. Thus, the integral PSE approximation of the viscous term $\nabla^2 \boldsymbol{\alpha}^p = \int_{V_p} \Delta_\varepsilon \boldsymbol{\omega}$ reads:

$$\nabla^2 \boldsymbol{\alpha}^p = \varepsilon^{-2} \sum_q [V_p \boldsymbol{\alpha}_q - V_q \boldsymbol{\alpha}_p]\, \eta_\varepsilon(\boldsymbol{r}_q - \boldsymbol{r}_p) \tag{131}$$

Winkelmans provides some examples of kernel functions with spherical symmetry, $\tilde{\eta}(r) = \eta(\boldsymbol{r})$, with $r = |\boldsymbol{r}|$. Some of them are listed here, with the respective regularization function $\tilde{\zeta}(r)$, which must satisfy $\tilde{\eta}(r) = -\frac{\tilde{\zeta}'(r)}{r}$. In DUST, the high-order algebraic kernel function is currently used in the PSE method:

$$\tilde{\zeta}(r) = \frac{15}{8\pi} \frac{1}{(1+r^2)^{7/2}} \quad \rightarrow \quad \tilde{\eta}(r) = \frac{105}{8\pi} \frac{1}{(1+r^2)^{9/2}} \tag{132}$$

2.2.10 *Generation of the vortex particles*

The vortex particles are generated from wake-vortex lattices. At each time step, the wake panels are advected downstream and another panel is generated; after a prescribed n_{wp} number of time steps of

existence of a panel, the panel is converted into a particle, generating after the initial transient a wake with n_{wp} rows of panels attached to the trailing edges followed by a particle wake. It is possible to obtain a complete particle wake by prescribing $n_{wp} = 1$, which leaves only the first row of implicit panels, necessary to enforce the Kutta condition.

When each panel is converted into a particle, the intensity comes from the integration of the vorticity contribution from the sides of the vortex lattice element. However, since each vortex-ring panel has a constant intensity and represents a closed path, the simple integration of its sides would lead to a zero value. For this reason, the actual intensity of each side due to the superposition of the sides of the neighboring panels is considered, and the contribution of each side is distributed to the neighboring particles that are about to be generated, as shown in Fig. 23. While the neighboring elements in the direction normal to the flow are converted simultaneously and their contribution is shared with the respective particles, the previous panel in the streamwise direction was already converted the previous time-step. For this reason, each particle is formed with the contribution of the two lateral sides of the panel, the rearmost side and a line vortex left by the previous panel, while the present panel leaves the foremost side as a line vortex attached to the following panel. The contribution from each lateral side can be defined as:

$$\boldsymbol{J}_s = \begin{cases} \int_s \frac{1}{2}(\mu_{w,i} - \mu_{w,i_n})\,d\boldsymbol{l} & \text{if neighbour } i_n \text{ present} \\ \int_s \mu_{w,i}\,d\boldsymbol{l} & \text{if neighbour } i_n \text{ absent} \end{cases} \tag{133}$$

where s is one side and the contribution is the whole vorticity of the side if no neighbor is present or half of the net contribution of the side. The other half will be assigned to the neighboring particle. The end-side contribution is as follows.

$$\boldsymbol{J}_E = \int_s (\mu_{w,i} - \mu_{i_{end}})\,d\boldsymbol{l} \tag{134}$$

where $\mu_{i_{end}}$ is the intensity of the corresponding end line vortex, and thus defining $\boldsymbol{J}_L, \boldsymbol{J}_R$ the contribution from the two lateral sided, the intensity of the generated particle is:

$$\boldsymbol{\alpha}_{i_p} = \boldsymbol{J}_L + \boldsymbol{J}_R + \boldsymbol{J}_E \tag{135}$$

The newly generated particle is placed in the center of the previous panel and then begins to evolve according to (121). Both the calculation of the intensity and the initial placement are approximations; however, they do not introduce a significant error, while using a more complex distribution of particles to describe a converted panel does not lead to a noticeable increase in accuracy.

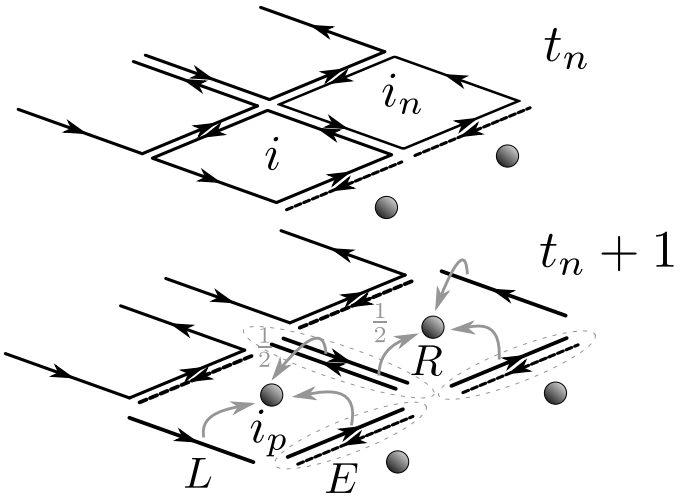


Figure 23: Schematics of an example of conversion from panels to particles

2.3 IMPLEMENTATION OF THE CO-SIMULATION SCHEME BETWEEN MBDYN AND DUST

2.3.1 *Theorical background*

Communications between DUST and MBDyn are managed by preCICE (Precise Code Interaction Coupling Environment), a coupling library for partitioned multiphysics simulations, originally developed for fluid-structure interaction and conjugate heat transfer simulations. preCICE offers methods for transient equations coupling, communication means, and data mapping schemes. It is written in C++ and offers additional bindings for C, Fortran, MATLAB, and Python. preCICE (`https://github.com/precice/`) is a free software released under the LGPL3 license. Although MBDyn uses its own C++, C, Fortran, and Python Application Programming Interface (API) to communicate with external software without any further modification of the C++ source code, no API was already available in DUST. Thus, new Fortran modules that collect all classes, subroutines, and functions required by the adapter for preCICE were implemented to support coupling with DUST.

A new adapter was implemented to support the communication of all kinematic variables (position, orientation, velocity, and angular velocity), plus the forces and moments that act on the nodes of a model MBDyn exposed through an external structural force element. Figure 24 shows the communication and information exchange scheme managed through the adapters for the two solvers.

The interface between structural and aerodynamic grids is obtained as a weighted average of the anisotropic distance between the nodes of the two grids and is used for motion interpolation and consistent force and moment reduction. Figure 26 shows the q nodes of the structural grid, namely Q_q. The centers and vertices of each aero-

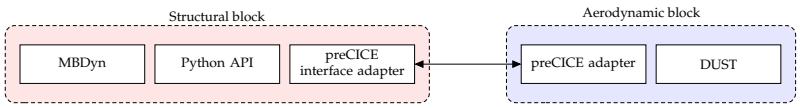


Figure 24: Scheme of the communication managed through adapters for MBDyn and DUST.

dynamic mesh are, respectively, P_e and P_p, where e and p are the corresponding indices.

The kinematic variables ϕ_p, of a point p positioned on the aerodynamic surface of a DUST component, are evaluated as the following weighted average:

$$\phi_p = \sum_q w_{pq} \phi_q \tag{136}$$

where ϕ_q is the same kinematic variable associated with the qth structural node of the MBDyn model.

In general, the weights w_{pq} are a set of non-negative real numbers that satisfy the normalization conditions.

$$\sum_q w_{pq} = 1 \qquad \forall p \tag{137}$$

since they define the weighted average of the variables associated with the structural nodes q in the aerodynamic nodes p. These coefficients are proportional to the inverse distance of the anisotropic norm of the vectors $\|(P_p - Q_q)\|$. As an example, using the local coordinates in the reference configuration $\boldsymbol{r}_{pq}$, the norm of these vectors can be defined as

$$\|(P_p - Q_q)\|^2 := \boldsymbol{R}_{pq}^T \, \mathbf{W} \, \boldsymbol{R}_{pq} \tag{138}$$

where $\mathbf{W}$ is a positive semidefinite matrix that provides an anisotropy degree of freedom in defining the norm [166]:

$$\mathbf{W} = \begin{bmatrix} 0.001 & 0 & 0 \\ 0 & 1. & 0 \\ 0 & 0 & 0.001 \end{bmatrix} \tag{139}$$

and $\boldsymbol{R}_{pq}$ is the rotation matrix that rotates the structural coordinates from the MBDyn reference frame to the DUST component one. These weights are also used to interpolate rotations, using the orientation vector parameterization $\boldsymbol{\varphi}_{pq}$.

2.3.1.1 *Automatic generation of the $\boldsymbol{R}_{pq}$ matrix*

In DUST, the aerodynamic component is defined reference system:

- The X-axis goes from the leading edge to the trailing edge.

- The Y-axis is along the aerodynamic center axis and goes from the root to tip.
- The Z-axis is determined by the right-hand rule.

while in MBDyn, the generic beam component is placed in the reference system:

- The X-axis is along the feathering or elastic axis of the beam and goes from the root of the beam to the tip.
- The Y-axis coincides with the section chord from the trailing edge to the leading edge.
- The Z-axis is determined by the right-hand rule.

To pass from the local MBDyn beam reference frame to the DUST wind axis reference frame, the following rotation matrices are used:

- Rotation matrix from the DUST reference frame to the MBDyn reference frame:

$$\mathbf{R}_b = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{140}$$

- Twist rotation of the MBDyn node around the beam axis x[1].

$$\mathbf{R}_\theta = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \tag{141}$$

- Sweep rotation of the MBDyn node. The sweep is taken to be positive for a rotation about $-z$.

$$\mathbf{R}_\lambda = \begin{bmatrix} \cos\lambda & -\sin\lambda & 0 \\ \sin\lambda & \cos\lambda & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{142}$$

- Dihedral rotation of the MBDyn node.

$$\mathbf{R}_\delta = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\delta & \sin\delta \\ 0 & -\sin\delta & \cos\delta \end{bmatrix} \tag{143}$$

1 In the case of a rotor blade, this matrix is equal to the identity, as the aerodynamic twist has already been integrated to build the structural beam. Then, the twist of each region will be zero, since it will be taken from the MBDyn node orientation

Then, the generic rotation matrix for a right component is:

$$\mathbf{R}_{\text{right}} = \mathbf{R}_b \mathbf{R}_\theta \mathbf{R}_\lambda \mathbf{R}_\delta \tag{144}$$

The generic rotation matrix for a left component is

$$\mathbf{R}_{\text{left}} = \mathbf{R}_b \mathbf{R}_\theta \mathbf{R}_\lambda^T \mathbf{R}_\delta^T \tag{145}$$

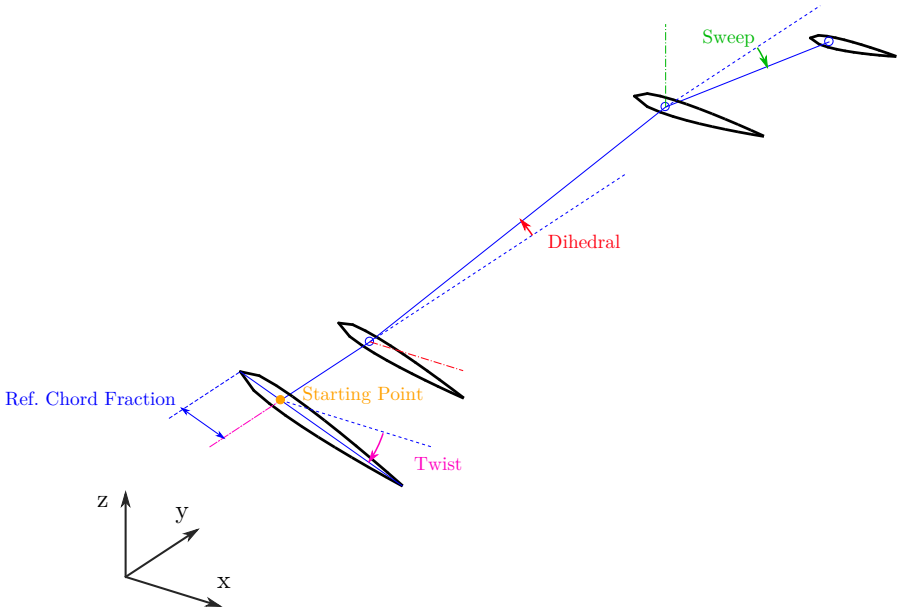


Figure 25: DUST component reference system

2.3.1.2 *Kinematic and Load Variables*

The position of a point P in the global reference frame g of the aerodynamic surface is evaluated as

$$(P_p - O)^g = \sum_{Q \in I_P} w_{pq} \left\{ (Q_q - O)^g + \mathbf{R}_Q^{r \to g} (P_p - Q_q) \right\} \tag{146}$$

where $Q \in I_p$ indicates the subset of structural points Q_q that belong to the aerodynamic component I_p, $(Q_q - O)^g$ is the distance from the origin of the Q_q structural point, and $\mathbf{R}_Q^{r \to g}(P_p - Q_q)$ rotates in global coordinates the distance between the aerodynamic point and the structural one. Its angular velocity $\boldsymbol{\omega}_P$ and velocity $\boldsymbol{v}_P$, respectively, are

$$\boldsymbol{\omega}_P = \sum_{Q \in I_P} w_{pq} \boldsymbol{\omega}_Q \qquad \boldsymbol{v}_P = \sum_{Q \in I_P} w_{pq} \left\{ \boldsymbol{v}_Q + \boldsymbol{\omega}_Q \times (P - Q) \right\} \tag{147}$$

Aerodynamic forces and moments are evaluated at the evaluation points P_e located in the center of each panel and then transferred

For the component of the lifting line, the loads are evaluated at the center of the lifting line, which is placed at 25% of the chord

to the structural nodes using the sum of forces and the transport of moments as follows:

$$\boldsymbol{f}_Q = \sum_{e \in J_Q} w_{qe}\, \boldsymbol{f}_e \quad \boldsymbol{m}_Q = \sum_{e \in J_Q} w_{qe} \{\boldsymbol{m}_e + (P_e - Q_q) \times \boldsymbol{f}_e\} \tag{148}$$

where $e \in J_Q$ indicates the subset of evaluation points that belong to each sub-component J_Q, and the weights w_{qe} are calculated using Equations (138) and (146), by computing the distance between each structural node and the evaluation point.

It can be shown that these expressions of the loads are consistent with the expressions of the kinematic variables. The power of the loads acting on the structure, W_s, is equal to the power of the loads acting on the aerodynamic surface, W_a, which is equivalent to saying that the virtual work performed by the aerodynamic forces is equal to the structural: $\delta W_s = \delta W_a$.

$$W_s = \sum_Q \left\{ \boldsymbol{f}_Q \cdot \boldsymbol{v}_Q + \boldsymbol{m}_Q \cdot \boldsymbol{\omega}_Q \right\} = \tag{149}$$

$$= \sum_Q w_{eq} \left\{ \sum_{e \in J_Q} \boldsymbol{f}_e \cdot \boldsymbol{v}_Q + \sum_{e \in J_Q} [\boldsymbol{m}_e + (P_e - Q) \times \boldsymbol{f}_e] \cdot \boldsymbol{\omega}_Q \right\} = \tag{150}$$

$$= \sum_Q \sum_{e \in J_Q} w_{eq} \left\{ \boldsymbol{f}_e \cdot \boldsymbol{v}_Q + [\boldsymbol{m}_e + (P_e - Q) \times \boldsymbol{f}_e] \cdot \boldsymbol{\omega}_Q \right\} = \tag{151}$$

$$= \sum_e \sum_{Q \in I_e} w_{eq} \left\{ \boldsymbol{f}_e \cdot \boldsymbol{v}_Q + [\boldsymbol{m}_e + (P_e - Q) \times \boldsymbol{f}_e] \cdot \boldsymbol{\omega}_Q \right\} = \tag{152}$$

$$= \sum_e \left\{ \boldsymbol{f}_e \cdot \sum_{Q \in I_e} w_{eq} \left[\boldsymbol{v}_Q + \boldsymbol{\omega}_Q \times (P_e - Q) \right] + \boldsymbol{m}_e \cdot \sum_{Q \in I_e} w_{eq}\, \boldsymbol{\omega}_Q \right\} = \tag{153}$$

$$= \sum_e \{ \boldsymbol{f}_e \cdot \boldsymbol{v}_e + \boldsymbol{m}_e \cdot \boldsymbol{\omega}_e \} = W_a \tag{154}$$

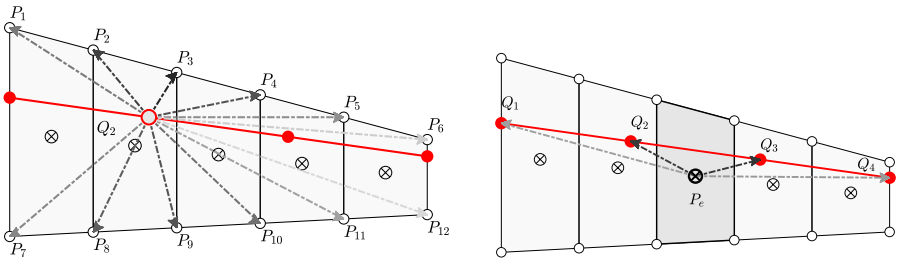


Figure 26: Scheme for motion interpolation (**left**) and force and moment transfer (**right**). Structural points are represented by red dots. Nodes of the aerodynamic mesh and panels centers are represented with plain dot and crosses, respectively.

2.3.2 *Implementation*

Figure 27 shows the flow of information during the coupled simulation between the two solvers for an implicit tight serial scheme. First, the object `precice` of class `t_precice` is declared to handle a coupled simulation through the preCICE library. This object is used to manage both data communication and to update coupled components of the aerodynamic model. Then, the instance of DUST participating in the coupled simulation is created, reading the XML preCICE configuration file. After some preliminary operations, the mesh used to couple the codes is defined, and the fields involved in the communication are initialized. The initialization of the Fast Multiple Kernels, wake, and linear system follows. Before the time loop starts, communication is first established between the coupled codes. The time loop starts with the update of DUST's explicit aerodynamic elements (lifting lines and actuator disks). A checkpoint of the exchanged fields is stored, to be reloaded during sub-iterations of preCICE's implicit coupling. DUST then receives the kinematic variables of the structural nodes from external software MBDyn, and updates the surfaces of the coupled components and the near-field wake elements. The linear system is then updated and solved, calculating the strengths of the vortexes of the surface panels and the vortex lattice elements.

The solution of the non-linear vortex lattice and lifting line problems follows. The aerodynamic forces and moments are reduced to the nodes of the interface between the aerodynamic and structural meshes and sent to MBDyn. The convergence check on the kinematics variables follows. If convergence is not reached, the checkpoint fields are reloaded and a new subiteration begins. If convergence is attained, the time step is finalized, saving the status and updating the wake and the geometry of the uncoupled components for the next time step. At each internal preCICE iteration, the location of the trailing edge point is updated consistently to correctly position the leading edge of the wake and then correctly position the released particle.

2.3.3 *Hinged surfaces modeling*

Modeling the deflection of a control surface is essential for the simulation of aircraft maneuvers and for trimming in the aircraft in free-free condition. In MBDyn, the deflection of a control surface can be easily modeled as a rigid/deformable body properly constrained to the fixed part of the vehicle to allow only its relative rotation about the hinge axis. In DUST, the possibility of including a control surface in the aerodynamic mesh has only recently been introduced. In the following, the description of the model implemented for hinged sur-

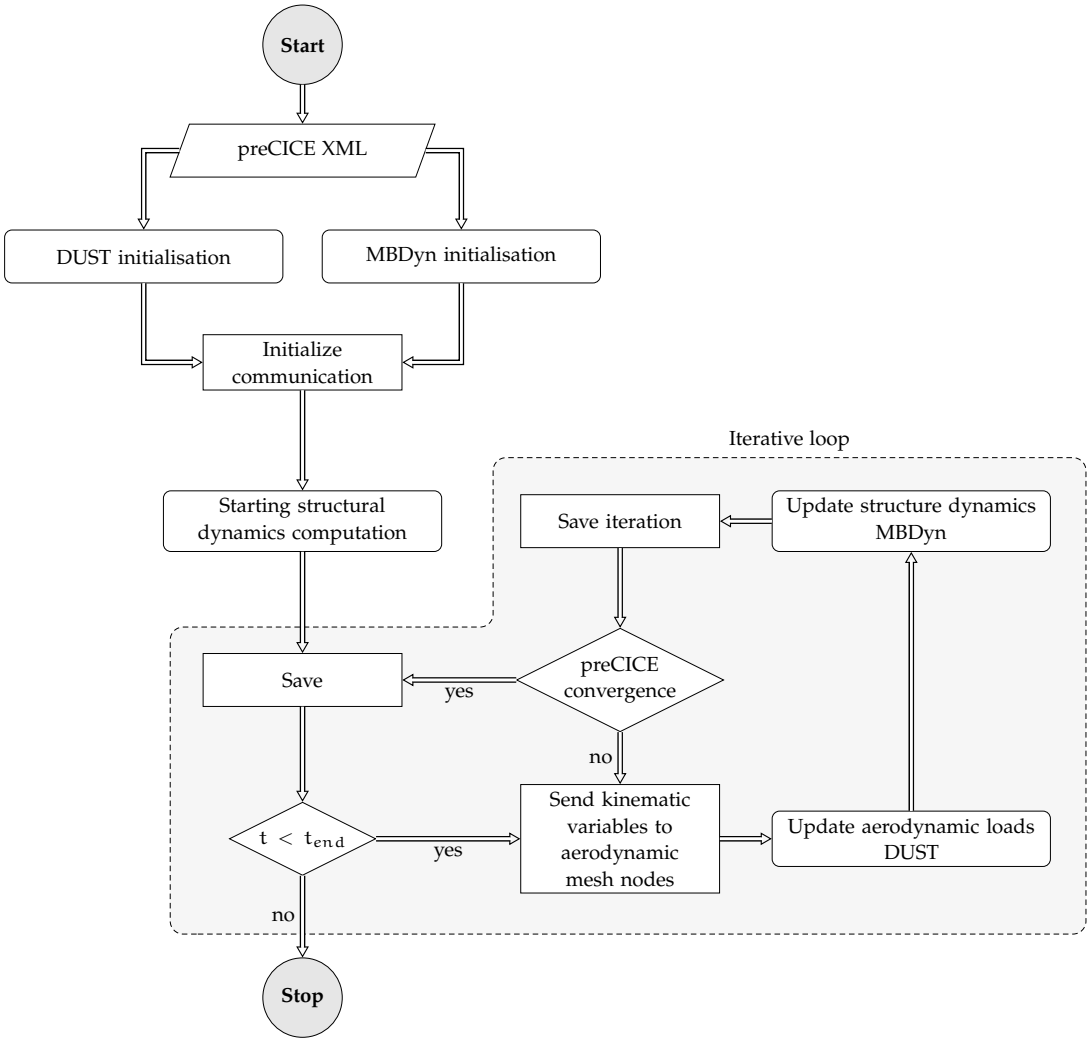


Figure 27: Flowchart of the implicit communication managed by preCICE between DUST and MBDyn [38].

faces in DUST is first introduced with a two-dimensional example and then extended to three-dimensional deformable components.

As outlined by the scheme in Fig. 28 (left), in a two-dimensional problem the control surface can be defined in the local reference frame of the component, by means of the hinge axis position H, the chordwise direction ξ and a blending region $[-u, u]$ introduced to avoid irregularities in the mesh as the surface rotates at an angle θ. Furthermore, in the 2D modeling, it is assumed that the rotation axis $\hat{\boldsymbol{h}}$ is orthogonal to the plane of the airfoil.

As shown in Fig. 28, an orthonormal reference frame for the hinge is defined with the origin in H and the axes $\hat{\boldsymbol{\xi}}$, $\hat{\boldsymbol{\eta}} = \hat{\boldsymbol{h}} \times \hat{\boldsymbol{\xi}}$. The position of a point with respect to this reference frame is as follows.

$$\boldsymbol{r} = \xi\,\hat{\boldsymbol{\xi}} + \eta\,\hat{\boldsymbol{\eta}} + h\,\hat{\boldsymbol{h}} \tag{155}$$

Three regions are defined using the coordinates based on this reference frame:

1. $\xi \leqslant -u$: no influence on control surface rotation;
2. $\xi \geqslant u$: rigid rotation on the hinge:

$$\Delta\boldsymbol{r} = \sin\theta\hat{\boldsymbol{h}} \times \boldsymbol{r} + (1 - \cos\theta)\hat{\boldsymbol{h}} \times \hat{\boldsymbol{h}} \times \boldsymbol{r} \tag{156}$$

3. $-u \leqslant \xi \leqslant u$: blending region to avoid irregularities, defined as an arc of a circle whose center is located at point C and whose radius is:

$$\overline{CC'} = \frac{\overline{C'H}}{\tan\dfrac{\theta}{2}} \qquad \text{where} \quad \overline{C'H} = u \tag{157}$$

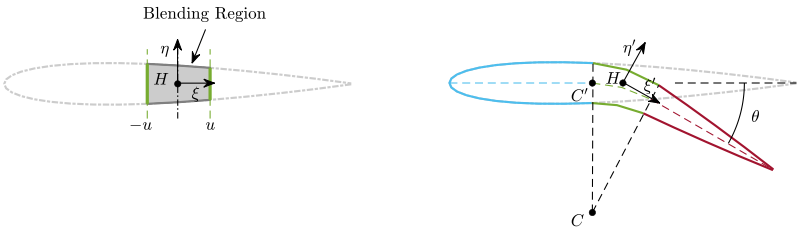


Figure 28: Scheme of the two-dimensional hinged surface configuration

In a three-dimensional problem, the reference configuration of a control surface for a generic swept wing is defined in the wind axis reference frame of the component, as shown in Fig. 29.

Therefore, the aerodynamic sections involved in the deflection of the control surface are those that satisfy condition $y(A) < y(P) <$

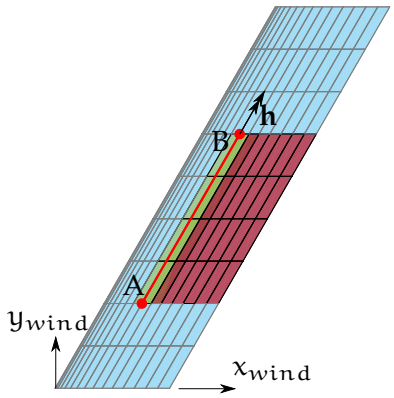


Figure 29: Hinge reference system for a swept wing.

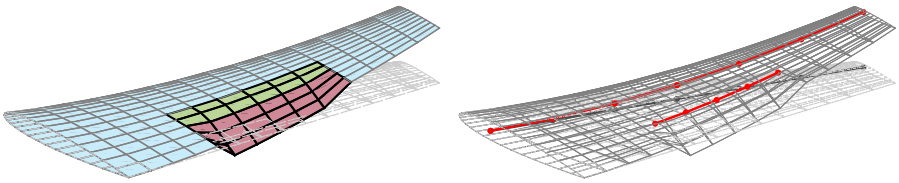


Figure 30: Control surface deflection for a deformable wing.

$y(B)$, where $y(P)$ is the ordinate of the aerodynamic mesh point $P_i - th$ expressed in the wind reference system. As in the 2D case, one can define the three regions for each stripe identified at the previous point. The y coordinate of the origin of the sectional reference frame is determined by linear interpolation between points A and B.

When the movable surface is coupled with a structural component in MBDyn, the orientation of the hinge in the DUST nodes comes from the orientation of the nodes of the MBDyn model. The rotation axis is defined as $\hat{\mathbf{h}} = \frac{(B-A)}{\|B-A\|}$. Each point of the movable surface is linked to the hinge nodes, which obtains its kinematic variables as a weighted average of the motion induced by the rotation of the hinge nodes. Weights w_{ph} are evaluated using only the h components of the vectors connecting the control surface points to the hinge nodes.

The weights w_{ph} are then combined with the structural interpolation weight w to allow deformation of the structure, as shown in Fig. 30. The weights are calculated for each point of the aerodynamic mesh, while the weights w_{ph} are specifically used to impose the rigid deflection of the moving surface.

3

GENERAL VALIDATION AND VERIFICATION

3.1 INTRODUCTION

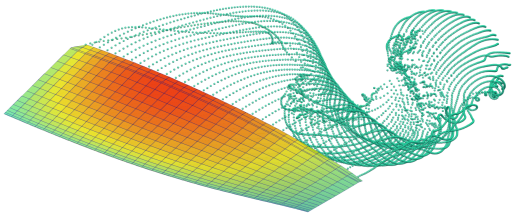


Figure 31: Goland Wing in flutter: DUST wake visualization

The implemented codes are validated considering several test cases [38, 39, 169, 171]. The first validation is provided by analyzing Goland wing, a numerical test case that was widely used in the literature as a benchmark for flutter predictions. This validated the coupling procedure between DUST and MBDyn. The second groups of test cases are related to the validation of the nonlinear vortex lattice element. Three cases are considered:

1. A fixed finite wing with a non-symmetric profile: the validation is made by comparing the nonlinear vortex lattice against the lifting line element and the CFD results.
2. A propeller operating at different inflow conditions and angular velocities. Comparisons are made with the lifting line and the experimental results.
3. XV-15 rotor equipped with metallic blades: here the validation is made comparing the experimental result and the CFD results in hover condition, advanced flight and axial flow.

3.2 DUST-MBDYN COUPLING: GOLAND WING

This section presents the application and validation of the coupled software to the Goland wing [71], which is widely used in the literature to test and validate aeroelastic codes. This low aspect ratio ($AR \approx 3.33$) is also interesting for highlighting the impact on flutter calculations of 2D and 3D aerodynamic models. Figure 32 shows the layout of the problem. EA indicates the elastic axis; CG indicates the

center of gravity axis. U_∞ is the free-stream velocity. All relevant geometrical and structural properties are reported in Table 2. They have been obtained from [146].

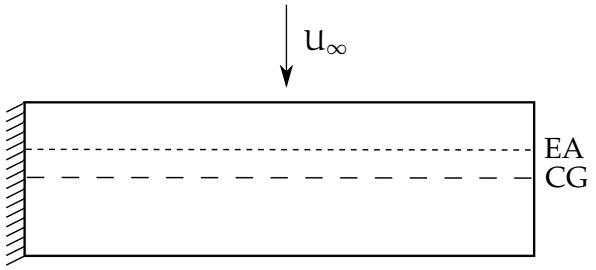


Figure 32: Goland's wing model layout

Table 2: Goland wing properties and flight conditions [146].

Wing properties			
Half span	6.096 m	Inertia/unit span[1]	8.64 kg m
Chord, c	1.8288 m	Mass/unit span	35.71 $kg\,m^{-1}$
Elastic axis	$0.33 \cdot c$	Torsional stiffness	0.99×10^6 $N\,m^2$
Center of gravity	$0.43 \cdot c$	Bending stiffness	9.77×10^6 $N\,m^2$
Flight condition			
Air density	1.020 $kg\,m^{-3}$	AoA Perturbation	0.05 deg

Two aerodynamic meshes are considered. In the first case, the wing is modeled as a flat plate using linear and nonlinear vortex lattice elements, while in the second case it is modeled with surface panels reproducing its geometrical shape and thickness. The results of a convergence analysis on the computed aerodynamic loads indicated the need to use 30 elements in the span-wise direction to obtain a correct spatial discretisation for both models. The flat mesh vortex lattice requires 30 uniform divisions in the chordwise direction, whereas the surface panel model requires 30 divisions for the lower and upper sides of the wing, with half-cosine-refined distribution at leading edge. The structural model built in MBDyn consists of four beams based on a C^0 beam discretisation based on the concept of finite volume proposed in [68]. A satisfactory number of beam elements is obtained by satisfying a convergence requirement in the first four modes of the wing, as indicated by comparing the frequencies of the first four normal vibration modes of the wing computed by MBDyn with respect to Goland's work [71] and NASTRAN shown in Table 3.

Tight coupled time-marching simulations are performed using a time discretization of 0.001 s and considering for the evolution of the wake particles a fourteen-chord-long box behind the wing, resulting in a developed wake made of about three thousand vortex

Table 3: Comparison of the first four natural frequencies computed for the Goland wing [71].

	Goland [71] Hz	NASTRAN Hz	MBDyn Hz
1st Bending	7.66	7.66	7.66
1st Torsion	15.24	15.24	15.21
2nd Bending	38.80	38.59	38.54
2nd Torsion	55.33	54.84	54.79

particles. The computation time to perform a time-marched coupled simulation with a total duration of one second, using a workstation equipped with an Intel® Core™ i9-9980XE processor running at a base frequency of 3.00 GHz, with 18 physical cores and 2 threads for each core, was about 9 and 13 min for the linear and non-linear vortex lattice simulation, and about 29 min for surface panel.

To study the flutter instability, a non-zero angle of attack non-zero of 0.05 deg is introduced as a perturbation, as in reference [134]. The frequency and damping of the response are identified from the time history of wingtip deflection using the matrix pencil estimation (MPE) method [80]. Figure 33 presents the z displacement of the last external structural node in the onset of the flutter calculated with the SP model. The red line corresponds to a stable damped response, whereas the blue line shows the condition of incipient flutter, in which a constant-amplitude free oscillation is reached. The green line shows the unstable response at a speed greater than the flutter speed. A corresponding representation of the deformed mesh associated with the bending-torsion instability and the related distribution of the pressure coefficient is shown in Figure 33 (left). These results indicate the ability of the coupled simulation to correctly capture fixed-wing flutter.

Further deatails about this method are given in Section 7.2.2

Table 4 reports a comparison of the speed and frequency of flutter calculated by several independent authors. The results computed with the coupled code agree quite well with those obtained by similar codes using 3D aerodynamic models [134, 146, 198]. In detail, the discrepancy with the results obtained with the same MBDyn structural model, but using its built-in aerodynamics based on two-dimensional unsteady strip theory, indicates the superior capability of the coupled code in the investigation of aeroelastic problems. The comparison of the results from the literature reported in Table 4, obtained with 2D and 3D models, confirms the need for a three-dimensional aerodynamic model for a correct and realistic flutter analysis of low aspect ratio wings.

To evaluate the capabilities of the coupled code using the different aerodynamic models employed, Figure 34 presents the frequency

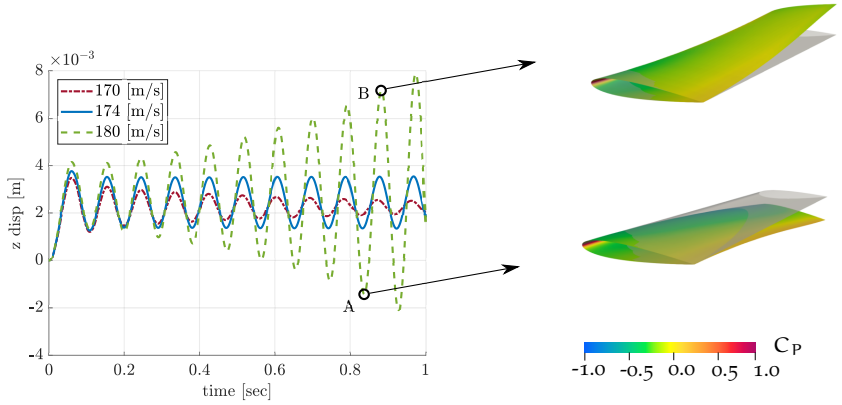
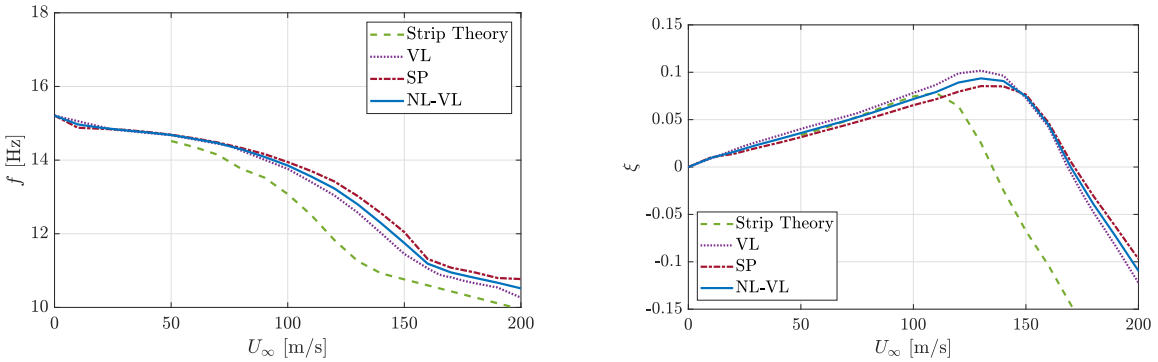


Figure 33: Time history of the Goland wing-tip deflection evaluated with surface panels aerodynamic mesh at three different wind speeds.

Table 4: Comparison of flutter speed and frequency computed for Goland wing.

Author	*Model*	V_f m s^{-1}	f_f Hz
Goland [71]	Analytical	137.2	11.25
Patil et al. [146]	Intrinsic beam + strip theory	135.6	11.17
Present Work	MBDyn's built-in strip theory	135.1	11.07
Wang et al. [198]	ZAERO	174.3	-
Wang et al. [198]	Intrinsic beam + UVLM	163.8	-
Murua et al. [134]	SHARP, Displacement beam + UVLM	165.0	10.98
Present Work	DUST (VL)-MBDyn	168.2	10.84
Present Work	DUST (NL-VL)-MBDyn	174.2	11.06
Present Work	DUST (SP)-MBDyn	174.2	11.06

and damping of the torsional mode of the first beam of the wing as functions of the free-stream velocity.



(a) Frequency (b) Damping

Figure 34: Frequency f and damping ratio ξ of the first torsional modes vs speed U_∞ for Goland's wing. Coupled DUST-MBDyn simulations results (VL, NL-VL and SP mesh) and MB-Dyn results with 2D strip theory aerodynamic model

The numerical results of the coupled simulations using a panel mesh (SP) show slightly higher aerodynamic damping compared to those obtained using VL. A predicted flutter speed increase of approximately 3.7% is observed (Figure 34 right). However, the fairly limited differences in the results of the two models indicate that, for simple configurations, without complex aerodynamic interaction between bodies, a vortex lattice mesh is more convenient than surface panels, as computational cost is significantly reduced.

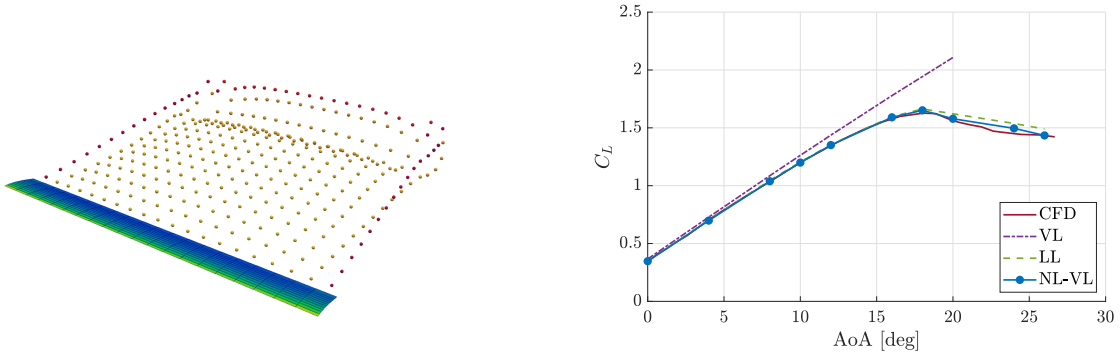
3.3 NON-LINEAR VORTEX LATTICE

The present section describes the results of the validation of the novel non-linear vortex lattice element tested on both fixed-wing and rotary-wing benchmark cases available from the literature for aerodynamic and aeroelastic problems.

3.3.1 *Finite Wing*

Firstly, a simple finite wing benchmark case is considered to test the correct implementation of the method. In particular, the test case consists of a rectangular wing with an aspect ratio equal to 12, a constant chord of 1 m and a constant NACA4415 airfoil section. Figure 35b shows the comparison of the wing lift coefficient as a function of the angle of attack calculated by high-fidelity CFD simulations by Hosangadi et al. [154] and by DUST using both the LL and VL elements already implemented and the novel non-linear vortex lattice method (NL-VL). The effectiveness of the non-linear vortex-lattice method is evident, as shown by the remarkable agreement with the curves

obtained using the lifting-line approach implemented in DUST and with high-fidelity CFD simulations, both considering viscous contributions for airloads evaluation. This result confirms the correctness of the novel implementation based on the integration of the viscous aerodynamic 2D data and shows the capabilities of the method to capture stalled and post-stalled conditions with respect to the purely potential VL model.



(a) Wake visualization

(b) Lift coefficient (C_L) as function of angle of attack

Figure 35: Comparison of the NACA 4415 wing lift coefficient (C_L) as function of angle of attack: high-fidelity CFD simulations results from [154] (CFD), DUST vortex lattice (VL), lifting line (LL) and non-linear vortex lattice (NL-VL) simulations results.

3.3.2 *Propeller*

As second test case, the results of the Workshop for Integrated Propeller Prediction (WIPP) [79] are compared with the DUST simulations. The WIPP model represents a useful test case for validating aerodynamic codes, since an intense experimental campaign was carried out, the results of which constitute a powered wind tunnel test database. The tests were carried out at the Lockheed Martin Low Speed Wind Tunnel (LSWT) in Marietta, Georgia, USA, as part of the NASA/Armstrong X-57 research program [79]. The model consisted of a 40.5% scale semi-span wing designed for the wind tunnel test campaign to be mounted on the LSWT external balance.

The span, from the tunnel floor to the top of the nacelle, is equal to 1.772 m, with a root chord of 0.295 m and a tip chord of 0.218 m. The wing has a sweep at the leading edge of 1.9°. The model incorporates a non-metric boundary layer splitter plate 0.163 m high, which is mounted on the tunnel floor. The model is equipped with an existing C-130 four-blade propeller with a diameter of 0.411 m and a blade pitch angle of 38°. The propeller was also extensively used in previous activities to test a 10% scale C-130 aircraft model. The propeller is mounted on a nacelle at the tip of the wing. The nacelle is 0.613 m long with a maximum diameter of 0.12 m. The accurate geometries

necessary for the construction of the numerical model are extracted by scanning a digital version of the wind tunnel model.

Figure 36 shows the comparison of the propeller thrust coefficient as a function of the RPM for three different conditions of the Mach number in the free stream using the lifting line and the non-linear vortex lattice.

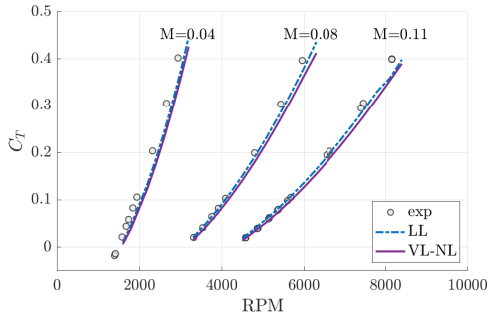


Figure 36: Comparison of the propeller thrust coefficient as a function of angular velocity for different free-stream Mach number.

Very good agreement with the experimental data is found for all three free-stream Mach numbers considered for both elements against the experimental results, even if for the non-linear vortex lattice the thrust is slightly under predicted.

A global visualization of the flow field computed by DUST around the complete WIPP model is provided in Fig. 37, showing the Q-criterion isosurface colored by the non-dimensional free-stream velocity component and the contours of the pressure coefficient on the model surface. The flow representation clearly shows the DUST capability to capture the interaction of the helical structure generated by the propeller blade tip vortices with the wing-nacelle surface. Moreover, the DUST numerical solution also appreciates a secondary helical vortical structure generated from the blades with the trailing edge interacting with the nacelle surface. Further details about this

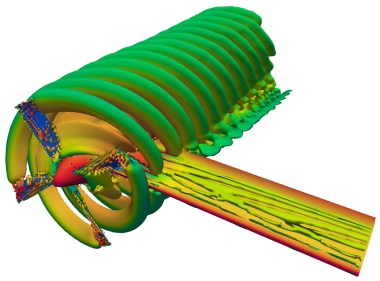


Figure 37: Visualization of the flow field computed by DUST around the complete WIPP model: Q-criterion iso-surface colored by the non-dimensional free-stream velocity component and contours of pressure coefficient on the model surface M_∞=0.11, C_T=0.4, AoA=0°

test case, and in particular related to the DUST propeller-wing interaction capability, can be found in [170].

3.3.3 *Rotor*

A third benchmark case, this time in the field of rotorcraft, consisting of the simulation of the proprotor of the XV-15 tiltrotor with metal blades in different flight conditions, that is, hover condition, forward flight in helicopter mode and airplane mode. The main rotor data and the aerodynamic profiles are provided in Table 5, and Table 6.

For the present test case, the three propeller blades were modeled as rigid blades using non-linear vortex lattice elements only. The tabulated aerodynamic performances of the propeller airfoils were taken from [62]. DUST simulations were performed considering a propeller revolution length of 10 with a discretization time of 5° of the azimuthal angle of the blade. The computational time required to complete the simulation of the rotor configuration was about 8 minutes using a workstation with a Dual Intel Xeon Gold 6230R @ 2.10Ghz with 104 core processor.

In the following, the results of DUST simulations obtained using the NL-VL approach are compared with the recent results of high-fidelity numerical simulations obtained by Jia et al. [83] using a Detached Eddy Simulation (DES) approach and with the experimental data collected in the test campaigns described by Felker et al. [62] and Betzina [25].

Table 5: XV-15 metallic blade rotor: main data

Parameter	Value	Unit
Blade	3	
Solidity	0.0891	
Radius (R)	3.81	m
Precone β	2.5	deg
Chord	0.3556	m
Twist	45	deg
Nominal speed (Ω)	589	RPM

3.3.3.1 *Hover flight condition*

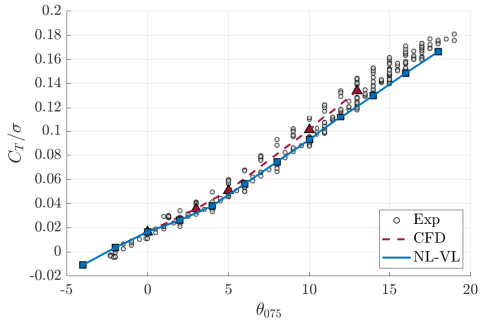
Figures 38a and 38b show the comparison of the rotor thrust coefficient (C_T) as a function, respectively, of the collective angle of the blade and of the rotor torque coefficient(C_Q), while Fig. 38c shows

Table 6: XV-15 metallic blade rotor: profiles

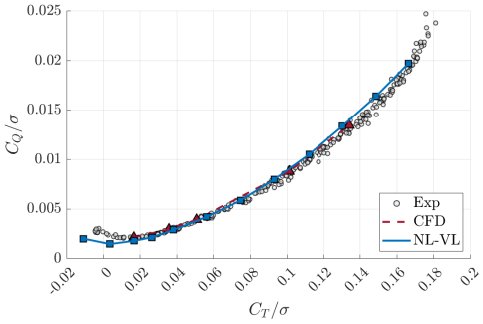
Profile	start (r/R)	end (r/R)
NACA 64-935	0.09	0.13
NACA 64-528	0.13	0.34
NACA 64-118	0.34	0.655
NACA 64-(1.5)12	0.655	0.9
NACA 64-208	0.9	1

the comparison of the rotor figure of merit (FM). These coefficients are calculated as follows:

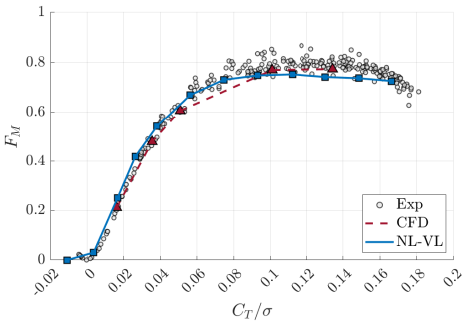
$$C_T = \frac{T}{\rho\pi\Omega^2 R^4} \qquad C_Q = \frac{Q}{\rho\pi\Omega^2 R^5} \qquad F_M = \frac{C_T^{3/2}}{C_Q\sqrt{2}} \tag{158}$$



(a) C_T/σ vs collective angle



(b) C_Q/σ vs C_T/σ



(c) F_M vs C_T/σ

Figure 38: Comparison of the aerodynamic performances for the XV-15 proprotor in hover. Experimental data taken from [62] (Exp), numerical data from [83] (CFD) and DUST (NL-VL)

The performance curves obtained with DUST quite well resume the behavior of the experimental data in the entire range of collective blade angles tested. Moreover, the curve comparison shows that the DUST approach provides capabilities similar to those of a DES approach in terms of aerodynamic performance evaluation for such a

case, but requires a much lower amount of computational effort. In particular, the figure of merit comparison, shown in Fig. 38c, clearly indicates very good agreement with experiments of the overall performance of the rotor in hover calculated by DUST, particularly at low C_T/σ. Moreover, the quite lower computational cost required by the DUST approach enabled one to perform a quite higher amount of simulations with respect to high-fidelity CFD, thus covering with a finer step of collective angle the whole operational range described by the experimental curve.

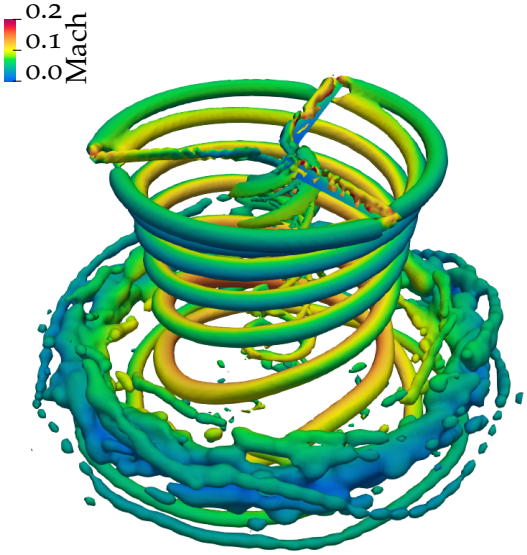


Figure 39: Wake visualization of the XV-15 proprotor in hover by means of isosurfaces of the Q-criterion computed by DUST colored by Mach number.

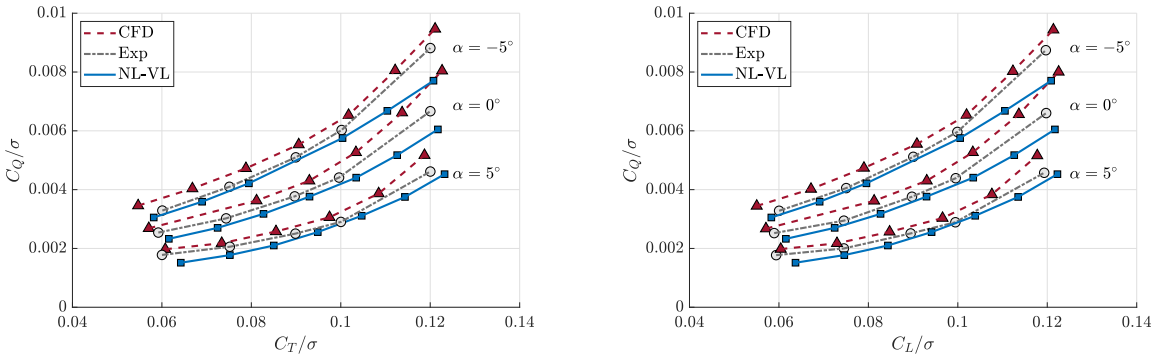
A flow field representation of the present test condition is presented in Fig. 39, showing the helical vortical structure of the proprotor wake in hover computed by DUST highlighted by the iso-surfaces of Q-criterion.

3.3.3.2 *Forward flight condition*

In forward flight conditions, helicopter mode configurations of the XV-15 proprotor are investigated with DUST considering three shaft angle attitudes, that is, $\alpha = -5°$, $\alpha = 0°$ and $\alpha = 5°$, at advance ratio 0.17. This choice allowed, analogously to what was done in [83], to investigate DUST capabilities in both propulsive and descending forward flight conditions.

Figures 40a and 40b show the comparison of the rotor torque coefficient (C_Q) as a function, respectively, of the rotor thrust coefficient(C_T) and of the lift coefficient (C_L).

A quite good agreement with experiments is found also for this flight condition for the proprotor aerodynamic performance calcu-



(a) C_Q/σ vs C_T/σ (b) C_Q/σ vs C_L/σ

Figure 40: Comparison of the aerodynamic performances for the XV-15 proprotor in forward flight. Experimental data taken from [25] (Exp), numerical data from [83] (CFD) and DUST (NL-VL)

lated by DUST. In particular, the DUST results slightly underestimate the experimental performance curves in the entire range of collective blade angles considered. However, the discrepancies of DUST results from the experimental data are relatively lower with respect to the CFD simulation results, particularly in the lower range of collective angles tested. However, in a higher thrust or lift coefficient range, larger discrepancies are found with respect to the experimental data for the DUST representation of proprotor aerodynamic performance. This behavior could be related to the limitation of the DUST approach in reproducing accurately high blade loading conditions characterized by more important viscous effects and separated flow regions.

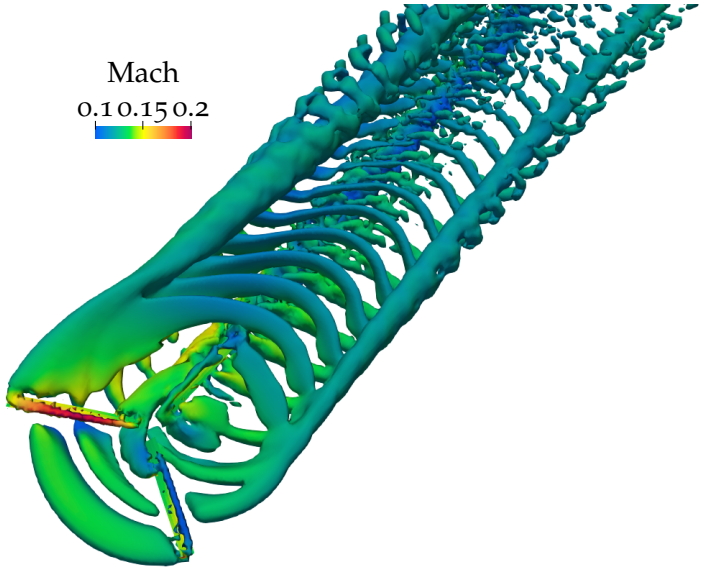


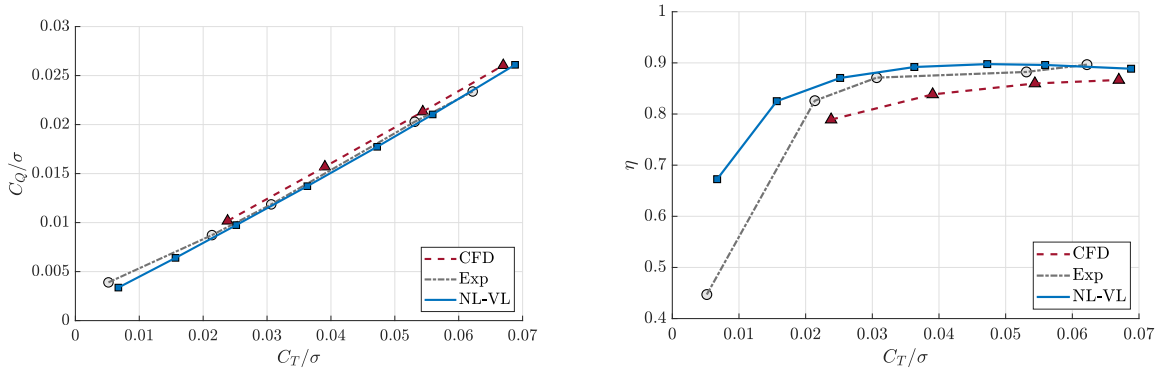
Figure 41: Wake visualization of the XV-15 proprotor in advanced flight at 5° collective, and α =5° by means of iso-surfaces of Q-criterion computed by DUST colored by Mach number.

In fact, these flight conditions are also characterized by consistent aerodynamic interactional effects, as shown by the flow visualization presented in Fig. 41 highlighting the interaction of the tip vortices with the downstream blades.

3.3.3.3 *Airplane Mode flight condition*

The aircraft mode configurations of the XV-15 proprotor are simulated using DUST at an advance ratio of 0.337 for different collective blade pitch angles. The DUST capabilities to reproduce the aerodynamic performance of the proprotor in this flight condition are evaluated by comparing the torque coefficient (C_Q) and the propulsive efficiency (η) as a function of the thrust coefficient (C_T); see, respectively, Figs. 42a and 42b. The propulsive efficiency (η) is calculated as follows:

$$\eta = \frac{C_T V_\infty}{C_Q \Omega R} \tag{159}$$



(a) C_Q/σ vs C_T/σ (b) C_Q/σ vs C_L/σ

Figure 42: Comparison of the aerodynamic performances for the XV-15 proprotor in airplane mode flight. Experimental data taken from [25] (Exp), numerical data from [83] (CFD) and DUST (NL-VL)

Very good agreement is found between the DUST simulation results and the experimental data for these flight conditions. In particular, the discrepancies with respect to experimental curves exposed by the results of DUST simulations are quite lower than those obtained by high-fidelity CFD simulations, thus confirming the suitability of the DUST approach for an accurate evaluation of propeller aerodynamic performance. In fact, a slight overestimation of the propulsive efficiency evaluated by the experiments can be observed from the results of the DUST simulation over almost the entire range of rotor thrust conditions tested, while a higher discrepancy of the order of 20% is found only for the lowest blade load condition tested.

The flow field representation of an airplane mode flight condition computed by DUST is shown in Fig. 43 by means of isosurfaces of the Q-criterion, highlighting a quite coherent helical vortical structure

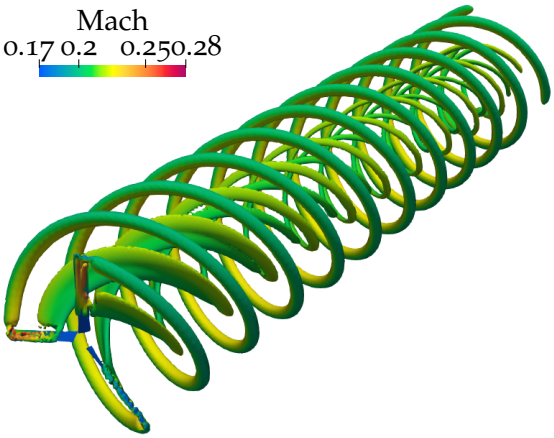


Figure 43: Wake visualization of the XV-15 proprotor in airplane mode by means of isosurfaces of Q-criterion computed by DUST colored by Mach number.

of the proprotor wake without interactions due to the free-stream velocity dragging characterizing this flight condition.

Part II

NUMERICAL MODEL AND RESULTS

4

AEROELASTIC ROTOR MODEL

4.1 INTRODUCTION

In the present chapter, the detailed aeroelastic model of the XV-15 rotor with Advanced Technology Blade (ATB) is treated. Starting from a detailed description of the original data and their pre-processing in order to translate them from the CAMRAD-JA input to MBDyn-DUST, the rotor dynamics and aerodynamic performance under hover conditions are validated against the experimental results [18, 31, 135, 136]; finally, the aeroelastic performance and stability in axial flow are analyzed.

The rotor model is made up of three main components:

1. Blade-yoke stick model.
2. Detailed flexible dual-path control chain.
3. Drive system.

4.2 ROTOR CONTROL CHAIN

The multibody model of the XV-15 proprotor is made up of ATB blades. The rotor is stiff-in-plane with a gimballed hub. In MBDyn, the universal gimbal joint has been modeled as an ideal homokinetic joint, neglecting the 2/rev components caused by rotor flapping [28].

Based on the information provided by [31], the layout of a control chain representative of that of the XV-15 proprotor shown in Fig. 44 was modeled. The flow chart in Fig. 46 details the blade pitch control system implemented in MBDyn, where the role of each component is

- `Nacelle`: This node represents the actual connection between the extremity of the nacelle and the rotor; when the isolated rotor is analyzed, this node is grounded.
- `Control`: this is the node to which controls (collective and cyclic) are prescribed. To decouple the longitudinal and lateral cyclic input, the node is defined in a reference system that rotates around the shaft by the angle $\psi_{sp} = \tan^{-1}(x_{sp}/y_{sp})$, where x_{sp} and y_{sp} are the in-plane components of the point where the pitch link is connected to the swashplate. The real XV-15 did not have lateral cyclic command, since lateral movements were commanded by a differential collective; however, for the scope of this model, it was advisable to have both lateral and longitudinal cyclic commands.

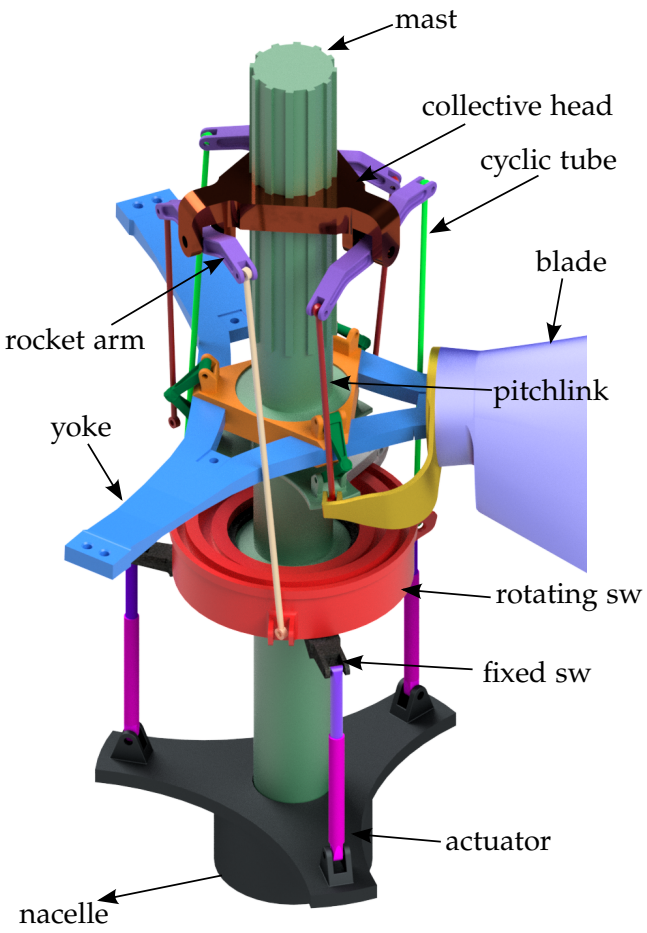


Figure 44: Layout of the XV-15 proprotor control chain

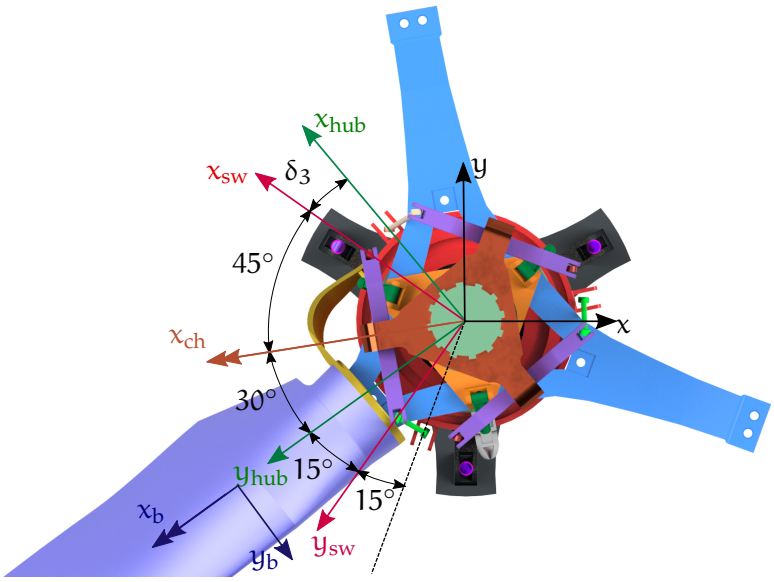


Figure 45: Layout of the XV-15 proprotor control chain: top view

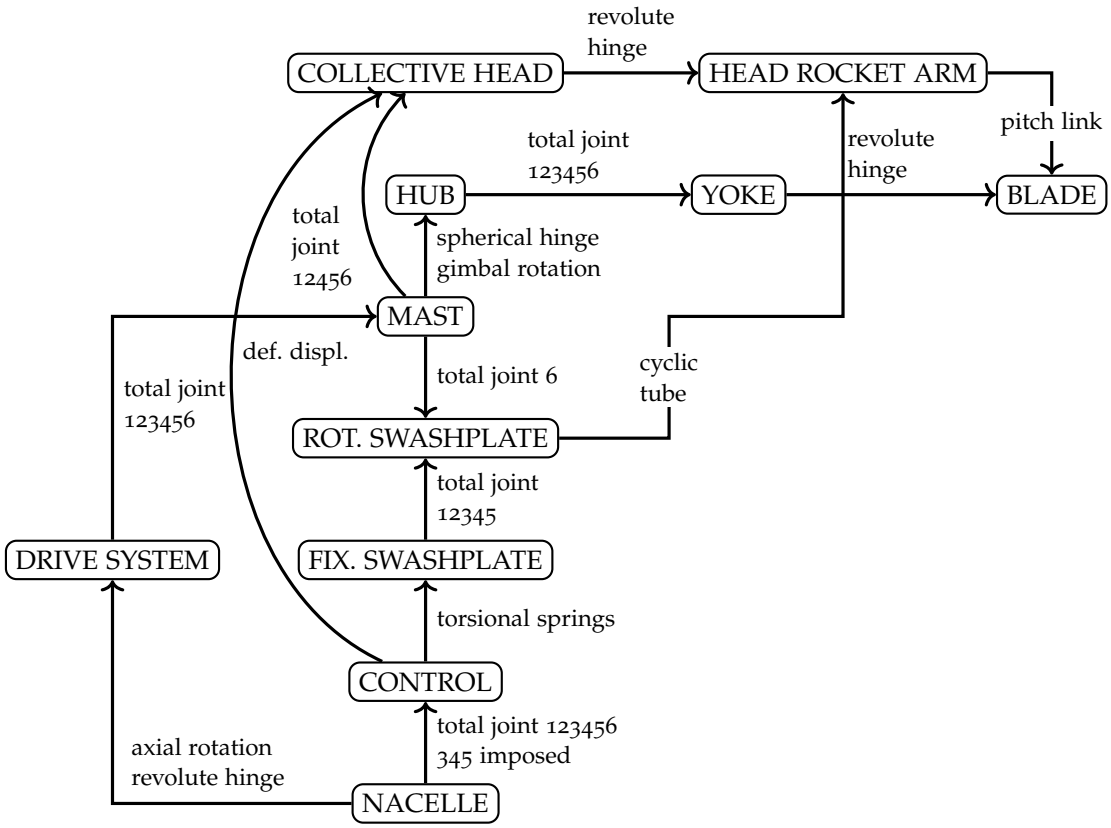


Figure 46: Flowchart of the MBDyn model of the XV-15 proprotor, particularly showing the individual blade pitch control system components and their connections for the dual control path.

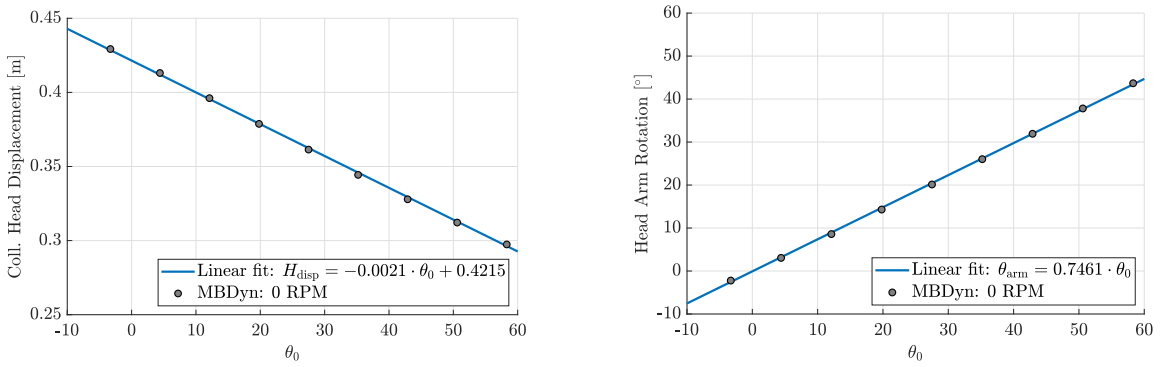
- `Fixed Swashplate`: This node represents the non-rotating portion of the swashplate. Its in-plane displacements and rotation about the shaft axis are restricted to the `Control` node. To account for the flexibility of the control chain, it is connected to the `Control` node by two torsional springs that were tuned to match the pitch mode frequency at 2.89/rev [31].
- `Rotating Swashplate`: This node is connected to the `Fixed Swashplate` by a revolute hinge and to the `Mast` node (defined later) by a joint that makes it rotate along with the shaft itself.
- `Collective Head`: This node is connected to the `Control` node by a deformable spring along the shaft axis, to account for the flexibility of the collective control path; it is placed in the rotating reference system.
- `Head Rocket Arm`: This node is connected to the `Collective Head` node by a revolute hinge and to the rotating swashplate by means of a rod (cyclic tube). Then, through the pitch link, collective and cyclic commands are transmitted to the blade.
- `Drive system`: This node represents the link between the rotor model and the drive system complex.
- `Mast`: This node drives the `Hub` (defined later) and the `Rotating Swashplate` nodes. It is connected to the `Nacelle` node by means of a revolute hinge, whose angular velocity is prescribed.
- `Hub`: This node is restricted to the `Mast` node by a spherical hinge and a gimbal joint [118]: the combination of these two joints represents an ideal constant velocity joint. To increase the first gimbal frequency to the 1.02/rev value, two torsional springs were added between the `Hub` and `Mast` node.
- `Yoke`: This component represents the flexible beam that connects the hub to the blade. The blade-to-yoke connection is obtained by using a single load path through a revolute hinge.
- `Blade`: This element is modeled as a flexible beam, is attached to the yoke through the connection of the pitch bearing, and the angle of inclination is imposed by the pitch link rod.

All rotor data were taken from the original CAMRAD-JA model presented in [31], here reported for clarity in Table 7. The complete list can be found in Table 21. Figure 47 shows the rotor collective kinematics for an imposed rotation of the collective at 0 RPM. An almost linear behavior is found for both collective head displacement ($R^2 = 98.7\%$) and head rocket arm rotation ($R^2 = 96.5\%$). Thus, to save computational time, these two linear relations will be adopted in order to start the rotor in an almost trimmed condition: the blade

Table 7: XV-15 main rotor data [31].

Rotor data		
Number of blades	3	
Solidity	0.103	
Radius	3.81	m
Precone	2.5	deg
Pitch-flap coupling, δ_3	−15	deg
Sweep	0.8870	deg
Droop	0.4617	deg
Chord	0.41	m
Helicopter speed (HP)	601	RPM
Airplane speed (AP)	480.8	RPM

root is oriented by an initial θ_0 angle, and consequently the reference systems of collective head and head rocket arm are placed.



(a) Rotor collective head displacement

(b) Head rocket arm rotation

Figure 47: ATB rotor control chain displacement and rotation for an imposed collective angle at 0 RPM

4.3 BLADE-YOKE STICK MODEL

The blade-yoke model is composed of an aerodynamic and structural part. The aerodynamic part is modeled both in MBDyn, using the built-in bidimensional aerodynamic beam, and in DUST using the non-linear vortex lattice element.

Instead, the structural part is modeled only in MBDyn using the finite-volume beam element, translating the input from the original CAMRAD-JA model presented in [31]. In the following section, a detailed description of the beam data preprocessing is carried out to facilitate the model reproduction.

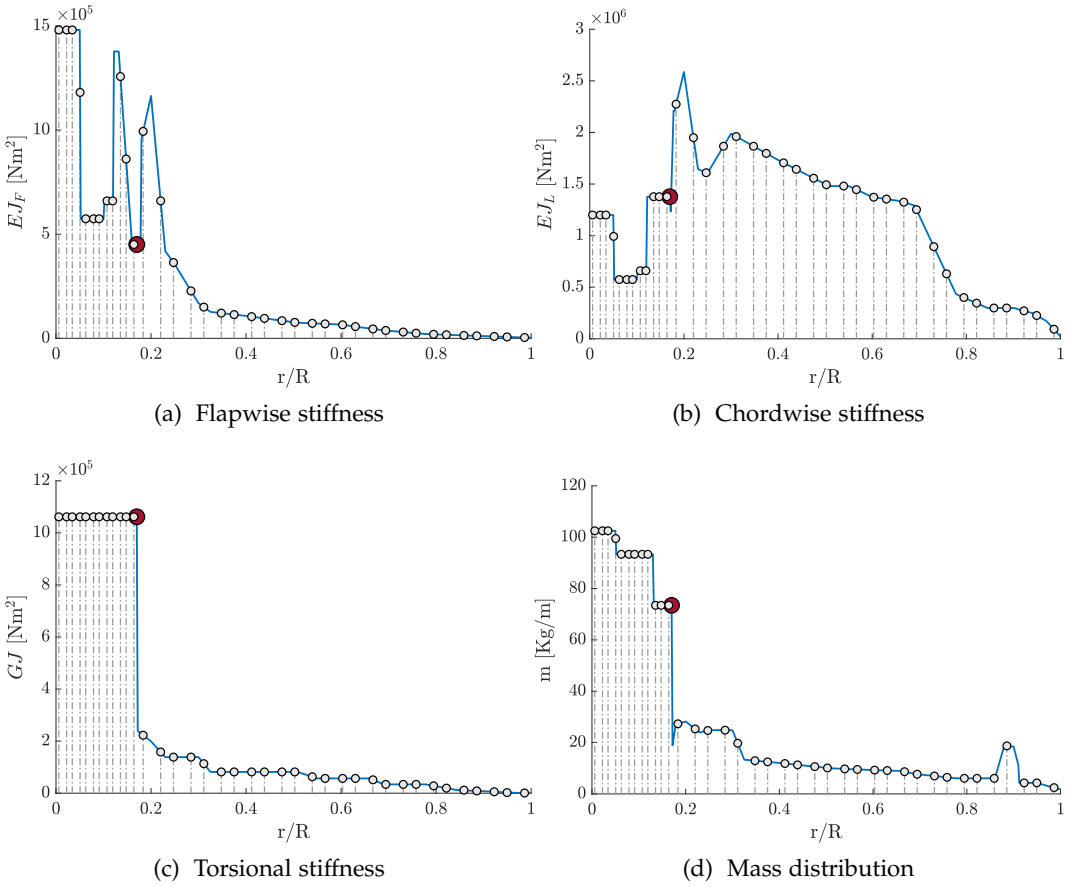
4.3.1 *CAMRAD-JA input data*

In CAMRAD-JA the properties of the aeroelastic beam are defined with respect to the following reference system.

- The X-axis coincides with the section chord and goes from the leading edge to the trailing edge.
- The Y-axis is along the beam elastic axis and goes for beam root to tip.
- The Z-axis is determined by the right and rule.
- The origin is the point of intersection between the section plane and the elastic axis and lies on the chord.

In CAMRAD-JA the following inputs are available:

- Root geometry:
  - *CONE*: precone angle β (positive up)
  - *DROOP*: droop angle δ (positive downwards from precone)
  - *SWEEP*: sweep angle Λ (positive aft)
- Structural and Inertial Data:
  - *MRI*: number of radial station
  - *TWISTI*: blade twist θ_{tw}
  - *XI*: offset of center of gravity aft of the elastic axis
  - *XC*: offset of tension center aft of the elastic axis
  - *KP2*: polar radius of gyration about the elastic axis: $KP2 = \int_A E(y^2 + z^2)dA \,/EA$ where EA is the axial stiffness.
  - *EIZZ*: flapwise bending stiffness: $\int_A Ez^2 dA$
  - *EIXX*: chordwise bending stiffness: $\int_A E(x - XC)^2 dA$
  - *ITHETA*: section moment of inertia about elastic axis $I_\theta = \int_A (y^2 + z^2)dm$
  - *GJ*: torsional stiffness
- Aerodynamic data:
  - *CHORD*: blade chord
  - *TWISTA*: blade aerodynamic twist
  - *XA*: offset of the aerodynamic center aft of the elastic axis.



(a) Flapwise stiffness

(b) Chordwise stiffness

(c) Torsional stiffness

(d) Mass distribution

Figure 48: Structural data: dotted lines correspond to the beam evaluation section, red dot corresponds to the pitch-bearing location

4.3.2 *MBDyn-DUST input data*

In MBDyn-DUST the properties of the aeroelastic beam are automatically generated from the standard format E50. In this form, each section is defined with respect to the following reference system.

- The X-axis coincides with the section chord and goes from the leading edge to the trailing edge.
- The Y-axis lies in the section plane and is positive from the bottom surface to the top surface.
- The Z-axis is determined by the right and the rule and goes from the blade tip to root.
- The origin is the point of intersection between the section plane and the feathering axis.

The following inputs are available:

- *MASS*: mass per unit length
- *XCG*: center of gravity X position
- *YCG*: center of gravity Y position
- *ROTAPI*: inertia principal axis rotation deg
- *JX*: mass moment of inertia per unit length about the centre of gravity
- *JY*: mass moment of inertia per unit length about the centre of gravity
- *JP*: mass polar moment of inertia per unit length about the centre of gravity
- *EA*: axial Stiffness
- *XNA*: elastic axis center X position
- *YNA*: elastic axis center Y position
- *ROTANE*: elastic axis rotation
- *EJX*: flapwise bending stiffness about the elastic axis
- *EJY*: chordwise bending stiffness about the elastic axis
- *GJ*: torsional stiffness about the tension centre
- *XCT*: tension center X position
- *YCT*: tension center Y position

To reproduce the entire blade properties, a rotation equal to the section aerodynamic twist about the feathering axis (opposite to the Z axis) has to be applied for each section.

4.3.3 *Transformation from CAMRAD-JA to* E50 *format*

In the E50 table format the blade offsets are given with respect to the feathering axis, since this axis is the only one that remains always straight. Instead, in CAMRAD-JA all offsets are given with respect to the elastic axis: This axis starts on the pitch bearing $\mathbf{P}_B$ and is rotated by a sweep-droop rotation. Let us define some rotation matrices to transfer the offsets from the elastic-axis reference frame to the hub reference frame.

- Rotation matrix around the local x-axis for the precone angle β:

$$\mathbf{R}_\beta = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\beta) & -\sin(\beta) \\ 0 & \sin(\beta) & \cos(\beta) \end{bmatrix} \tag{160}$$

- Rotation matrix around the local z-axis for the sweep angle λ:

$$\mathbf{R}_\lambda = \begin{bmatrix} \cos(\lambda) & \sin(\lambda) & 0 \\ -\sin(\lambda) & \cos(\lambda) & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{161}$$

- Rotation matrix around the local y-axis for the droop angle δ:

$$\mathbf{R}_\lambda = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\delta) & \sin(\delta) \\ 0 & -\sin(\delta) & \cos(\delta) \end{bmatrix} \tag{162}$$

- Rotation matrix around the local x-axis for the twist angle θ:

$$\mathbf{R}_\theta = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix} \tag{163}$$

The coordinates of the feathering axis in the hub reference frame are the following:

$$\mathbf{f}_A^{\mathcal{H}} = \mathbf{R}_\beta \cdot \mathbf{r} \tag{164}$$

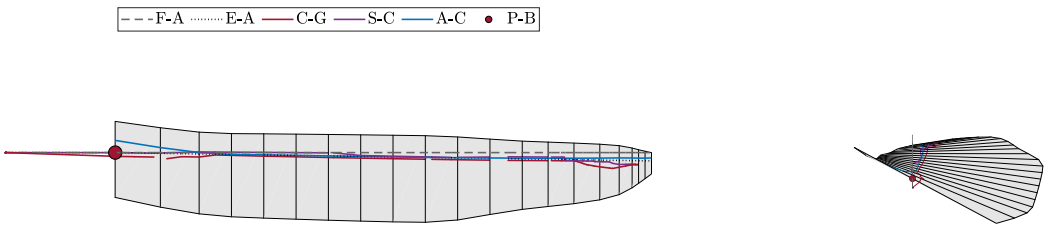
where the vector $\mathbf{r}$ contains the radial station where the characteristics are defined. Then the pitch bearing coordinates in the hub reference system are:

$$\mathbf{r}_{pb}^{\mathcal{H}} = \begin{cases} x_{pb}^{\mathcal{H}} = 0 \\ y_{pb}^{\mathcal{H}} = r_{pb} \cdot \text{Radius} \\ z_{pb}^{\mathcal{H}} = y_{pb} \cdot \tan(\beta) \end{cases} \tag{165}$$

Finally, the location of the elastic axis ($\mathbf{r}^{\mathcal{H}}_{ea}$), center of gravity $\mathbf{r}^{\mathcal{H}}_{cg}$, the shear center $\mathbf{r}^{\mathcal{H}}_{ct}$, and the aerodynamic center $\mathbf{r}^{\mathcal{H}}_{ca}$ can be expressed as the hub coordinate.

$$\begin{aligned}
\mathbf{r}^{\mathcal{H}}_{ea} &= \mathbf{R}_{\lambda} \cdot \mathbf{R}_{\delta} \cdot \left(\mathbf{f}^{\mathcal{H}}_{A} - \mathbf{r}^{\mathcal{H}}_{pb}\right) + \mathbf{r}^{\mathcal{H}}_{pb} \\
\mathbf{r}^{\mathcal{H}}_{cg_i} &= \mathbf{r}^{\mathcal{H}}_{ea} + \mathbf{R}_{\theta_i} \cdot (\mathbf{XI}) \\
\mathbf{r}^{\mathcal{H}}_{ct_i} &= \mathbf{r}^{\mathcal{H}}_{ea} + \mathbf{R}_{\theta_i} \cdot (\mathbf{XC}) \\
\mathbf{r}^{\mathcal{H}}_{ca_i} &= \mathbf{r}^{\mathcal{H}}_{ea} + \mathbf{R}_{\theta_i} \cdot (\mathbf{XA})
\end{aligned} \tag{166}$$

In Fig. 49, the location of the blade axis is shown.



(a) Top view (b) Chord view

Figure 49: blade principal axis definition

Let us now transform the coordinates from the hub reference frame ($\mathcal{H}$) to section one $\mathcal{S}$: In the format E50, the EA, CG, and CT offsets are given in a reference frame that takes into account the aerodynamic twist angle. The orientation of the elastic axis (*ROTAN*) is then the difference between the aerodynamic twist and the structural twist. AC offset is given in an untwisted reference frame. To correctly build the section, the aerodynamic center is first placed and then the correct twist is applied to the section.

$$\begin{aligned}
\mathbf{r}^{\mathcal{S}}_{ca} &= \mathbf{R}^{T}_{\beta} \cdot \mathbf{r}^{\mathcal{H}}_{ca} \\
\mathbf{r}^{\mathcal{S}}_{ea_i} &= \mathbf{R}^{T}_{\theta a_i} \cdot \mathbf{R}^{T}_{\beta} \cdot \mathbf{r}^{\mathcal{H}}_{ea} \\
\mathbf{r}^{\mathcal{S}}_{cg_i} &= \mathbf{R}^{T}_{\theta a_i} \cdot \mathbf{R}^{T}_{\beta} \cdot \mathbf{r}^{\mathcal{H}}_{cg} \\
\mathbf{r}^{\mathcal{S}}_{ct_i} &= \mathbf{R}^{T}_{\theta a_i} \cdot \mathbf{R}^{T}_{\beta} \cdot \mathbf{r}^{\mathcal{H}}_{ct} \\
\text{rotan} &= \theta_{s_i} - \theta_{a_i}
\end{aligned} \tag{167}$$

To automatically build the parametric DUST mesh, the sweep and dihedral law must be calculated from the location of the aerodynamic center. For the $j-th$ region, between sections $i+1$ and i, they are equal to:

$$\begin{aligned}
\Lambda^{\text{dust}}_{j} &= \tan^{-1}\left(\frac{\mathbf{r}^{\mathcal{S}}_{ca^{i+1,1}} - \mathbf{r}^{\mathcal{S}}_{ca^{i,1}}}{\mathbf{r}^{\mathcal{S}}_{ca^{i+1,2}} - \mathbf{r}^{\mathcal{S}}_{ca^{i,2}}}\right) \\
\delta^{\text{dust}}_{j} &= \tan^{-1}\left(\frac{\mathbf{r}^{\mathcal{S}}_{ca^{i+1,3}} - \mathbf{r}^{\mathcal{S}}_{ca^{i,3}}}{\mathbf{r}^{\mathcal{S}}_{ca^{i+1,2}} - \mathbf{r}^{\mathcal{S}}_{ca^{i,2}}}\right)
\end{aligned} \tag{168}$$

The twist will then be taken directly, through the interpolation matrix, from the orientation of the MBDyn node, which incorporates the aerodynamic twist.

4.3.4 *MBDyn beam modeling*

Blades and yoke are modeled using the MBDyn finite-volume beam element [19, 68]. A total of 12 beam elements were used for each blade and 5 for each yoke arm. This number was defined in order to have convergence in the rotor mode frequencies.

4.3.4.1 *Stiffness matrix*

In MBDyn, the beam stiffness matrix is derived with respect to the following reference system:

- The X-axis is along the beam feathering axis and goes from the beam root to the tip.
- The Y-axis coincides with the section chord from the trailing edge to the leading edge.
- The Z-axis is determined by the right and rule.
- The origin is the point of intersection between the section plane and the feathering axis.

When considering an isotropic beam section, the constitutive matrix 6×6 can always be written in terms of elementary stiffness and geometrical properties. In isotropic uniform beam sections, the properties can be split into two groups: those related to shear stress and strain and those related to axial stress and strain. The stiffness properties of the beam are evaluated at two stations: $P_e = \pm\ell/(2\sqrt{3})$ from the mid-node.

This is done by means of a linear interpolation between the two closest station in the E50 *table*

The axial properties can be transformed from the elastic reference frame to the section reference frame by means of the following rotor-translation transformation.

$$\mathbf{K}_{ax} = \mathbf{T}_{ax}\mathbf{R}_{ax}\begin{bmatrix} EA_e & 0 & 0 \\ 0 & EJY_e & 0 \\ 0 & 0 & EJZ_e \end{bmatrix}\mathbf{R}_{ax}^{T}\mathbf{T}_{ax}^{T} \tag{169}$$

In CAMRAD-JA, any value for axial stiffness is provided. It can be proved that this stiffness is equal to:

$$EA = \frac{EIZZ + EIXX}{KP2 - XC^2} \tag{170}$$

Here, the rotation matrix is given by angle $\alpha_e = \text{rotan}$:

$$\mathbf{R}_{ax} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\alpha_e) & -\sin(\alpha_e) \\ 0 & \sin(\alpha_e) & \cos(\alpha_e) \end{bmatrix} \tag{171}$$

and the translation matrix is:

$$\mathbf{T}_{ax} = \begin{bmatrix} 1 & 0 & 0 \\ z_{na_e} & 1 & 0 \\ -y_{na_e} & 0 & 1 \end{bmatrix} \tag{172}$$

The same transformation can be applied to the shear properties:

$$\mathbf{K}_{sh} = \mathbf{T}_{sh}\mathbf{R}_{sh} \begin{bmatrix} GAY_e & 0 & 0 \\ 0 & GAZ_e & 0 \\ 0 & 0 & GJ_e \end{bmatrix} \mathbf{R}_{sh}^T \mathbf{T}_{sh}^T \tag{173}$$

Since CAMRAD-JA employs a Euler-Bernoulli beam mode, any value is provided for the shear stiffness. This value is assumed to be equal to:

$$GAY = GAZ = GA = \frac{EA}{2(1+\nu)} \quad \nu = 0.33 \tag{174}$$

Here, the rotation matrix is the same as the axial one: $\mathbf{R}_{sh} = \mathbf{R}_{ax}$, and the translation matrix is

$$\mathbf{T}_{sh} = \begin{bmatrix} 1 & 0 & 0 \\ z_{ct_e} & 1 & 0 \\ -y_{ct_e} & 0 & 1 \end{bmatrix} \tag{175}$$

Finally, the fully coupled stiffness matrix 6×6 can then be obtained as

$$\mathbf{K} = \begin{bmatrix} K_{ax_{11}} & 0 & 0 & 0 & K_{ax_{12}} & K_{ax_{13}} \\ 0 & K_{sh_{11}} & K_{sh_{12}} & K_{sh_{13}} & 0 & 0 \\ 0 & K_{sh_{21}} & K_{sh_{22}} & K_{sh_{23}} & 0 & 0 \\ 0 & K_{sh_{31}} & K_{sh_{32}} & K_{sh_{33}} & 0 & 0 \\ K_{ax_{21}} & 0 & 0 & 0 & K_{ax_{22}} & K_{ax_{23}} \\ K_{ax_{31}} & 0 & 0 & 0 & K_{ax_{32}} & K_{ax_{33}} \end{bmatrix} \tag{176}$$

The list of stiffness properties of the XV-15 rotor blade can be found in Table 23.

4.3.4.2 *MBDyn inertial properties*

The inertial properties of each beam are introduced by lumped masses placed at each MBDyn beam node defined in the following reference system:

- The X-axis is along the elastic axis of the beam and goes from the root of the beam to the tip.
- The Y-axis is rotated by an angle $\texttt{rotapi} = \texttt{rotan}$ with respect to the Y section axis.
- The Z-axis is determined by the right and rule.
- The origin is in the center of mass of the body at a distance X_{CG}, Y_{CG}, Z_{CG} from the center of the beam section.

The mass lumping is described in the following passages:

- Each beam is divided into three segments: the first and third segments have a length of $1/2(1 - 1/\sqrt{3})\ell_i$, while the second (the mid-length) has a length of $(1\sqrt{3})\ell_i$, where ℓ_i is the beam's length.
- Mass and inertia are obtained by exploiting the trapezoidal integration rule (considering x as the beam axis):

$$M^{i,i+1} = \int_{s^i}^{s^{i+1}} m_{2D}\, dx = \frac{(m_{2D}^i + m_{2D}^{i+1})(s^{i+1} - s^i)}{2} \tag{177}$$

$$I_{xx}^{i,i+1} = \int_{s^i}^{s^{i+1}} I_{xx}^{2D}\, dx \tag{178}$$

$$I_{yy}^{i,i+1} = \int_{s^i}^{s^{i+1}} I_{yy}^{2D}\, dx + \frac{M^{i,i+1}(s^{i+1} - s^i)^2}{12} \tag{179}$$

$$I_{zz}^{i,i+1} = \int_{s^i}^{s^{i+1}} I_{zz}^{2D}\, dx + \frac{M^{i,i+1}(s^{i+1} - s^i)^2}{12} \tag{180}$$

  where m_{2D}^i and $I_{jj}^{2D^i}$ are the density of the section and the moment of sectional inertia around the axis jj evaluated at the station s^i.

- The position of the center of mass of the body is determined as follows.

$$\mathbf{P}_{CG}^{i,i+1} = \frac{m_{2D}^{i+1} \cdot \mathbf{P}_{CG}^{i+1} + m_{2D}^i \cdot \mathbf{P}_{CG}^i}{m_{2D}^i + m_{2D}^{i+1}} \tag{181}$$

The sectional moments of inertia (I_{yy}^{2D}, I_{zz}^{2D}) are not provided by the original CAMRAD-JA model, so their value is determined by assuming that:

$$\begin{cases} I_{xx}^{2D} = I_{yy}^{2D} + I_{zz}^{2D} \\ \dfrac{I_{yy}^{2D}}{I_{zz}^{2D}} = \dfrac{EI_{yy}}{EI_{zz}} \end{cases} \rightarrow \begin{cases} I_{zz}^{2D} = I_{xx}^{2D} - I_{yy}^{2D} \\ I_{yy}^{2D} = \dfrac{I_{yy}^{2D}}{1 + \dfrac{EI_{yy}}{EI_{zz}}} \end{cases} \tag{182}$$

where I_{xx}^{2D} is the sectional polar moment of inertia with respect to the center of gravity:

$$I_{xx}^{2D} = I_\theta - m \cdot XC^2 \tag{183}$$

The list of inertial properties of the XV-15 rotor blade can be found in Table 22.

4.3.5 *Aerodynamic modeling*

To model the blade aerodynamics, two levels of fidelity were used. The low-fidelity model is based on the 2D strip theory and is available in MBDyn; instead, the mid-fidelity model is based on the DUST non-linear vortex lattice element, using 35×5 elements 50. The number and radial discretisation of the aerodynamic panels have been determined after convergence on hover loads. Each aerodynamic panel can incorporate aerodynamic twist variations and profile transitions. The four airfoils are placed along the blade with a non-smooth transition, following the distribution presented in Table 8, where the blade aerodynamics was modeled considering the Blade Element/Momentum Theory with a Glauert model [103] for the induced velocity with the following corrections, according to the original CAMRAD-JA model: The list of aerodynamics center definitions for the XV-15 rotor blade can be found in Table 24.

- Hover correction factor = 1.2
- Forward flight correction factor = 2.0
- Memory factor = 0.9
- Tip loss factor = 0.995

Table 8: XV-15 ATB Blade airfoil distribution [31].

Airfoil data		
Profile	Start	End
V43030-1p58	0.17	0.20
VR7-3	0.20	0.84
VR8mod	0.84	0.95
VR8	0.95	1

The viscous and compressibility effects are incorporated by means of the airfoil table `.c81`. A 3D map of these coefficients is provided in Fig. 51.

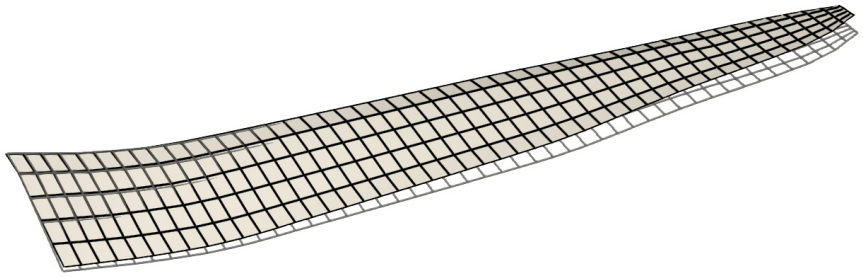
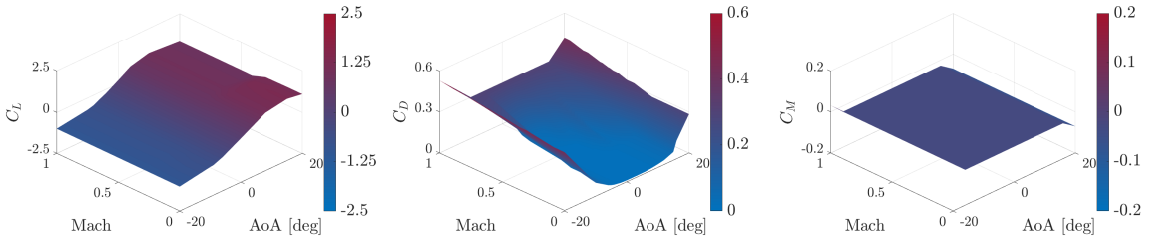
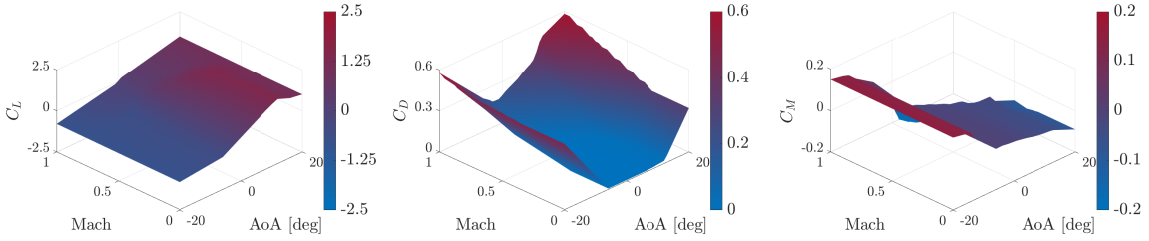


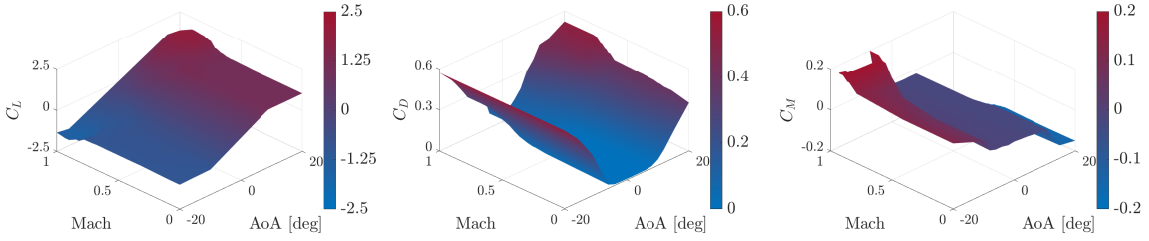
Figure 50: ATB blade aerodynamic mesh: deformed and undeformed view



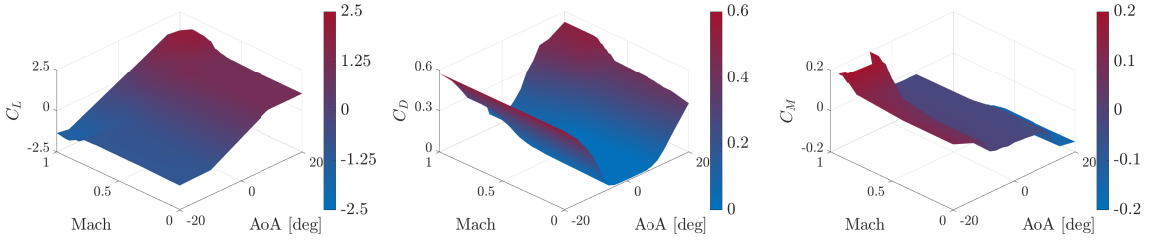
(a) V43 030



(b) VR7-3



(c) VR8mod



(d) VR8

Figure 51: Blade airfoil aerodynamic coefficients: C_L, C_D and C_M for the 4 different profiles

4.3.6 *Clockwise rotor*

The previous discussion applies only for a counter-clockwise rotating rotor; few modifications are necessary to obtain a symmetric rotor that rotates clockwise. The reference system of the counter-clockwise is maintained, while the chord-wise offset of the center of mass, aerodynamic center, and elastic axis changes its sign, as well as the twist rotations:

$$\begin{aligned} y_{cg}^{left} &= -y_{cg}^{right} \\ y_{ea}^{left} &= -y_{ea}^{right} \\ y_{ac}^{left} &= -y_{ac}^{right} \\ \theta_a^{left} &= \theta_a^{right} \end{aligned} \tag{184}$$

To verify the correct modeling of the symmetry conditions, two analyses were performed. In the first analysis, the two blades were clamped and subjected to a uniform free flow velocity of $20\,m\,s^{-1}$. The flap and lag displacements are shown in Fig. 52 and the difference in displacement between the two blades is on the order of 10×10^{-9} mm The

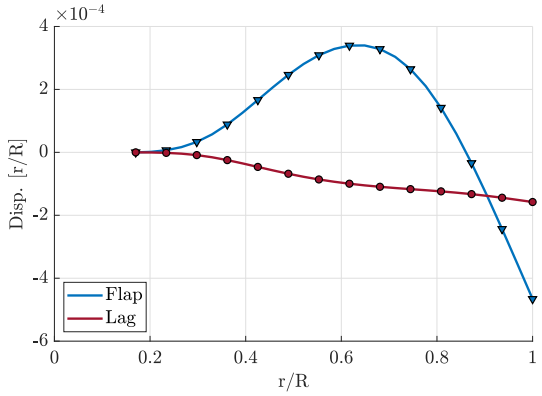


Figure 52: Symmetric check: clamped blade deformation. Dots: left blade, lines: right blade

second analysis was done comparing the hover performance of the two rotors. The results are shown in Table 9.

Table 9: Errors between left and right rotor in hover

θ_0	ΔC_T	ΔC_Q	ΔFM
-2	8.73×10^{-10}	-3.7×10^{-9}	2.504×10^{-8}
0	4.44×10^{-9}	-4.8×10^{-9}	3.2×10^{-7}
4	1.47×10^{-7}	1.42×10^{-8}	9.766×10^{-6}

4.4 DRIVE SYSTEM

A simplified model of a symmetric drive train system is proposed to represent the XV-15 engine drive train. Its scheme is shown in Fig. 53. The XV-15 engine model is derived from [31]: The engines are approximated as lumped rotational inertia, whereas the rotor nodes do not have any inertia, since it is already included in the MBDyn models of the rotors. The two gearboxes are joined by an interconnecting shaft modeled with a torsional spring. The stiffness value of the torsional spring and the inertia of the engine are reported in [31].

Each gearbox is modeled by two gimbal joints phased by 90° that impose the same angular velocity through each shaft. This arrangement provides redundancy to the loss of one engine, since it permits power transfer for transient conditions and provides rotational speed synchronization.

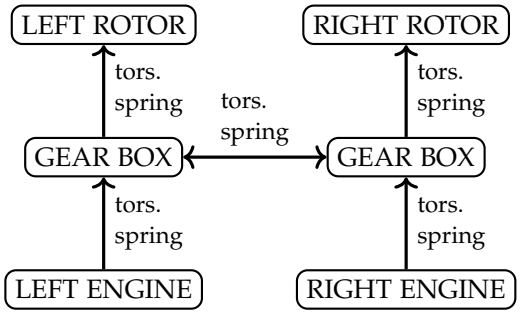


Figure 53: Flowchart of the MBDyn model of the XV-15 drive system

The validation was carried out by comparing the frequency of the three modes of the system, as shown in Table 10.

Table 10: Drive system modes frequency (Hz) comparison

Mode	[136]	MBDyn	Diff.
First anti-symmetric	4.50	4.50	0.18%
First symmetric	13.98	13.99	0.05%
Second anti-symmetric	16.62	16.59	−0.19%

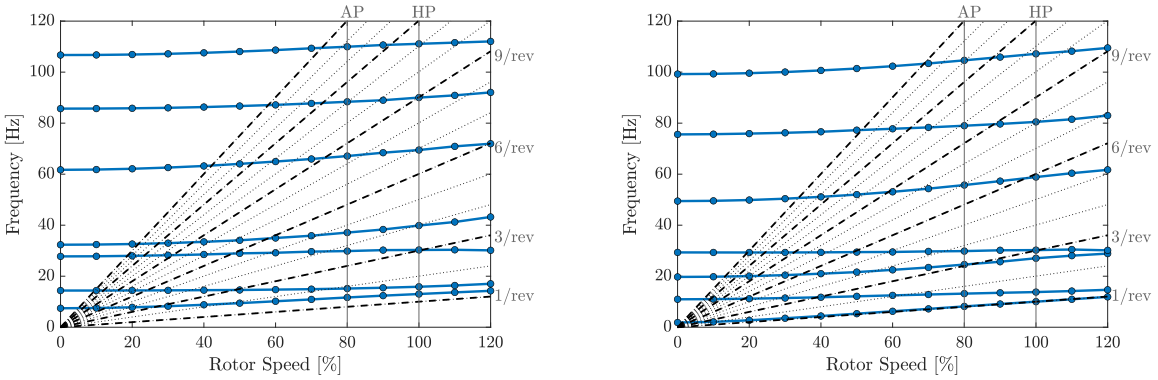
4.5 ROTOR DYNAMICS

4.5.1 *Rotor Campbell diagrams*

The dynamic behavior of the isolated rotor in vacuum is validated by means of the Campbell diagram. On these diagrams, the rotational frequency of each mode shape is tracked against the rotor speed. Each Campbell diagram has been calculated considering dif-

ferent rotor speeds and evaluating the eigenvalues using the Arpack library [102] and the eigenvalue card present in MBDyn [112]. Figures 54a and 54b show the results considering a collective angle of 12° for collective modes and cyclic modes, respectively. The same results considering a collective of 40° are reported in Figs. 55a and 55b. Collective modes are those in which all blades move in the same direction simultaneously, whereas cyclic modes are those in which the blades move in an antisymmetric manner. The flap mode is referred to a beam bending mode where the predominant component of the eigenvector is the one that is out of plane, whereas the lag mode has the predominant component of the eigenvector that is the in-plane one, finally the torsion mode is defined as the mode which has the dominant component of the eigenvector in the rotation around the blade axis.

A detailed comparison between the CAMRAD-JA and MBDyn models is presented in Tables 12 and 11. Good agreement is visible in both the collective and the cyclic modes. The most noticeable difference can be seen in the first torsion mode. This can be explained by the fact that in the CAMRAD-JA the control chain is modeled as a conventional helicopter-like control chain with a single control path, whereas in the MBDyn model it is modeled with a dual load path. The pitch mode is very close to the 3/rev. When it was flown in forward helicopter mode, this 3/rev aerodynamic excitation coupled with the natural frequency of the system produces high structural loads [109].



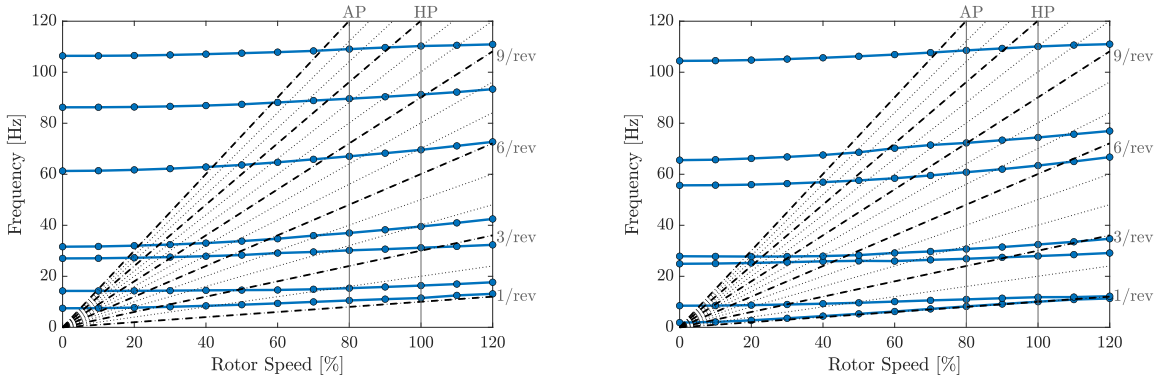
(a) Collective Modes

(b) Cyclic modes

Figure 54: Rotor Campbell diagram in vacuum at collective pitch of 12°

4.5.2 *Rotor mode shape*

To better underline the effects the collective angle has on the mode shapes, the first collective lag and flap modes are shown in Figs. 56a and 56b, using a single-blade model, in which the gimbal is locked and both the drive system and the control chain are considered rigid.



(a) Collective Modes

(b) Cyclic modes

Figure 55: Rotor Campbell diagram in vacuum at collective pitch of 40°

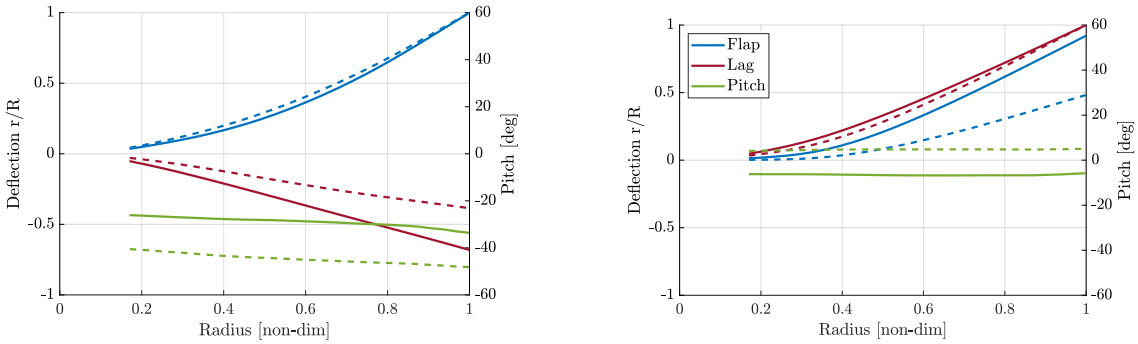
Table 11: Frequency (Hz) comparison in non-rotating frame of the first 4 rotor modes in vacuum at 12° collective in HP mode: β_0, β^-, β^+ are the collective, regressive and progressive mode respectively

	CAMRAD-JA			MBDyn		
Mode	β_0	β^-	β^+	β_0	β^-	β^+
Gimbal	–	0.19	20.22	–	0.15	20.19
First Lag	12.68	3.18	23.22	12.81	3.38	23.43
First Flap	16.03	17.90	37.93	16.24	18.02	38.10
First torsion	30.39	20.85	40.88	30.92	20.23	40.41

Table 12: Frequency (Hz) comparison in non-rotating frame of the first 4 rotor modes in vacuum at 40° collective in AP mode: β_0, β^-, β^+ are the collective, regressive and progressive mode respectively

	CAMRAD-JA			MBDyn		
Mode	β_0	β^-	β^+	β_0	β^-	β^+
Gimbal	–	0.24	16.27	–	0.05	16.25
First Lag	10.19	2.42	18.45	10.53	2.83	19.10
First Flap	15.94	22.85	38.88	15.35	22.57	38.83
First Torsion	28.87	20.44	36.47	30.75	18.74	35.72

It is evident that an increase in collective angle reduces the modal participation of the cross-component of the mode: When the collective moves from 12° to 40°, both the lag and the flap components are reduced in the flap and lag modes, respectively. The pitch component increases with the collective in the flap mode, whereas it changes the sign in the lag mode. This effect can be explained by the change in the chordwise position of the blade center of mass that occurs when the collective increases.



(a) Flap mode

(b) Lag mode

Figure 56: Collective modes at 100% RPM. Continuous line: $\theta_{75} = 12°$, dotted lines: $\theta_{75} = 40°$

Figures 57 and 58 analyze the effects that flexibility of the control chain and of the drive system have on the firsts 4 collective modes. For these analyses, the drive is simplified to a single torsional spring and torsional inertia, tuned to match the first drive system mode. Eigenanalyses are performed in vacuum in HP mode for a collective angle of 12°.

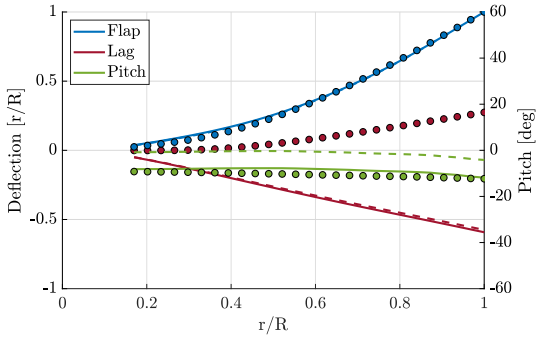
The flap component of the collective flap is not affected by these additional flexible degrees of freedom, whereas the drive system flexibility greatly affects the lag component of the mode, changing its sign, while the control chain flexibility affects the pitch component.

In the collective lag mode, the main differences are produced by the presence of the drive system, in particular in the pitch and flap component. Furthermore, while the other modes maintain their frequency, in this case the frequency of the lag mode increases from 12.8Hz to 48Hz when the (simplified) drive system is added.

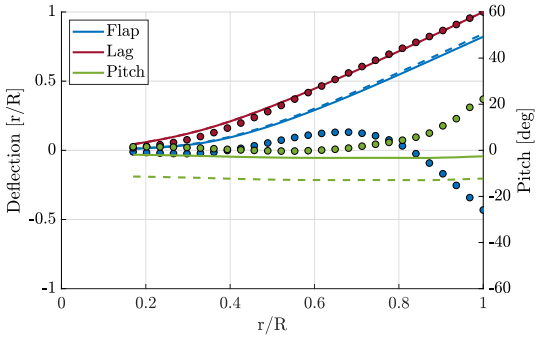
The pitch mode is absent when the control chain is rigid, since the pitch degree of freedom is locked, while introducing the drive system flexibility almost cancels the flap and lag components that were present when only the flexibility of the control chain was considered. The presence of these two components in the pitch mode is due to the dual load control chain that mixes the collective pitch mode with the cyclic one.

Finally, in the 1^st collective beam mode, the introduction of the flexibility of the control chain has a huge effect on the pitch component,

while the flexibility of the drive system increases the lag component in the mode.

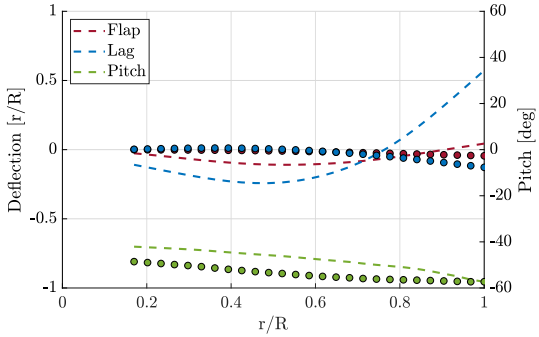


(a) Collective flap

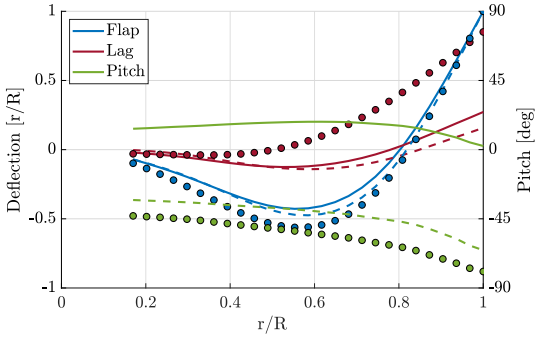


(b) Collective lag

Figure 57: Collective modes at 100% RPM $\theta_{75} = 12°$. Continuous line: rigid chain, dotted lines: flexible chain, dots: flexible chain, 1^{st} drive system mode



(a) Collective pitch



(b) Collective flapwise beam

Figure 58: Collective modes at 100% RPM $\theta_{75} = 12°$. Continuous line: rigid chain, dotted lines: flexible chain, dots: flexible chain, 1^{st} drive system mode

4.6 ROTOR AEROELASTIC ANALYSIS

The capability of the model to predict rotor performance, loads, and stability in hover- and airplane-mode conditions is now treated. In both conditions, the results obtained from the low-fidelity MBDyn model are compared with the mid-fidelity MBDyn-DUST model and, for which the hover condition is concerned, with the experimental results of OARF [18]. In the following analysis, the convection of the signs for the internal moments will be:

This convention is coherent with the CAMRAD-JA sign convention

- Torsion moment: positive for the pitch-up rotation ($+Rx_b$)
- Flapwise moment: positive for a positive cone rotation ($-Ry_b$)
- Chordwise moment: positive for a rotation from the trailing edge to the leading edge ($+Rz_b$)

Rotation around the MBDyn local beam axis

4.6.1 *Hover condition*

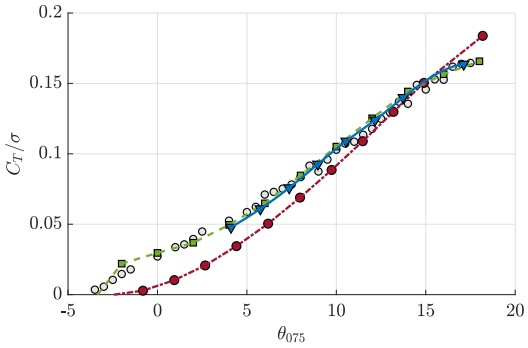
The following simulations are performed with a time-marching simulation using a time step of 120 per rotor revolution. Ten revolutions are considered and the results are obtained by averaging the loads and displacement of the last revolution. For all hover analyzes, the drive system is considered rigid. For each point, the computational time required to complete the simulation of the rotor configuration was approximately 8 minutes for the DUST alone case and 16 for the coupled case, using a workstation with a Dual Intel Xeon Gold 6230R @ 2.10Ghz with a processor of 104. Instead, the MBDyn alone model is capable of running on a normal laptop computer in about 1 minutes. The numerical results are compared with the experimental run 48 of the OARF test campaign.

Figure 59a shows the comparison of the rotor thrust coefficient (C_T/σ) as a function of the collective angle of the blade evaluated in 75% of the blade, Fig. 59b compares the rotor torque coefficient (C_Q/σ) as a function of the rotor thrust coefficient(C_T/σ), while Fig. 59c shows the comparison of the rotor figure of merit (FM).

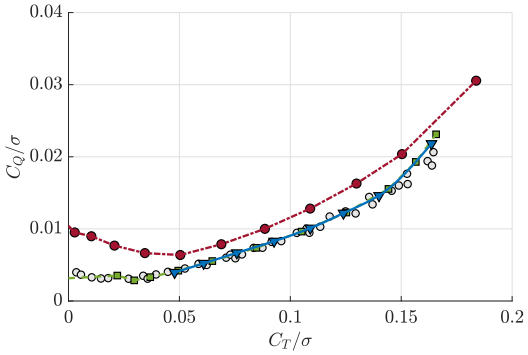
The performance curves obtained with both the DUST alone and the coupled MBDyn-DUST models quite well resume the behavior of the experimental data in the entire range of blade collective angles tested. In particular, the model can correctly capture the rotor stall that begins to occur for a collective angle greater than 15°. In contrast, the MBDyn alone model is not able to correctly capture hover performances: in particular, the slope of the $C_T/\sigma - \theta_{075}$ curve is steeper compared to the experimental results and the DUST one, and no stall is predicted. For what concerns the polar curve, MBDyn overestimates the torque, and this prediction is further amplified when we compare the figure of merit of the rotor.

Figures 60 and 61 compare the chord and flap moments in 0.06% R (hub section) and in 0.10% R (yoke section) with the experimental results of the run 48. The chord-wise moment trend is well captured by both models; however, the MBDyn model slightly overpredicts this moment with respect to the experimental and coupled model results. Both models capture the linear behavior of the flap moment, even if the slope of the curve is steeper with respect to the experimental results. This difference can be due to a control-chain modeling that is not identical to the real one: this can lead to different vertical load sharing between the blade-to-hub and blade-to-pitch-link paths, which eventually cause the difference in the flap-wise moment. This difference is more significant at the hub station, whereas at the yoke station the flap moment is well captured. On the contrary, the chord-wise moment is better captured at the hub station.

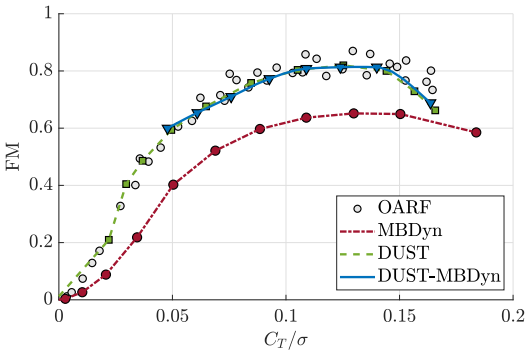
Figures 62, 63 and 64 compare the deformation of the blade-yoke feathering axis for a collective range between $\theta_0 = 6° - 18°$ for the



(a) θ_{075}

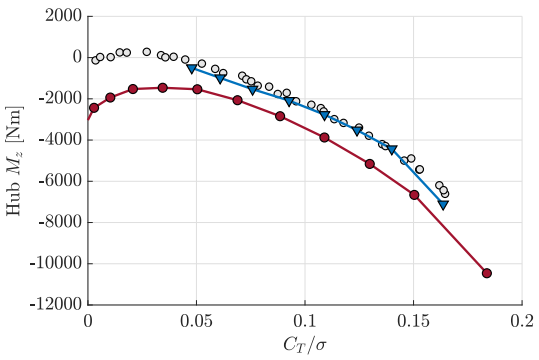


(b) C_Q/σ vs C_T/σ

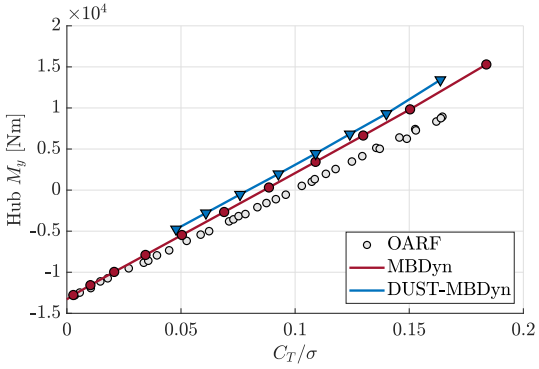


(c) F_M vs C_T/σ

Figure 59: ATB rotor hover performances

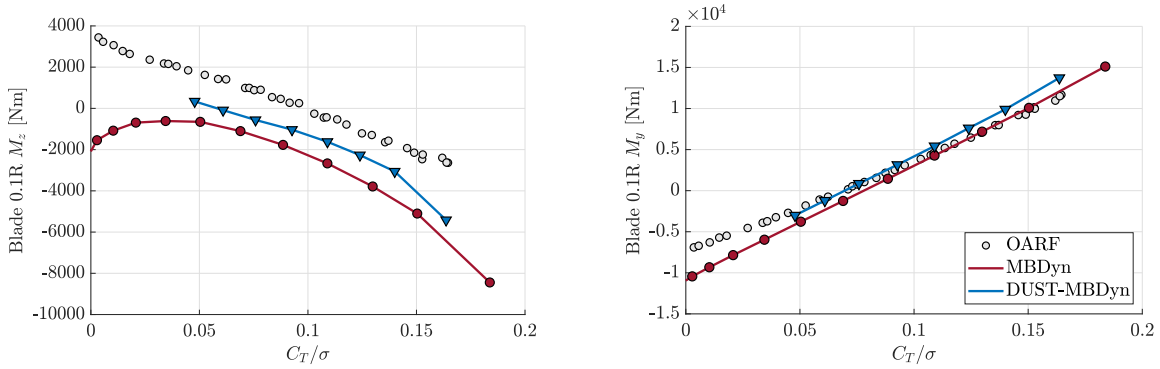


(a) Chordwise moment



(b) Flapwise moment

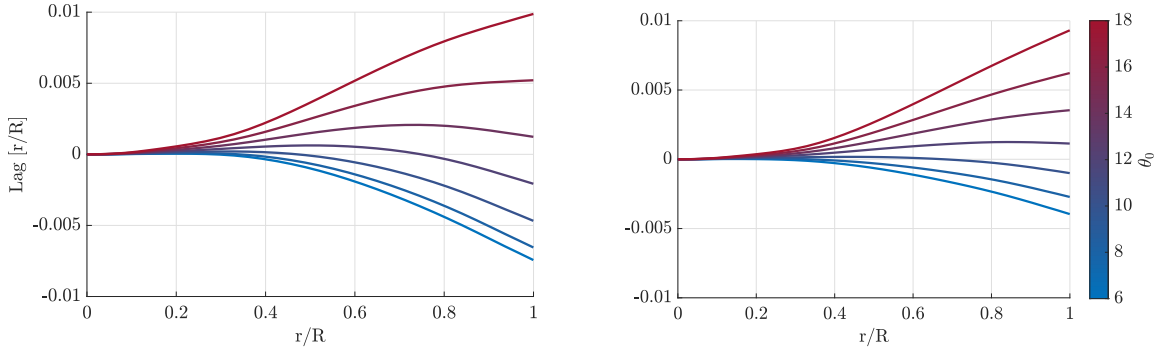
Figure 60: Hub moments vs C_T/σ



(a) Chordwise moment (b) Flapwise moment

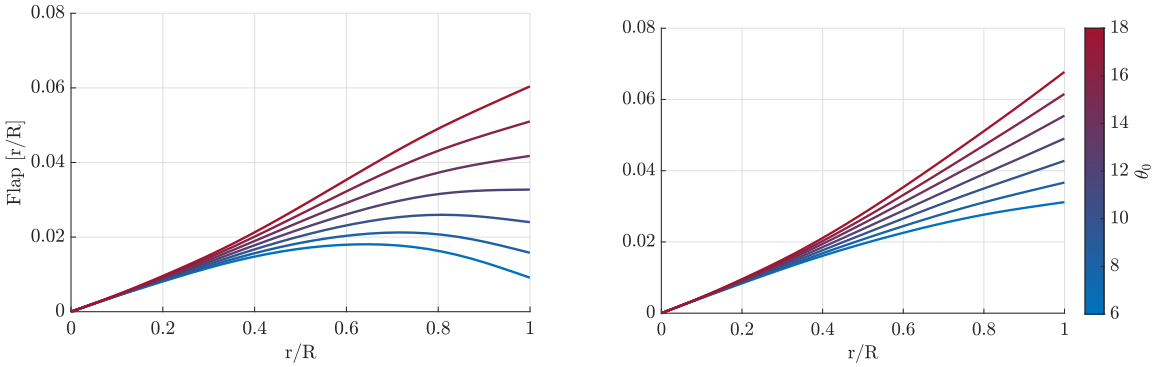
Figure 61: Blade 0.1R moments vs C_T/σ

MBDyn model and DUST. The deformations are plotted in the hub reference frame with the internal force sign convention. From both flap and lag deformation, it can be observed that the MBDyn model produces a larger deformation compared to the coupled one. In particular, the tip of the blade is more bent in the MBDyn model, whereas the coupled one tends to have a more straight behavior. From the pitch deformation, which is obtained by subtracting the pitch angle of the initial section from the final one: $\Delta\theta = \theta_f - \theta_0$, it is clear that the coupled model tends to torque the blade section with respect to the MBDyn one. Furthermore, the significant jump in pitch rotation at the pitch-bearing location (0.17R) means that control chain is highly deformable: At $\theta_0 = 18°$, the collective angle loss due to the control chain for the coupled model is approximately 3.6°. Instead, for the same collective angle, the collective loss due to blade torsion is less than 1°. Figure 65 shows the difference between the imposed collective angle and the measured angle. The trend is almost linear for both the MBDyn model and the coupled one; however, the slope of the curve is steeper for the MBDyn model ($\theta_{075} = 0.873\theta_0 - 1.2044$) than for the coupled model ($\theta_{075} = 0.8138\theta_0 - 0.8186$).



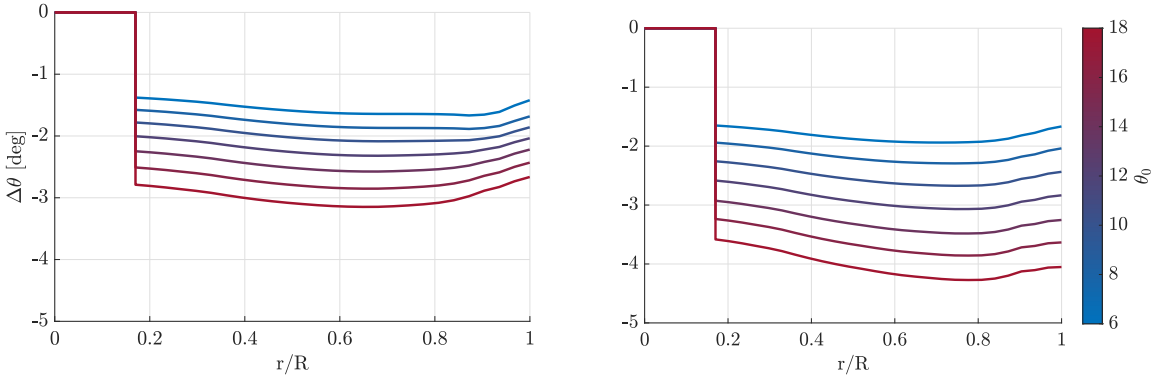
(a) MBDyn (b) DUST-MBDyn

Figure 62: Lag deformation



(a) MBDyn (b) DUST-MBDyn

Figure 63: Flap deformation



(a) MBDyn (b) DUST-MBDyn

Figure 64: Pitch deformation

Figures 66, 67 and 68 compare the chord, flap and torsional moment of the blade for a collective range between $\theta_0 = 6° - 18°$ for the MBDyn model and DUST. Internal moments are plotted in the local beam-axis reference system. The same trend is found for both flapwise and chordwise bending moments; however, a larger chordwise bending moment is found for the MBDyn model, instead of the larger flapwise moment, it is found in the coupled model. For both internal moments, a jump is visible at the pitch-bearing location: the difference in load will be carried out by the pitch link under axial load. Finally, the torsional moment has an almost linear behavior in both cases in the blade, ranging from the pitch bearing to 0.85% R, after which the torsion moment goes to zero. The coupled model exhibits a greater torsional moment compared to the MBDyn model.

Figures 69 and 70 compare the lift coefficients found along the blade in hover for a collective range between $\theta_0 = 6° - 18°$. The lift coefficient on the MBDyn is constantly decreasing along the blade span; this behavior is due to the Glauert prescribed inflow model; instead, in the coupled model a bump at the tip of the blade is clearly visible: this effect can be explained by the presence of the vortexes at the tip

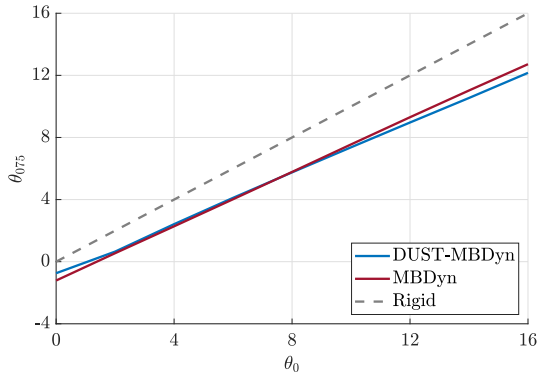
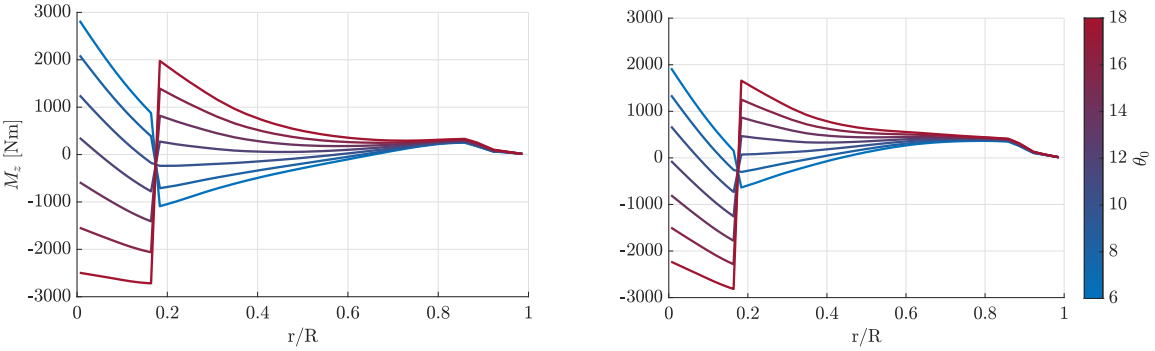
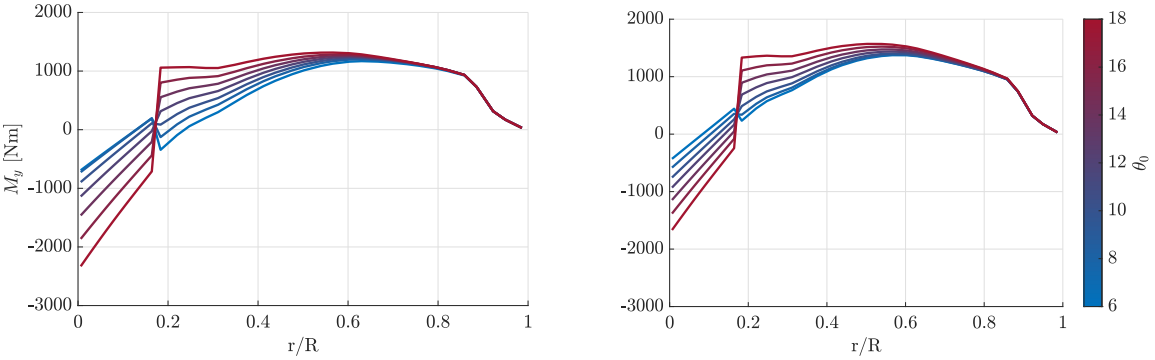


Figure 65: Imposed collective angle θ_0 vs measured θ_{075}



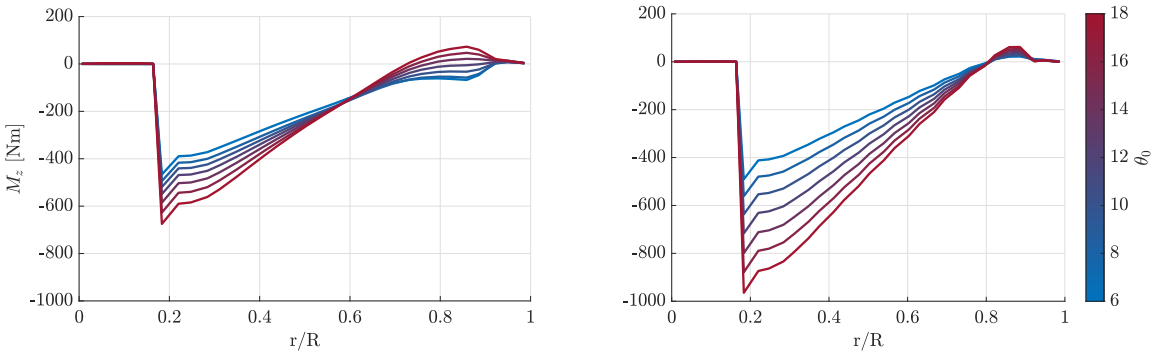
(a) MBDyn (b) DUST-MBDyn

Figure 66: Chordwise bending moment along the blade



(a) MBDyn (b) DUST-MBDyn

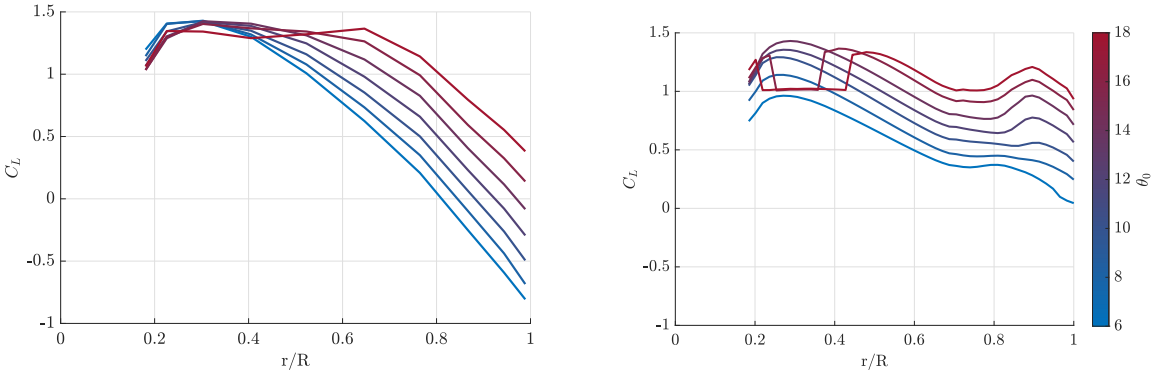
Figure 67: Flapwise bending moment along the blade



(a) MBDyn (b) DUST-MBDyn

Figure 68: Torsional moment along the blade

of the blade that, interacting with the blades, causes a local increase in the angle of attack and therefore in the lift coefficient. For higher collective angles, both models predict a partial stall of the blades in the inner sections.

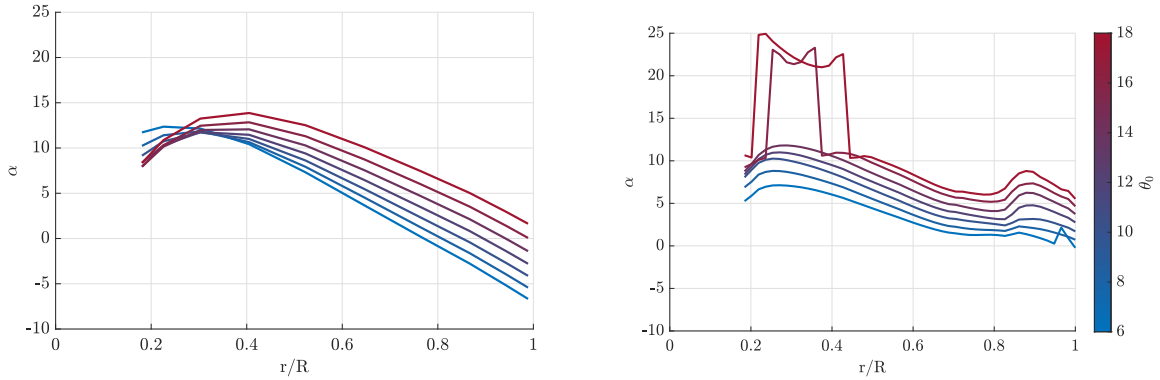


(a) MBDyn (b) DUST-MBDyn

Figure 69: Lift coefficients (C_L) along the blade

Finally, for the coupled model, the wake velocity profile was measured for different values of thrust coefficients, and the results were compared with the experimental one. In the experimental campaign, a wake rake consisting of pitot-static tubes 22 was mounted behind the rotor disc plane at the station corresponding to the upper surface of the wing. The purpose of the rake was to measure the isolated rotor slipstream velocities and angles under different operating conditions of the rotor and to use these data to understand the structure of the rotor slipstream and the wing download and its distribution. A great correlation was found; this is due to the fact that VPM prescribes the conservation of vorticity.

Figure 72 shows wake visualization using isosurfaces of a q criterion obtained with the coupled model. Two helicoidal vortical structures are present: one that originates from the blade tips and the other



(a) MBDyn (b) DUST-MBDyn

Figure 70: Angle of attack (α) along the blade

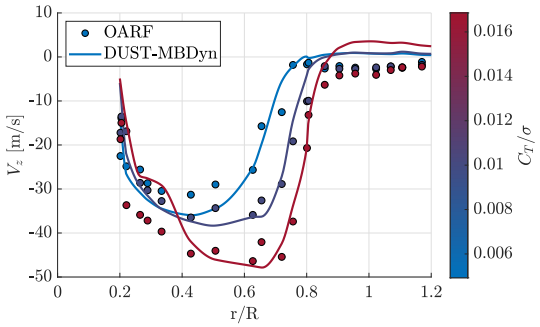


Figure 71: Wake velocity measurements of the DUST-MBDyn model: comparison with experimental results

that start at the blade root. At the bottom, a toroidal vortical structure is visible that originates from the impulsive starting of the rotor.

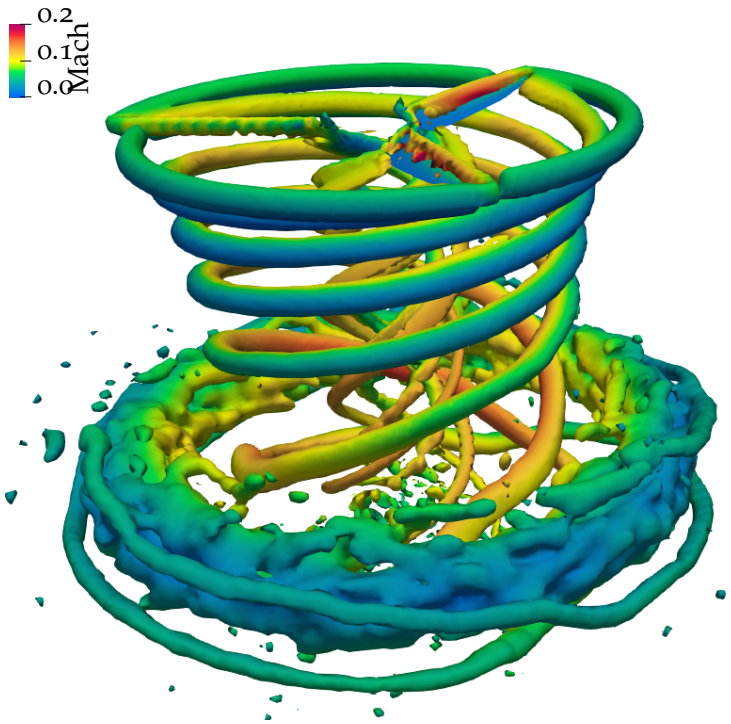


Figure 72: Hover wake visualization by means of a q-criterion (Q=500) vorticity iso-surfaces, colored by local Mach value, for a collective angle of $\theta_0 = 12°$

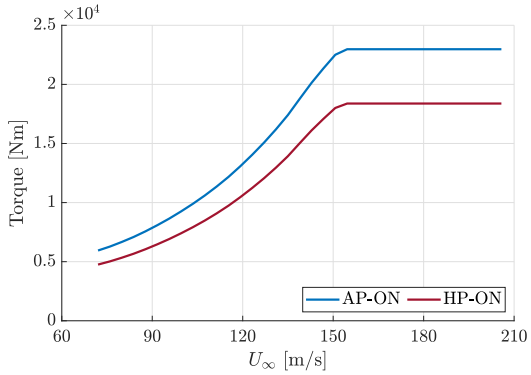
4.6.2 *Airplane Mode*

In this section, the performance and stability of the rotor are evaluated under axial flow conditions. Unlike the hover case, in which the rotor is not trimmed but instead the rotor speed and the collective are imposed, in airplane mode the rotor is trimmed to maintain the prescribed angular velocity. Power-on trim is achieved by starting from an initial guess of the collective angle and then applying the desired torque on the engine side. Engine torque is constantly increased with airstream speed until maximum torque is reached, and then a constant torque trim is maintained to represent steady powered descent. The desired rotor speed is maintained using an integral controller since the only objective of the controller is to guarantee a null steady-state error that sets the required swashplate collective as a function of the errors on the integral of the rotor angular velocity. The power-off condition is achieved by simply setting the applied torque to zero. Since only one rotor is considered, the dynamics of the drive train is reduced to only one torsional spring, whose stiffness is adjusted to

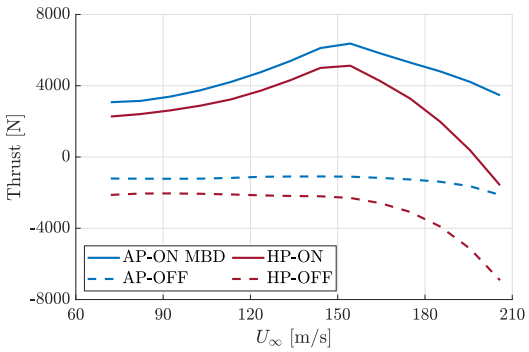
match the first symmetric drive system mode. Four different subconditions were evaluated:

1. Power-on, AP mode: the engine torque is prescribed and the rotor speed is maintained at 80% RPM.
2. Power-off, AP mode: the engine torque is set to zero (the engine failed) and the rotor speed is maintained at 80% RPM.
3. Power-on, HP mode: the engine torque is prescribed and the rotor speed is 80% RPM.
4. Power-off, HP mode: the engine torque is set to zero (the engine failed), and the rotor speed is maintained at 100% RPM.

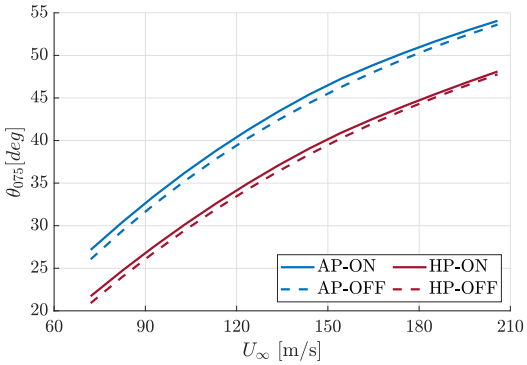
For each configuration 14 wind speeds U_∞ were considered, ranging from 140 knts to 400 knts with an increment of 20 knts from one point to the other. The following simulations are carried out with a time-marching simulation using a time step of 120 step per rotor revolution, and the simulation ends when the error between the measured angular velocity and the prescribed one is below 0.1%, usually this requires 20 rotor revolutions.



(a) Torque vs U_∞



(b) Thrust vs U_∞

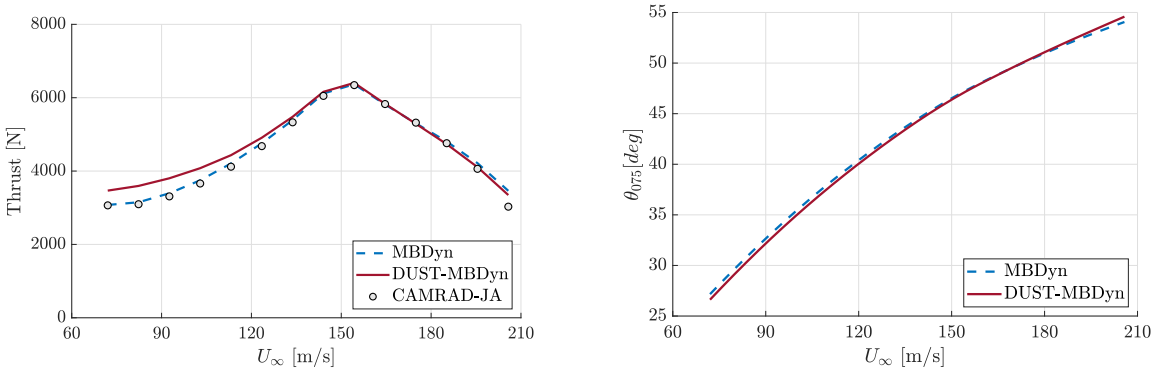


(c) Collective vs U_∞

Figure 73: Trim results in airplane mode for the MBDyn model

Figure 73 shows the MBDyn rotor trim results (thrust and collective) and the applied torque for all four configurations. In figure 73b

it is clear that increasing the rotor speed has a decremental effect on the rotor thrust, simply because the applied torque is 20% less than the one in airplane mode. The windmill condition produces a negative thrust, even if the difference in collective angle is less than 2°. This suggests that to control the blade pitch, a governor controller is of paramount importance, since small variations in collective angle can lead to huge thrust variations.

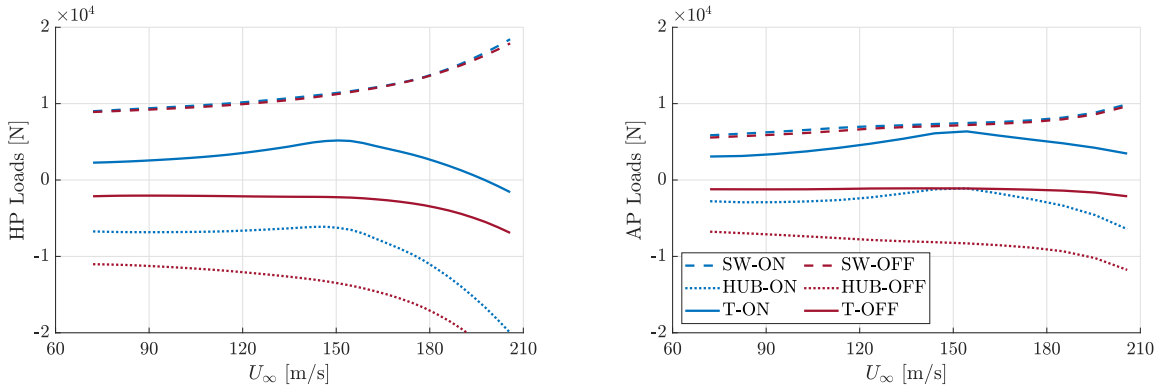


(a) Thrust

(b) Collective θ_{075}

Figure 74: Thrust-Collective vs U_∞: MBDyn vs DUST-MBDyn

Figure 74 compares the results of the MBDyn rotor trim (thrust and collective) of the configuration 1 with that obtained from the coupled model and the CAMRAD-JA model. No significant differences were found when comparing the thrust of the MBDyn model with the CAMRAD-JA model, since both use the same Glauert inflow model with the same corrections. Instead, when comparing this result with the coupled model, it is evident that the thrust predicted by this model is higher at low speed, even if the difference in collective angle is small. However, the difference tends to decrease as the wind speed increases. This difference can be explained by looking at the helicoidal structure of the wake: for low speed, the tip vortexes are closer to each other, and this vortex-interaction effect cannot be captured by a simple model that uses a prescribed inflow. The interaction between vortexes becomes less and less strong as the wind speed increases because the helicoidal wake is stretched out. Figure 75 compares the sharing of the thrust between vertical force transmitted at the hub and swashplate for all four configurations using the MBDyn model. It is evident that the swashplate and hub are more loaded in helicopter mode, even if the total thrust produced is less than that of an airplane mode. Passing from the powered condition to the windmill hugely affects the compressive hub load, whereas the swashplate does not fall this difference (it is slightly less loaded in windmill condition). This difference can be explained by the fact that the engine is directly connected to the hub through the shaft, whereas the swash-



(a) HP mode (b) AP mode

Figure 75: Loads sharing: power on vs power off

plate is connected to the pitch links, which control the blade pitch that does not change from one condition to the other.

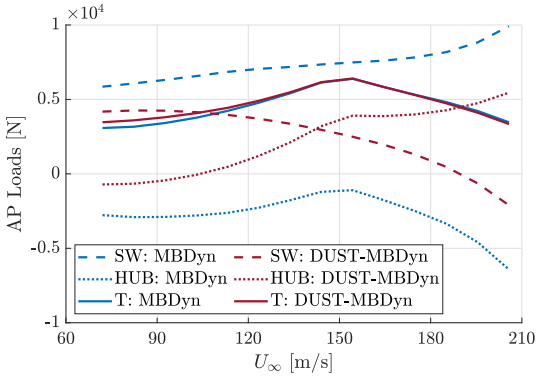
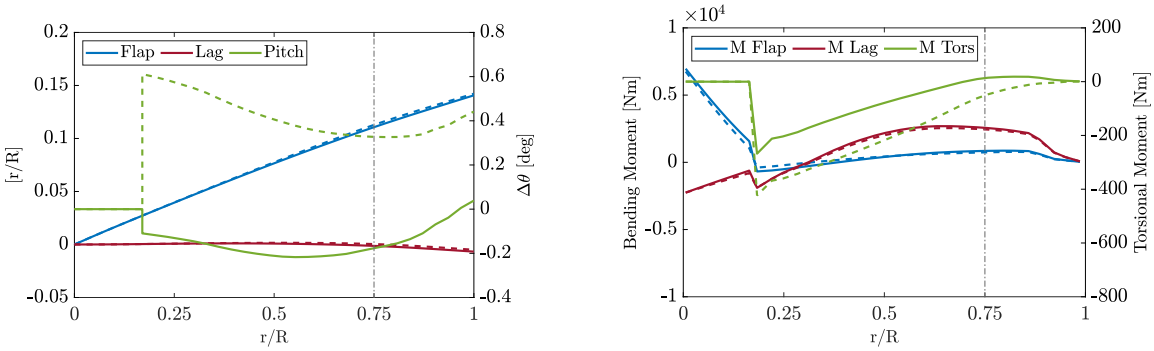


Figure 76: Loads sharing: MBDyn vs DUST-MBDyn

Finally, 76 compares the thrust partitioning for the configuration 1 with respect to the MBDyn model and the coupled one. In this case, the differences between the two models are significant: for the MBDyn model, the load of the washplate increases with wind speed; instead, the opposite trend is found for the coupled model. The hub load in MBDyn almost follows exactly the thrust trend, with a huge increase (in modulus) during the descent phase; instead, the coupled model predicts an increase in this load that has a parabolic shape in the first phase, then during the descent phase, it tends to have a flatter shape.

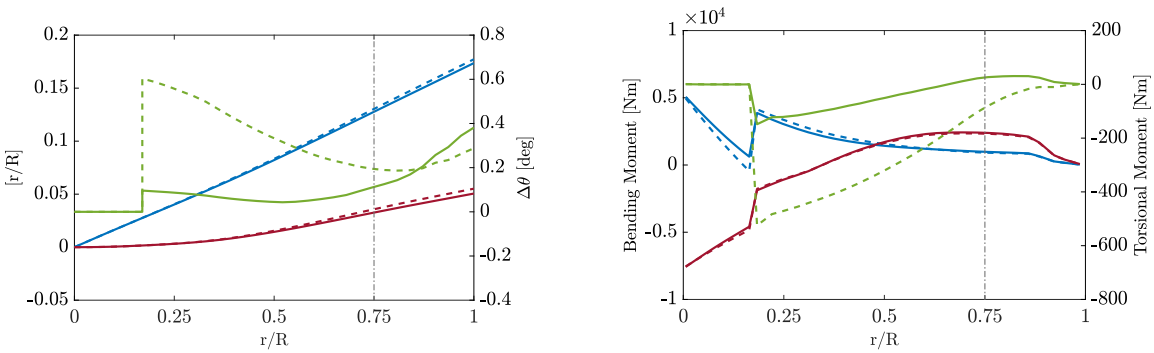
To better understand the origin of these differences, since at first glance the two models appeared almost identical in terms of collective and thrust, internal moments and deformations are analyzed for three different air speeds: 160, 300, and 400 knts. The sign convention is the same as that used for the hover case. The results are shown in figures 77, 78, and 79. It is clear that at low speed and at cruise speed the flap and lag deformations are almost identical between

the two models: the coupled one predicts slightly less deformation; moreover, the flap-wise and chord-wise bending moments are almost identical. Instead, a significant difference is found for the pitch deformation and, therefore, for the torsional moment. On the contrary to the hover case, here the coupled model predicts a smaller torsional moment, that is, reducing as the wind speed increases. At the highest speed, the torsional trends are opposite between the two models, which leads to a higher control chain deformation in the MBDyn model. In this case, a quite significant difference is also found in the flap and lag deformation components, which is mainly due to structural pitch-flap and pitch-lag structural couplings.



(a) Deformations (b) Internal moments

Figure 77: $U_\infty = 160$ knts ($82.3\,\mathrm{m\,s^{-1}}$): dotted MBDyn, continuous DUST-MBDyn

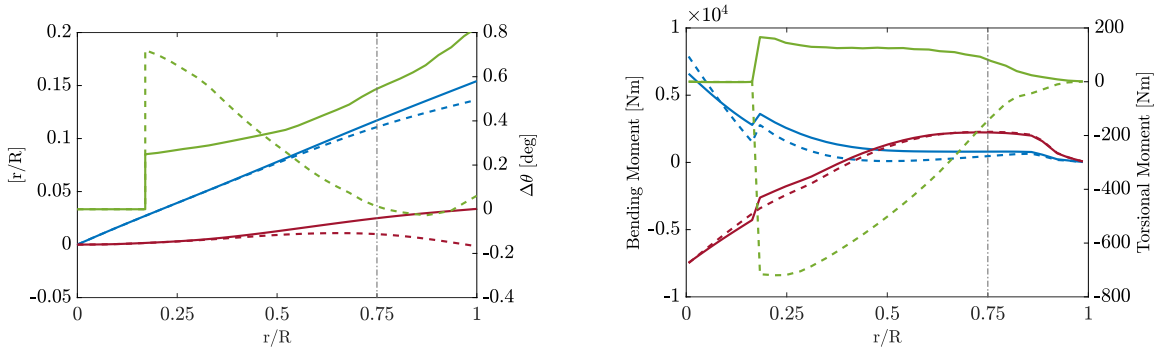


(a) Deformations (b) Internal moments

Figure 78: $U_\infty = 300$ knts ($154.3\,\mathrm{m\,s^{-1}}$): dotted MBDyn, continuous DUST-MBDyn

4.6.2.1 *Rotor stability in airplane mode*

Finally the rotor stability analysis has been conducted in the MBDyn model using the library Arpack for all 4 configurations. Each eigenanalysis is performed after reaching the trim-point condition, and the inflow states are neglected.



(a) Deformations

(b) Internal moments

Figure 79: $U_\infty = 400$ knts ($205.8\,m\,s^{-1}$): dotted MBDyn, continuous DUST-MBDyn

The first three rotor modes were considered: regressive gimbal Ω^-, cyclic regressive lag ξ^-, and cone β. The following equations are used to bring the cyclic eigenvalues from the rotating reference frame to the fixed one:

$$\tilde{\omega}_n = \frac{1}{2\pi}\sqrt{\mathbb{R}e(\lambda_i)^2 + \mathbb{I}m(\lambda_i \pm \Omega)^2} \tag{185}$$

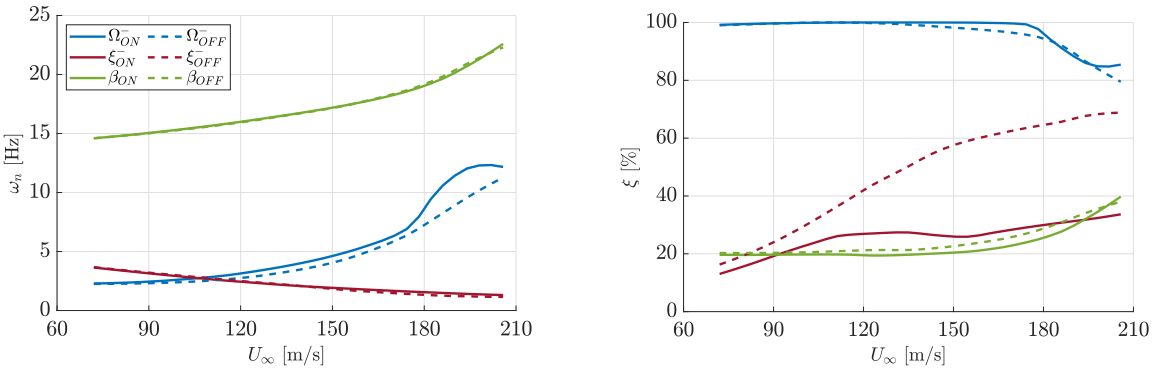
$$\tilde{\xi} = -\frac{\mathbb{R}e(\lambda_i)}{\sqrt{\mathbb{R}e(\lambda_i)^2 + \mathbb{I}m(\lambda_i \pm \Omega)^2}} \tag{186}$$

Here Ω is the angular velocity of the rotor, and $\mathbb{R}e(\lambda_i)$ and $\mathbb{I}m(\lambda_i)$ are, respectively, the real and imaginary parts of the eigenvalue. In the case of a collective mode, its value is identical in both fixed and rotating reference frames.

Figure 80 compares the power and windmill configurations in AP mode using a flexible control chain. In terms of frequency trend, the engine torque does not play a significant role; instead, a significant difference can be observed in the trend of cyclical lag damping, in which the windmill configuration exhibits a much higher damping ratio.

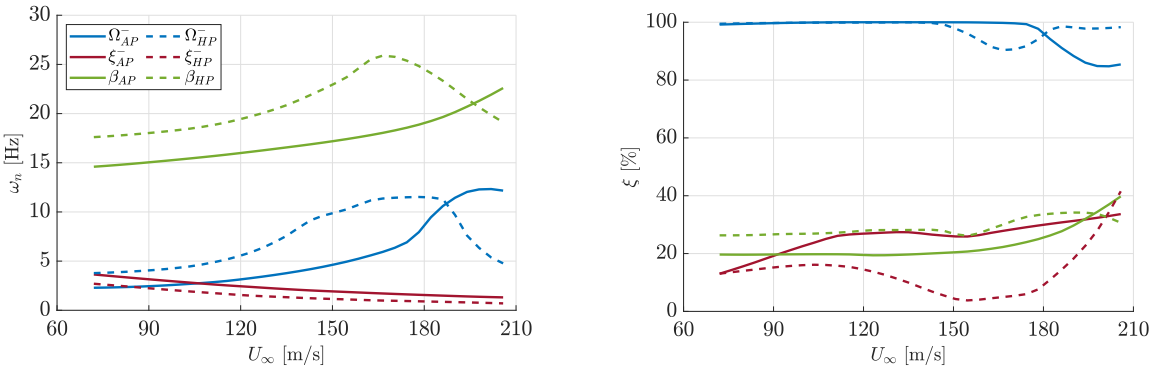
Figure 81 compares the effect of rotor speed in the powered configuration using a flexible control chain. For what concerns the frequency trend, the cyclic lag increases its natural frequency with the rotor speed, whereas both the regressive gimbal and the cone mode reduce their natural frequency with the rotor speed. Looking at the damping trend, the cyclic lag in helicopter mode has a decrease in the damping ratio to a value close to 2% in $154.8\,m\,s^{-1}$. Regarding the gimbal mode, both the natural frequency and the damping ratio are shifted to the right as the rotor speed increases.

Finally, figures 82, 83, 84, and 85 investigate the effects that a flexible control chain plays on the rotor stability for the four configurations. For all four configurations, the natural frequency of the cone mode of the flexible control chain has a bumpy trend with a maximum around $170\,m\,s^{-1}$ for the HP rotor speed; instead, with a



(a) Frequency (b) Damping ratio

Figure 80: Eigenmodes comparison in AP mode: power on vs power off

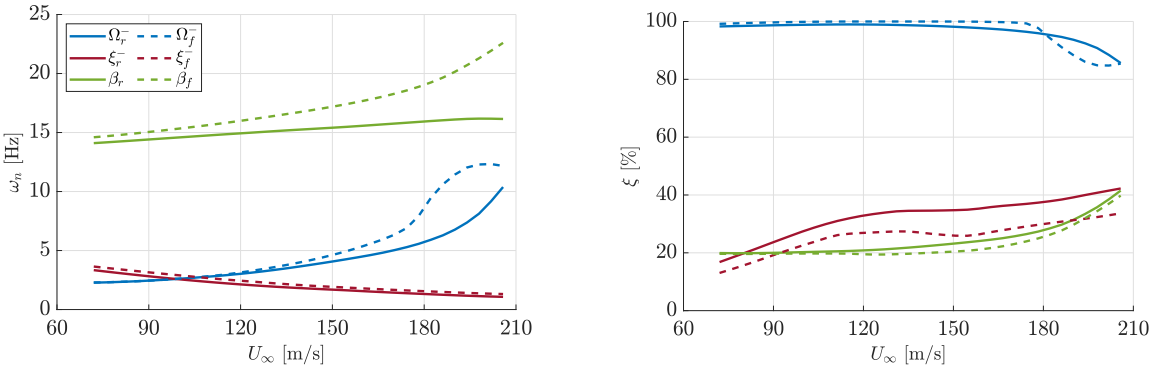


(a) Frequency (b) Damping ratio

Figure 81: Eigenmodes comparison in power on: AP mode vs HP mode

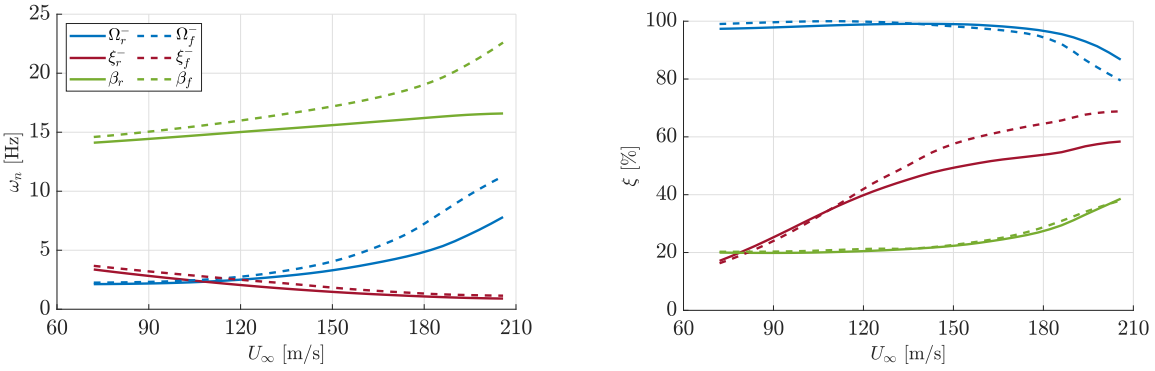
rigid control chain this mode tends to have an almost linear increasing trend. The same consideration can be taken for the regressive gimbal mode. Instead, the control chain slightly increases the natural frequency of the regressive lag mode without altering its trend.

The damping trends show that the cyclic lag mode tends to degrade its stability when the flexible chain is used in the powered configurations, and the opposite happens to the windmill condition. In the AP mode, the effect of the control chain is negligible for the damping of the cone and the gimbal mode; instead, in the HP, the cone mode with a rigid chain tends to increase its value to a maximum around $170\,\mathrm{m\,s^{-1}}$.



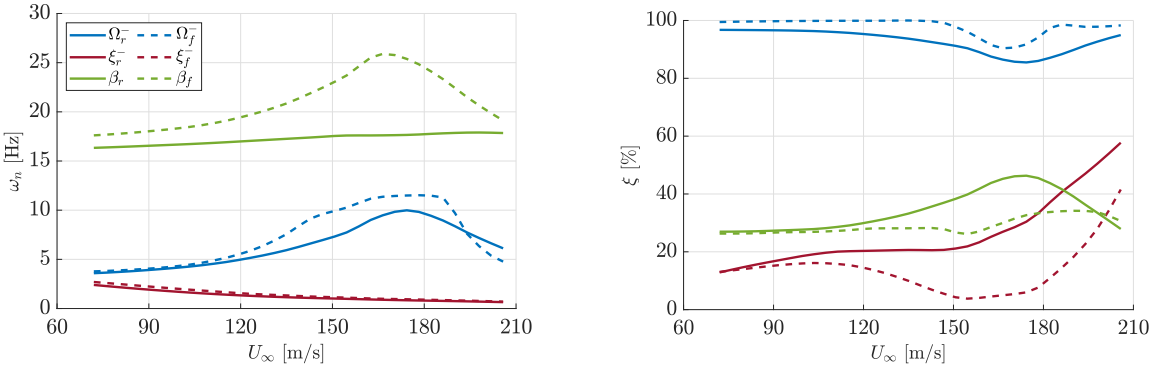
(a) Frequency (b) Damping ratio

Figure 82: Eigenmodes comparison in AP mode and power on: flexibility effect



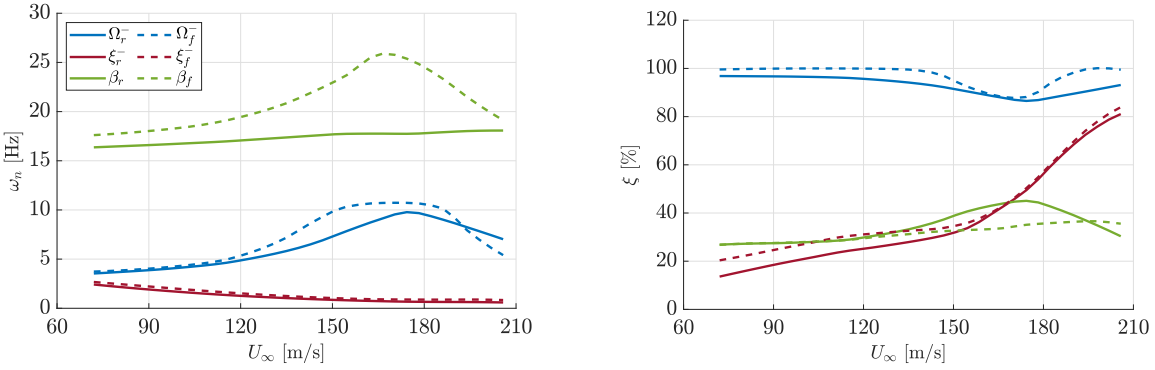
(a) Frequency (b) Damping ratio

Figure 83: Eigenmodes comparison in AP mode and power off: flexibility effect



(a) Frequency (b) Damping ratio

Figure 84: Eigenmodes comparison in HP mode and power on: flexibility effect



(a) Frequency (b) Damping ratio

Figure 85: Eigenmodes comparison in HP mode and power off: flexibility effect

5 FIXED AIRFRAME MODEL

5.1 INTRODUCTION

In the present chapter, the aeroelastic model of the XV-15 fixed airframe model is treated. The model has been developed based on the original Finite Element (FEM) stick model proposed by Acree [5]. Regarding the original model, the MBDyn-DUST model the control surfaces to allow the ability to trim the aircraft in free-free conditions.

Figure 86 shows the model of the XV-15 airframe during a roll maneuver.

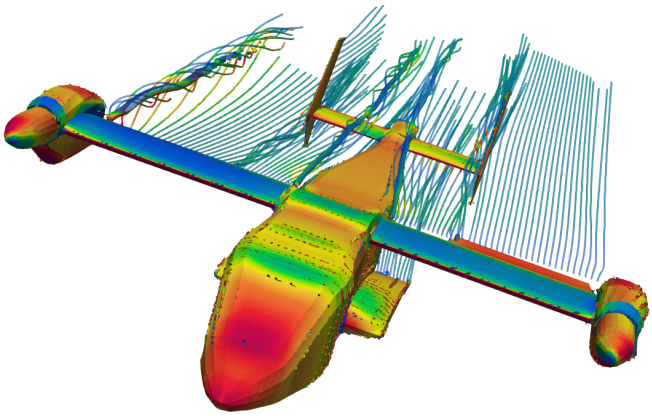


Figure 86: XV-15 MBDyn-DUST airframe during a roll maneuver

5.2 FIXED AIRFRAME MODEL

5.2.1 *Structural model*

The fixed airframe model is composed of six main components:

1. Flexible wings.
2. Rigid fuselage.
3. Rigid empennages.
4. Nacelle tilt mechanism.

5. Control surfaces: elevator, rudder, flaps, and flaperons.

6. Pilot's cockpit: seat and control inceptors (collective and cyclic).

The model is developed to reproduce the fundamental frequencies and mode shapes of the full aircraft in terms of the wing-nacelle part. The stiffness and inertial data are taken from the Finite Element Model (FEM) model presented in [5]. The model is tuned to capture the six lowest normal modes of the wing-nacelle system: symmetric/antisymmetric wing bending (SWB/AWB), symmetric/antisymmetric wing chord (SWC/AWC), and symmetric/antisymmetric wing torsion (SWT/AWC).

In MBDyn, the wing model is made of 5 finite-volume three-node beam elements and a set of lumped masses as for blades and yoke. The nacelle is considered a rigid body, connected to the wing by deformable joints that represent the flexibility of the downstop attachment. The stiffness of the downstop varies as a function of the conversion angle ϕ; it is calculated following the approach presented in [135]. The list of inertial and stiffness properties in the format E50 can be found in Tables 26 and 27, while the characteristics of rigid bodies are listed in Table 28. Control surfaces are modeled as rigid bodies, with associated polar inertia, and attached to the fixed-wing part by a statically determined constraint that is a combination of a spherical and inline joint: this model choice corresponds to a perfectly isostatic structure.

The fuselage is modeled as a rigid body that carries the pilot seat, landing gears, horizontal and vertical empennages, and the wing-nacelle system. A representation of the stick model is given in Fig. 87

The dynamics validation is performed by comparing the frequency and hub modal displacements of the first three symmetric and antisymmetric wing modes in the downstop-ON condition, with respect to the original model proposed in [5]. The resulting frequencies are reported in Table 13, whereas Figures 88,89 and 90 show the MBDyn mode shapes and Figures 91, 92 and 93 compare the modal displacement of the hub between the original FEM model and the MBDyn one.

Figure 94 shows the frequency trend of the symmetric and antisymmetric wing modes as functions of the angle of the nacelle. The SWB mode is the only one that maintains its frequency almost constant over the range of the maneuver, whereas the others change in a non-linear way. This is mainly due to the change in position of the center-of-mass of the rotor-nacelle system. Furthermore, the change in the stiffness of the downstop with respect to the angle of the nacelle contributes to the frequency change. In particular, the SWT mode increases its frequency by almost 1 Hz when the downstop locking mechanism is activated.

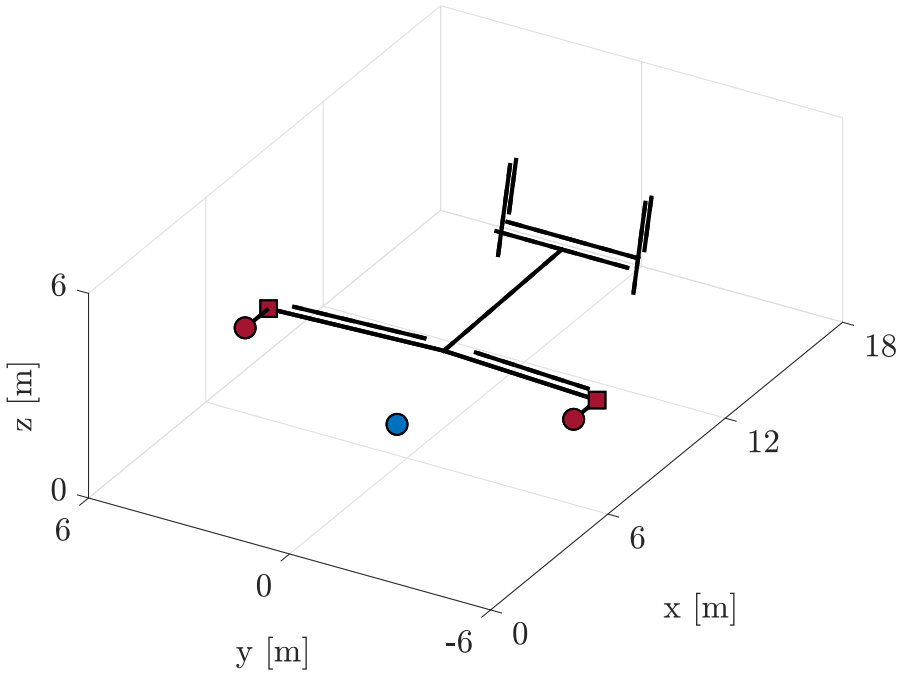
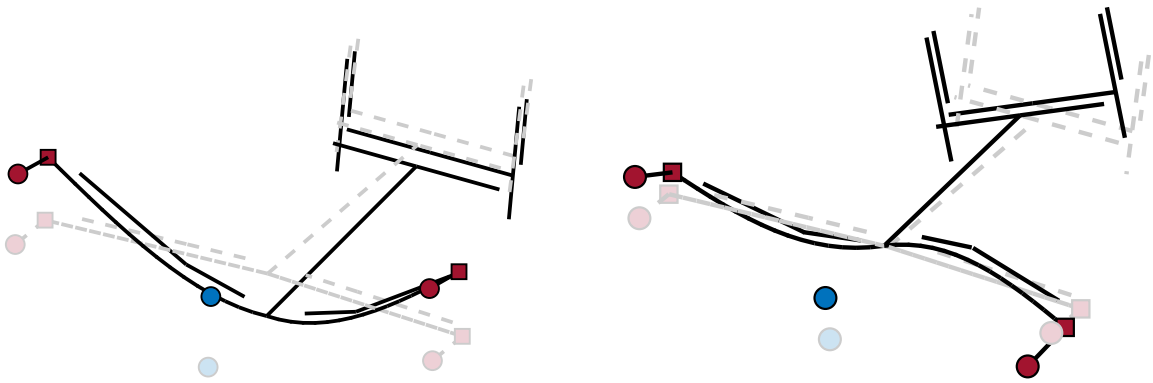


Figure 87: XV-15 stick model

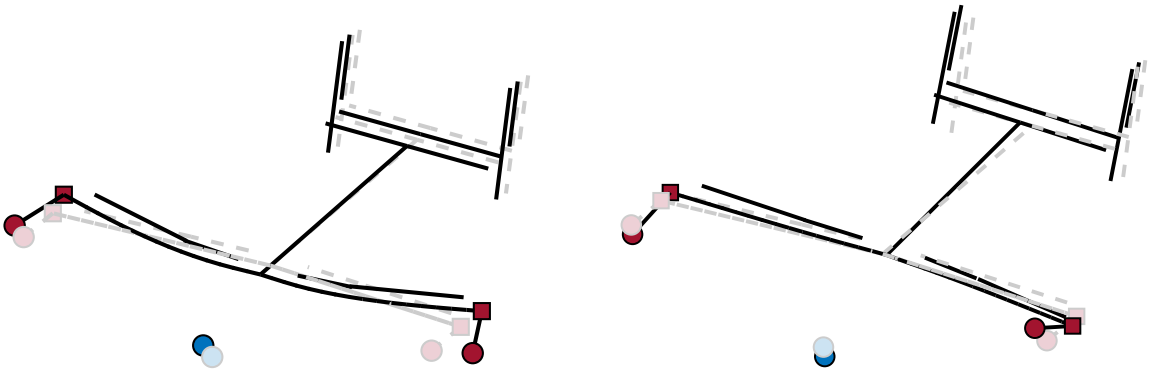
Table 13: Airframe modes frequency [Hz] comparison

Mode	Ref. [5]	MBDyn	Diff.
Symmetric beam	3.33	3.34	0.24%
Symmetric chord	6.32	6.13	−3.08%
Symmetric torsion	8.38	8.31	−0.81%
Anti-symmetric beam	6.43	6.22	−3.36%
Anti-symmetric chord	8.76	8.80	0.51%
Anti-symmetric torsion	7.15	7.02	−1.81%



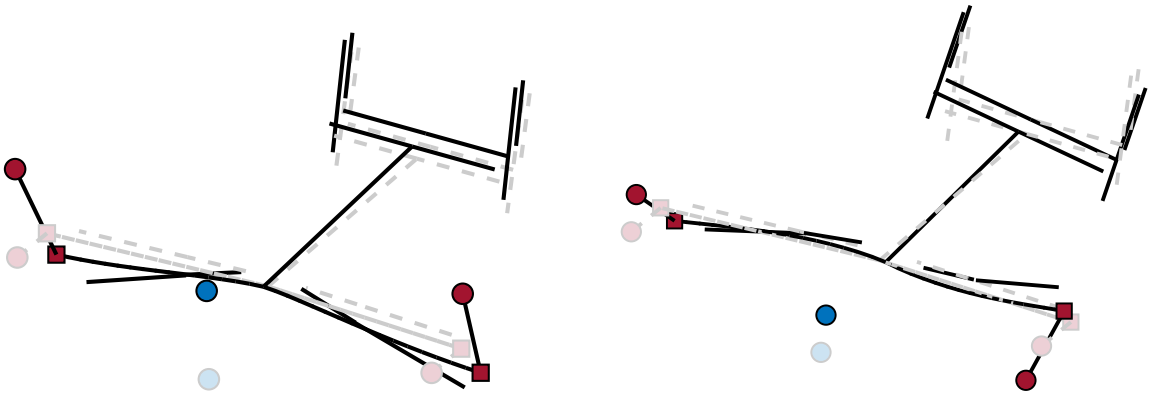
(a) Symmetric (b) Anti-symmetric

Figure 88: Airframe: bending modes



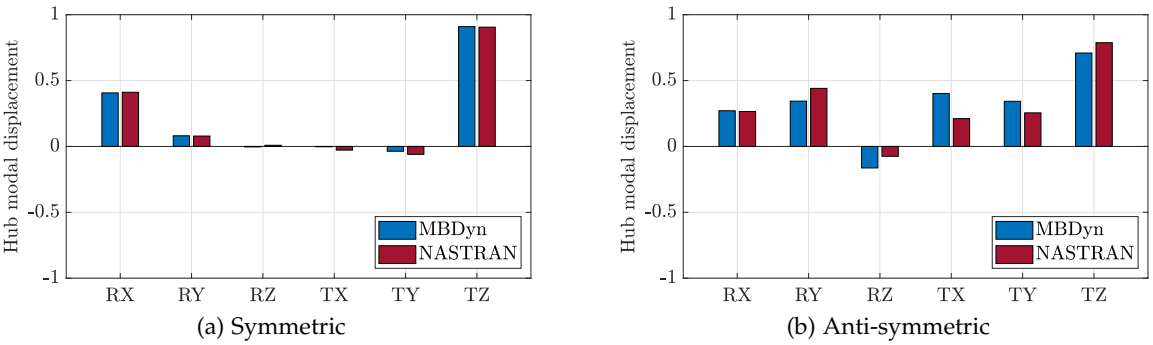
(a) Symmetric (b) Anti-symmetric

Figure 89: Airframe: chord modes



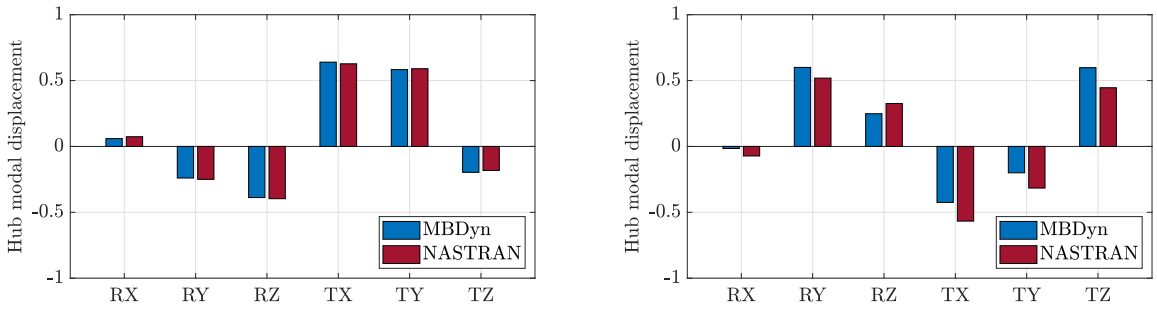
(a) Symmetric (b) Anti-symmetric

Figure 90: Airframe: torsion modes



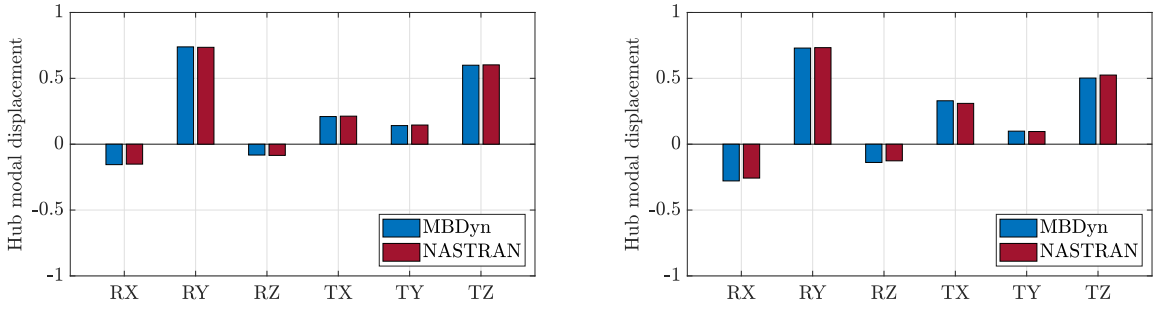
(a) Symmetric (b) Anti-symmetric

Figure 91: Hub modal displacement: bending modes



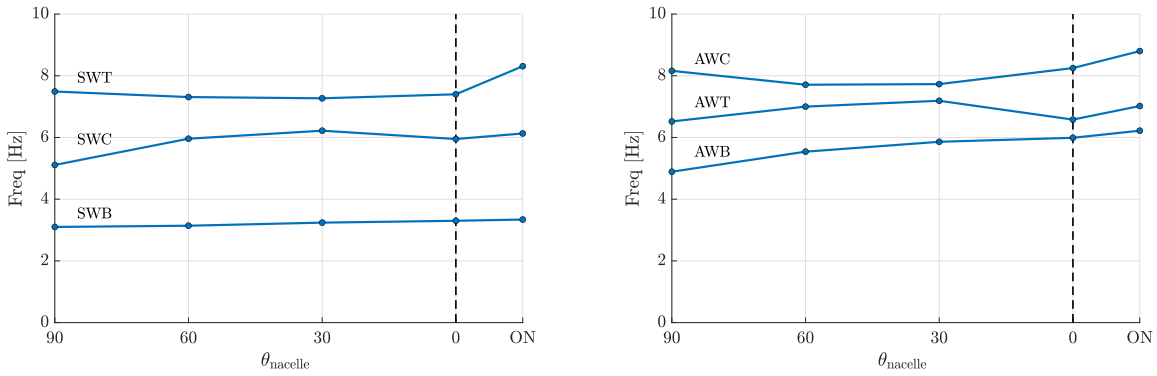
(a) Symmetric (b) Anti-symmetric

Figure 92: Hub modal displacement: chord modes



(a) Symmetric (b) Anti-symmetric

Figure 93: Hub modal displacement: torsion modes



(a) Symmetric (b) Anti-symmetric

Figure 94: Airframe modes frequency vs nacelle conversion angle

5.2.2 *Aerodynamic model*

The numerical model of the XV-15 tiltrotor is built considering the full scale dimensions and components of the aircraft As for the rotor case, the aerodynamic is modelled in two ways: a simplify model adopts the build-in MBDyn aerodynamic body, whereas a more sophisticated one is introduced by employing the DUST vortex lattice and panel elements coupled with MBDyn.

In case MBDyn, the control surfaces are models as simple lifting line elements placed in tandem with respect to the fixed profile, while in DUST they are introduced by chordwise mesh deformation. The introduction of control surfaces is necessary to perform aeroelastic trim analysis, flutter stability analysis, and transient maneuvers of the entire aircraft. In particular, the DUST model includes the fuselage, the horizontal and vertical tailplanes, the wing equipped with control surfaces (flap and flaperon), and the two proprotors with the corresponding nacelles. The main geometric characteristics of the entire vehicle, including the airfoils, are reported in Table 14.

Table 14: Geometric characteristics of the XV-15 [108].

	Wing	Horizontal Tail	Vertical Tail
Airfoil	NACA 64A223	NACA 64 015	NACA 0009
Span	9.8 m	3.91 m	2.34 m
Mean aerodynamic chord	1.60 m	1.20 m	1.13 m
Sweep (c/4)	$-6.5°$	$0°$	$31.6°$
Dihedral	$2.0°$	$0°$	-
Incidence	$3.0°$	$0°$	$0°$
	Flap	Flaperon	
Span along hinge line	1.30 m	2.40 m	
Chord/Wing chord	0.25	0.25	
Maximum deflection	$75°$	$47°$	

The aerodynamic model uses different types of aerodynamic elements: and surface panels for all other aerodynamic components: 2514 elements are used for the fuselage, 1620 for the tail planes, 835 for each nacelle and 3500 for the wings. The aerodynamic mesh of the complete XV-15 tiltrotor is shown in Figure 95.

A faster but simplified linear vortex-lattice model that models only the lifting surfaces (wing and tailplane) has been used to tune the trimming procedure and the autopilot control gain 5.3. This model uses 10×20 elements for the wing and 8×10 elements for the vertical and horizontal tail planes. In this model, the bluff bodies are

neglected. In Fig. 95, the complete DUST aerodynamic mesh of the XV-15 aeroelastic model is presented.

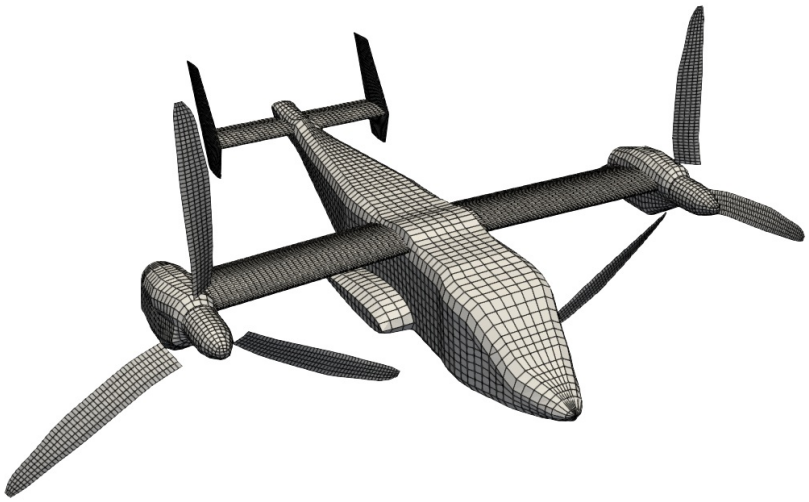


Figure 95: XV-15 aerodynamic mesh of the refined model

The precision and convergence of the aerodynamic mesh are validated by comparing the lift coefficient of the wing-nacelle complex with the experimental results of Fergurson [61]. The model neglects the presence of the fuselage, which is replaced by the continuation of the wing, and for the vortex lattice mesh, the nacelle is also neglected, and to compensate for its effect, the wing span is elongated up the rotor axis. In Fig. 96, the lift coefficient curves are compared using a free-stream velocity of $U_\infty = 140$ Knts. The surface panels agree well with the experimental data, while the linear vortex lattice element is estimated to have a loss on lift coefficient of 7%.

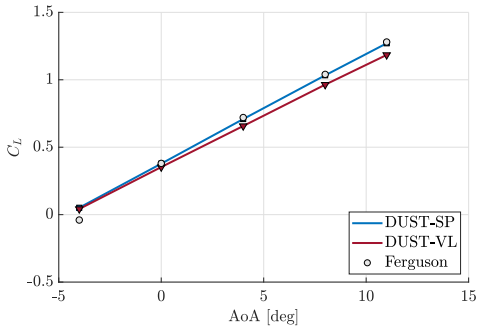


Figure 96: Wing-Nacelle lift coefficient: comparison between the wind tunnel data of Fergurson [61] and the surface panel and vortex lattice DUST mesh.

From a structural point of view, the control surfaces are modeled as a rigid body connected to two nodes that represent the hinges. The first node is constrained by a spherical hinge, while the second is constrained by an inline joint. The rotation of the control surface is

obtained by imposing a rotation on the second constraint. The combination of these three joints makes the constraint statically determined. The line connecting the two nodes identifies the axis of rotation of the movable surface; the aerodynamic mesh is deformed according to the methodology illustrated in Section 2.3.3.

5.3 TRIM PROBLEM OF THE FULL AIRCRAFT

Considering an almost symmetric flight condition in airplane mode, the trimming problem consists in computing the elevator deflection δ_{el}, the pitch angle of the aircraft θ and the collective pitch angle θ_0. Since the trim condition searched is a straight flight condition, it is possible to decouple the rotor trim from the aircraft trim.

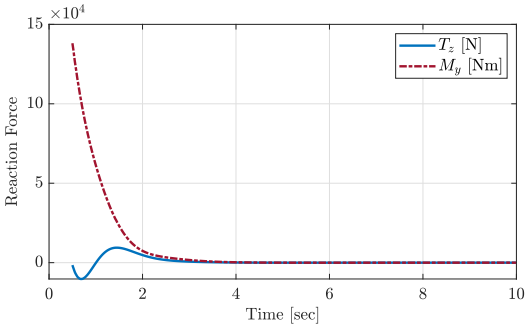
5.3.1 *Aircraft Trim*

Since the condition is symmetric, the problem reduces to finding the deflection and pitch angle of the elevator that guarantee longitudinal equilibrium. The trim problem in the coupled simulation is initialized by setting an initial elevator deflection and pitch angle of the entire aircraft. Then, to automatically find the static trim point, a PID regulator is introduced in the simulation with the aim of bringing to zero the vertical reaction force F_z and the pitching moment M_y of the total joint located at the attachment between the wing and the fuselage, by computing δ_E and θ. This point is chosen because it is sufficiently close to the center of mass. Since the objective of the controller is to bring the reaction forces to zero, the most important term in the controller is the integrator $\mathbf{K_I}$:

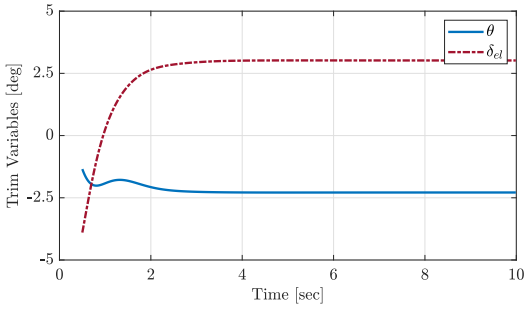
$$\begin{Bmatrix} \theta \\ \delta_{el} \end{Bmatrix} = \begin{bmatrix} K_{I\theta} & 0 \\ 0 & K_{I\delta} \end{bmatrix} \begin{Bmatrix} \int T_z \\ \int M_y \end{Bmatrix} \tag{187}$$

A simplified model is used to compute the static trim curves: the lifting surfaces are modeled thought vortex lattice elements with movable surface. In the structural model, the rotors are simplified with a lumped mass element; instead, the flexibility of the wing is considered. To avoid a possible interaction between the controller and structural dynamics, a Butterworth second-order low-pass filter with a cutting frequency $\omega_c = 1\,\mathrm{Hz}$ is placed between the reaction force measure and the regulator. The results of the simulation are shown in Figs. 97b–97a for a wind speed of $164.4\,\mathrm{m\,s^{-1}}$. Figure 97c shows the resulting trim variables obtained as a function of velocity using a simplified DUST vortex lattice model coupled with the elastic MBDyn aircraft frame and for model MBDyn alone. Note that the trim values in this case are not compared to the trim point of the real aircraft, since the computation has been performed using a very sim-

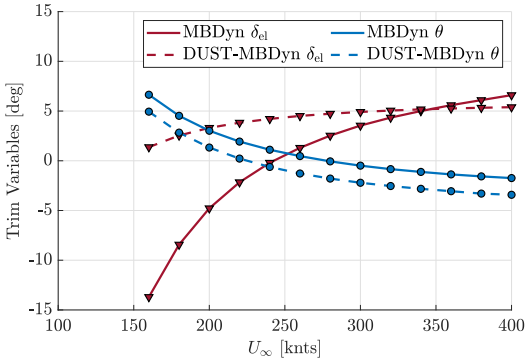
plified aerodynamic model, neglecting the influence of the rotors and the presence of the bluff bodies.



(a) Time history of the reaction forces



(b) Time history of the trim variables



(c) Trim variables as function of U_∞

Figure 97: Static trim results

5.3.2 *Autopilot*

The trim curves shown in Fig. 97a are used only to obtain the trimmed initial condition, whereas during the simulation the task of keeping the level flight is also required when the flight velocity is changed or when the rotorcraft is excited by external disturbances. As depicted in Fig. 98 the elevator deflection is therefore the sum of two distinct contributions, the trim contribution δ_t and the autopilot contribution δ_p. The autopilot control law is designed starting from the following longitudinal dynamics equilibrium equation for the airframe where the degrees of freedom are altitude z and attitude θ:

$$m(\ddot{z} + d_{cg}\ddot{\theta}) - L_w - L_t + mg = 0 \tag{188}$$

$$I_{yy}\ddot{\theta} + md_{cg}\ddot{z} + L_w(d_w + d_{cg}) - M_w - L_t(d_t - d_{cg}) - M_t = 0 \tag{189}$$

Where m and I_{yy} are the mass of the aircraft and the pitch inertia moment, d_{cg}, d_t, d_w is the distance between the pole and the center of mass, the aerodynamic center of the wing and the aerodynamic center of the tail, L_w and L_t are the lift associated with the wing and horizontal tail plane, M_w and M_t are the aerodynamic moment of

the wing and tail plane. Aerodynamic forces are modeled with the classical strip theory model with a Prandlt-Glauert compressibility correction.

$$L = \frac{\frac{1}{2}\rho U_\infty^2 S C_{L_\alpha} \alpha}{\sqrt{1 - Ma^2}} \tag{190}$$

where C_{L_α} is the slope of the lift coefficient curve and α the angle of attack of the considered part, wing or tail. In the context of this work, the aerodynamic characteristics $C_{L_\alpha}^w$ and $C_{L_\alpha}^t$, for the wing and tail, respectively, are calculated with DUST, considering the three-dimensional effects and the aerodynamic interaction between the wake of the wing and the horizontal tail.

Angle of attack α can be written in linearized form as

$$\alpha(\theta, \dot{\theta}, \dot{z}) = \alpha_0 - \theta - \frac{\dot{z} + d_w \dot{\theta}}{U_\infty} + \eta \delta_{el} \tag{191}$$

Inserting into equation 189 the expression of equations 190 and 191 and casting the system in state space with the elevator deflection input δ_{el} and the output the altitude measure, we obtain the following.

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\delta_{el} \tag{192}$$

$$y = \mathbf{C}\mathbf{x} \tag{193}$$

where $\mathbf{x} = [\dot{z}\ \dot{\theta}\ z\ \theta]^T$ and $y = z$

In Fig. 99 the aircraft transfer function is plotted for different airstream velocities. To avoid a possible adverse interaction between structural dynamics and the autopilot, the vertical position measurement z is filtered with low pass using a second-order Butterworth filter with a 1 Hz cut frequency for a wind speed of $144\,m\,s^{-1}$. Therefore, the state increases as: $\mathbf{x} = [\mathbf{x}^T \mathbf{x_f}^T]^T$.

To compensate for the initial trimming error of the airplane, due to the flexibility of the wing and the presence of the rotors, an integral action in the autopilot is mandatory. The system is therefore augmented as:

$$\begin{Bmatrix} \dot{\mathbf{x}} \\ \dot{x}_i \end{Bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{o} \\ \mathbf{C} & 0 \end{bmatrix} \begin{Bmatrix} \mathbf{x} \\ x_i \end{Bmatrix} + \begin{bmatrix} \mathbf{B} \\ 0 \end{bmatrix} \delta_{el} + \mathbf{w} \tag{194}$$

$$y_i = \begin{bmatrix} \mathbf{o} & 1 \end{bmatrix} \begin{Bmatrix} \mathbf{x} \\ x_i \end{Bmatrix} + v \tag{195}$$

Here, $\mathbf{w}$ and v are Gaussian white noises for the process and measurement. The Linera quadratic Gaussian (LQG) regulator is designed to minimize the cost function:

$$J = \int_t \begin{Bmatrix} \mathbf{x}^T & x_i \end{Bmatrix} \mathbf{Q} \begin{Bmatrix} \mathbf{x} \\ x_i \end{Bmatrix} + \delta_{el} R \delta_{el} \tag{196}$$

Here, **Q** and **R** are diagonal weight matrices that are needed to adjust autopilot performance. The LQG regulator is designed to solve the continuous Riccati equation. The list of parameters used to configure the optimal control can be found in Table 29.

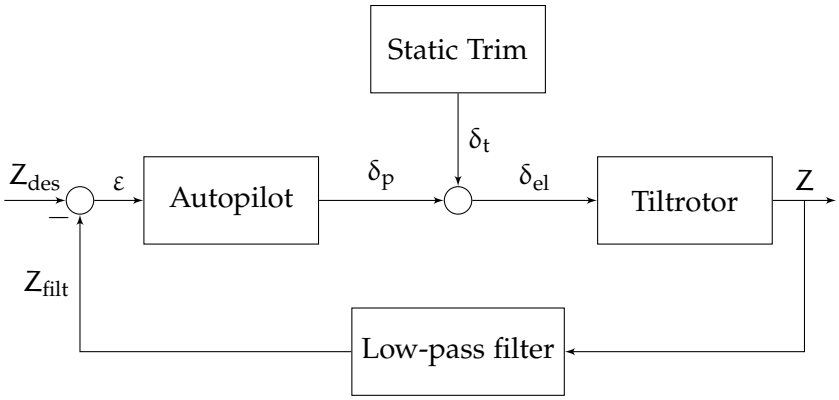


Figure 98: Tiltrotor autopilot loop

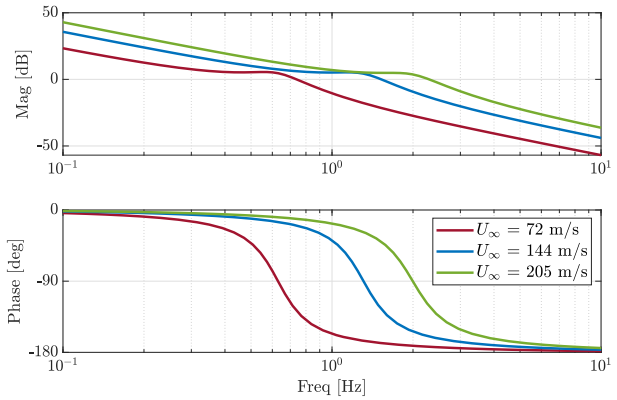


Figure 99: Rigid tiltrotor longitudinal dynamics: Bode frequency response

5.3.2.1 *Stability Augmentation System*

A stability augmentation system (SAS) is introduced to improve the damping ratio of the short period mode in the airplane mode. The control strategy consists of the introduction of a third contribution to the deflection of the elevator that is proportional to the pitch of the tilrotor q:

$$\delta_{SAS} = K_{SAS} q \tag{197}$$

To limit the action of the SAS to only the short period mode, without altering the other aeroelastic modes, a second-order low-pass filter with a cutting frequency of 2 Hz is introduced.

5.3.3 *Control performances*

The control performances in airplane mode is tested, considering a cosine-shaped gust to excite the aircraft. The gust parameters are chosen following the certification rules CS25.341. Three different gust gradients H are used:

1. $H = 9\text{m}$
2. $H = 20\text{m}$
3. $H = 107\text{m}$

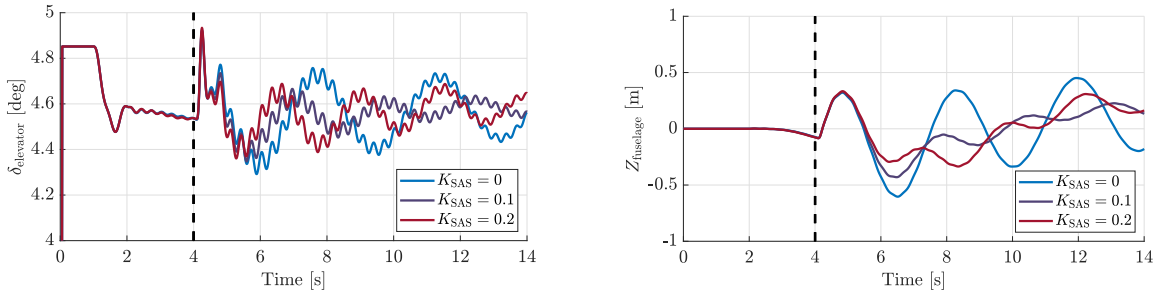
for 4 different velocities:

1. $V_A = 200$ Knts
2. $V_C = 260$ Knts
3. $V_D = 300$ Knts
4. $1.20 \cdot V_D = 380$ Knts

The stability of the control system is then evaluated using three different gain values for $K_{SAS} = [0.0, 0.1, 0.2]$ for both MBDyn and coupled models. The short period modal parameters are identified by looking at the aircraft pitch θ time response, using the Matrix Pencil Identification (MPE) method, detailed in Section 7.2.2.

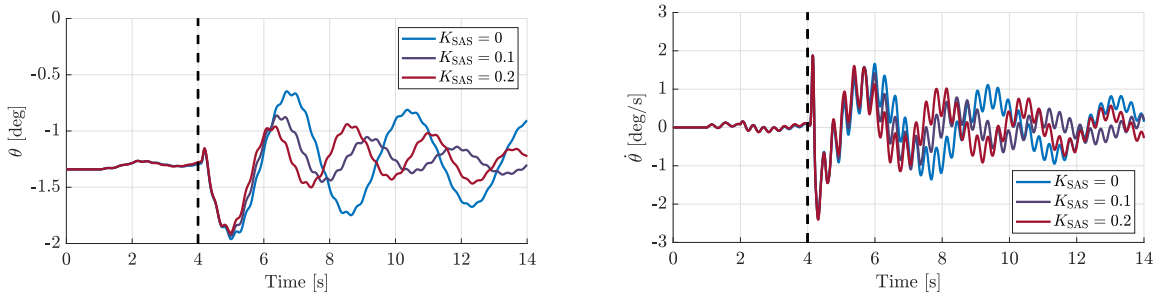
Figures 100 show the time response of the elevator deflection δ_{el} and the variation of the aircraft fuselage node $Z_{fuselage}$, whereas fig. 101 show the pitch and pitch rate for different SAS control gains. Simulations are carried out using the simplified DUST-MBDyn model at V_C with a $H = 9\text{m}$ gust gradient. From the pitch time response 101a, it is clear that increasing the proportional gain value of SAS has the effect of increasing the frequency of the short-period mode, while the damping ratio has a maximum value for $K_{SAS} = 0.1$. Since the chosen gust gradient is the shortest, this kind of gust is able to excite not only the short period mode, but also produces a dynamics response on the aircraft that is able to excite the first symmetric bending mode: the dynamic response is clearly visible in the pitch rate time responses, which in the short period mode is superimposed on a higher frequency one, and from Fig. 102b an identical bending mode response is observed for all three gains, thanks to the low-pass filter.

Figures 103, 104, and 105 show the effect that different gust gradients have on the response of the aircraft to freezing the free stream velocity at V_C and with $K_{SAS} = 0.1$. In particular, the longest gust produces the highest acceleration on the fuselage node, reaching the aircraft limit load factors ($n_{max} = 3g$), however, the $H = 20\text{m}$ produces the highest bending loads, as the frequency of the cosine function is very close to the 1^{st} flap bending mode. Regarding the short period



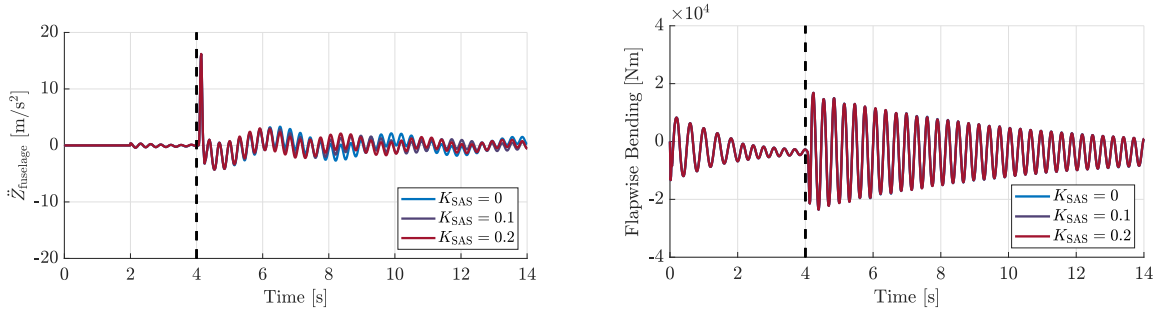
(a) Elevator deflection (b) Altitude variation

Figure 100: Gust response, SAS effect: control input-output



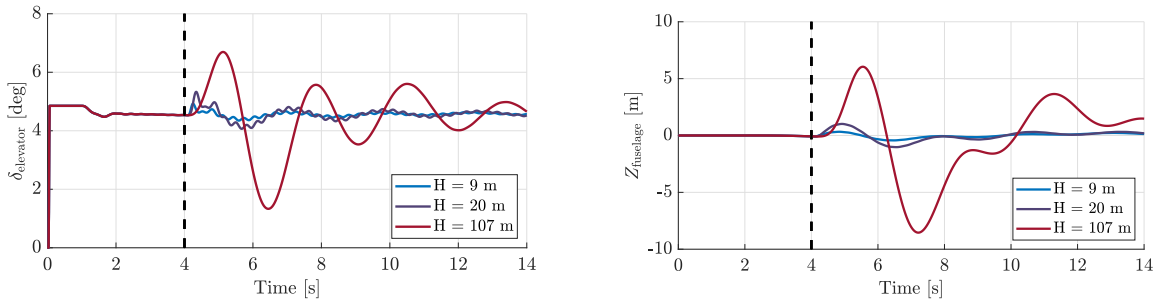
(a) Pitch θ (b) Pitch rate $q = \dot{\theta}$

Figure 101: Gust response, SAS effect: pitch angle and rate



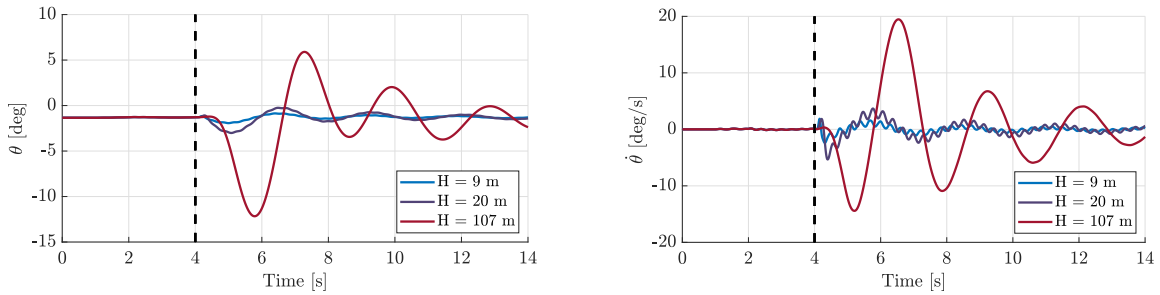
(a) Fuselage vertical acceleration (b) Flapwise bending moment

Figure 102: Gust response, SAS effect: vertical accelerations and bending moment



(a) Elevator deflection (b) Altitude variation

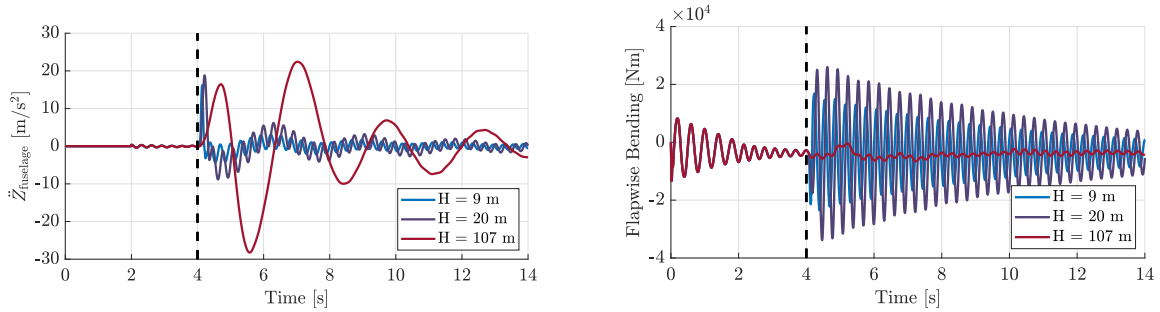
Figure 103: Gust response, gradient effect: control input-output



(a) Pitch θ

(b) Pitch rate $q = \dot{\theta}$

Figure 104: Gust response gradient effect: pitch angle and rate



(a) Fuselage vertical acceleration

(b) Flapwise bending moment

Figure 105: Gust response gradient effect: vertical accelerations and bending moment

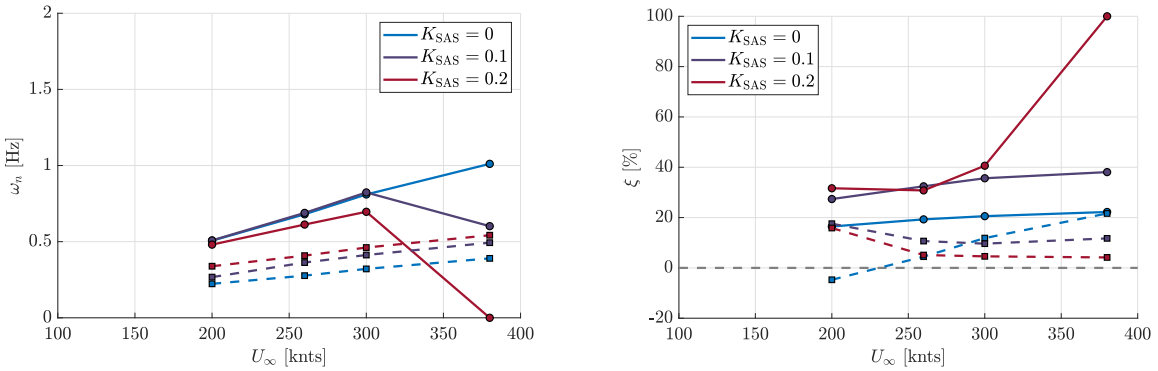
mode, from all three gusts the same modal parameters are obtained, this imply an almost linear response in the short period mode.

Figure 106 compares the frequencies and damping value against the wind speed for the MBDyn model and the coupled one for different gains of the stability augmentation system. It is evident that the MBDyn model has a short period frequency that is almost double that of the coupled model. This huge difference can be explained by looking at the rough approximation of the short-period frequency ω_{sp}, which is:

$$\omega_{sp} = \sqrt{\frac{C_{M_\alpha}}{I_{yy}}} \tag{198}$$

Since the structural model is exactly the same, the pitching inertia is the same for both models; therefore, the pitching stiffness coefficient for the MBDyn model (C_{M_α}) is roughly 4 times that of the coupled model. This could be explained by the fact that MBDyn totally neglects the aerodynamic interference between the lifting surfaces; therefore, every surface (in this case the horizontal tail plane) is always 100% effective.

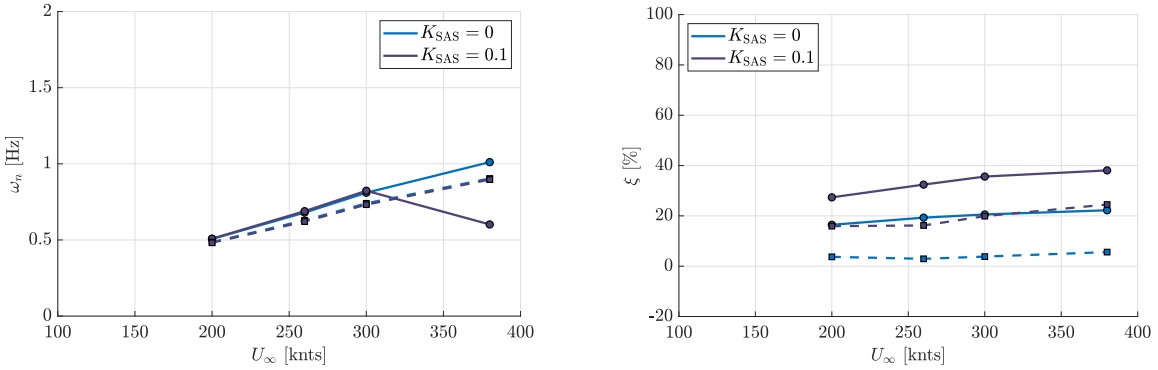
Figure 107 analized the effect that the presence of the rotor produces on the short mode. In this cases the comparison is made between the MBDyn model adopting two values for the stability aug-



(a) Frequency (b) Damping

Figure 106: Short period mode for the isolated airframe : comparison between MBDyn model (continuous lines) and coupled model (dashed lines)

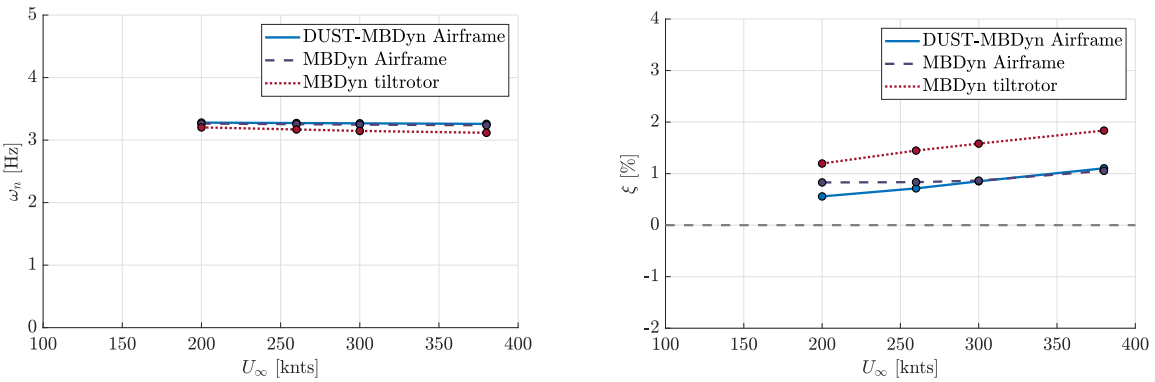
mentation system gain. In both cases, it is clear that the presence of the rotors has a negative impact on the damping of this mode, since the damping ratio is reduced by a factor of 15% for both K_{SAS}. This result could be explained by the gyroscopic loads produced by the rotors, which tend to destabilize the system. This result remarks on the importance of having a comprehensive model, especially when dealing with low-frequency flight dynamics modes.



(a) Frequency (b) Damping

Figure 107: Tiltrotor short period mode: comparison between MBDyn isolated airframe model (continuos lines) and complete flexible MBDyn tiltrotor model (dashed lines)

Finally, in Fig. 108 the bending modal parameters are identified using the matrix pencil identification method after excitation of the model with a short cosine-shaped gust of length $H = 9m$. In general, the difference between the aiframe model and the coupled one is minimal in terms of frequency and damping, instead of adding the rotors produces a slight increase in the damping value of about 0.7%.



(a) Frequency

(b) Damping

Figure 108: Tiltrotor bending mode excited by a $H = 9$ gust load for a $K_{SAS} = 0.1$: comparison between DUST-MBDyn isolated airframe model (continuos lines), MBDyn isolated airframe (dashed lines) complete flexible MBDyn tiltrotor model (dotted lines)

6

PILOT BIOMECHANICAL MODEL

6.1 INTRODUCTION

To investigate the effects on the tiltrotor performance of the pilot biomechanics, a multibody model of the pilot upper body, consisting of a detailed biomechanical model of the torso and upper limbs, has been developed in several years of research by the authors' research group, using the free, general-purpose multibody solver MBDyn [119]. The focus of the modeling is on the estimation of the biomechanical feedthrough at the cyclic and collective control inceptors; therefore, the model includes the upper body (head, torso, upper limbs, part of the thighs), the collective and cyclic inceptors and the related balancing and trim components, and the seat. A detailed description of the model is provided in the following sections, which are an expansion of Ref. [209].

Due to their load periodicity, rotating wing aircraft represents an intrinsically harsh vibration environment. Therefore, pilots and passengers' bodies are generally subject to a significant vibratory excitation during flight, with twofold consequences: On the one hand, both pilots and passengers can feel discomfort and, in severe cases, also suffer health-related problems [67, 72, 78]; on the other hand, in the case of pilots, discomfort is also associated with a degradation of handling qualities. Furthermore, vibrations are transmitted to the command inceptors, a phenomenon that is known as biodynamic feedthrough[149, 150, 162].

Both discomfort and biodynamic feedthrough depend on the human body vibratory response. Therefore, efficient estimation of the latter is of great importance, especially in the early stage of aircraft design. This aspect is particularly evident when the wide variability of the parameters that influence the biomechanical behavior of the human body is taken into account: mechanical properties and geometry above all, which further depend on anthropometric parameters like age, gender, weight, and stature. The confidence in which design choices are made in the early stages of the design is directly related to the robustness with which the estimation of the dynamical behavior of the vehicle and its occupants can be carried out.

First-principles approaches, like multibody modeling, are to be preferred in this context, precisely because of their ability to account for all the different sources of variability in physics-based fashion. Fully parametrized biomechanical multibody models of the human body can support the evaluation of the fitness of design choices with re-

spect to comfort and the robustness of the design with respect to the insurgence of adverse interaction phenomena.

This chapter presents multibody modeling of the human upper body with the purpose of enhancing the understanding of the human-machine interaction between the rotating-wing aircraft and its occupants. The spine model is the main focus of the presented work: it has been incrementally developed in the last several years[116], and has received a great deal of attention toward its thorough validation and its conjunction with the upper limbs model, also developed using MBDyn, which was originally presented in [124].

6.2 HUMAN SPINE BIOMECHANICAL MODELS

Approaches to numerical modeling of human body dynamical behavior, in the context of evaluating human-machine interactions in both comfort and handling-qualities-related aspects, vary significantly in previous experiences reported in the literature. However, they can be broadly divided into three categories: Lumped parameter models (LPM), finite element models (FEM), and multibody dynamics (MBD) models.

6.2.1 *Lumped Parameter Models*

Lumped parameter models make use of elementary mechanical elements, such as lumped masses and linear viscoelastic elements. The main advantage of these models is their low computational cost and ease of parameter identification. The core of LPM is the identification of the human body as a simplified mechanical system. The parameters are tuned to fit the biomechanical characteristics of a specific group of subjects. On the other hand, one disadvantage of LPM is that they are not consistent with the actual parameters of human anatomy and biodynamics, so they may not fully reflect the response of each part of the human body, and they are intrinsically difficult, or impossible, to extend outside the domain of parameter values upon which they are developed and validated. Moreover, they are often used to describe a uni-directional dynamic response of the human body because of compact expression and effective performance of the models only requiring as parameters the equivalent mass, stiffness, and damping.

One of the first lumped models in this context was proposed in [44]. From the measurement of the mechanical impedance of eight people with different heights, weights, and ages, a linear model of 1 degree of freedom (DOF) was proposed. In [180], a 1 DOF nonlinear model was proposed and applied to the analysis of the dynamic response of human bodies during helicopter landing. Ref. [197] presented a 4 DOF series-to-parallel linear model (shown in Fig. 109c)

to describe the dynamic response of seated occupants subjected to vibrations. This non-linear model was optimized in [2] using a genetic algorithm (GA). In [29], a measure of the vertical driving point mechanical impedance of the seated vehicle drivers was proposed in the range between 0.625–10 Hz; a 4 DOF linear model was derived, as shown in Fig. 109d, whose parameters were estimated to satisfy both the measured driving point mechanical impedance and the seat-to-head transmissibility (STHT). The four degrees of freedom are associated with the vertical displacement of the heads and neck masses, upper torso, viscerae, and lower torso. An optimization of the model presented in [29] was carried out in [57] using an average-weighted genetic algorithm. Several other lumped parameter models were proposed: The interested reader is referred to Ref. [104] for a detailed review of the subject.

In [16], a methodology is proposed to systematically identify the best configuration or structure of a 4 DOF human vibration model and identify its parameters. The results of this optimization are shown in Fig.109(a-b). The models were calibrated using the frequency response functions recommended by ISO 5982 [179].

6.2.2 *Finite Element Models*

Finite element models are more complete compared to LPM models. First, they respect the geometry of the human body; furthermore, they can be parametrized to represent individuals with different properties. Moreover, the output of the analysis is far richer: for example, in a frequency response analysis the acceleration and the internal load of each vertebra of the spine can be measured. The two main drawbacks are the higher computational cost and the difficult identification of the mechanical parameters of the model, compared to LPMs.

FEM models can be divided into two categories: discrete and continuous. The discrete models treat the spine as a structure made of rigid elements, representing the vertebral bodies, that are connected through deformable elements representing the intervertebral discs. In contrast, continuum models treat the spine as a homogeneous beam.

In [188], each vertebra was modeled separately in a discrete non-linear model with 8 DOF. However, the model is restricted to only the axial direction. In [142], a discrete parameter model of the human body was developed in a variety of impact situations; in detail, the pilot ejection problem was thoroughly investigated. These two models were restricted to one- or two-dimensional behavior and did not consider the interaction of the spine with other parts of the torso, such as the rib cage and the viscerae.

In [24], a discrete three-dimensional mathematical model of the human spine, torso, and head was developed. This model is based on a small-strain, large-displacement formulation. In [23], visceral masses,

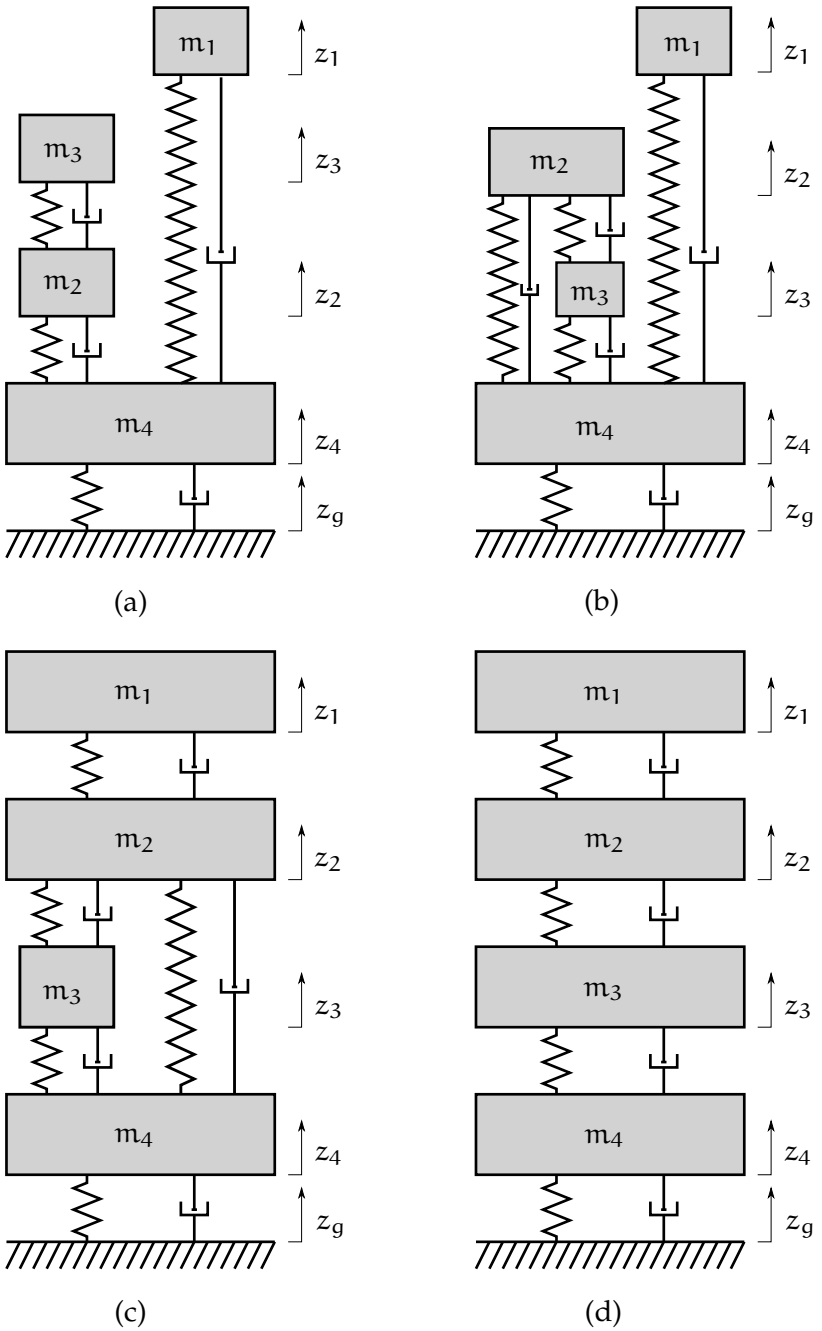


Figure 109: Lumped parameters models: (a, b) are the $4DOF_{12-6}$ and $4DOF_{14-9}$ of Ref. [16]; (c) is the model of Refs. [2, 197]; (d) is that of Refs. [29, 57]. The model elements can be generally associated with the structures of head and neck, upper torso, viscerae, and lower torso; however, in some cases they do not have a strict physical meaning, and therefore cannot be associated with specific anatomical structures.

dampings, and stiffnesses, and visceral modeling were tuned to better match the experimental results. The visceral wall can, in fact, be viewed as a secondary path of force transmission between the seat and the upper torso: in the proposed model, forces are transmitted through the rigid link that connects the highest visceral element to T10. The model presented in [93], an evolution of the one proposed in [23], is a 2D FEM model, which allows movements only in the sagittal plane. It models the spine, viscera, head, pelvis, and buttock tissues, using beam, spring, and mass elements. The geometric and mechanical parameters are based on those presented in [23], but some of them have been modified to match the mode shapes obtained by experimental work [94]. The model is entirely linear; it includes 134 elements and 87 master degrees of freedom.

In [186], the authors proposed a spine FEM model inspired by that of [93] (cf. Fig. 110), for the purpose of evaluating the comfort of the rotorcraft: with respect to the model presented in [93], the proposed model can capture the full 3D behavior of the spine with respect to a nominal reference condition.

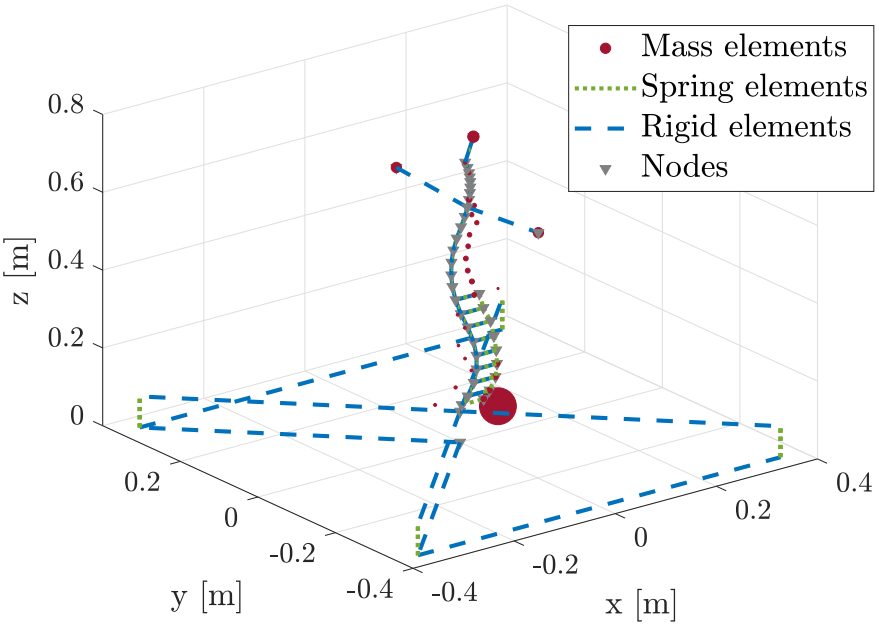


Figure 110: Finite element spine model proposed in [186].

6.2.3 *Multibody Models*

Multibody Dynamics (MBD) adds flexibility to LPM with ease of constraint formulation. Additionally, flexible elements include and extend the capabilities of FEM models. The great advantage of MBD is

that it can capture effects related to nonlinearities, especially those originating from 3D geometry.

In [208], a multibody spine model with 10 DOF was presented. This model was restricted to the sagittal plane; the cervical and thoracic sections were expressed as a single rigid body.

In [53], a two-dimensional 20 DOF model was proposed that can be used to study vertical and fore-aft vibrations. Parameters were optimized using a genetic algorithm.

A full multibody model of the spine was proposed in [195] for an accurate assessment of seated body vibration, but also this model was also limited in the sagittal plane. In [196] the model was improved to capture the three-dimensional motion of the spine.

The advantages and disadvantages of each type of model are summarized in Table 15.

Table 15: Pros and cons of the three types of spine models

Model	Advantage	Disadvantage
LPM	Low computational cost Ease of parameter identification	Single-axis model Not consistent with anatomy Limited output
FEM	3D model Consistent with anatomy Richer output	Higher computational cost Parameters are difficult to identify
MBD	3D model Can capture nonlinearities Ease of constraint formulation	Parameter are difficult to identify

6.3 UPPER BODY MODEL

The proposed multibody upper body model consists of a detailed model of the torso, of the upper limbs, and their interface with the surrounding environment, typically an aircraft or rotorcraft cockpit.

6.3.1 *Torso model*

The torso model comprises 34 rigid bodies associated with nodes (that is, entities possessing degrees of freedom) placed in the correspondence of vertebrae from S1 to C1, the head, and of 8 visceral masses elastically connected to vertebrae from S1 to T10. The model

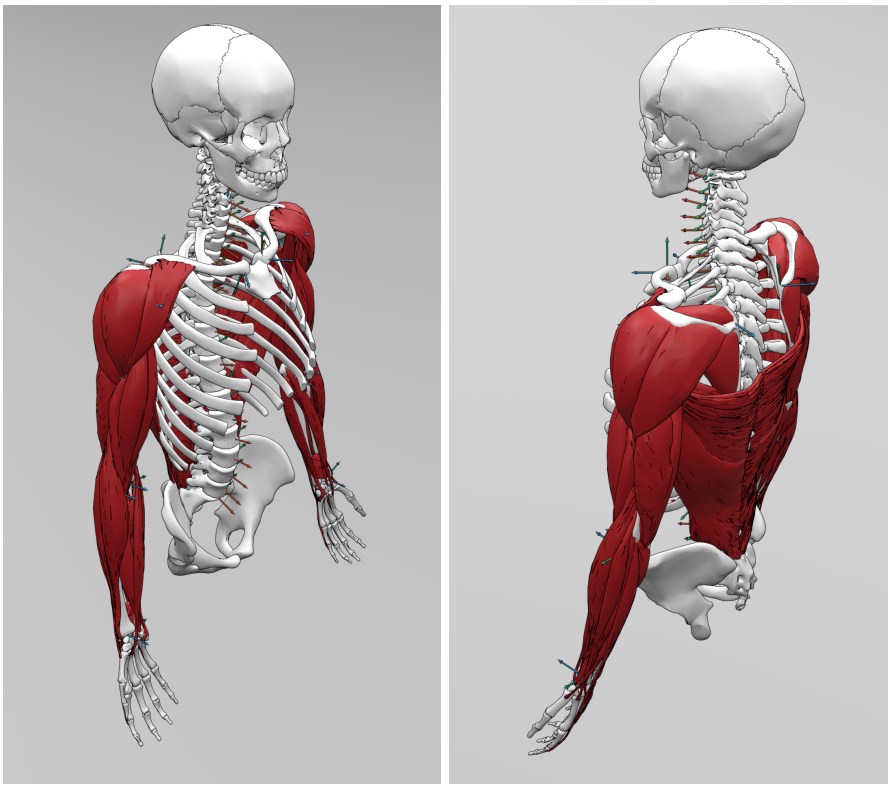


Figure 111: Upper body multibody model.

follows the concept originally proposed in 1997 by Kitazaki and Griffin [93], in turn, based on the database provided in 1978 by Privitzer and Belytschko [23].

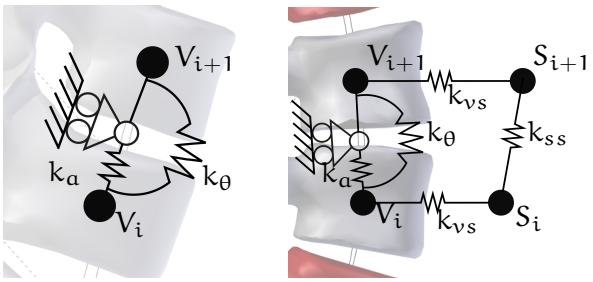


Figure 112: Algebraic and deformable constraints connecting vertebrae nodes, indicated with V, and viscerae nodes, indicated with S.

Each node of the vertebral section is located in the center of the corresponding vertebral body, with the axis z aligned with the local tangent to the curve described by the longitudinal axis of the spine, the axis y directed laterally and the axis x pointing anteriorly (Fig. 113).

The spine itself is composed of 25 vertebral nodes connected by 24 viscoelastic 3D deformable elements that act on both relative displacements and rotations between adjacent nodes. These are represented in a simplified way in Fig. 112 by the linear and angular springs of the elastic constants k_a and k_θ. Linear viscoelastic constitutive laws

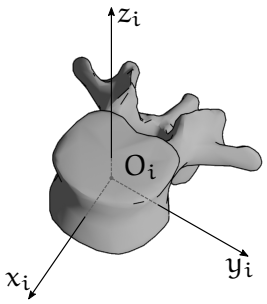


Figure 113: Each node related to a vertebral section is located in the center of the corresponding vertebra's body, with axis z_i along the local tangent to the spine axis curve and axis y_i pointing laterally.

relate the strains of the deformable elements to their internal forces and moments. The head and sacrum (S1) are modeled as rigid bodies and are, respectively, connected to vertebrae C1 and L5 by linear viscoelastic elements.

When the analysis is restricted to the sagittal plane, the vertebral nodes are connected to each other by algebraic constraints that limit their relative degrees of freedom to sliding along the spine axis and rotating about the lateral axis. Otherwise, when the model is used for three-dimensional simulations, all relative rotation degrees of freedom of the vertebral nodes are unconstrained, whereas the translation constraints remain in place (cf. Fig. 112) The intervertebral constraints mimic the behavior of the anatomical joints: the superior and inferior faces of the vertebrae bodies are joined by a symphysis joint, allowing all relative rotations and relative displacements permitted by the deformations of the fibrocartilaginous disc that lies between them; however, the intervertebral relative displacement in the transverse plane is constrained by a facet joint, a synovial joint that acts on the posterior processes.

The upper and lower visceral masses are rigidly connected to the vertebrae T10 and S1, while the other ones are connected to the respective vertebra node, in the section between T11 and L5, by kinematic constraints that allow only their relative displacement in the sagittal plane. Linear viscoelastic elements act on the relative viscera-vertebra displacements along the local x_i and z_i directions (cf. Figs. 113,112). Visceral nodes do not possess rotational degrees of freedom: as such, the visceral masses are considered as point masses. The elastic elements are depicted as linear springs in Fig. 112: k_{vs} denotes the elastic constants of the springs connecting a vertebral node and a visceral node, while k_{ss} denotes the elastic constants of the springs connecting two visceral nodes.

Detailed modeling of the muscular structures of the torso has been avoided, since the additional modeling effort would not have been justified for the application cases for which the model is currently used. Torso muscles of both rotorcraft pilots and passengers are involved

in postural control, i.e. typically they act to maintain a steady reference condition. The fully detailed modeling would allow us to greatly extend the range of validity of the model, especially with regard to poses that deviate considerably from the reference one (sitting-rest position). It would, however, also make the model underdetermined due to overactuation, and special techniques would have to be applied to estimate the equivalent impedance at each joint. Although similar experiences have been carried out successfully by the authors, in particular with respect to the impedance of the upper extremities at the control initiators [122, 123], it was concluded that for this particular case the simpler approach that relies on lumped viscoelastic elements is sufficient for targeted analyzes.

The pelvic region is modeled taking into account the compliance of the buttock tissue, using two viscoelastic elements that connect the sacrum with a node representing the mean interface point between the buttocks and the resting surface. A single rigid body, which also accounts for a third of the tights mass, in addition to those of the buttocks and pelvic structures, is connected to the pelvis node.

Geometry and inertial parameters are adapted to represent a generic subject that has the desired anthropometric characteristics of age, gender, stature, and weight, as detailed in the next section.

Table 32 lists the stiffness and damping values for a reference subject of 34 years, 84 kg and 1.783 m. The values K_a and C_a are associated with the axial springs and dampers that act in the z direction; K_{b_i} and C_{b_i} are associated with the rotational springs and dampers that act on the axis i. Table 33 lists the mass value, the moments of inertia and the position of each node for the reference subject.

Therefore, the total number of degrees of freedom of the system, before constraints are enforced, is 228. After the enforcement of the constraint, 103 degrees of freedom remain.

6.3.2 *Upper limbs model*

The spine multibody model has been joined with the upper limb multibody model developed in [124, 125] based on [151].

The model comprises 4 nodes (and the associated rigid bodies) per limb, associated with the humerus, radius, ulna and hand. The total number of unconstrained degrees of freedom is 24. The hand is condensed into a single rigid body since the applications that are simulated usually involve gripping tasks, in which the dynamics of the fingers can be disregarded. The nodes are connected by ideal algebraic constraints that model the following articular joints.

- the *Glenohumeral* joint (spherical hinge);
- the *Radiohumeral* joint (spherical hinge);
- the *Humeroulnar* joint (revolute hinge);

- the *Radioumeral* joint (point-on-line);
- the *Radiocarpal* joint (Cardano hinge)

The application of algebraic constraints eliminates 17 degrees of freedom, resulting in a 7 degree-of-freedom system. Each limb is actuated by 25 muscle fascicles, modeled by active viscoelastic rod elements. Therefore, the force F_i exerted by the muscle i -th depends on the distance ℓ between the origin and the insertion points, rigidly attached to the respective nodes, on its derivative $\dot{\ell}$ and on the non-dimensional activation coefficient a, ranging from 0 to 1:

$$F_i = F_{0,i}\left(\ell, \dot{\ell}, a\right) = F_{0,i}\left(f_1(\ell) f_2(\dot{\ell}) a + f_3(\ell)\right) \tag{199}$$

where the definitions of $f_1(\ell)$, $f_2(\dot{\ell})$ and $f_3(\ell)$ follow the simplified Hill-type model [200] presented in [151].

Since the applications shown in this work involve tasks in which the elevation angle of the humerus is consistently kept at very low values, the shoulder complex is not modeled in detail, disregarding the clavicle and the scapula. This is the case of piloting tasks: the expected effect of the scapula and clavicle motion on the shoulder kinematics is very limited. The glenohumeral joint is represented in the simplified scheme by a spherical joint located in the glenoid fossa. The interested reader is referred to the cited literature for further details regarding the upper limbs model.

6.3.3 *Seat Model*

The complete upper body model has been coupled with the seat model proposed in [37]. The seat is modeled as a single rigid body, suspended to the airframe floor through a linear viscoelastic element, tuned to limit the transfer of loads to the helicopter seat occupant. The cushion is modeled as a unilateral, non-linear viscoelastic element having the following constitutive law:

$$F(\epsilon) = \frac{a}{3}\left(1 - \tanh(b\epsilon)\right)\epsilon^3 \tag{200}$$

where a and b are positive real coefficients and ϵ is the deformation of the viscoelastic element. This unilateral constitutive law is chosen to have a null viscoelastic force when the element is undergoing positive (tensile) straining and an exponentially increasing reaction force to negative (compressive) deformation. The backrest is modeled using five unilateral viscoelastic elements, which link the vertebrae from T_4 to T8 to the backrest node. These elements share the same constitutive law as the cushion. The backrest node is rigidly connected to the seat node. A schematic representation of the seat model is shown in Fig. 114.

The flight control inceptors can be held in an arbitrary position in indifferent or stable static equilibrium by the mechanical action of

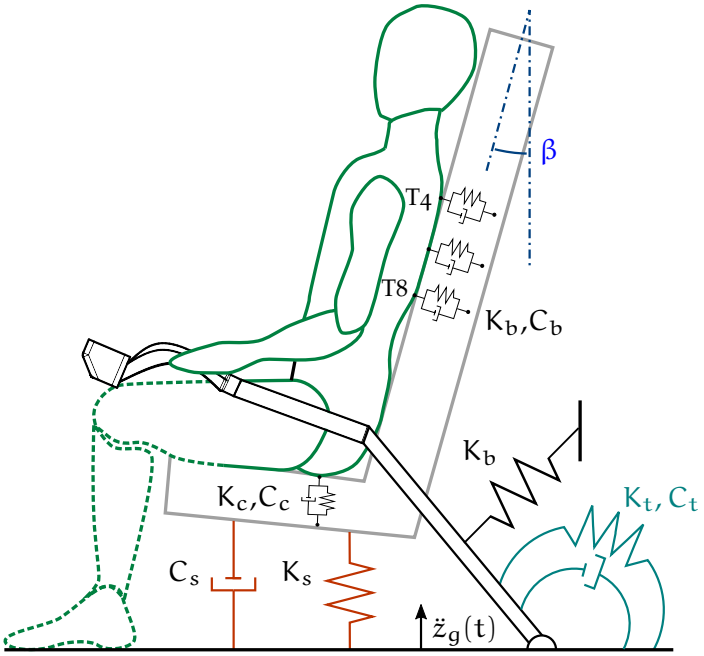


Figure 114: Seat constraint model: the seat is connected to the rotorcraft floor trough a viscoelastic suspension of stiffness K_s and damping C_s. The elastic coefficients K_c and K_b of the cushion and the backrest can be easily found differentiating the internal force expression of eq. (200). Damping coefficients C_c and C_b are proportional to the stiffness coefficients.

two viscoelastic elements, the *balancing* and the *trim* springs shown in Fig. 115: the viscoelastic restoring moment can significantly alter the feedthrough of the pilot. The parameters of the control inceptors used in this work are reported in Table 31.

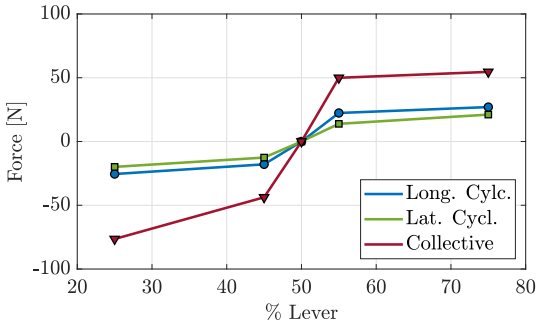


Figure 115: Trim spring stiffness nonlinear behavior

6.3.4 *Model Parametrization*

The model is mainly intended for the estimation of the vibratory response of the upper body of rotorcraft passengers and pilots, thus focusing on dynamics in the 1–30 Hz frequency range. In this context, the model should be able to reproduce the *average*, or *most plausible* subject possessing a certain set of anthropometric characteristics, while it is not intended to be purely subject specific. Parameterization is based on an anthropometric dataset $\mathbf{s}$ that comprises age a, Body Mass Index BMI, stature h, and sex g:

$$\mathbf{s} = \begin{Bmatrix} a & \text{BMI} & h & g \end{Bmatrix}^T \tag{201}$$

The statistical model proposed by Shi et al. [175] is used to generate the most plausible rib cage geometry associated with the anthropometric parameters. The model was established by analyzing computed tomography thorax scans from 89 subjects, divided into 8 age groups and of both sexes. Due to image segmentation, the positions of 464 landmarks were identified on one side of each subject's rib cage (Fig. 116). The correlations between the landmark positions and the anthropometric parameters were studied through a Principal Component Analysis (PCA), following the procedure outlined by Allen et al. [11].

After generating the rib cage landmarks, a bounding box is fitted to them, and their dimensions x, y, z are compared to those of the reference subject of the work of Kitazaki and Griffin [93], x_0, y_0, z_0, to produce three scaling coefficients:

$$\lambda_x = \frac{x}{x_0} \qquad \lambda_y = \frac{y}{y_0} \qquad \lambda_z = \frac{z}{z_0} \tag{202}$$

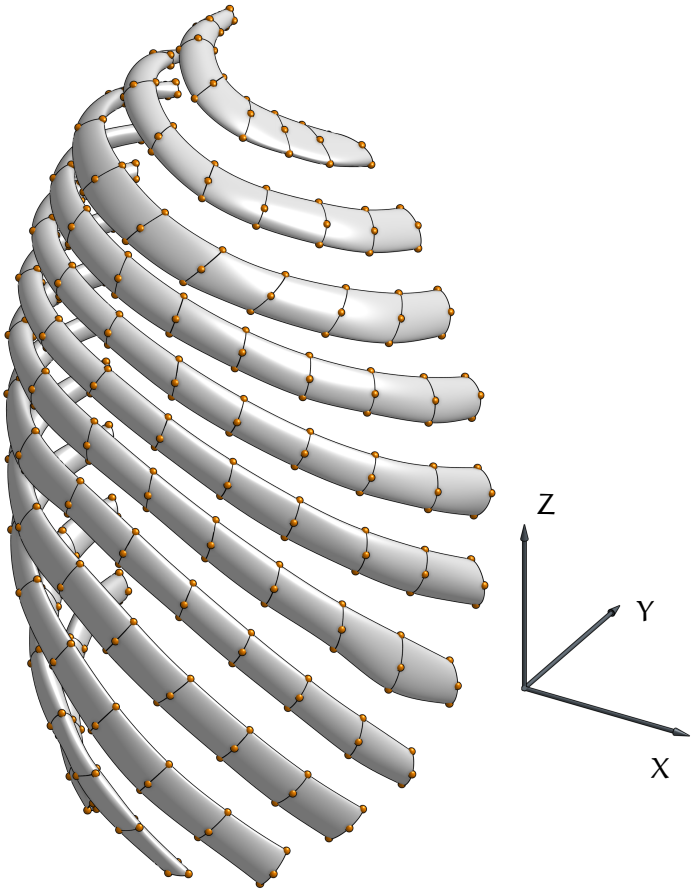


Figure 116: Example of the generated ribcage geometry from the landmarks point cloud, using the model proposed by Shi et al. [175]. The spheres are located the locations predicted by the statistical ribcage model. Interpolating surfaces have been added for clarity, but are not used in the multibody model generation.

Since the anthropometric characteristics of the modeled subject are not reported in [93], they were estimated a posteriori using a Sequential Quadratic Programming optimization algorithm. Find the set $\mathbf{s}_0$ that minimizes the mean square error of the distances between the estimated landmarks that lie closer to the transverse processes of the vertebrae and their corresponding locations estimated from the data of Kitazaki and Griffin. The resulting reference subject is a 34 year old man, 1.78 m tall and weighting 84 kg, thus having a BMI of approximately 26.5.

Using as a reference the normal configuration found in [93], the initial positions of the nodes for the modeled subject are obtained by directly scaling their components with the coefficients λ_i of Eq. (202).

6.3.4.1 *Mass Scaling*

Each rigid body of the upper limb model, comprising bone and associated soft tissue, is scaled with the total weight using linear regression relationships identified in large survey studies [35, 36] (cf. Appendix D.1). According to [50], the mass of the trunk M_t of the subject represents 68 % of the total body mass M_b, excluding the mass of the upper extremities M_l:

$$M_t = (M_b - M_l) \cdot 0.68 \tag{203}$$

The masses of the viscerae M_v are estimated as 20 % of the total body mass, excluding the upper limbs:

$$M_v = (M_b - M_l) \cdot 0.20 \tag{204}$$

Then, the ratio between each trunk and visceral mass element and the total trunk and visceral mass is defined as follows:

$$T_i = \frac{M_{t_i}}{\sum M_{t_i}} \tag{205a}$$

$$V_i = \frac{M_{v_i}}{\sum M_{v_i}} \tag{205b}$$

where the values provided in [93] are used for M_{t_i} and M_{v_i}. The scaled value of each mass element is then obtained by multiplying the mass ratios T_i and V_i by the total mass of the trunk and viscerae.

6.3.4.2 *Scaling of Stiffness and Damping Coefficients*

The parameters of the constitutive laws of intervertebral elements are scaled as follows:

1. The ratio between each stiffness or damping coefficient and the corresponding mass element is evaluated, using the parameters of the model in [93].

2. The value of this ratio is then multiplied by the corresponding mass scaled for the specific subject.

3. The resulting stiffness coefficients are modified using a second order polynomial $\mathcal{P}(\mathrm{BMI})$ (Cf. Appendix D.1) that correlates the BMI with the first resonance frequency, and the damping factor that corresponds to this peak,

$$K_i = \mathcal{P}(\mathrm{BMI}) \frac{k_{i_{KG}}}{m_{i_{KG}}} M_i \tag{206}$$

where $k_{i_{KG}}$ and $m_{i_{KG}}$ are the stiffness and mass parameters of the Kitazaki and Griffin model [93].

6.4 SOLUTION STRATEGIES

The multibody models require specialized solution strategies to cope with their different levels of indeterminacy. Indeed, the upper limbs model is both kinematically underdetermined and overactuated. The spine model is also kinematically underdetermined [124, 213].

The kinematic underdeterminacy of the upper limbs model is evidenced by the fact that—referring to a single limb—when the motion of the hand is full prescribed, one degree of freedom still remains: namely, it is possible to perform the same task with different angles of humerus elevation. The spine model presesents an even higher degree of underdeterminacy if only the motion of the head is prescribed: the same head trajectory is obtainable by a wide variety of spine configurations, thanks to the comparably high number of allowed relative vertebral configurations.

The upper-limb model ovearctuation is, in contrast, due to the fact that usually multiple muscle fascicles act about the same articular joint in numerous agonist/antagonist pairs. The total torque produced around the articular joint is therefore a result of the Central Nervous System (CNS) actuation strategy, which usually leverages muscular synergies. It is often postulated [54, 212] that the central nervous system (CNS) acts to optimize energy expenditure to achieve different goals: minimization of total activation, minimization of metabolic cost, and maximization of equivalent impedance of the limb are some possible examples.

For typical applications, that is, the estimation of the vibratory response of helicopter pilots and occupants for comfort assessment and the evaluation of proneness to rotorcraft-pilot couplings of existing and concept aircraft layouts, the general solution strategy is therefore made of several cascaded simulation steps:

1. From the desired anthropometric dataset, the model geometry and structural parameters of the model are generated according to the procedure outlined in Section 6.3.4.

2. An underdetermined inverse kinematics analysis is performed, in which both the pose of the spine and of the limbs are estimated imposing the motion on the hands and on the head.

3. An inverse dynamics analysis problem is solve, to yield estimates of the activations of the upper limbs muscles.

4. A direct dynamics analysis aimed at estimating the effects of the active and reflexive muscular forces in the upper limbs, and muscular intervertebral moments is carried out.

5. As an optional step, a generalized eigenanalysis, directly performed on Differential-Algebraic Equations (DAE) system, in descriptor form, to extract Reduced Order Models (ROMs).

Steps 2–5 will be briefly described hereinafter, highlighting the most important aspects and referring to the related literature for further information.

6.4.1 *Inverse Kinematics*

To obtain a square problem in kinematic inversion when computing the pose of the system, a series of static problems is established, in which *ergonomy* springs act on redundant degrees of freedom [64]. The stiffness coefficients of the springs act as penalty coefficients for the motion of the degrees of freedom to which they are connected. For example, they can be crafted to minimize the norm of the internal bending moment in the sagittal plane due to the weight or the norm of the deviation with respect to a nominal reference condition.

The inverse kinematics problem can be stated, following the approach outlined in [64], directly at the configuration level, as the problem of finding the static equilibrium configuration of the system, augmented by a set of dummy or ergonomy springs, with respect to the imposed motion and the algebraic constraints. Therefore, the equivalent problem can be stated as the constrained minimization of the springs' elastic energy.

$$\begin{aligned} J_{\mathbf{x}} = \frac{1}{2} \left(\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{ergo}}\right)^{\mathsf{T}} \mathbf{K}(\boldsymbol{\theta}) \left(\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{ergo}}\right) \\ + \boldsymbol{\lambda}^{\mathsf{T}} \boldsymbol{\phi} + \boldsymbol{\mu}^{\mathsf{T}} \left(\boldsymbol{\psi} - \boldsymbol{\alpha}(t)\right) \end{aligned} \tag{207}$$

Resulting

$$\boldsymbol{\theta}'^{\mathsf{T}}_{\mathbf{x}} \mathbf{K}(\boldsymbol{\theta}) \left(\boldsymbol{\theta} - \boldsymbol{\theta}_{\text{ergo}}\right) + \boldsymbol{\phi}'^{\mathsf{T}}_{\mathbf{x}} \boldsymbol{\lambda} + \boldsymbol{\psi}'^{\mathsf{T}}_{\mathbf{x}} \boldsymbol{\mu} = \mathbf{0} \tag{208}$$

$$\boldsymbol{\phi} = \mathbf{0} \tag{209}$$

$$\boldsymbol{\psi} = \boldsymbol{\alpha}(t) \tag{210}$$

where the algebraic constraints equations are $\boldsymbol{\phi}(\mathbf{x}) = \mathbf{0}$, the imposed displacements $\boldsymbol{\psi}(\mathbf{x}) = \boldsymbol{\alpha}(t)$ and $\boldsymbol{\theta}(\mathbf{x})$ represent the angles of the artic-

ular joint. Stiffness $\mathbf{K}(\boldsymbol{\theta})$ can be crafted to penalize configurations in which the angles differ significantly from the intermediate ones:

$$\theta_{ergo} = \frac{\theta_{max} - \theta_{min}}{2} \tag{211}$$

The choice is made under the assumption that intermediate configurations are the most ergonomic [124]. A high-degree polynomial function of $\boldsymbol{\theta}$, centered around $\boldsymbol{\theta}_{ergo}$ is used.

In the case of the spine model, the same intervertebral viscoelastic elements described in Section 6.3 are also used as ergonomy elements. Figure 117 shows three successive vertebral node configurations during an inverse kinematic analysis. In particular, the initial, intermediate, and final configurations are shown: the spine is initially assembled using the reference coordinates in Table 15. The head is then rigidly rotated to align the local z-axis of the associated node with respect to the global z-axis. When the desired head configuration is reached, the strain of the intervertebral elements is stored. In subsequent analyses, it is added to the same elements as the value of prestrain.

The solution is completed at the velocity and acceleration levels by solving two other constrained optimization problems, in which the weighted distance of the solution from the reference $\mathbf{x}_{ref}$, obtained by numerical differentiation of the one at the configuration level, is minimized. In both cases, the mass matrix $\mathbf{M}$ is used for weighting.

At the velocity level, the problem is therefore stated as the minimization of the

$$\begin{aligned} J_{\dot{x}} = &\frac{1}{2}\left(\dot{\mathbf{x}} - \dot{\mathbf{x}}_{ref}\right)^{T} \mathbf{M}\left(\dot{\mathbf{x}} - \dot{\mathbf{x}}_{ref}\right) \\ &+ \boldsymbol{\lambda}^{'} \boldsymbol{\phi}_{/\mathbf{x}} \dot{\mathbf{x}} + \boldsymbol{\mu}^{'}\left(\boldsymbol{\psi}_{/\mathbf{x}} \dot{\mathbf{x}} - \dot{\boldsymbol{\alpha}}(t)\right) \end{aligned} \tag{212}$$

with respect to $\dot{\mathbf{x}}$, $\boldsymbol{\lambda}'$, and $\boldsymbol{\mu}'$, which yield

$$\begin{aligned} \mathbf{M}\left(\dot{\mathbf{x}} - \dot{\mathbf{x}}_{ref}\right) + \boldsymbol{\phi}_{/\mathbf{x}}^{T} \boldsymbol{\lambda}^{'} + \boldsymbol{\psi}_{/\mathbf{x}}^{T} \boldsymbol{\mu}^{'} &= \mathbf{0} \\ \boldsymbol{\phi}_{/\mathbf{x}} \dot{\mathbf{x}} &= \mathbf{0} \\ \boldsymbol{\psi}_{/\mathbf{x}} \dot{\mathbf{x}} &= \dot{\boldsymbol{\alpha}}(t) \end{aligned} \tag{213}$$

The acceleration problem is analogous.

6.4.2 *Upper limbs inverse dynamics and muscular activations*

Once the inverse kinematics problem has been solved, the total resulting muscular torques about the articular joints of the upper limbs can be computed using an inverse dynamic analysis. This step is needed to cope with the upper limbs overactuation: each joint torque is produced by multiple muscles, usually acting together. The problem is fully determined, in this case, as can be stated by finding the torques

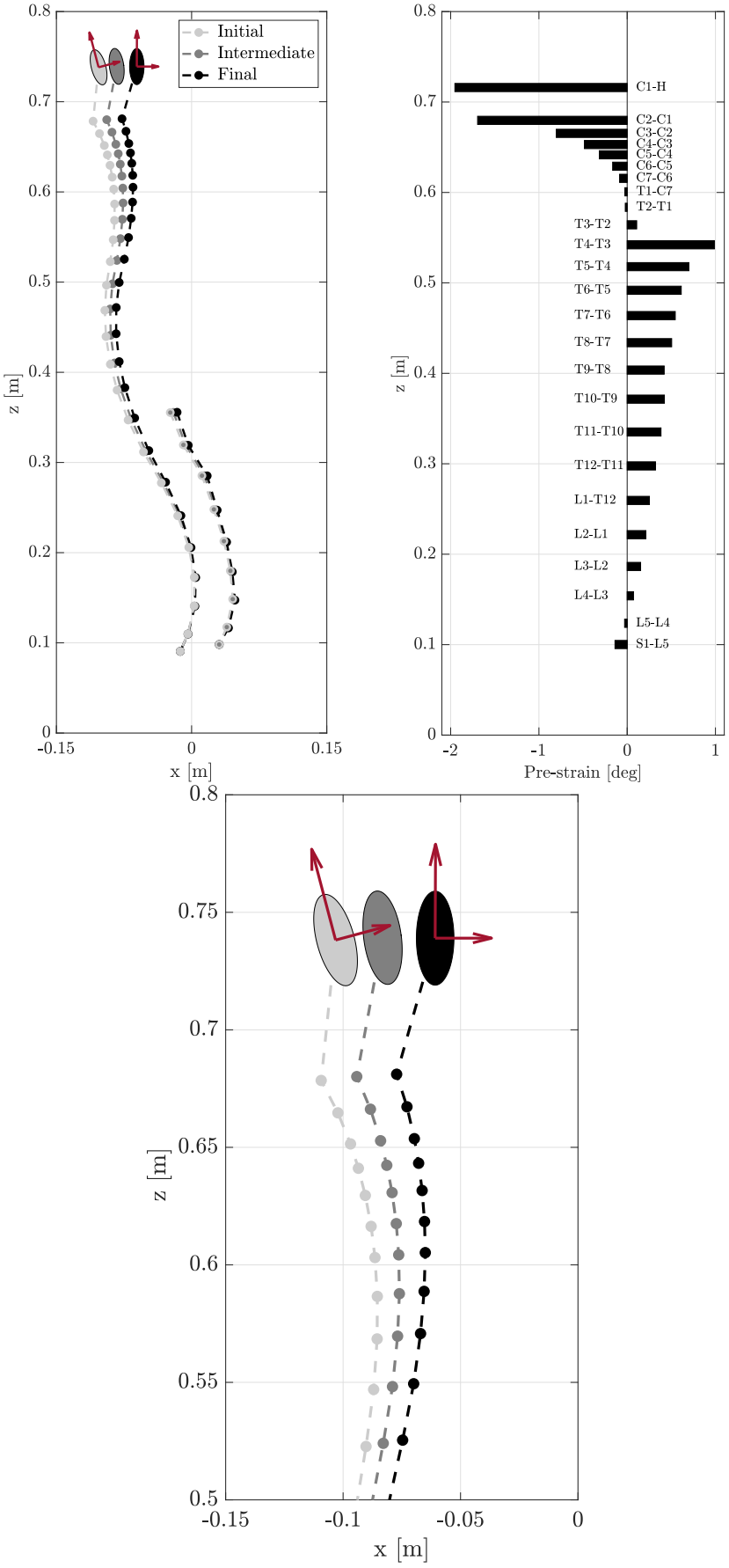


Figure 117: Inverse kinematics simulation of the spine for a backrest reclined by 10°.

$\mathbf{c}$ and the reaction forces' Lagrange multipliers $\boldsymbol{\lambda}$ that, at each instant of time of interest, are able to produce the desired motion $\mathbf{x}(t)$, $\dot{\mathbf{x}}(t)$, $\ddot{\mathbf{x}}(t)$:

$$\mathbf{M}\ddot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \dot{\mathbf{x}}, t) + \boldsymbol{\theta}_{/\mathbf{x}}^{\mathsf{T}}\mathbf{c} + \boldsymbol{\phi}_{/\mathbf{x}}^{\mathsf{T}}\boldsymbol{\lambda} \tag{214}$$

yielding the linear and square problem,

$$\begin{bmatrix} \boldsymbol{\phi}_{/\mathbf{x}}^{\mathsf{T}} & \boldsymbol{\theta}_{/\mathbf{x}}^{\mathsf{T}} \end{bmatrix} \begin{Bmatrix} \boldsymbol{\lambda} \\ \mathbf{c} \end{Bmatrix} = \mathbf{f}(\mathbf{x}, \dot{\mathbf{x}}, t) - \mathbf{M}\ddot{\mathbf{x}} \tag{215}$$

where $\boldsymbol{\theta}_{/\mathbf{x}}$ is the Jacobian of the joint coordinates $\boldsymbol{\theta}(\mathbf{x})$ with respect to the generalized coordinates $\mathbf{x}$.

When the joint torques of the upper limbs are known, the muscular activations required to produce them can be computed. The problem is again underdetermined:

$$\mathbf{c} = (\boldsymbol{\theta}_{/\mathbf{x}}^{+})^{\mathsf{T}}\mathbf{B}\mathbf{F}_0\left(\mathbf{F}_{12}\mathbf{a} + \mathbf{F}_3\right) \tag{216}$$

where $\mathbf{B}$ is the matrix of the moment arms of the muscle forces with respect to the nodes, and $(\cdot)^{+}$ denotes pseudo-inversion. Muscle forces, as clearly shown by Eq. (199), are made up of an active and a passive part. The diagonal matrix $\mathbf{F}_0$ contains the maximum isometric contraction forces of the muscles of the upper extremities. The matrix $\mathbf{F}_{12}$, also diagonal, contains the products of the functions $f_1(\ell_i)$ and $f_2(\dot{\ell}_i)$. They respectively represent the ratio, with respect to its peak isometric force, between the maximum force that the i-th muscle can produce at the current length ℓ_i and the elongation velocity $\dot{\ell}_i$. The matrix $\mathbf{F}_3$ is diagonal as well; its elements are the values of the function $f_3(\ell_i)$, representing the relationship between the passive force of the i-th muscle, including the contribution of the tendon, at current length ℓ_i and the maximum isometric contraction force.

A formal solution to the problem could be found with a pseudoinverting matrix $(\boldsymbol{\theta}_{/\mathbf{x}}^{+})^{\mathsf{T}}\mathbf{B}\mathbf{F}_0\mathbf{F}_{12}$; however, such a solution would not guarantee that the resulting activations are limited in the admissible range $0 \leqslant a_i \leqslant 1$, necessary to ensure that the force produced by each muscle is purely contractile and does not exceed the maximum value at the current length and contraction velocity, as dictated by the nature of the mechanics of the skeletal muscles [200]. Therefore, an optimization problem, formulated as the minimization of a cost function $J(\mathbf{a})$ that can be adapted to fit the specific simulated scenario, is instead solved, subject to the constraints of satisfying the equivalence of Eq. (216), and the activation bounds.

Another consequence of the indeterminacy of the problem is that the computed activation level can be increased by any contribution that does not alter the total joint torques. Such contributions have been labeled by the authors Torque-Less Activation Modes (TLAMs): they can be useful, for example, to modulate the equivalent impedance of the limbs at the control inceptors [124], as a function of the estimated workload or stress state of the pilot.

6.4.3 *Direct Dynamics*

When the activations in the upper limb muscles are known, a direct analysis of the desired scenario can be performed. The intervertebral stiffness and damping coefficients of Table 32 already take into account the muscular impedance *active* (i.e., caused by proprioception), and are kept constant throughout the simulations. The latter assumption can be justified, noting that the deviation from the reference condition introduced by the prestrain estimated in the inverse kinematics step is generally limited for the applications for which the model has been designed (cf. Fig. 117).

Upper-limb muscle activations estimated in the inverse dynamics step, and the subsequent optimization problem, are augmented during the direct dynamics phase with a contribution proportional to the muscles' length variation and elongation velocity, with respect to the reference values estimated during the inverse kinematics step. For the single muscle, this contribution, which models the *reflexive* part of the muscular activation behaviour [122, 182] , is

$$a_r = k_p (\ell - \ell_{ref}) + k_d \dot{\ell} \tag{217}$$

where ℓ_{ref} is the reference muscle length in the pose estimated in the inverse kinematics step and k_p,k_d are proportional and derivative gains. Therefore, the total activation of each muscle bundle in the upper limb is

$$\mathbf{a}(t) = \mathbf{a}_0 + \mathbf{K}_{TLAM}\mathbf{V}_{TLAM}\mathbf{b} + \mathbf{a}_r \tag{218}$$

where $\mathbf{a}_0$ is the reference activation, $\mathbf{a}_r$ is the reflexive part, and a linear combination $\mathbf{b}$ of TLAMs is introduced, scaled by the gain coefficients $\mathbf{K}_{TLAM}$.

6.4.4 *ROM Extraction*

The extraction of Reduced Order Models is based on the direct linearization of the full DAE problem [112]

$$\begin{aligned} \mathbf{M}\dot{\mathbf{x}} &= \mathbf{p} \\ \dot{\mathbf{p}} + \phi^T_{/\mathbf{x}}\boldsymbol{\lambda} &= \mathbf{f}'(\dot{\mathbf{x}}, \mathbf{x}, t) \\ \phi(\mathbf{x}) &= \mathbf{0} \end{aligned} \tag{219}$$

where the first block row is the definition of the momenta $\mathbf{p}$, the second block row represents the generalized equilibria, with $\mathbf{f}' = \mathbf{f}(\dot{\mathbf{x}}, \mathbf{x}, t) + (\mathbf{M}\dot{\mathbf{x}})_{/\mathbf{x}}$ and $\mathbf{f}$ being the applied force vector. The third block row contains the algebraic constraints' equations. The linearization yields the descriptor form a state-space system.

$$\begin{bmatrix} \mathbf{M} & \mathbf{0} & \mathbf{0} \\ \mathbf{C} & \mathbf{I} & \phi^T_{/\mathbf{x}} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{Bmatrix} \dot{\mathbf{x}} \\ \dot{\mathbf{p}} \\ \dot{\boldsymbol{\lambda}} \end{Bmatrix} = \begin{bmatrix} \mathbf{0} & \mathbf{I} & \mathbf{0} \\ -\mathbf{K} & -\mathbf{D} & \mathbf{0} \\ \phi^T_{/\mathbf{x}} & \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{Bmatrix} \mathbf{x} \\ \mathbf{p} \\ \boldsymbol{\lambda} \end{Bmatrix} \tag{220}$$

with $\mathbf{K} = (\mathbf{M}\ddot{\mathbf{x}} - \mathbf{f}')_{/\mathbf{x}}$ and $\mathbf{D} = -\mathbf{f}'_{/\dot{\mathbf{x}}}$ The eigenvalues and eigenvectors of the regular matrix pencil characterizing the homogeneous solution of (220) can be found, for example, through a generalized Schur or QZ, the decomposition [130, 181] of the pencil matrices. The system can then be reduced to the ODE form, eliminating the algebraic state variables in a staggered manner. More details can be found in [167].

6.5 COMPARISON WITH EXPERIMENTAL RESULTS

The biodynamic response of human subjects seated to whole-body vibrations of different types and magnitudes has been widely investigated, often in conjunction with the role played by seat design and posture. The transmissibility of vibrations from the seat to the head is therefore a natural choice for the validation for models of the upper body dynamics.

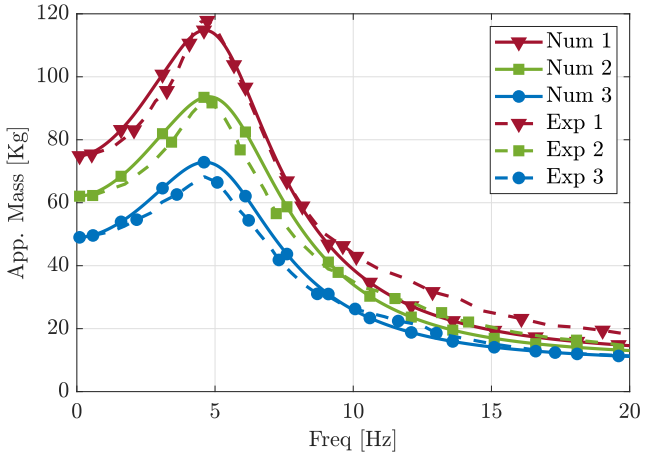


Figure 118: Comparison between the experimental apparent mass from [189] and the one resulting from the MBD

In [59] the results of the measurements are presented with respect to the vertical apparent mass (AM) of 60 seated subjects, including 24 men, 24 women and 12 children. The measurements were made using random acceleration as input at frequencies up to 20 Hz. In [29], the mechanical impedance at the driving point was evaluated using 7 male subjects seated on a rigid seat and subjected to ten different acceleration excitations. In [94], eight subjects were exposed to random vertical vibration while maintaining three different postures on a rigid seat without backrest. The modes in the frequency bandwidth under 10 Hz were extracted. The principal resonance of the human upper body was observed at approximately 5 Hz. These experimental results were used to validate the FEM model presented in [93].

In [189], the AM of 80 seated adults (41 males and 39 females) was measured at frequencies between 0.6–20 Hz. In [110], the AM and STHT were investigated by exposing the body to vibrations applied

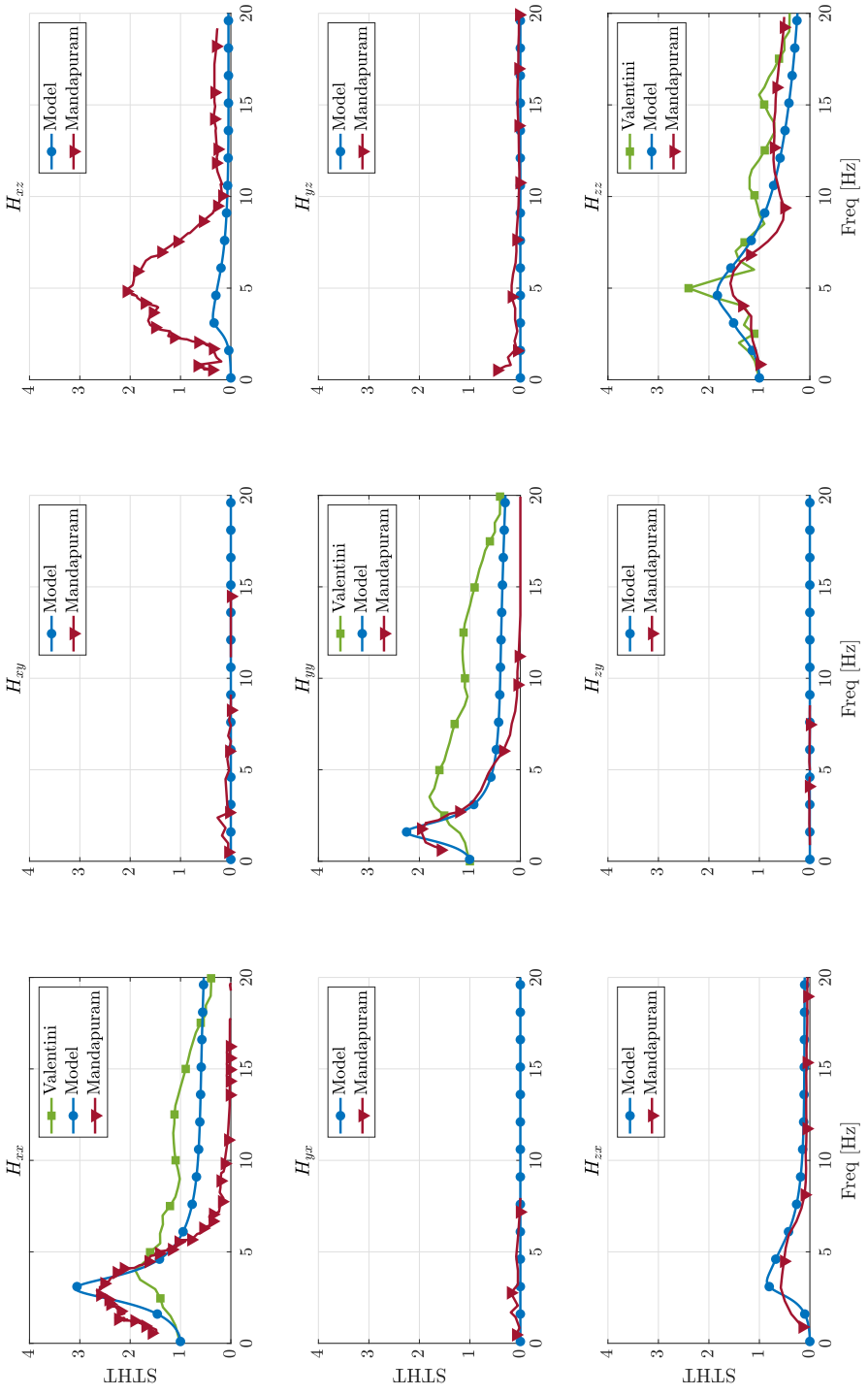


Figure 119: Seat to head transmissibility (STHT) in all directions: comparison with experimental results of Mandapuram [110] and numerical results from Valentini [196].

in the fore, lateral, and vertical directions. ISO [179] defined the ideal ranges of AM and STHT magnitudes and the phase responses of the subject seated without back support and exposed to vertical vibration of magnitude up to $5\,\mathrm{m\,s^{-2}}$. A synthesis of the reported data on the biodynamic responses of the human body exposed to vibrations in different directions and the associated experimental conditions can be found in [163].

The validity of the scaling procedure of the presented model has been tested by comparing the experimental apparent mass between 0-20 Hz of three subjects reported in [189].

The corresponding anthropometric parameters are reported in Table 16.

Table 16: Anthropometric parameters used to validate the scaling procedure

Model	Age	Weight	Height	BMI
	Years	kg	m	$\mathrm{kg\,m^{-2}}$
1	33.7	54.88	1.591	21.681
2	33.7	69.5	1.71	23.768
3	33.7	83.9	1.823	25.246

The apparent mass is calculated from the ratio of the cross-spectral density between the force F_{zs} and the acceleration in the seat A_{z_s}, to the power spectral density of the acceleration in the seat. The H_1 estimator is obtained:

$$H_1 = \frac{F_{z_s} \cdot A^*_{z_s}}{A_{z_s} \cdot A^*_{z_s}} \tag{221}$$

where the exponent $(\cdot)^*$ indicates complex conjugation. The three experimental and numerical apparent mass curves are compared in Fig. 118. The model correlates well with the experimental results. Both the static mass and the first resonance frequency are well captured.

The model proposed here is intended as a complete three-dimensional non-linear model of the upper body. As a consequence, to validate its behavior along the three axes, seat-to-head transmissibility (STHT) up to a frequency of 20 Hz has been computed and compared with experimental data from [110] and with results from the multibody model proposed in [196]. The STHT is computed as the ratio between the cross-spectral density of the accelerations at the head A_{h_i} and at the buttock A_{b_j}, and the power-spectral density of the acceleration at the buttock:

$$H_{ij} = \frac{A_{h_i} \cdot A^*_{b_j}}{A_{b_j} \cdot A^*_{b_j}} \tag{222}$$

The results are shown in Fig. 119. The model correlates relatively well with the experimental results for frequencies up to 10 Hz with respect

to same-axis excitation and measurement, whereas non-negligible differences can be observed in the off-axis term H_{xz}, i.e., in the head longitudinal response to a vertical acceleration input. Work is currently being done by the authors to understand the origin of the discrepancy, that is probably due to the distribution of stiffness and damping coefficients in the intervertebral elements' constitutive laws. It should be noted, however, that the *off-axis* response of the human upper body has been shown to be characterized by a relatively high level of variability [215].

6.5.1 *Biodynamic Feedthrough*

The complete upper body model of the reference subject has been used to perform a frequency sweep analysis in the frequency range of 0.5–12.5 Hz, evaluating the frequency response function (FRF) between vertical acceleration of the cockpit floor and rotation of the collective and cyclic levers, usually termed biodynamic feedthrough (BDFT). In Eq. (223), θ_i is the i-th control input (collective lever rotation, cyclic lever rotation in longitudinal or lateral direction), θ_0 is the trim angle of the lever, and $\ddot{z}_g$ is the acceleration of the floor imposed in the vertical direction:

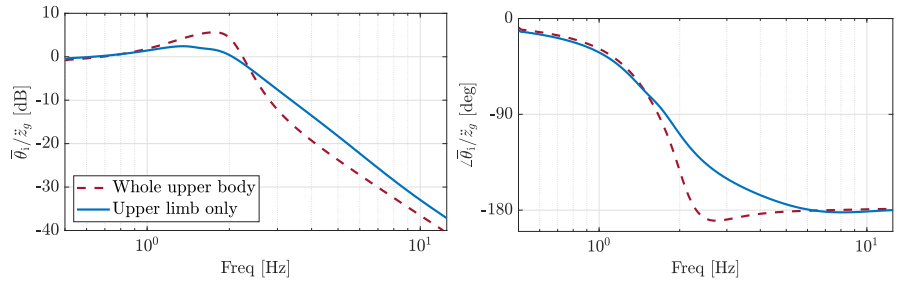
$$\text{BDFT} = \frac{\theta_i/\theta_0}{\ddot{z}_g} = \frac{\overline{\theta}_i}{\ddot{z}_g} \tag{223}$$

The plots in Fig. 120 compare the FRFs of the complete upper body model with those of a model consisting of only the upper limbs, for a vertical acceleration input of 0.2 g.

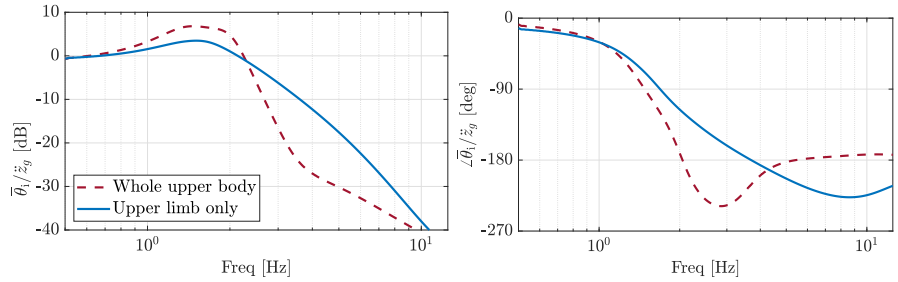
Table 17: Resonance frequency, amplitude and damping factor for the models with and without the spine.

	Upper limb only			Whole upper body		
	Freq [Hz]	Mag [dB]	Damp.	Freq Hz	Mag [dB]	Damp.
Collective	1.33	1.33	0.56	1.75	1.92	0.28
Cycl. lat.	1.49	1.50	0.40	1.45	2.20	0.35
Cycl. long.	1.55	2.63	0.24	1.56	3.04	0.29

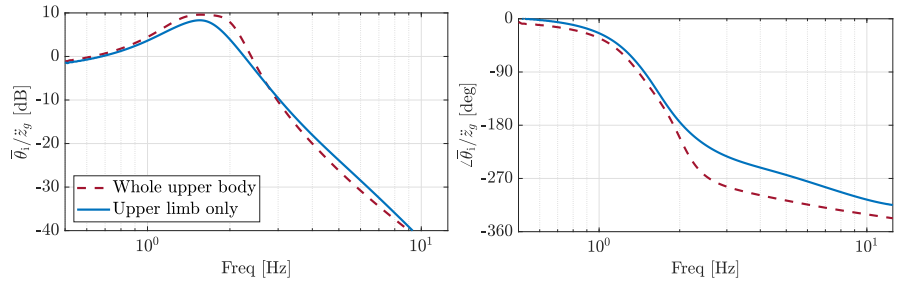
Table 17 presents the resonance frequency and amplitude values for the two models. As one can see from the three FRFs, when the complete dynamics of the spine is considered, the magnitude of the resonance peak increases, in particular in the response of the collective control inceptor. This inceptor, as one would expect, is the most sensitive to vertical accelerations because it rotates in the vertical plane and acts on the rotorcraft heave-axis control. When considering the



(a) Collective

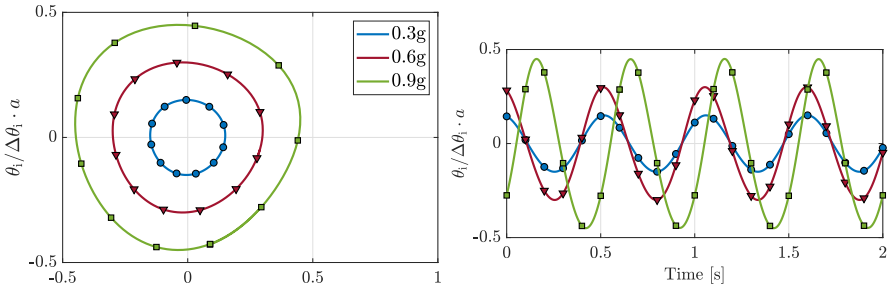


(b) Lateral Cyclic

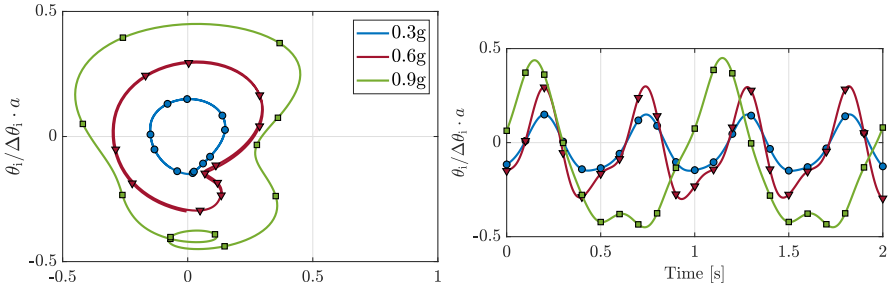


(c) Longitudinal Cyclic

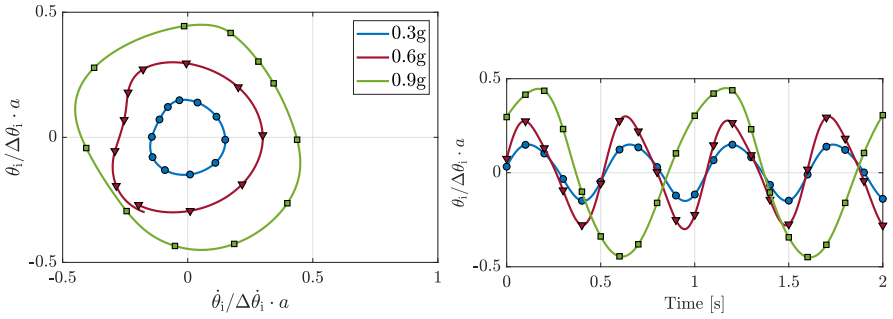
Figure 120: BDFT of the pilot: comparison of upper limb and upper body model transmissibility.



(a) Collective

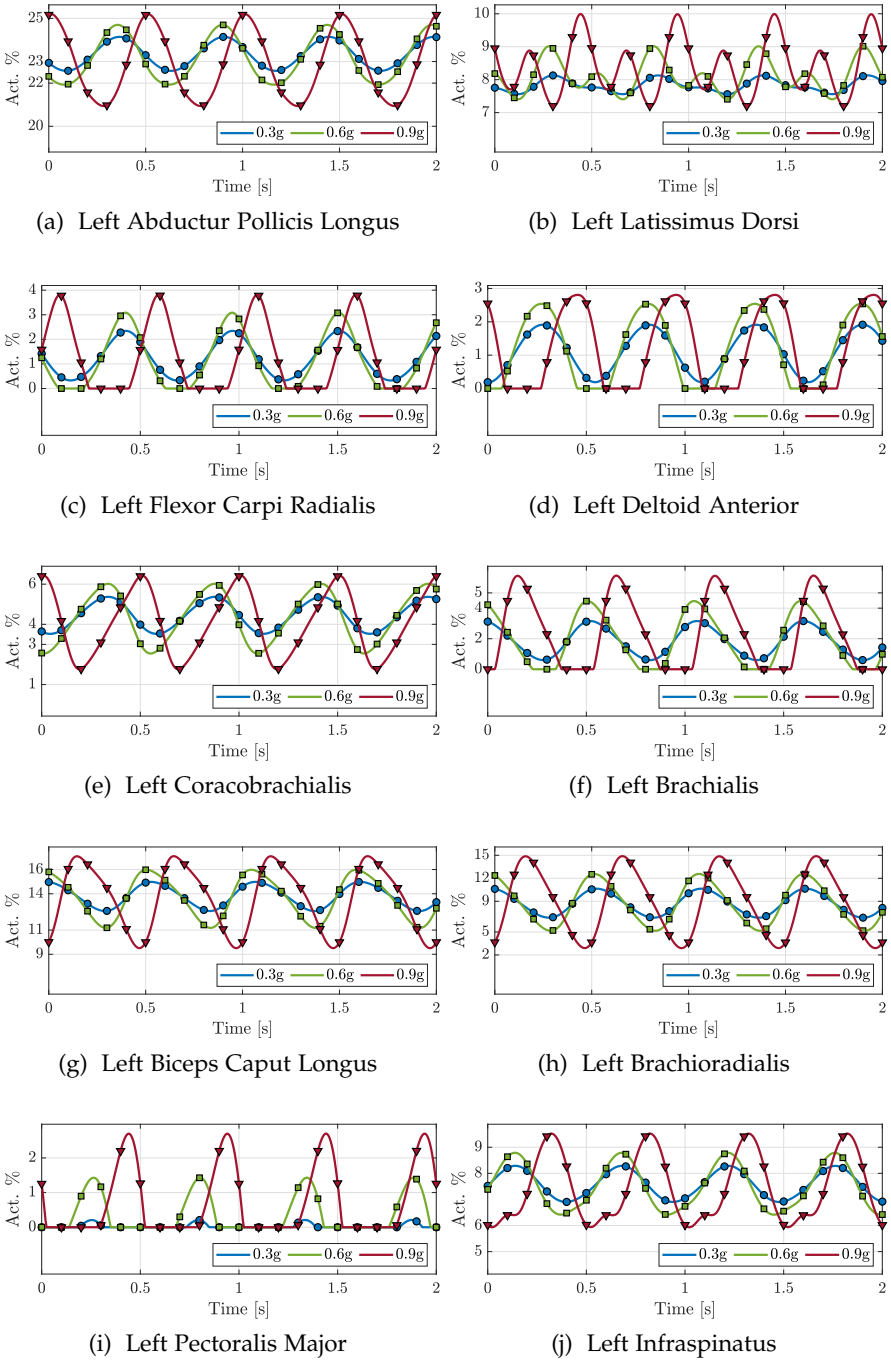


(b) Lateral Cyclic



(c) Lateral Cyclic

Figure 121: Phase diagrams and time history of the collective and cyclic rotation response to heave acceleration input applied to the cockpit floor, at different excitation magnitude a, in $\mathrm{m\,s^{-2}}$.



(a) Left Abductur Pollicis Longus

(b) Left Latissimus Dorsi

(c) Left Flexor Carpi Radialis

(d) Left Deltoid Anterior

(e) Left Coracobrachialis

(f) Left Brachialis

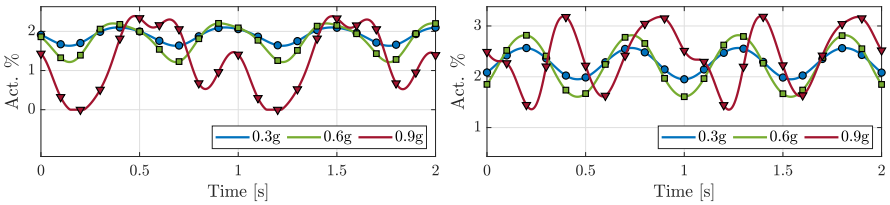
(g) Left Biceps Caput Longus

(h) Left Brachioradialis

(i) Left Pectoralis Major

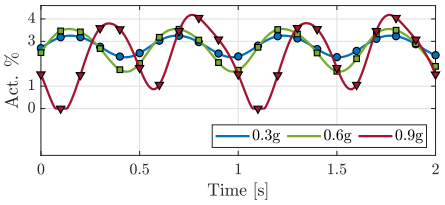
(j) Left Infraspinatus

Figure 122: Muscular activations on the left arm for cockpit floor vertical acceleration of 0.3 g, 0.6 g and 0.9 g at a frequency of 2 Hz.

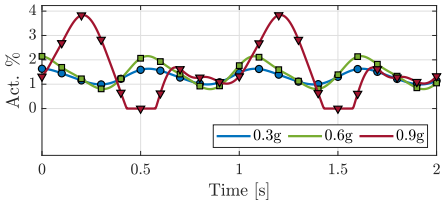


(a) Right Abductur Pollicis Longus

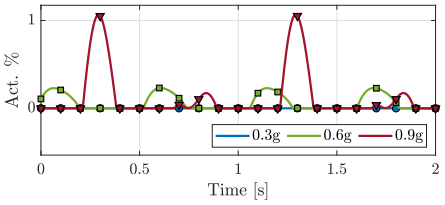
(b) Right Latissimus Dorsi



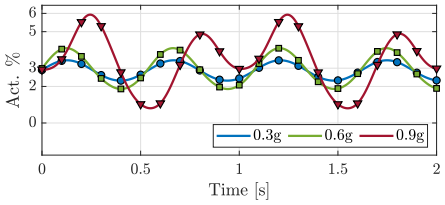
(c) Right Flexor Carpi Radialis



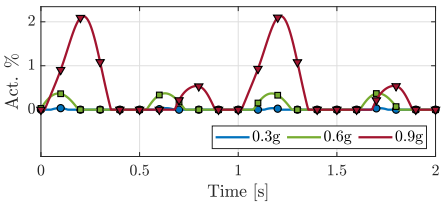
(d) Right Deltoid Anterior



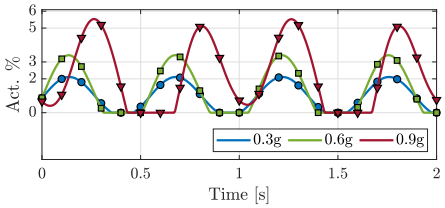
(e) Right Coracobrachialis



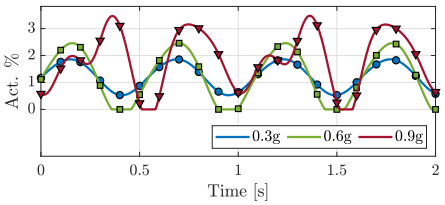
(f) Right Brachialis



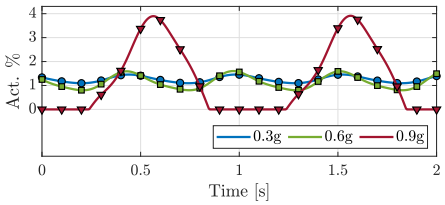
(g) Right Biceps Caput Longus



(h) Right Brachioradialis

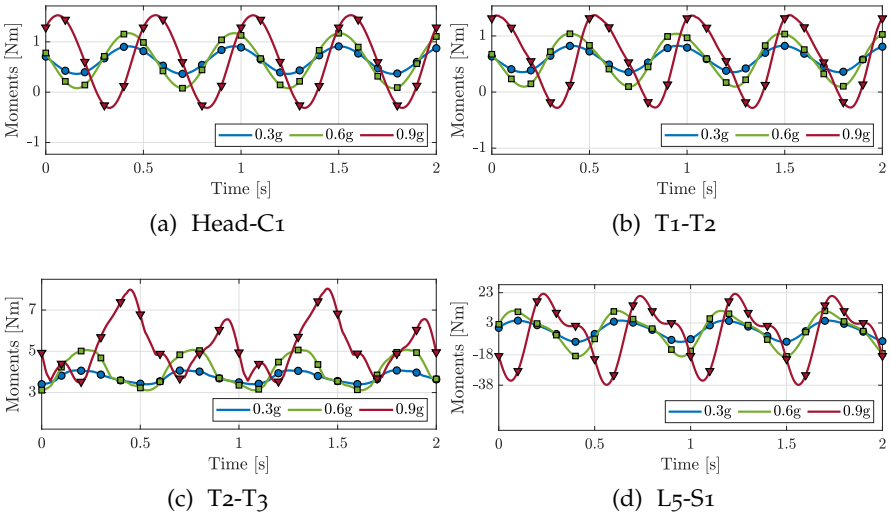


(i) Right Pectoralis Major



(j) Right Infraspinatus

Figure 123: Muscular activations on the right arm for cockpit floor vertical acceleration of 0.3 g, 0.6 g and 0.9 g at a frequency of 2 Hz.



(a) Head-C1 (b) T1-T2

(c) T2-T3 (d) L5-S1

Figure 124: Intervertebral moments around the medial axis for cockpit floor vertical acceleration of 0.3 g, 0.6 g and 0.9 g at a frequency of 2 Hz.

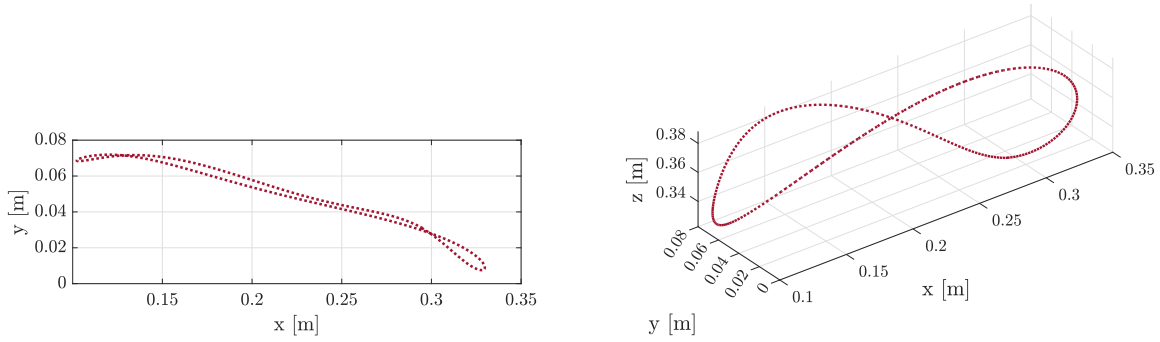
complete spine model, the resonance frequency of the collective response increases by 33 %.

A small rotation of the cyclic control inceptor, caused by the heave motion of the cockpit, has been observed. In both longitudinal and lateral cases, the amplitude of the resonance peak is amplified when the spine dynamics is introduced; the resonance frequency increases approximately 10 % for the lateral cyclic response and approximately 18 % for the longitudinal cyclic response.

The phase diagram of each lever's rotation angle is plotted in Fig. 121. The ratio between instantaneous command rotation θ_i and the deviation with respect to the trim configuration $\Delta\theta_i$ is plotted against the analogous ratio on the command angular velocities. Both quantities are scaled with respect to the input acceleration level a. The results are obtained by imposing a sinusoidal excitation on the cockpit, at 2.00 Hz, increasing the amplitude of the input vertical acceleration from 0.3 g to 0.6 g and 0.9 g. This frequency has been selected because it is very close to the resonance frequency for both collective and longitudinal cyclic dynamics.

A non-linear dependence between the magnitude of the forcing term and the response can be observed in the plots of Fig. 121. This non-linear behavior is more pronounced in the cyclic response, whereas the collective only marginally departs from linearity.

Figures 122 and 123 show several muscular activations in the left and right limbs, respectively, imposing on the floor of the cockpit sinusoidal vertical accelerations at 2 Hz. From these graphs, it is evident that the maximum required activations increase with the amplitude of the forcing term. Furthermore, as expected, the activations



(a) 2D trajectory

(b) 3D trajectory

Figure 125: Trajectory of the cyclic command inceptor's grip, traced in the xy-plane and in the $3D$-space, in response to an harmonic acceleration input applied to the cockpit floor in the heave direction of 0.9 g at 2 Hz.

required for the left limb are much higher than those required for the right limb. For example, *Abductur Pollicis Longus* is the most active muscle in the left limb, with a maximum required activation of 0.26 at 0.9 g, whereas the maximum activation of the same muscle in the right limb is 0.03; an analogous consideration can be taken for *Biceps Caput Longus*. The *Latissimus Dorsi* shows a non-linear behavior as the amplitude increases: in particular, for the left limb, a second-harmonic component can be clearly observed for an excitation amplitude of 0.6 g. On the right limb, for an excitation amplitude of 0.9 g, the activations show sub-harmonic contributions at half the frequency of the excitation. This effect is most notable, for example, in the *Deltoid anterior* and in the *Infraspinatus*. For excitation amplitudes of 0.3 g and 0.6 g the activations show a nearly sinusoidal response. This lower frequency component is related to a compound rotation of the cyclic lever about the longitudinal and lateral axes (Cf. Fig. 125), that is present at this specific frequency, while it is not noted at neighboring ones: it is not exhibited, for example, at tests conducted with an excitation frequency of 1.75 Hz. The phenomenon also shows a strong dependence on the angle of reclination of the backrest. A moderate angle of reclination (about 10° is typical in helicopters) will suppress it. In any case, it should be noted that the test conditions were particularly harsh: a heave vibration, with magnitude close to 1 g, sustained for several seconds, can definitely be regarded rare in normal operation of the rotorcraft. Furthermore, no retention devices, that is, the so-called *trim* springs, were introduced in the flight control inceptors: the inceptors were left in the *trim release* configuration, in which only a balancing element was active, counteracting the effect of weight but offering no restoring moment.

Figure 124 shows some intervertebral moments on the medial axis between the two adjacent vertebrae for four periods of excitation at 2 Hz. As for the activation case, the moments increase as the forcing

amplitude increases, but in particular for the moments between *L5–S1* and *T2–T3* the increment is non-linear.

More analyzes on the combination of different parameters related to cockpit layout and different pilot models can be found in [43, 111, 210].

7

COMPREHENSIVE STABILITY ANALYSIS

7.1 INTRODUCTION

In this chapter, the stability analysis of the entire tiltrotor model is considered. In the first part, the modal identification techniques are introduced; then they are applied to the complete tiltrotor model in order to asses its whirl flutter stability margins. In this phase, the MBDyn model is considered, since the objective is to determine the influence that operating parameters, such as the speed of the rotor, the engine failure, and the stiffness of the downstop, have on the stability of the whole tiltrotor.

Finally, a high-speed gust response is analyzed: the coupled model is compared with the MBDyn alone model. To have a fully comprehensive analysis, the detailed pilot multibody model is inserted into the loop. The goal of this final simulation is to show the capability of the proposed software tools and mid-fidelity comprehensive model to retrieve from a single simulation a huge amount of output: from the detailed aeroelastic loads on the different components of the tiltrotor to the pilot involuntary action on the control inceptors. The huge advantage of this comprehensive and monolithic approach is that now the machine can be seen since the preliminary design phase as a whole complex dynamical system, and not as a series of subcomponents. Figure 126 shows the wake visualization of the XV-15 model.

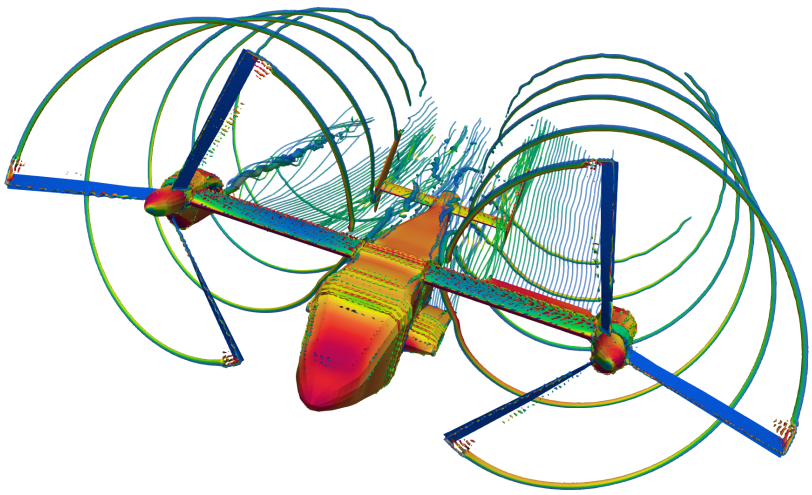


Figure 126: XV-15 MBDyn-DUST wake visualization

7.2 IDENTIFICATION TECHNIQUES

Vibration testing of structures are extremely useful for a variety of reasons: to determine the natural frequencies of a structure, to verify analytical models, or to monitor the condition of a structure under various loading conditions. Operational Modal Analysis OMA has the great advantage of extracting these informations from structure response measurements only, in contrast to Experimental Modal Analysis (EMA), where input measurements are also required. For this reason, OMA techniques, although derived from traditional input-output modal identification procedures, have been developed in a stochastic framework where the input is assumed to be unknown. This characteristic allows us to perform vibration tests with lower economic cost and easily apply them to large structures, such as buildings or wind turbines. However, this results in an increase in complexity and a difficulty in obtaining good results. When identifying, two main approaches are available: using time-domain or frequency-domain methods. The first are based on the analysis of response time histories or correlation functions. The seconds aim at extracting the modal parameters of the system from spectral density functions.

Time-domain algorithms are the most employed in the OMA field: a huge variety of techniques have been developed over the years. These methods appeared to be the most effective in correctly evaluating the modal parameters of the systems. However, when analyzing the system's free responses at each time, the contribution of each mode has to be considered, making it impossible to reduce the rank of the problem isolating each mode individually. However, in the frequency domain, when modes are well spaced, their contributions are limited to a single portion of the power spectral density matrix, allowing for a natural modal decomposition. Frequency domain techniques, however, suffer from errors in signal processing, such as leakage.

7.2.1 *Linear Time Invariant (LTI) systems time-domain identification methods*

A wide variety of techniques have been developed to evaluate the modal parameters of the system from the output-only recorded data. In addition to the distinction presented in the previous paragraph based on the domain in which the algorithms operate, the methods differ in the aspects of data reduction, mathematical modeling, and signal processing. In general, OMA time-domain methods can be divided into the following categories:

- Methods based on computing autocorrelation and cross-correlation of time signals and assuming the results as the system impulse response. Time-domain EMA techniques are then applied to the

obtained signals to retrieve the modal parameters. Among these there is the polyreference complex exponential method [10], the Ibrahim Time Domain method [81], Eigensystem Realization Algorithm [90] and the Matrix Pencil Estimation method [80] that is used in this work.

- Methods based on discrete-time data models. The most used are the vector AutoRegressive Moving Average(ARMAV) models consisting of fitting a model to the measured signals [14].
- Stochastic Subspace Methods. The most famous is the Stochastic Subspace Identification (SSI) method in both its data-driven and covariance-driven versions [85].

In this work, the Matrix Pencil Estimation method is used to identify the modal parameters, therefore in the following section a brief description of his working principles is presented.

7.2.2 *Matrix Pencil Estimation method*

The Matrix Pencil Estimation method (MPE), was designed by Hua [80] to estimate the parameters of exponentially damped or undamped sinusoids in noise. The method was tested by the authors only on simplified problems, for example, a signal composed of two sinusoids and background white noise, and it proved to be less sensitive to noise than the classical Prony method [76]. Favale [157] further proposed a version of the multiple input algorithm, who also tested it on more complex problems, such as tiltrotor whirl flutter analysis.

The MPE method is able to estimate the modal parameters of a system from its free responses to an external input. Note that the method provides consistent results only if the sampled response is composed of a free decay and no external unknown forces are applied to the system.

A general response signal $y(t)$ is made up of two contributions: the free decay of the system $x(t)$ and unknown noise $n(t)$. The signal in discrete time can be written as:

$$y(k) = x(k) + n(k),\ k = 0, 1, \ldots, N-1 \tag{224}$$

where N is the total number of samples obtained using a sampling frequency $f = 1/dt$. From the above hypothesis, the response signal of the system in discrete time can be approximated as a sum of complex exponentials.

$$y(k \cdot dt) \approx \sum_{i=1}^{M} R_i z_i^k + n(k \cdot dt),\ k = 0, 1, \ldots, N-1 \tag{225}$$

where M is the order of the system and z_i is equal to:

$$z_i = e^{s_i dt} = e^{\left(-\omega_i \zeta_i + j\omega_i \sqrt{1-\zeta_i^2}\right)} \tag{226}$$

The free response of the system can be described by its modal parameters: the complex amplitude R_i and the poles of the system s_i with frequencies ω_i and damping ratios ζ_i. The aim is then to find these parameters from the system-recorded response. The algorithm is now presented for the noiseless signal but can be extended to the noisy one [80].

Starting from the noiseless signal $x(k)$, a rectangular Hankel matrix with dimensions $(N-L)\times(L+1)$ can be created.

$$\boldsymbol{X} = \begin{bmatrix} x(0) & x(1) & \dots & X(L) \\ x(1) & x(2) & \dots & X(L+1) \\ \vdots & \vdots & \vdots & \vdots \\ x(N-L-1) & x(N-1) & \dots & X(N-1) \end{bmatrix} \tag{227}$$

where L is the pencil parameter, which should be less than $(N-1)$. Two matrices can be created excluding the first and last columns of X:

$$\boldsymbol{X}_1 = \boldsymbol{X}(:, 1:L) \tag{228}$$

$$\boldsymbol{X}_2 = \boldsymbol{X}(:, 2:L+1) \tag{229}$$

The matrices $\boldsymbol{X}_1$ and $\boldsymbol{X}_2$ can be rewritten as follows.

$$\boldsymbol{X}_1 = \boldsymbol{Z}_1 \boldsymbol{R} \boldsymbol{Z}_0 \boldsymbol{Z}_2 \tag{230}$$

$$\boldsymbol{X}_2 = \boldsymbol{Z}_1 \boldsymbol{R} \boldsymbol{Z}_2 \tag{231}$$

Where the matrices $\boldsymbol{R}$,$\boldsymbol{Z}_0$,$\boldsymbol{Z}_1$, and $\boldsymbol{Z}_2$ are functions of the complex amplitudes R_i and the poles of the system z_i:

$$\boldsymbol{R} = \mathrm{diag}\{R_1, R_2, \dots, R_M\} \tag{232}$$

$$\boldsymbol{Z}_0 = \mathrm{diag}\{z_1, z_2, \dots, z_M\} \tag{233}$$

$$\boldsymbol{Z}_1 = \begin{bmatrix} 1 & 1 & \dots & 1 \\ z_1 & z_2 & \dots & z_M \\ \vdots & \vdots & \vdots & \vdots \\ z_1^{N-L-1} & z_2^{N-L-1} & \dots & z_M^{N-L-1} \end{bmatrix} \tag{234}$$

$$\boldsymbol{Z}_2 = \begin{bmatrix} z_1^{L-1} & z_1^{L-2} & \dots & 1 \\ z_2^{L-1} & z_2^{L-2} & \dots & 1 \\ \vdots & \vdots & & \vdots \\ z_M^{L-1} & z_M^{L-2} & \dots & 1 \end{bmatrix} \tag{235}$$

Consequently, from 230 and 231:

$$\boldsymbol{X}_2 - \lambda \boldsymbol{X}_1 = \boldsymbol{Z}_1 \boldsymbol{R} \left(\boldsymbol{Z}_0 - \lambda \boldsymbol{I}\right) \boldsymbol{Z}_2 \tag{236}$$

It can be demonstrated that M is the rank of the matrix pencil if $M \leqslant L \leqslant N-M$ and the diagonal elements of $\boldsymbol{Z}_0$ are the eigenvalues

of the matrix pair $\boldsymbol{X}_2$ and $\boldsymbol{X}_1$. Finding the system's poles reduces to solving the following eigenvalue problem:

$$\left(\boldsymbol{X}_1^{\mathrm{H}}\boldsymbol{X}_1\right)^{-1}\boldsymbol{X}_1^{\mathrm{H}}\boldsymbol{X}_2 - \lambda\boldsymbol{I} \tag{237}$$

The same procedure described here is applicable for signals disturbed by noise. Hua [80] suggested that to obtain the best results, the parameter L should be chosen as high as possible, but limited between N/3 and N/2. A fundamental step after having created the Hankel block matrix of the collected responses $\boldsymbol{Y}$ is to perform a singular value decomposition.

$$\boldsymbol{Y} = \boldsymbol{U}\boldsymbol{\Xi}\boldsymbol{V}^{\mathrm{H}} = \boldsymbol{U}\begin{bmatrix}\sigma_1 & & \\ & \ddots & \\ & & \sigma_{L+1}\end{bmatrix}\boldsymbol{V}^{\mathrm{H}} \tag{238}$$

Singular values σ_i are real ordered non-negative numbers. The first, larger singular values, are the most representative of the analyzed system. A new set of filtered matrices can be obtained by truncating the matrices to the first dominant values. From this analysis, it is possible to determine, in the first approximation, the order of the system M: when using very high signal-to-noise ratio data, the first M singular values can be of orders of magnitude larger than the others. However, when very noisy signals are considered, the singular values relative to effective system contribution do not stand out significantly among all the others. A possible approach in these cases is, after having normalized the singular values, to compute their first derivative and identify the order corresponding to its maximum value. The matrices $\boldsymbol{\Xi}$ and $\boldsymbol{V}$ are truncated to obtain the reduced matrices $\boldsymbol{\Xi}'$ and $\boldsymbol{V}'$. Then it is possible to define the following.

$$\boldsymbol{Y}_1 = \boldsymbol{U}\boldsymbol{\Xi}'\boldsymbol{V}_1'^{\mathrm{H}} \tag{239}$$

$$\boldsymbol{Y}_2 = \boldsymbol{U}\boldsymbol{\Xi}'\boldsymbol{V}_2'^{\mathrm{H}} \tag{240}$$

where $\boldsymbol{V}_1'$ and $\boldsymbol{V}_2'$ are obtained similarly to 228 and 229 removing the first and last rows of the matrix $\boldsymbol{V}'$.

Finally, the values z_i can be computed by solving the following eigenvalue problem:

$$\boldsymbol{Y}_1^{+}\boldsymbol{Y}_2 - \lambda\boldsymbol{I} \tag{241}$$

Furthermore, using a least-squares problem, complex amplitudes R_i can be computed:

$$\begin{bmatrix} y(0) \\ y(1) \\ \vdots \\ y(N-1)\end{bmatrix} = \begin{bmatrix} 1 & 1 & \dots & 1 \\ z_1 & z_2 & \dots & z_M \\ \vdots & \vdots & \vdots & \vdots \\ z_1^{N-1} & z_2^{N-1} & \dots & z_M^{N-1}\end{bmatrix}\begin{bmatrix} R_1 \\ R_2 \\ \vdots \\ R_M\end{bmatrix} \tag{242}$$

The poles of the system are finally recovered, inverting the equation 226:

$$s_i = \frac{\ln(z_i)}{dt} \tag{243}$$

The previous discussion was conducted considering a single input response; however, the algorithm can easily be extended to multiple input channels considering a matrix of responses $\boldsymbol{y}$ instead of a column array. The Hankel matrix generated $\boldsymbol{Y}$ will be the result of the concatenation of Hankel matrices bmY_j obtained by considering separately each input. The resulting matrix will have dimension $N_y(N-L) \times (L+1)$ where N_y is the selected number of channels used in the identification process. Extension to multiple input channels is particularly useful for increasing the quality of the results obtained.

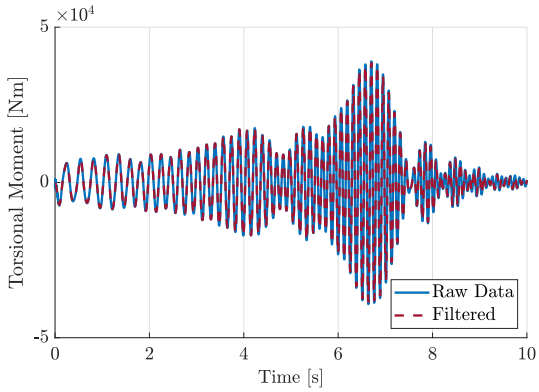
7.3 TILTROTOR FREQUENCY RESPONSE FUNCTION

To tune the excitation parameters, such as the excitation frequency amplitude and the control surface, needed for the free decay time response, two frequency sweep simulations are performed.

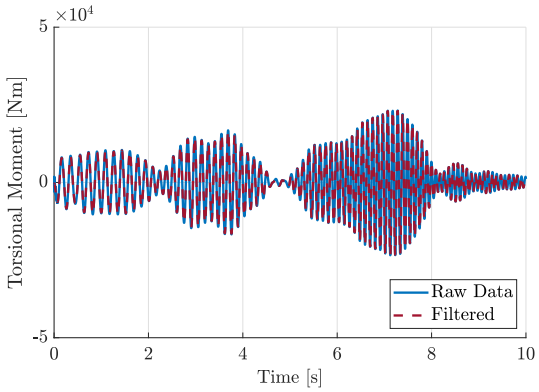
Following the strategy adopted by [3] during flight tests, symmetric modes are excited when the left and right flaperons are driven in phase, whereas antisymmetric modes are excited when the flaperons are driven in the opposite phase. A chirp function varying from 3 Hz to 9 Hz for the duration of 18 s is applied to the flaperon to excite the symmetric mode, whereas to excite the antisymmetric modes the chirp function varies between from 5 Hz to 9 Hz for the duration of 10 s.

The starting amplitude is of 7.5° of flaperon motion, which is quadratically decreased with frequency, to have a constant excitation power during the frequency sweep. To eliminate rigid body movements, the internal moments of the wings are used as measurement points. The time history is filtered with a Butterworth second-order low-pass filter with a cutting frequency $\omega_c = 10$ Hz. The filtered signals are converted from time domain into frequency response data by empirical transfer function estimation. The comparison between the original time response and the filtered time response of the internal moments of the wing at the root is shown in Figs. 127, 128, and 129.

The estimated Frequency Response Function (FRF) is then used to identify a state-space model of the tiltrotor vibration response. A value of 12 000 frequency points was used for the response calculation. The response function is fitted with an iterative nonlinear least-squares refinement, based on the Levenberg-Marquardt search method, of the model's parameters. The comparison between the original FRF of the internal moments of the wings at the root and the identified model is shown in Figs. 130, 131,132.

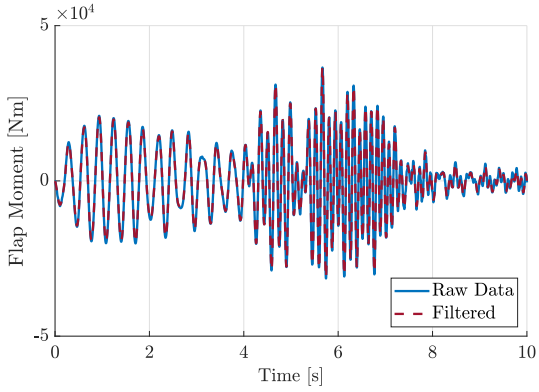


(a) Symmetric

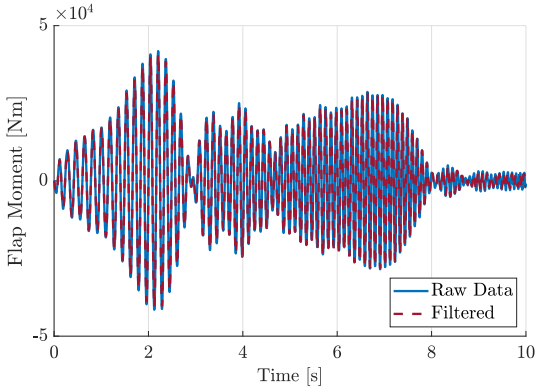


(b) Antisymmetric

Figure 127: Wing torsional moment time response

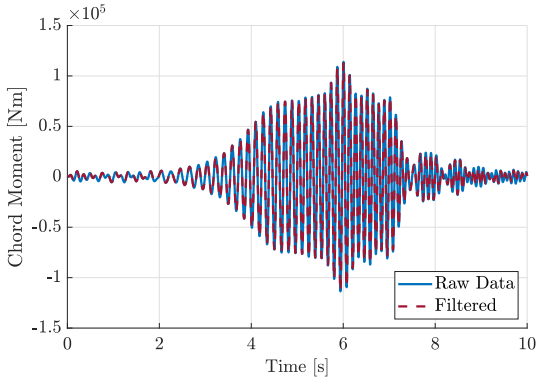


(a) Symmetric

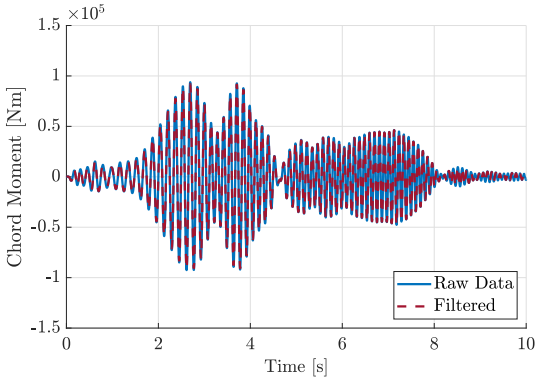


(b) Antisymmetric

Figure 128: Wing flapwise bending moment time response

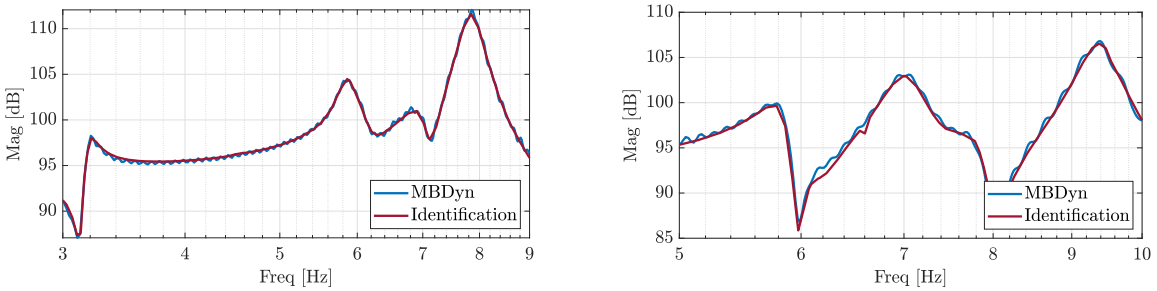


(a) Symmetric



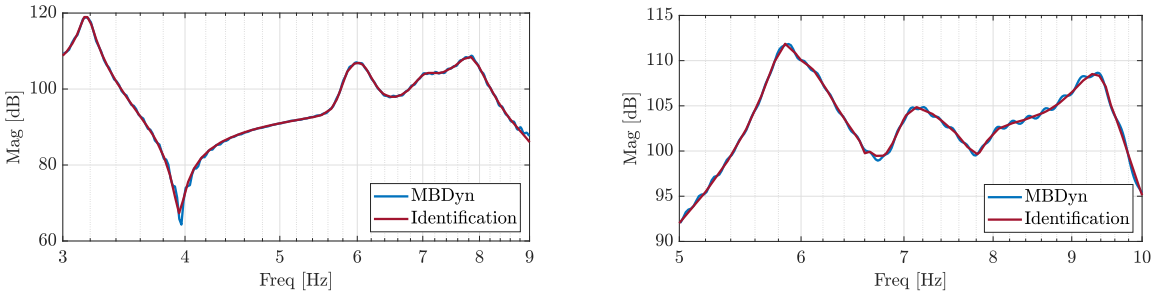
(b) Antisymmetric

Figure 129: Wing chordwise bending moment time response



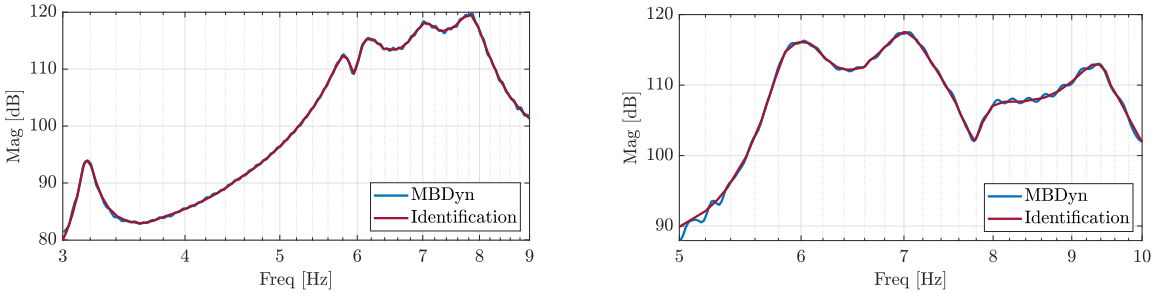
(a) Symmetric (b) Anti-symmetric

Figure 130: Wing torsional moment transfer function



(a) Symmetric (b) Anti-symmetric

Figure 131: Wing flapwise bending moment transfer function



(a) Symmetric (b) Anti-symmetric

Figure 132: Wing chordwise bending moment transfer function

7.3.1 *Stabilization diagrams*

Finally, the stabilization diagram is used to distinguish physical and spurious modes among all those identified by the algorithm used. It is usually used in combination with the previously presented stochastic subspace identification; however, in this work, following the example of Favale et al. [157], it is also used in the matrix pencil estimate method and in frequency response identification.

It consists of a plot of different model orders versus the poles' frequencies identified at each order. When performing an identification process, the order of the system or the number of modes in the response is unknown and is usually not easily identifiable. A practical way of estimating it could be to evaluate the number of peaks in the Complex Mode Indicator Function (CMIF), computed as the dominant singular values of power spectral density matrix of the system responses at each frequency. However, if the model order is overestimated, spurious poles, not related to effective mode contribution in the considered signals, may appear leading to erroneous results. On the other hand, physical poles should always be present regardless of the selected model order. When creating a stabilization diagram, a fixed number of increasing orders is set, and to distinguish real and spurious poles, the consistency of modal parameters are checked with respect to different criteria and threshold: typically, frequency ω and damping ratio ζ absolute variation and eigenvectors consistency using Modal Assurance Criterion are inspected.

All poles that conform to the introduced relations are defined as stable poles and are eligible to be selected as physical poles. For varying system orders, these poles should align approximately on vertical line with the same frequency. A threshold, defined as a percentage of the maximum iteration number used in the creation of the diagram, can be defined to more effectively identify the correct modal parameters and obtain a first estimate of their uncertainties. For example, when a column of stable poles is composed of a number approximately equal to the employed iterations, it could be confidently stated that those are related to a physical mode of the system. Figure 133 shows the stabilization diagrams obtained from the frequency sweep simulation.

The symmetric bending mode is easily identifiable in the first column of the symmetric stabilization diagram, and its frequency is almost identical to the frequency resulting from the airframe eigenanalysis. This is also confirmed by looking at Fig. 131a, where the dominant peak is found at 3.30 Hz.

Symmetric chordwise mode is not easily identifiable since multiple peaks are present in the chordwise transfer function (Fig. 132a) however, by looking at the hub modal components of the chord modes (Fig. 92a), it can be noted that a non-negligible flapwise bending com-

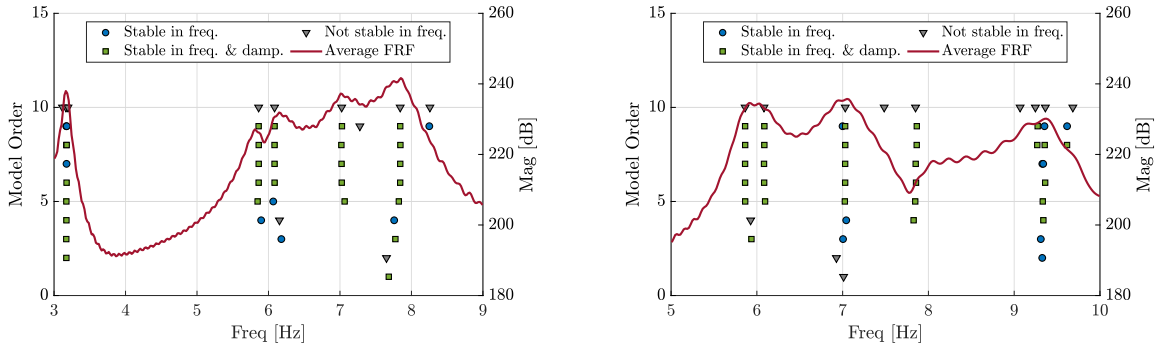
ponent is present (TZ) therefore the second peak at 6.0 Hz can be associated with the chord mode.

The symmetric torsion mode is then associated with the last column of the stabilization diagram, and this is also confirmed by the corresponding torsion moment transfer function (Fig. 130).

The remaining columns in the symmetric stabilization diagram can be associated with the antisymmetric airframe that are slightly excited due to the presence of the intershaft spring in the drive system, as it is visible locking to its symmetric transfer function shown in Fig. 134a.

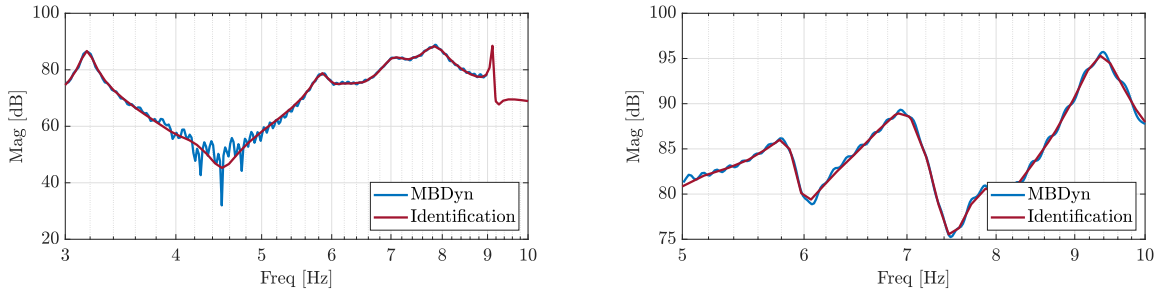
The antisymmetric flap-wise bending mode is found in the first two columns of the antisymmetric stabilization diagram 133b. This is also confirmed by looking at the frequency response function of the antisymmetric flap-wise bending moment 131b.

The chord-wise antisymmetric mode is identified in the third columns of the antisymmetric stabilization diagram, whereas the fifth column identifies the antisymmetric torsional mode. The frequency of this mode is almost 1 Hz higher with respect to the one found in the in vacuo eigenanalysis: this is due to the modal participation of the intershaft drive system in the mode shown in Fig.134b.



(a) Symmetric (b) Anti-symmetric

Figure 133: Wing stabilization diagram



(a) Symmetric (b) Anti-symmetric

Figure 134: Drive-system intershaft spring transfer function

7.4 WHIRL-FLUTTER STABILITY ANALYSIS

In this section, the whirl flutter stability analysis of the MBDyn tiltrotor model is investigated. Six different operational configurations, shown in Table 18, have been analyzed to cover the entire envelope of the XV-15 tiltrotor. The first configuration represents the nominal operating condition of the airplane mode, whereas the second configuration differs from the first by the fact that the engines are inoperative (windmilling). Configurations three and four represent the final part of the conversion corridor: the downstop is engaged, but the rotor is still in helicopter mode. Configurations five and six differ from the previous two by the fact that the downstop mechanism is not locked and, therefore, its nominal stiffness is degraded.

Table 18: Whirl flutter test matrix

	Conf. 1	Conf. 2	Conf. 3	Conf. 4	Conf. 5	Conf. 6
Power	ON	OFF	ON	OFF	ON	OFF
Downstop	ON	ON	ON	ON	OFF	OFF
Mode	AP	AP	HP	HP	HP	HP
U_∞ [Knts]	$160-380$	$160-380$	$160-260$	$160-260$	$160-260$	$160-260$

For each configuration and speed, six different simulations are performed to better isolate the three symmetric and antisymmetric mode contributions in the recorded data. The simulation is composed of three different parts:

1. Startup phase: during this phase, the model is gradually trimmed with the trimming procedure described in Sect. 5.3. This phase can be further divided into four subphases:
   a) Grounded model: in this phase, the model is grounded in order to allow the initial aerodynamic transient to vanish.
   b) Stationary trim: in this phase, the integral trimming procedure is introduced.
   c) Autopilot: after having found the reaction forces close to zero, the ground constraint is released, leaving the aircraft in a free-free condition. Therefore, the autopilot plus the stability augmentation system is therefore used to maintain the altitude of the tiltrotor.
   d) Rotors trim: This phase consists in trimming the rotors by imposing a prescribed torque and maintaining the rotor speed by chaining the collective angle. This phase is conducted in parallel with respect to the other three.

2. Excitation phase: This phase is the most delicate for retrieving satisfactory results during the identification process. To optimize the frequency content of the collected responses, it is chosen to excite the system using sinusoidal input. The input is applied to the aileron at the estimated frequencies of the wing from the frequency sweep simulations. Particular care must be taken when selecting the input amplitude. Using a too small input could produce more noisy responses and consequently higher uncertainties in the identification process. On the other hand, a large input amplitude could lead to a convergence problem on the simulation and minor instabilities under conditions of reduced stability margin of the system. Additionally, a large input could also generate excessive loads on the structure. For these reasons, the maximum allowable input amplitude had to be fixed to a limit value for the first trim point. This initial value was fixed using a trial-and-error procedure taking into account all the above considerations.

3. Free response: In this final phase, the system is excited, and the simulation continues for a fixed time. This parameter must be chosen to record a long enough response. The lengths of the collected data are fundamental to achieve the required frequency resolution.

7.4.1 *Data selection and algorithm application*

The last part of the identification procedure consists of selecting the relevant system-free responses and applying the identification algorithm. During the selection, the most important aspect to consider is undoubtedly data quality: clean system free responses greatly increase the odds of obtaining satisfactory and reliable results. The ability to achieve the required data quality is strictly related to adequate system excitation, as discussed in previous subsection, and appropriate measurement channel selection. The latter is especially relevant when the system has closely spaced modes. In this case, in the selected data sets, the closely spaced modes should be clearly visible and distinguishable. Regarding this aspect, no particular issues are encountered during the identification process, the selected channel and system excitation produced almost single degree-of-freedom responses of the mode of interest with limited contamination of other modes.

In the simulated experiment, the collected responses are wings internal moments at 4 different stations along the wing span.

The algorithm implemented is classified as a Multi-Input Multi-Output (MIMO) algorithm, allowing us to use multiple data records as input and to identify different modes of the system. In general, considering various channels increases the identification accuracy. A

normalization procedure, as suggested in [157], is implemented to neutralize possible dissimilarities in signals of orders of magnitude and to be able to compare them more efficiently. However, if the data are homogeneous, a normalization procedure can help amplify noisy signals and reduce the weight of clean ones. Basic pre-processing, consisting of filtering and resampling, is applied to the data. Figure 135 shows the flap-wise bending moment response at the wing root obtained for configuration 1 at 300 Knts.

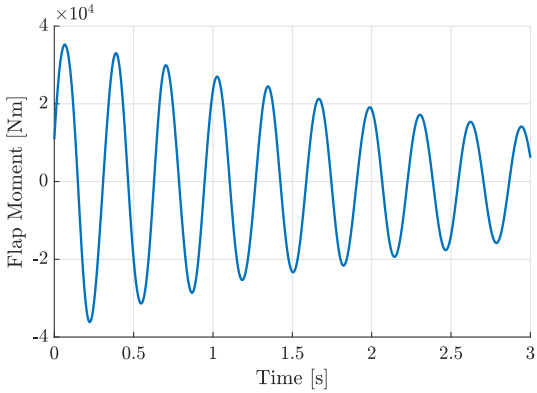


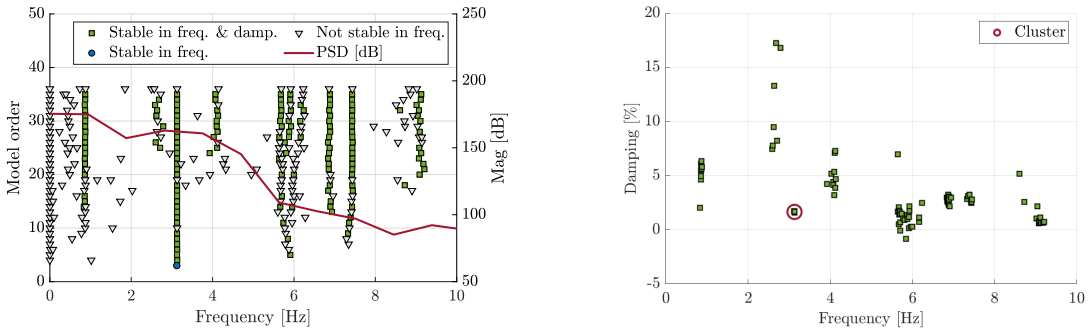
Figure 135: Flapwise bending response

As explained in 7.2.2, the algorithm estimates the order of the system from the relative magnitude of the principal values of the Hankel block matrix generated from the input data. This method is particularly effective, although it could not be sufficient for a low signal-to-noise ratio. The same procedure explained in [157] is then implemented. This methodology takes advantage of the stabilization diagram. In figure 136 the stabilization diagram obtained for the identification process of wing bending is shown.

The bending mode is visible as a peak in the complex mode indicator function plot, highlighted by second the vertical green line, and the identification algorithm is able to identify its frequency and damping with no particular issues, in fact, an evident column of stable poles is present at its frequency. The other wing modes and the short-period mode are also identifiable from the stabilization diagram, but they are much more spread out.

A simplified version of the strategy explained in [157, 172, 202] was also used to automate the extraction of modes. Due to their definition, stable poles tend to aggregate in clusters in the complex plane, allowing one to exploit cluster identification techniques to automate the whole process. The plot of stable poles in the complex plane or in the frequency-damping plane is usually called a Cluster Stabilization Diagram (CSD). Figure 136a shows the CSD for the case of identification of wing bends. From the figure it is clear that the poles associated with wing's bending are located in a cluster, identified by the red circle, with limited dispersion, granting high identification accuracy. To

further automate the procedure in the case of multiple results from the CSD, the desired mode was extracted by comparing its eigenvector and the one associated with the mode of interest identified at the previous wind speed.



(a) Stabilization diagram

(b) Cluster stabilization diagram

Figure 136: Symmetric bending stabilization diagram and clustering

7.4.2 *Whirl flutter results*

Table 19: Flutter velocities (in Knts) for each modes obtained by polynomial extrapolation

	Conf. 1	Conf. 2	Conf. 3	Conf. 4	Conf. 5	Conf. 6
SWB	695.80	578.8	-	344.43	277.54	-
SWC	376.50	356.07	301.89	338.32	240.21	235.39
SWT	424.79	364.21	-	376.93	332.66	321.45
AWB	430.84	365.22	295.32	294.43	363.49	312.39
AWC	372.91	368.22	292.78	279.18	289.09	279.60
AWT	435.23	413.02	339.01	-	384.23	342.46

The procedure described in the previous section is repeated 288 times to complete the test matrix of Table 18. For each simulation, the total simulation time was 10 s with a time step of 120 steps per rotor revolution. Table 19 summarizes the speed of flutter obtained for each configuration for all six modes.

Figures 137, 139, 141, 143, 145, 147 report frequency behavior for the symmetric and antisymmetric airframe modes considered. In all configurations, the torsion mode exhibits a decremental behavior as the free-stream velocity increases, whereas the other modes tend to maintain the original frequency. To better excite the torsion modes, the excitation frequency is adjusted for each wind speed using the frequency identified at the previous wind speed.

Figures 138, 140, 142, 144, 146, 148 report the damping trend for the six configurations. In the configuration 1 the unstable mode is the chordwise bending mode for both the symmetric and antisymmetric cases: the flutter point is found around 373 Knts. In the configuration 2, the instability appears almost simultaneously for the symmetric chord and torsion modes and for both the antisymmetric bending and the chord modes. Regarding the configuration 1, the overall damping trend of the symmetric torsion mode is reduced by a factor of 2%, while the other modes exhibit the same trends. An analogous consideration can be made for the antisymmetric bending mode.

Note that all simulations where conducted without any structural damping

In the configuration 3, the instability is found around 300 Knts. This means that increasing the rotor speed has decremental effects on overall flutter stability. In particular, the symmetric bending moment decreases its damping ratio by a factor of 1.5% with respect to airplane mode configurations. The flutter speed is further reduced when the engine fails, as in configuration 4, where the unstable mode is found to be the antisymmetric chord mode; however, with respect to configuration 3 the reduction in damping factor is not significant. Finally, as expected, disengaging the downstop, i.e. reducing the stiffness of the downstop spring, leads to a further degradation of the stability margins. In particular, the symmetric chord mode is marginally stable for the entire flutter envelope, while the reduction in the torsional frequency with respect to the previous configurations is 0.8 Hz as initially predicted by the in-vacuo eigenanalysis of the isolated aircraft.

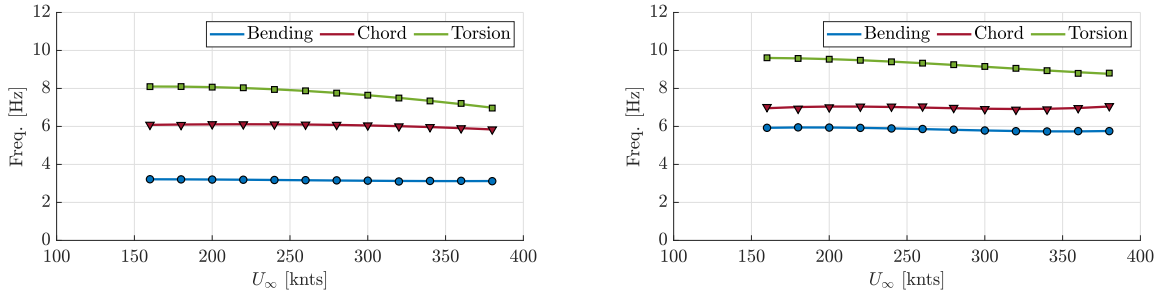
Obtained by polynomial interpolation

Lastly, it is important to remember that all analyzes were performed using a completely non-linear model in a time-marching simulation. This means that the response, especially at wind speeds closer to the flutter one, is far from the linear one. This effect is greatly shown in Fig. 149, where the phase diagram (vertical velocity vs. acceleration) of the hub center is obtained in the configurations 2 for the torsional mode is plotted. In particular, it is visible that for wind speed below the flutter one the response is almost linear: the orbits are almost circles that will be attracted at the center, while the wind speed above the flutter one the response is found to be highly nonlinear, with an orbit that will diverge from the center.

7.5 HIGH SPEED GUST RESPONSE

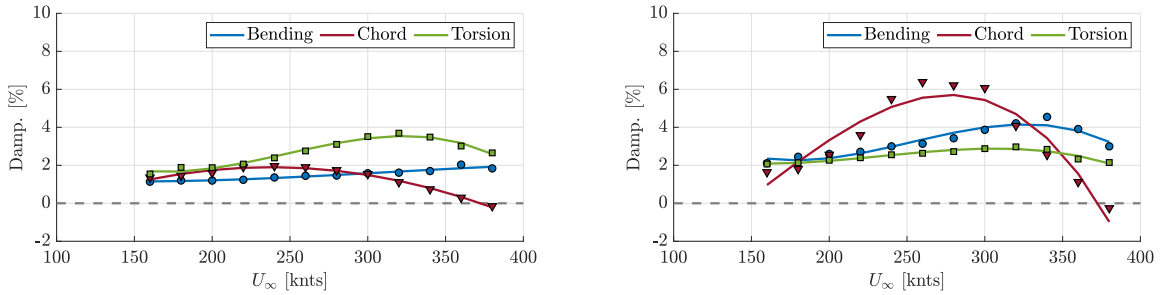
In this final section, all the sub-components that have been presented in the previous chapter, such as the complete MBDyn multibody tiltrotor model, coupled with the mid-fidelity vortex particle DUST model joined with the detailed upper-body pilot model presented in chapter 6.

As anticipated in section 7.1, the goal of this simulation is to show the capability of this model to extract a huge amount of multi-physics outputs from a single time marching simulation. In particular, the fol-



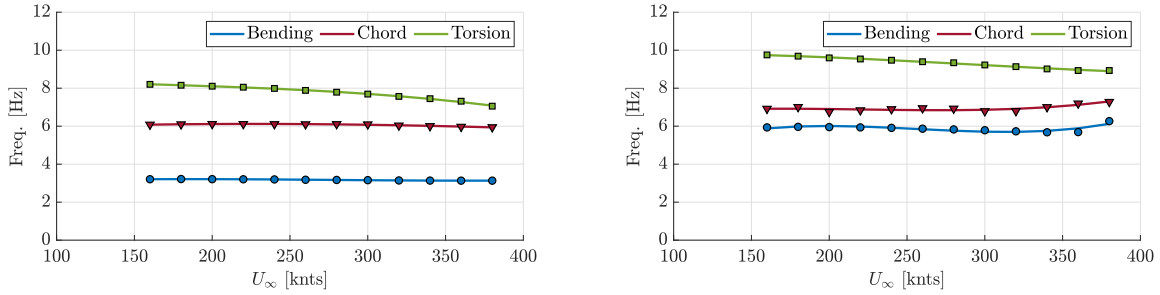
(a) Symmetric (b) Anti-symmetric

Figure 137: Configuration 1 (Power ON, AP mode, Downstop ON): frequency diagram



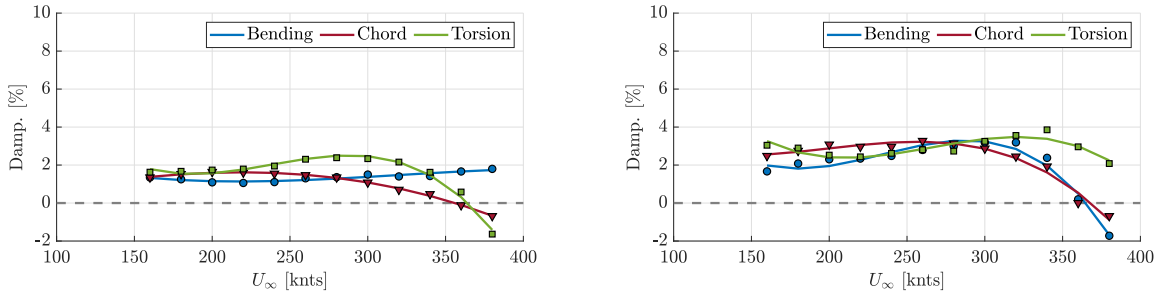
(a) Symmetric (b) Anti-symmetric

Figure 138: Configuration 1 (Power ON, AP mode, Downstop ON): damping diagram



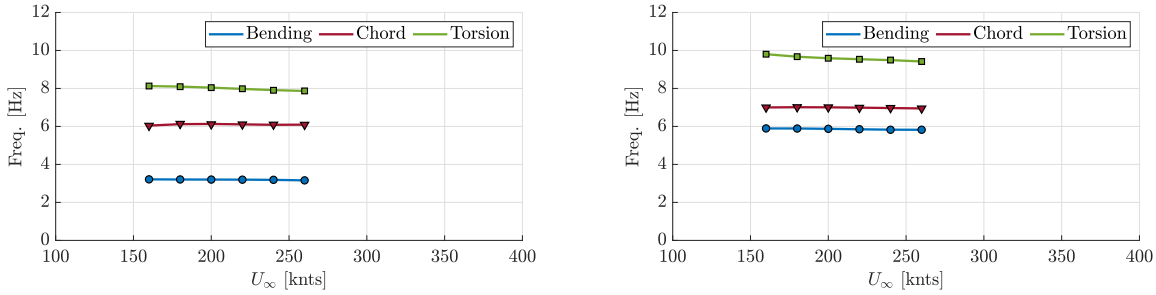
(a) Symmetric (b) Anti-symmetric

Figure 139: Configuration 2 (Power OFF, AP mode, Downstop ON): frequency diagram



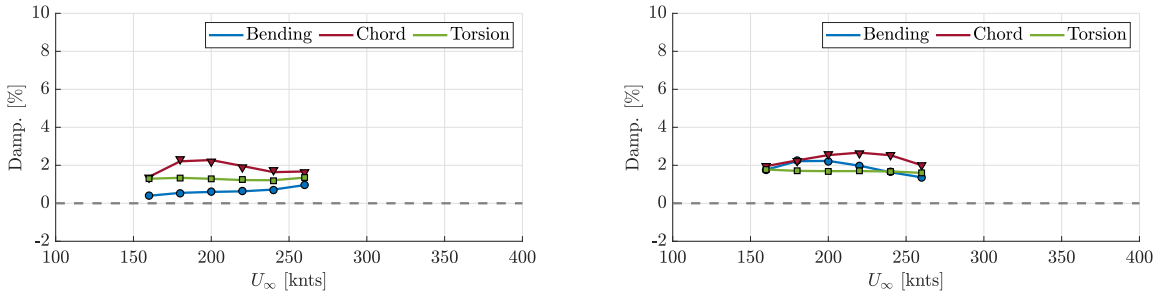
(a) Symmetric (b) Anti-symmetric

Figure 140: Configuration 2 (Power OFF, AP mode, Downstop ON): damping diagram



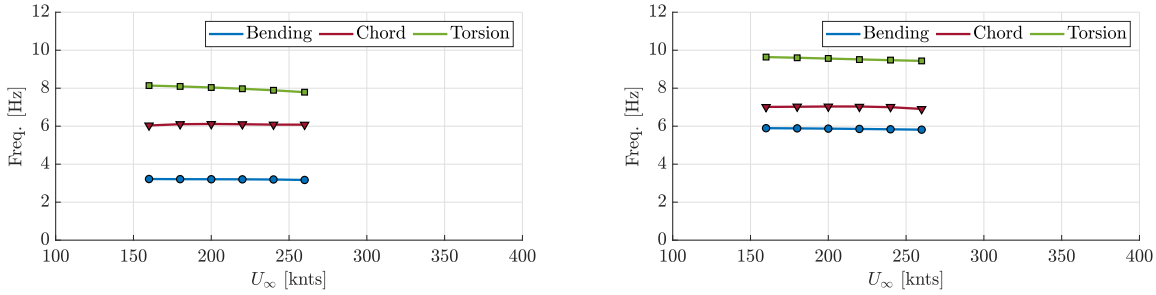
(a) Symmetric (b) Anti-symmetric

Figure 141: Configuration 3 (Power ON, HP mode, Downstop ON): frequency diagram



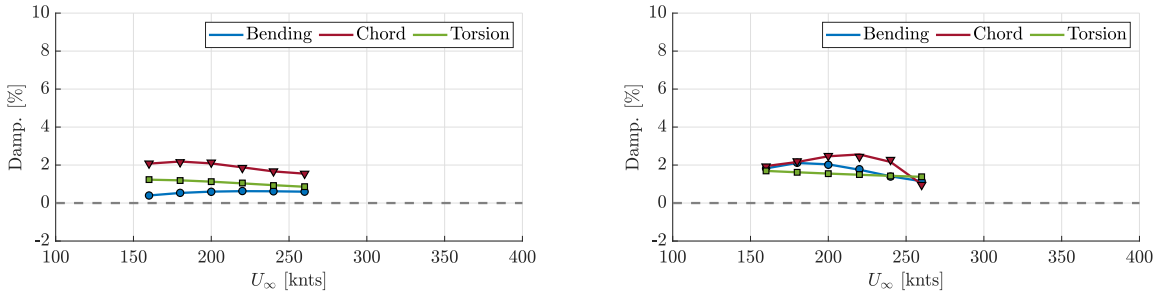
(a) Symmetric (b) Anti-symmetric

Figure 142: Configuration 3 (Power ON, HP mode, Downstop ON): damping diagram



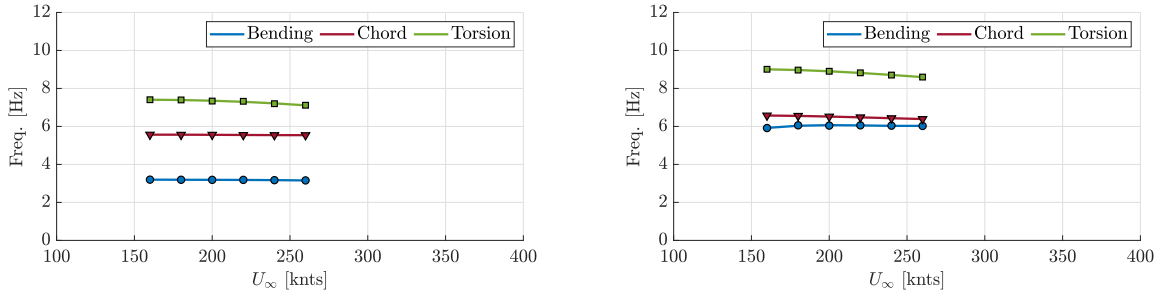
(a) Symmetric (b) Anti-symmetric

Figure 143: Configuration 4 (Power OFF, HP mode, Downstop ON): frequency diagram



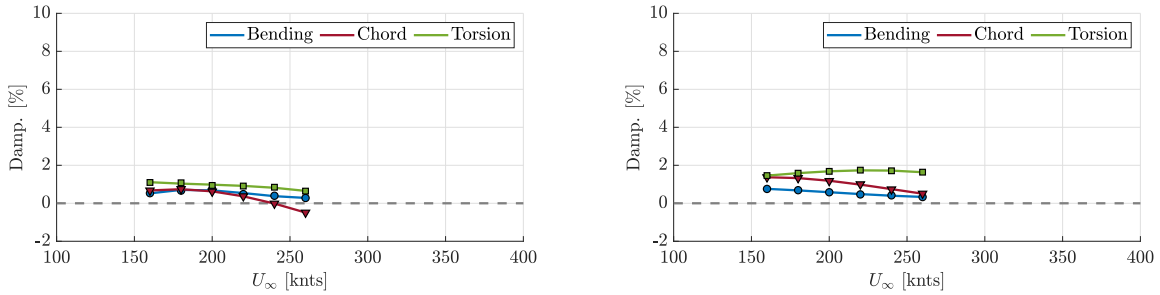
(a) Symmetric (b) Anti-symmetric

Figure 144: Configuration 4 (Power OFF, HP mode, Downstop ON): damping diagram



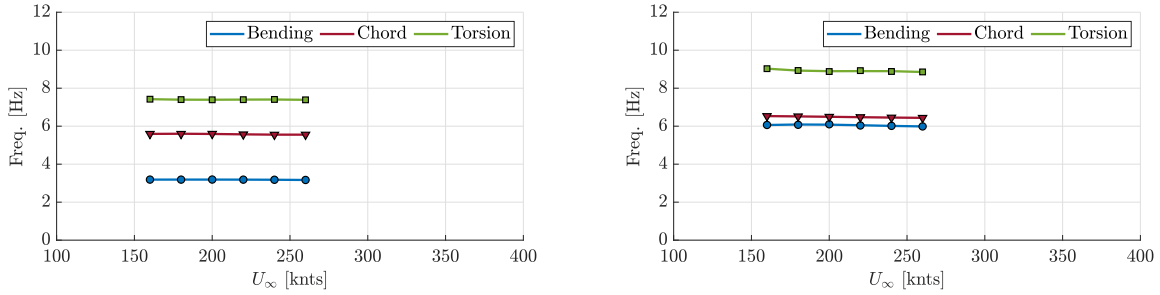
(a) Symmetric (b) Anti-symmetric

Figure 145: Configuration 5 (Power ON, HP mode, Downstop OFF): frequency diagram



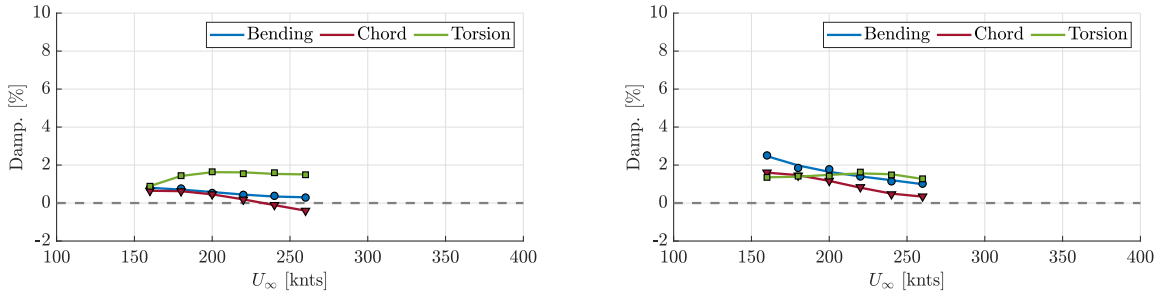
(a) Symmetric (b) Anti-symmetric

Figure 146: Configuration 5 (Power-ON, HP mode, Downstop OFF): damping diagram



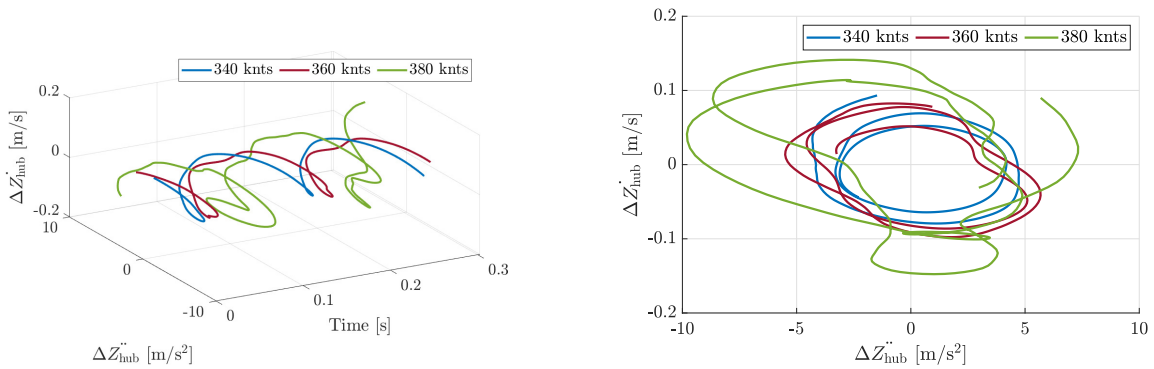
(a) Symmetric (b) Anti-symmetric

Figure 147: Configuration 6 (Power OFF, HP mode, Downstop OFF): frequency diagram



(a) Symmetric (b) Anti-symmetric

Figure 148: Configuration 6 (Power OFF, HP mode, Downstop OFF): damping diagram



(a) Time history

(b) Phase diagram

Figure 149: Configuration 2 (Power OFF, AP mode, Downstop ON): phase diagram

lowing sub-section will focus on the different time responses, resulting from two different aerodynamic modellings, that the vehicle will produce when it is hit by a vertical gust. The simulation parameters are resumed in Table 20. The pilot model is connected to the aeroelas-

Table 20: Simulation parameters

Parameter	Value	Unit
U_∞	360	Knts
H	20	m
U_{gust}	12	Knts
K_{SAS}	0.0	-
K_{cyclic}	1	-
Power	ON	-
Downstop	ON	-
Mode	AP	-
Trim spring	ON	-
Backrest inclination	10	deg
Inceptors levers	short	-

tic model using a built-in MBDyn socket that exchanges the required parameters between the two models. Therefore, two communication sockets are built:

- Seat kinematics: This socket takes the 3 positions and the 3 rotations that are retrieved by the seat node, located in the aeroelastic model, and sends them to the floor node of the pilot model. In this way, all 6 degrees of freedom are exposed in the coupling; however, communication can be restricted to fewer degrees of

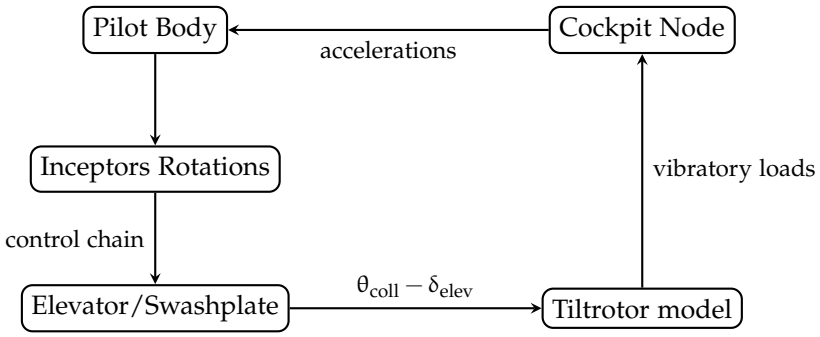


Figure 150: Pilot-in-the-loop communication scheme

freedom: for example, only the vertical displacement can be considered.

- Control inceptors: This socket takes the rotations of the inceptors levers, namely cyclic and collective, and sends them to the correspondent joint located in the aeroelastic model. The movements of the inceptors are then transmitted to the correspondent control surface, that is, longitudinal cyclic to the elevator, lateral cyclic to the ailerons, and collective to the engine torque, using an appropriate gear ratio. For this simulation, only the longitudinal cyclic is connected to the elevator, whereas the ailerons are locked and the collective is trimmed with a prescribed torque.

Control gear ratio are taken from [61, 143]

The simulation is performed using the same three-phase approach (trim, excitation, and free response) used in the whirl flutter analysis, with the difference that in this case the trim phase takes into account an additional input on the elevator deflection that depends on the longitudinal cyclic rotation angle. Note that the coupling is performed using only the kinematics variable (rotation of the control inceptors); therefore, the only force resistance on the control inceptors is produced only by the non-linear trim springs. Figure 150 summarizes the pilot coupling loop, while Fig. 151 shows the wake evolution during the trimming phase.

7.5.1 *Pilot's Response*

In this section, the pilot response is analyzed in the first two seconds after the gust. In particular, Fig. 152a shows the cockpit pitch attitude, and Fig. 152b shows the vertical acceleration extracted at the seat node, which is the result of the aeroelastic load. From the acceleration figure, it is clearly visible that the MBDyn model exhibits a stable pitch response, whereas the coupled model is unstable in the short-period mode. Moreover, coupled model's acceleration has a higher high-frequency content, since the wing-rotor interaction is captured by the vortex particle aerodynamics.

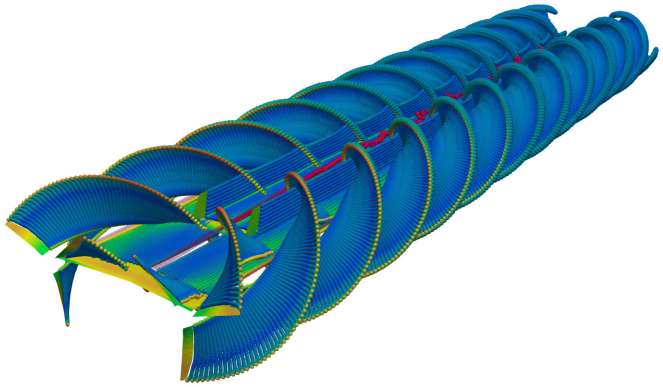
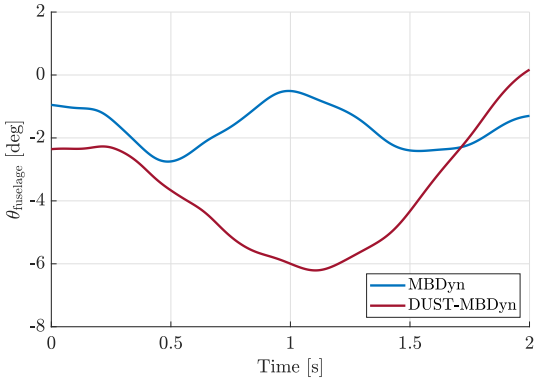
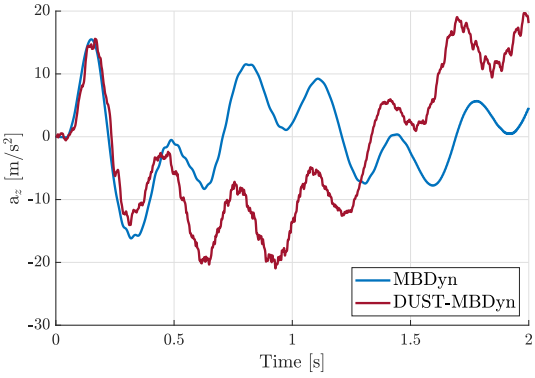


Figure 151: Simplified DUST-MBDyn model: wake evolution



(a) Cockpit pitch attitude



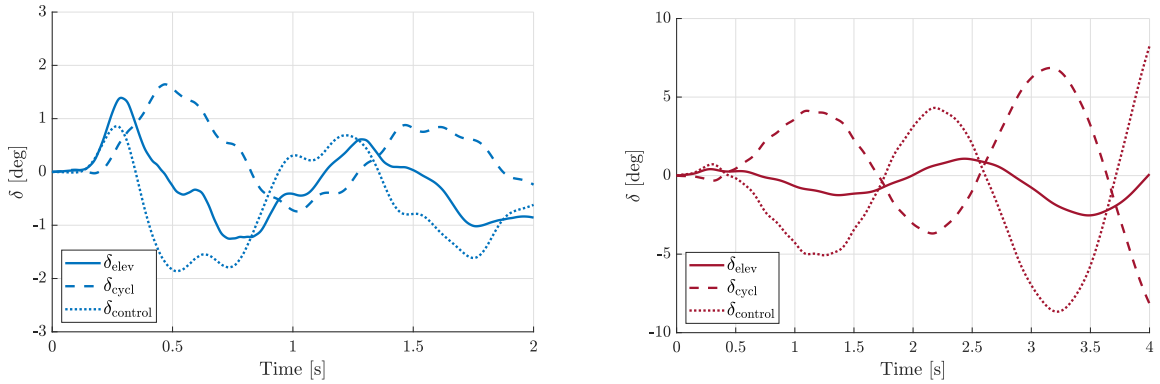
(b) Seat vertical acceleration

Figure 152: Cockpit global time history

Figure 153 shows the components of the elevator deflection for the MBDyn model and the coupled one. The deflection of the elevator can be decomposed as follows.

$$\delta_{elev} = \delta_{control} + \delta_{cycl} \tag{244}$$

Where δ_{elev} is the elevator deflection measured as the hinge rotation, δ_{cycl} is the longitudinal cyclic control inceptor rotation and $\delta_{control}$ is the equivalent rotation applied by the autopilot. When comparing the two responses, it is visible that, in both cases, the short period modes dominate the response; however, in the coupled one, much larger rotations are obtained in the coupled model: this is due to the fact that the short period frequency in this model is almost half with respect to the MBDyn model. This causes a larger vertical acceleration of the cockpit and, moreover, a larger cockpit pitch rotation that indeed causes a larger inceptor deflection. From the MBDyn model a stable response is visible: also in this case the response is dominated by the short period mode, however, higher frequency modes are present in the control inceport rotation caused by the coupling between the aiframe modes, the biomechanical modes, and the nonlinear trim spring.



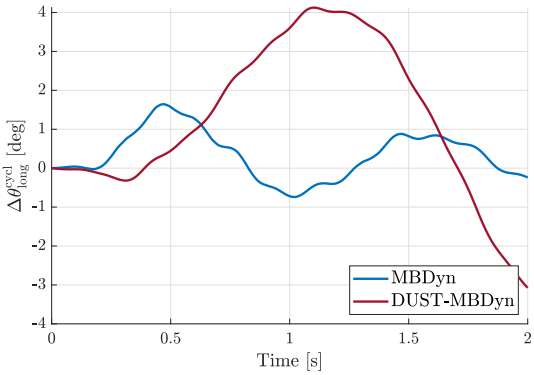
(a) MBDyn (b) DUST-MBDyn

Figure 153: Elevator rotations composition

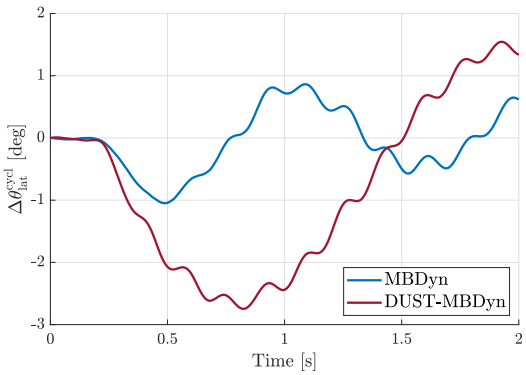
Figures 154 show the time responses of the rotation of the three control inceptors. All responses are plotted as differences with respect to the control trim $\Delta\theta = \theta - \theta_0$.

Since all six rigid degrees of freedom are free, the control response is not null for any of the inceptors, even though the aircraft is excited by a vertical gust. The cyclic components are excited mainly by the pitch response of the aircraft, while the collective is excited mainly by vertical movement. It is interesting to note, by comparing Fig.154c with Fig.152b that the pilot biomechanics acts as a low-pass filter, canceling all the high-frequency components that are present in the vertical seat acceleration of the coupled model.

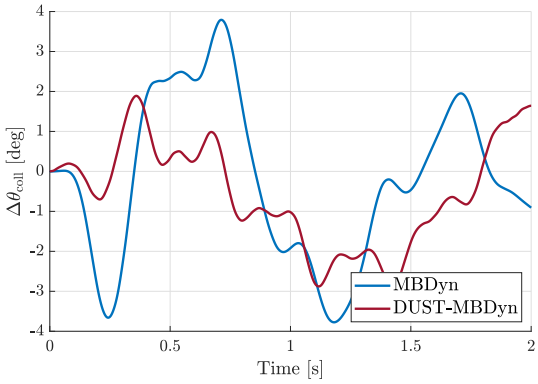
Figure 155 compares the phase diagram of the longitudinal cyclic response of the MBDyn model with respect to the coupled one. From



(a) Longitudinal cyclic



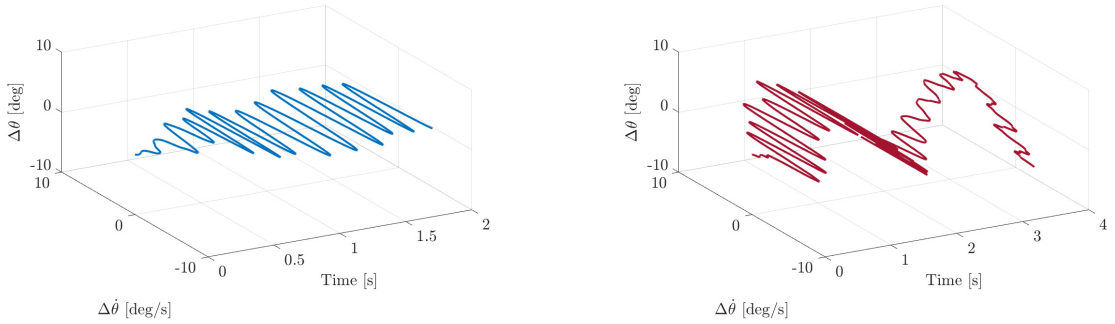
(b) Lateral cyclic



(c) Collective

Figure 154: Control inceptors rotation time responses

Fig. 155b it is clear that the angular velocity converges to a steady value, whereas the rotation component tends to diverge: this can be explained by the fact that the pilot biomechanical model damps the higher frequency content while amplifying the low-frequency oscillation (short-period mode).

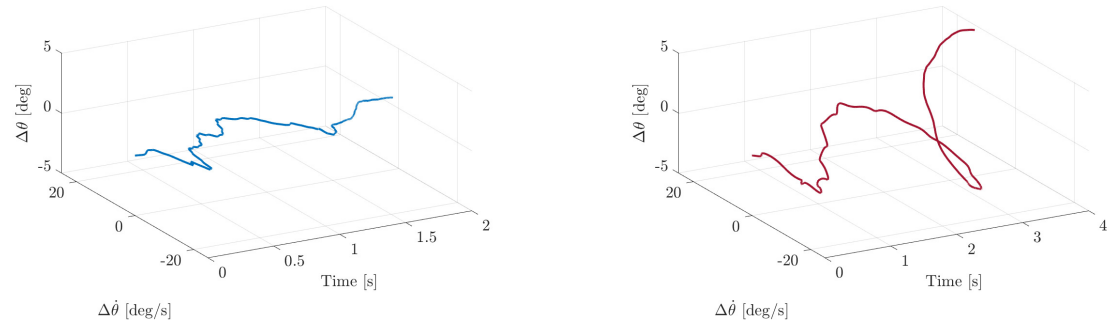


(a) MBDyn (b) DUST

Figure 155: Longitudinal cyclic phase diagram

Figure 156 compares the phase diagram of the lateral cyclic response of the MBDyn model with respect to the coupled model. Non-linear responses are observed in this case, caused by the presence of the non-linear trim spring. In particular, from 156b it is visible that after 2 s the response becomes more "linear", but unstable, since the amplitude of the oscillation has broken the trim stop.

Finally, Fig. 157 compares the phase diagram of the collective response of the MBDyn model with respect to the coupled one.

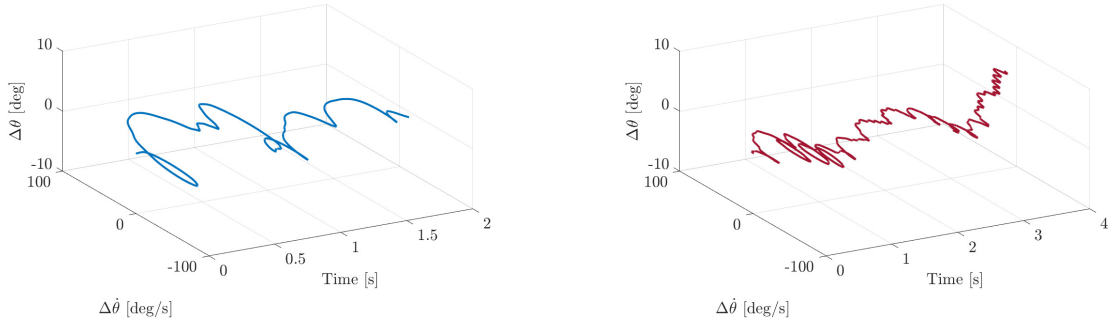


(a) MBDyn (b) DUST

Figure 156: Lateral cyclic phase diagram

7.5.2 *Internal moments: time history and frequency response*

In this last section, the internal moments of the wing, taken at the root of the wing, are compared taking into account both the time



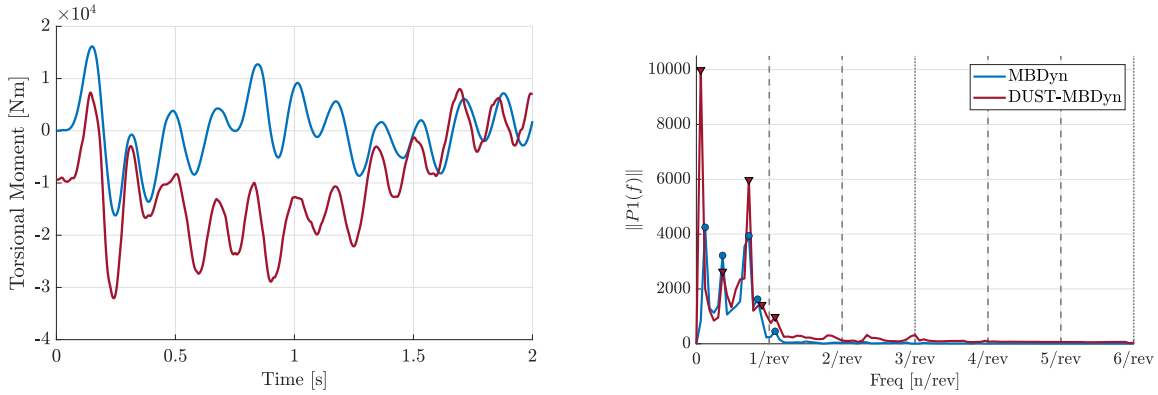
(a) MBDyn (b) DUST

Figure 157: Collective phase diagram

response and the relative frequency spectrum. To compute the frequency response, two short period oscillations are considered, which means a free response of 2 s for the MBDyn model and 4 s for the coupled one. Figure 158 shows the torsional moment-free response: from the frequency spectra three main peaks are visible: the first is associated with the short period mode, the second with the first symmetric flapping bending mode, while the third with the torsion mode. The frequency placement of the flexible modes is almost identical, which means that the responses of these modes are mainly dominated by the structural stiffness and inertial matrix, while the correspondent aerodynamic stiffness and inertial contribution have a limited impact of these modes.

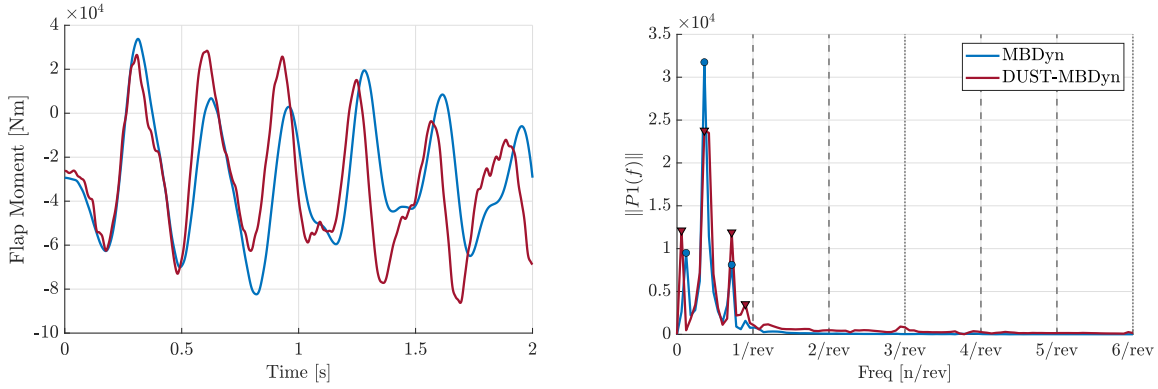
Analogous consideration can be made by looking at the flap-wise internal moment response reported in Fig. 159: In this case, the frequency spectrum is dominated by the bending mode with limited components of the short period and the torsional mode.

Finally, Fig. 158 shows the time response of the chordwise bending moment. The frequency content of the coupled model is closer to that of the MBDyn model. In particular, a 3/rev peak can be observed which is due to the interaction between the wake of the rotor and the wing. Also, in this case, the most dominant mode is the wing chord mode: the frequency placement in the coupled case is slightly right-shifted with respect to the MBDyn model.



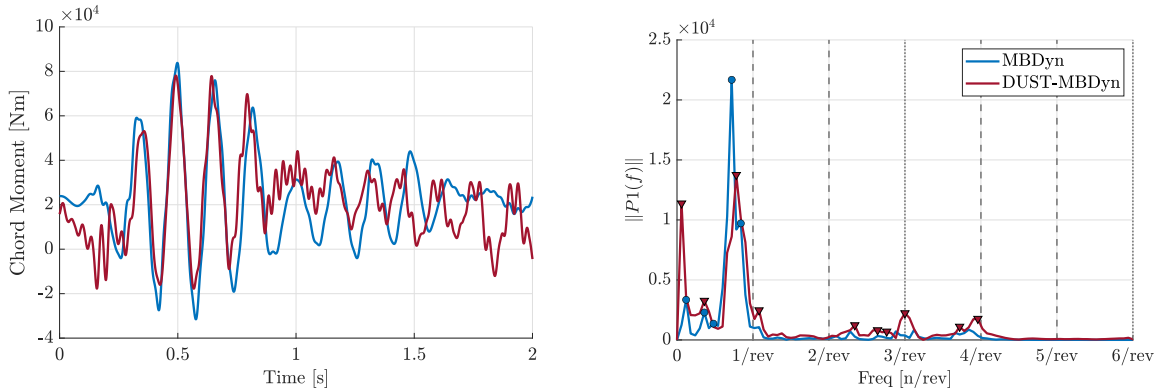
(a) Time response (b) Frequency response

Figure 158: Wing torsional moment



(a) Time response (b) Frequency response

Figure 159: Wing flapwise bending moment



(a) Time response (b) Frequency response

Figure 160: Wing chordwise bending moment

8

CONCLUSIONS AND FURTHER DEVELOPMENTS

The objective of this work was to develop, validate and test a comprehensive bio-aero-servo-elastic model Bell XV-15 tiltrotor equipped with ATB blade, adopting a co-simulation approach between a multibody solver coupled with a mid-fidelity aerodynamic solver. This model makes an advancement in modeling and simulation since, to the author's knowledge, is the first attempt that considers all the ingredients an aeroservoelastic in one single model and simulation. This allows the possibility of evaluating the interactional effects that each subcomponent has on the others from the preliminary project phase.

In the first part of the work, the aeroelastic numerical tool was presented, obtained by coupling a mid-fidelity aerodynamic solver with multibody dynamics software. It showed the advantages and limits of the proposed approach for the aeroelastic simulation of rotary wing vehicles. The resulting solver was thoroughly described, highlighting the key aspects of the mathematical formulation of both solvers, as well as the implementation details of their coupling. Furthermore, the capability to simulate movable surface deflection implemented in the aerodynamic solver was highlighted, due to their importance in simulating maneuvers and control studies, which represent essential aspects in the design of innovative air vehicles. The key advantage of the combination of a multibody structural model with a mid-fidelity numerical model is that it enabled accurate capture of aerodynamic interactional effects that characterize rotary-wing aircraft with limited computational cost compared to conventional, higher-fidelity CSD/CFD tools.

Moreover, to overcome the limitations of the state-of-the-art numerical elements, i.e. lifting lines and classical vortex lattice elements, related, respectively, to the occurrence of numerical instabilities and to the inability to capture viscous effects, a novel nonlinear vortex lattice element was implemented in the mid-fidelity aerodynamic code DUST.

The coupled code and the nonlinear vortex lattice elements were validated by comparing their results with numerical and experimental data available in the open literature for fixed- and rotary-wing problems. In detail, validation tests showed the importance of a more accurate description of aerodynamics to reproduce the dynamic behavior of a wedged wing in flutter condition. DUST nonlinear vortex lattice simulations performed over the XV-15 proprotor, equipped with a metallic blade, showed very good agreement with experimen-

tal data for different flight conditions ranging from hover to airplane mode. In particular, the results of DUST simulations showed the capability to obtain a degree of accuracy in terms of rotor aerodynamic performance similar to the high-fidelity CFD approach, but at a much lower computational cost. This result further highlighted the potential of the implemented approach for the design and investigation of rotorcraft configurations.

In the second part of the work, the XV-15 ATB blade model was presented, and each subcomponent was thoroughly validated against the available results and, from these, they were extended to allow validation with other analytical tools. Globally, a good correlation was reached in the dynamics analysis for the entire system. As highlighted in the hover aeroelastic comparison, the simple 2D aerodynamic MBDyn aerodynamic model is inadequate to match the experimental results; however, this limitation can be easily overcome by coupling the MBDyn's structural part to the Vortex Particle aerodynamic of DUST: the coupled model indeed showed a great correlation with respect to the experimental results in terms of rotor performances and wake measurement. In the axial-flow condition, where the disk loading is much lower, and the helicoidal wake is transported by the free stream velocity, the differences between the aerodynamic models greatly reduce.

The airframe part of the XV-15 model was presented and dynamically validated against the reference finite element model. Trim points were found using a simplified aerodynamic model of the airframe with an integral controller; then, the longitudinal free-free condition is controlled through optimal longitudinal control of the tiltrotor designed using a reduced-order model of the tiltrotor. The dynamic stability of the short-period mode was deeply investigated using a discrete gust as excitation: the stability margins in this mode are degraded when using the coupled model and when the aeroelastic rotors are added to the airframe model.

A detailed biomechanical model of the pilot was presented. This model was obtained by adding a detailed three-dimensional model of the spine to a previously formulated detailed model of the upper limbs. A complete model scaling procedure, based on the existing biomechanical properties database, was developed and validated against available experimental data. The purpose of the complete upper body model was to evaluate the biodynamics of helicopter pilots and occupants, for vibratory and rotorcraft-pilot coupling analysis. The proposed model showed a good correlation with the experimental data in terms of seat-to-head transmissibility.

Biodynamic feedthrough analysis of the complete upper body model showed that when the dynamics of the spine is considered, the frequency and magnitude of the first resonance peak increase for all control inceptors; the largest increase was observed for the collective

control inceptor. Furthermore, the cyclic control inceptor response departs from linearity when the amplitude of the acceleration imposed on the cockpit reaches 0.6-0.9 g at resonance.

In the last part, the whirl flutter aeroelastic stability of the entire aircraft was evaluated through a series of free decay numerical experiments. Modal parameters were identified with the matrix pencil identification method. To tune the excitation frequency, two frequency sweep analyzes were conducted and the modes were identified using a frequency domain approach. In this work, only the preliminary result of the analysis based on a rigid blade rotor model with simplified airframe aerodynamics has been presented. Finally, the detailed biomechanical pilot model was coupled with the aeroelastic tiltrotor model, and the response to a discrete cosine-shaped vertical gust was analyzed, highlighting the differences that the two aerodynamic models produce in response. This simulation however must be intended as a starting point towards a series of further analyses in which the control law of the tiltrotor will be designed taking into account the additional input that the pilot biomechanics introduces. Furthermore, the effects of pilot biomechanics will be deeply investigated when considering the hovering mode of the helicopter and the transition phase.

The outcome of this work opens up a novel scenario for the scientific and industrial community. Indeed, the significantly lower computational effort required by the coupled simulations with respect to conventional, higher-fidelity CSD tools coupled with URANS solvers, as well as the quite high level of accuracy that could be obtained by these simulations, indicate that the use of a mid-fidelity aerodynamic solver for coupled aerodynamics-multibody dynamics simulations represents a valuable instrument for performing the large number of aeroelastic analyses required in the preliminary design phase of innovative rotary-wing vehicles (eVTOL), not only in the rotorcraft field but also in other domains, for example, wind energy and turbomachinery applications.

Part III

APPENDIX

A

AERODYNAMIC INFLUENCE COEFFICIENT

Some Aerodynamic Influence Coefficients (AIC) are required for the discretization of both the equation of the potential part of the velocity field and the dynamical equation of the rotational part of the velocity field.

A.1 POINT SOURCES

Let consider a potential field with spherical symmetry and let the velocity of the points at distance R have the same intensity. The potential induced in $\boldsymbol{r}$ by a unit intensity point source located in $\boldsymbol{r}_0$ is

$$\varphi^s(\boldsymbol{r}) = -\frac{1}{4\pi}\frac{1}{|\boldsymbol{x}|} \tag{245}$$

being $\boldsymbol{x} := \boldsymbol{r} - \boldsymbol{r}_0$. The equation can be proved, since the Laplace equation in spherical coordinate for the potential must hold. The velocity induced in $\boldsymbol{r}$ by a unit intensity point source located in $\boldsymbol{r}_0$ is

$$\boldsymbol{u}^s(\boldsymbol{r}) = \boldsymbol{\nabla}\varphi^s(\boldsymbol{r}) = \frac{1}{4\pi}\frac{\boldsymbol{x}}{|\boldsymbol{x}|^3} \tag{246}$$

A.2 POINT DOUBLETS

Let consider the simultaneus presence of a source and a sink. The potential induced in $\boldsymbol{r}$ by a unit intensity point doublet located in $\boldsymbol{r}_0$ is

$$\varphi^d(\boldsymbol{r}) = -\frac{1}{4\pi}\frac{\hat{\boldsymbol{n}}\cdot\boldsymbol{x}}{|\boldsymbol{x}|^3} \tag{247}$$

The velocity induced in $\boldsymbol{r}$ by a unit intensity point doublet located in $\boldsymbol{r}_0$ is

$$\boldsymbol{u}^d(\boldsymbol{r}) = \boldsymbol{\nabla}\varphi^d(\boldsymbol{r}) = \frac{1}{4\pi}\left\{-\frac{\hat{\boldsymbol{n}}}{|\boldsymbol{x}|^3} + 3\hat{\boldsymbol{n}}\cdot\boldsymbol{x}\frac{\boldsymbol{x}}{|\boldsymbol{x}|^5}\right\} \tag{248}$$

The velocity gradient induced in $\boldsymbol{r}$ by a unit intensity point doublet located in $\boldsymbol{r}_0$ is

$$\boldsymbol{\nabla}\boldsymbol{u}^d(\boldsymbol{r}) = \frac{1}{4\pi}\left\{\frac{3}{|\boldsymbol{x}|^5}\left(\hat{\boldsymbol{n}}\otimes\boldsymbol{x} + \boldsymbol{x}\otimes\hat{\boldsymbol{n}} + \hat{\boldsymbol{n}}\cdot\boldsymbol{x}\boldsymbol{I}\right) - 15\hat{\boldsymbol{n}}\cdot\boldsymbol{x}\frac{\boldsymbol{x}\otimes\boldsymbol{x}}{|\boldsymbol{x}|^7}\right\} \tag{249}$$

A.3 SURFACE DOUBLETS

A.3.1 *Surface doublets: kinetic potential,* D

The potential induced in $\boldsymbol{r}_i$ (passive point) by the unit-strength intensity surface doublet S_k (active surface) is

$$D_{ik} = -\frac{1}{4\pi}\int_{S_k} \frac{\hat{\boldsymbol{n}}\cdot(\boldsymbol{r}_i - \boldsymbol{r})}{|\boldsymbol{r}_i - \boldsymbol{r}|^3} \tag{250}$$

The following Newman [138], $\Phi = -4\pi D_{ik}$ is recognized as the solid angle of the panel S_k, as seen from the passive point $\boldsymbol{r}_i$. From the Gauss-Bonnet theorem, the solid angle Φ is equal to

$$\Phi = \sum_{\ell=1}^{N_e} \beta_\ell - (N_e - 2)\pi \tag{251}$$

i.e., the sum of the angles β_ℓ of the projection of the edges N_e of the element on the unit sphere, centered in $\boldsymbol{r}_i$.

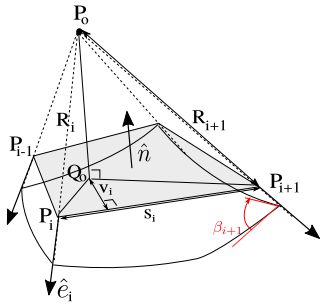


Figure 161: Definition of the quantities for the computation of the surface singularity AICs.

For each element, Φ can be computed with a loop over the vertices of an element, as follows.

- The unit vector aligned with the vector from the passive point P_0 to the vertex of the element is computed: $\boldsymbol{e_i} = (P_i - P_0)/|P_i - P_0|$
- normal projection of the edges $\boldsymbol{\ell}_i = P_{i+1} - P_i$, $\boldsymbol{\ell}_{i-1} = P_i - P_{i-1}$ w.r.t. $\boldsymbol{e_i}$ is computed

$$\begin{cases} \boldsymbol{a}_i = \boldsymbol{\ell}_i - \boldsymbol{\ell}_i \cdot \hat{\boldsymbol{e}}_i \hat{\boldsymbol{e}}_i \\ \boldsymbol{a}_{i-1} = -(\boldsymbol{\ell}_{i-1} - \boldsymbol{\ell}_{i-1} \cdot \hat{\boldsymbol{e}}_i \hat{\boldsymbol{e}}_i) \end{cases} \tag{252}$$

- the desired angle β_i, is the angle included between $\boldsymbol{a}_i$ and $\boldsymbol{a}_{i-1}$

$$\begin{cases} \sin\beta_i = \dfrac{\boldsymbol{a}_i \times \boldsymbol{a}_{i-1} \cdot \hat{\boldsymbol{e}}_i}{|\boldsymbol{a}_i||\boldsymbol{a}_{i-1}|} \\ \cos\beta_i = \dfrac{\boldsymbol{a}_i \cdot \boldsymbol{a}_{i-1}}{|\boldsymbol{a}_i||\boldsymbol{a}_{i-1}|} \end{cases} \tag{253}$$

and can be computed with the function `atan2`, providing only the numerators of the ratios appearing in $\sin\beta_i$ and $\cos\beta_i$.

A.4 LINE VORTICES

The velocity induced in $\boldsymbol{r}_i$ by the vortex ring ℓ, whose intensity is Γ, is calculated by means of the Biot-Savart law:

$$\boldsymbol{u}(\boldsymbol{r}_i) = -\frac{\Gamma}{4\pi}\oint_\ell \frac{\boldsymbol{r}_i - \boldsymbol{r}}{|\boldsymbol{r}_i - \boldsymbol{r}|^3} \times \hat{\boldsymbol{t}} \tag{254}$$

For a vortex ring, whose sides are straight lines, a simple analytical expression of the previous integral can be found. After dividing the vortex ring ℓ into segments ℓ_k, the induced velocity reads:

$$\boldsymbol{u}(\boldsymbol{r}_i) = -\frac{\Gamma}{4\pi}\sum_{k=1}^{N}\int_{\ell_k} \frac{\boldsymbol{r}_i - \boldsymbol{r}}{|\boldsymbol{r}_i - \boldsymbol{r}|^3} \times \hat{\boldsymbol{t}}_k = \frac{\Gamma}{4\pi}\boldsymbol{I}_k \tag{255}$$

having introduced the definition of $\boldsymbol{I}_k$ as the velocity induced by the k-th side with intensity -4π. It is possible to prove that $\boldsymbol{I}_k$ can be computed as:

$$\boldsymbol{I}_k = \frac{\boldsymbol{R}_k \times \boldsymbol{R}_{k+1}(R_k + R_{k+1})}{R_k R_{k+1}(R_k R_{k+1} + \boldsymbol{R}_k \cdot \boldsymbol{R}_{k+1})}\,, \tag{256}$$

being $\boldsymbol{R}_k = \boldsymbol{r} - \boldsymbol{P}_k$ the vector connecting the k-th vertex of the element with the passive point $\boldsymbol{r}_i$.

It can be shown [77] that the induced velocity of a doublet surface distribution of constant strength is equivalent to a doublet-vortex ring.

A.5 SURFACE SOURCES

A.5.1 *Surface sources: kinetic potential, S*

The potential induced in $\boldsymbol{r}_i$ (passive point) by the unit-strength intensity surface source S_k (active surface) is

$$S_{ik} = -\frac{1}{4\pi}\int_{S_k} \frac{1}{|\boldsymbol{r}_i - \boldsymbol{r}|} \tag{257}$$

The computation of the velocity potential of a constant-strength source surface distribution exploits the expression of the Φ term appearing in the computation of the doublet potential; see Sect.§A.3.1. Following [138], the integral $\Psi = -4\pi S$ is defined as the potential of a surface doublet of intensity -4π. It is possible to show

$$\frac{\partial\Psi}{\partial\zeta} = -\Phi \quad \text{and thus} \quad \Psi = -\int_\zeta^\infty \zeta' \mathrm{d}\Phi - \zeta\Phi \tag{258}$$

B

XV-15 ROTOR DATA

Table 21: XV-15 ATB blade main rotor data

Typology	Gimballed, Stiff in plane Rotor	
Nb	3	-
R	3.81	m
A	45.60	m^2
c (MAC)	0.41	m
SOLIDITY	0.103	-
RPM 80%	480.80	RPM
RPM 100%	601.00	RPM
VTIP 80%	191.83	$m\,s^{-1}$
VTIP 100%	239.79	$m\,s^{-1}$
EFLAP	0.00	r/R
ELAG	0.00	r/R
RFA	0.6477	r/R
KFLAP	0.00	$N\,m\,rad^{-1}$
KLAG	0.00	$N\,m\,rad^{-1}$
KTo	18 852.62	$N\,m\,rad^{-1}$
KTC	18 852.62	$N\,m\,rad^{-1}$
KTR	18 852.62	$N\,m\,rad^{-1}$
DELTA3G	−15.00	deg
CONE	2.50	deg
DROOP	0.46	deg
SWEEP	0.89	deg
PHIPH	−20.50	deg
PHIPL	0.00	deg
RPB	0.347	m
RPH	0.065	m
XPH	0.240	m
Kg	22 536.0	$N\,m\,rad^{-1}$
GDAMPC	25.000	%
K_{mast}	1 017 227.938	$N\,m\,rad^{-1}$
K_{engine}	2.0209×10^7	$N\,m\,rad^{-1}$
$K_{intershaft}$	381 121.4006	$N\,m\,rad^{-1}$
I_{rotor}	481.8	$kg\,m^2$
I_{engine}	315.9	$kg\,m^2$

Table 22: XV-15 ATB blade inertial data in E50 format: Z-axis goes from tip to root, X-axis from the leading edge to the trailing edge.

STA	WEIGHT	XCG	YCG	ROTAPI	JX	JY	JP	EA
mm	$kg\,mm^{-1}$	mm	mm	deg	kg mm	kg mm	kg mm	N
3810	0.001 79	69.077	−32.750	−0.07	0.0806	0.569	0.650	3.5×10^8
3752.85	0.002 50	68.567	−31.751	−0.06	0.2020	4.951	5.153	5.6×10^8
3714.75	0.002 95	68.222	−31.084	−0.04	0.4645	14.248	14.712	4×10^8
3695.7	0.003 21	68.049	−30.754	−0.04	0.4804	15.937	16.418	4.5×10^8
3657.6	0.003 75	75.552	−30.098	−0.05	0.5922	19.645	20.237	4.6×10^8
3619.5	0.004 29	83.087	−29.441	−0.04	0.6629	22.016	22.679	4.8×10^8
3581.4	0.004 29	90.614	−28.796	−0.04	0.6095	20.184	20.794	5.2×10^8
3562.35	0.004 29	88.374	−28.435	0.00	0.6670	21.538	22.205	5.2×10^8
3524.25	0.004 29	83.965	−27.733	0.08	0.7814	24.066	24.847	5.3×10^8
3474.72	0.004 29	77.375	−26.822	0.24	0.9287	26.942	27.871	5.5×10^8
3467.1	0.011 25	75.393	−26.661	0.31	0.6293	18.256	18.885	1.5×10^9
3436.62	0.016 90	70.624	−26.416	−0.13	0.5664	16.000	16.566	2.4×10^9
3429	0.018 41	69.479	−26.256	−0.06	0.5595	15.681	16.241	2.6×10^9
3409.95	0.018 51	63.167	−25.965	−0.12	0.8892	23.707	24.596	2.4×10^9
3402.33	0.018 55	60.664	−25.813	−0.05	0.9960	26.555	27.551	2.4×10^9
3371.85	0.018 72	50.767	−25.326	−0.11	1.4351	35.779	37.214	2.1×10^9
3364.23	0.018 76	48.318	−25.190	−0.04	1.5078	37.591	39.099	2.1×10^9
3356.61	0.018 80	45.879	−25.070	−0.08	1.5907	39.184	40.774	2×10^9
3348.99	0.018 18	44.672	−24.945	−0.05	1.6182	39.592	41.210	2×10^9
3303.27	0.010 53	45.263	−24.240	−0.17	1.5980	35.688	37.286	1.3×10^9
3295.65	0.009 26	45.535	−24.115	−0.10	1.5712	35.055	36.626	1.1×10^9
3284.22	0.007 35	46.175	−23.953	−0.22	1.5633	34.070	35.633	9.2×10^8
3276.6	0.006 07	46.328	−23.827	−0.15	1.5398	33.426	34.966	7.8×10^8
3219.45	0.006 07	46.225	−22.932	−0.01	1.8946	37.161	39.055	7×10^8
2964.18	0.006 07	45.721	−19.225	0.01	2.6023	54.596	57.198	6.9×10^8
2628.9	0.007 75	42.166	−14.711	0.09	3.8170	128.176	131.993	1.1×10^9
2514.6	0.008 93	40.565	−13.279	−0.05	4.5278	128.585	133.113	1.3×10^9
2286	0.009 29	37.239	−10.540	−0.05	6.1246	127.555	133.680	1.4×10^9
2095.5	0.009 64	33.974	−8.453	0.01	5.9214	123.280	129.201	1.7×10^9
1935.48	0.010 00	31.116	−6.829	0.04	6.5060	128.577	135.083	1.7×10^9
1524	0.012 05	22.450	−3.427	−0.30	8.3643	134.716	143.080	2.2×10^9
1238.25	0.013 32	15.277	−1.643	−0.22	8.8128	133.583	142.396	2.8×10^9
1143	0.024 82	28.101	−1.291	−0.31	14.412	170.314	184.726	4×10^9
1040.13	0.024 82	26.345	−0.883	−0.35	19.954	132.032	151.987	4.6×10^9
952.5	0.024 82	39.314	−0.715	−0.41	19.920	89.950	109.871	5.1×10^9
876.3	0.023 93	28.876	−0.344	−0.18	16.747	65.784	82.532	7.4×10^9
762	0.028 20	26.890	0.087	0.32	17.3272	38.493	55.820	2.1×10^{10}
685.8	0.027 13	25.560	0.211	0.48	1240.236	2859.658	4099.895	1.7×10^{10}
678.18	0.026 05	25.427	0.243	0.55	581.9406	2853.927	3435.868	1.4×10^{10}
655.32	0.018 89	25.029	0.204	0.46	386.8101	1059.355	1446.165	6.8×10^9
647.7	0.073 45	0.000	0.000	0.61	0.0000	0.000	0.000	1.6×10^{10}
609.6	0.073 45	0.000	0.000	0.00	0.0000	0.000	0.000	1.6×10^{10}
502.92	0.073 45	0.000	0.000	0.00	0.0000	0.000	0.000	2.4×10^{10}
495.3	0.093 33	0.000	0.000	0.00	0.0000	0.000	0.000	2.4×10^{10}
464.82	0.093 33	0.000	0.000	0.00	0.0000	0.000	0.000	2.4×10^{10}
457.2	0.093 33	0.000	0.000	0.00	0.0000	0.000	0.000	1.1×10^{10}
388.62	0.093 33	0.000	0.000	0.00	0.0000	0.000	0.000	1.2×10^{10}
381	0.093 33	0.000	0.000	0.00	0.0000	0.000	0.000	1×10^{10}
198.12	0.093 33	0.000	0.000	0.00	0.0000	0.000	0.000	1×10^{10}
190.5	0.102 49	0.000	0.000	0.00	0.0000	0.000	0.000	1.7×10^{10}
0	0.102 49	0.000	0.000	0.00	0.0000	0.000	0.000	1.7×10^{10}

Table 23: XV-15 blade stiffness data: Z-axis goes from tip to root, X-axis from the leading edge to the trailing edge.

STA	XNA	YNA	ROTAN	EJX (FLAP)	EJY (LAG)	GJ	XCT	YCT
mm	mm	mm	deg	$\mathrm{N\,mm^2}$	$\mathrm{N\,mm^2}$	$\mathrm{N\,mm^2}$	mm	mm
3810	44.4	−32.7	−0.075	2.870×10^9	2.025×10^{10}	1.149×10^8	69.077	−32.750
3752.85	43.9	−31.7	−0.06	4.047×10^9	9.918×10^{10}	7.687×10^8	68.567	−31.751
3714.75	43.6	−31.1	−0.0439	4.823×10^9	1.479×10^{11}	1.207×10^9	64.717	−31.082
3695.7	43.4	−30.7	−0.0427	5.195×10^9	1.723×10^{11}	1.422×10^9	63.820	−30.751
3657.6	43.1	−30.1	−0.0478	5.967×10^9	1.979×10^{11}	1.860×10^9	65.647	−30.089
3619.5	42.7	−29.4	−0.0428	6.744×10^9	2.240×10^{11}	2.298×10^9	66.830	−29.429
3581.4	42.3	−28.8	−0.0418	7.538×10^9	2.496×10^{11}	3.442×10^9	60.679	−28.773
3562.35	42.2	−28.4	0.0037	7.922×10^9	2.558×10^{11}	4.017×10^9	52.091	−28.436
3524.25	41.8	−27.8	0.0848	8.695×10^9	2.678×10^{11}	5.166×10^9	51.439	−27.780
3474.72	41.3	−27.0	0.2415	9.786×10^9	2.839×10^{11}	6.777×10^9	50.945	−26.932
3467.1	41.2	−26.8	0.3143	9.873×10^9	2.864×10^{11}	6.901×10^9	47.820	−26.812
3436.62	40.9	−26.3	−0.1264	1.056×10^{10}	2.984×10^{11}	7.563×10^9	47.509	−26.365
3429	40.8	−26.2	−0.0576	1.065×10^{10}	2.984×10^{11}	7.645×10^9	47.431	−26.233
3409.95	40.6	−25.9	−0.1206	1.119×10^{10}	2.984×10^{11}	8.141×10^9	47.234	−25.931
3402.33	40.6	−25.8	−0.0508	1.119×10^{10}	2.984×10^{11}	8.182×10^9	47.154	−25.800
3371.85	40.2	−25.3	−0.1145	1.197×10^{10}	2.984×10^{11}	8.926×10^9	46.835	−25.318
3364.23	40.2	−25.2	−0.0447	1.197×10^{10}	2.984×10^{11}	8.926×10^9	46.754	−25.188
3356.61	40.1	−25.1	−0.0799	1.211×10^{10}	2.984×10^{11}	9.050×10^9	46.673	−25.071
3348.99	40.0	−24.9	−0.0477	1.220×10^{10}	2.984×10^{11}	9.133×10^9	46.592	−24.946
3303.27	39.5	−24.2	−0.1674	1.336×10^{10}	2.984×10^{11}	1.025×10^{10}	46.098	−24.243
3295.65	39.4	−24.1	−0.0989	1.337×10^{10}	2.984×10^{11}	1.025×10^{10}	25.188	−24.080
3284.22	39.3	−23.9	−0.2196	1.369×10^{10}	2.984×10^{11}	1.058×10^{10}	25.205	−23.873
3276.6	39.2	−23.8	−0.148	1.374×10^{10}	2.984×10^{11}	1.177×10^{10}	25.159	−23.772
3219.45	38.6	−22.9	−0.0084	1.521×10^{10}	2.984×10^{11}	1.177×10^{10}	25.061	−22.929
2964.18	35.5	−19.2	0.014	2.066×10^{10}	4.335×10^{11}	3.386×10^{10}	24.374	−19.230
2628.9	31.3	−14.7	0.0916	3.846×10^{10}	1.291×10^{12}	3.386×10^{10}	29.238	−14.732
2514.6	29.7	−13.3	−0.0491	4.699×10^{10}	1.334×10^{12}	5.682×10^{10}	26.993	−13.267
2286	26.6	−10.5	−0.055	6.600×10^{10}	1.374×10^{12}	5.682×10^{10}	22.380	−10.526
2095.5	23.8	−8.5	0.0075	7.112×10^{10}	1.481×10^{12}	5.682×10^{10}	18.410	−8.455
1935.48	21.4	−6.8	0.045	7.492×10^{10}	1.481×10^{12}	8.178×10^{10}	2.918	−6.853
1524	14.9	−3.4	−0.3025	1.076×10^{11}	1.733×10^{12}	8.178×10^{10}	−0.311	−3.309
1238.25	10.2	−1.6	−0.2196	1.270×10^{11}	1.925×10^{12}	8.178×10^{10}	−2.774	−1.576
1143	8.6	−1.2	−0.3091	1.683×10^{11}	1.989×10^{12}	1.386×10^{11}	−3.635	−1.123
1040.13	6.8	−0.8	−0.3532	2.692×10^{11}	1.781×10^{12}	1.386×10^{11}	−0.689	−0.719
952.5	5.3	−0.5	−0.4074	3.553×10^{11}	1.604×10^{12}	1.386×10^{11}	1.810	−0.453
876.3	4.0	−0.3	−0.1822	4.186×10^{11}	1.644×10^{12}	1.386×10^{11}	3.976	−0.268
762	2.0	−0.1	0.3224	1.164×10^{12}	2.586×10^{12}	1.993×10^{11}	1.990	−0.056
685.8	0.7	0.0	0.4822	9.583×10^{11}	2.210×10^{12}	2.275×10^{11}	0.661	−0.002
678.18	0.5	0.0	0.5512	4.506×10^{11}	2.210×10^{12}	2.292×10^{11}	0.528	0.000
655.32	0.1	0.0	0.4581	4.506×10^{11}	1.234×10^{12}	2.402×10^{11}	0.129	0.001
647.7	0.0	0.0	0.6071	4.506×10^{11}	1.377×10^{12}	1.062×10^{12}	0.000	0.000
609.6	0.0	0.0	0.0000	4.506×10^{11}	1.377×10^{12}	1.062×10^{12}	0.000	0.000
502.92	0.0	0.0	0.0000	1.377×10^{12}	1.377×10^{12}	1.062×10^{12}	0.000	0.000
495.3	0.0	0.0	0.0000	1.377×10^{12}	1.377×10^{12}	1.062×10^{12}	0.000	0.000
464.82	0.0	0.0	0.0000	1.377×10^{12}	1.377×10^{12}	1.062×10^{12}	0.000	0.000
457.2	0.0	0.0	0.0000	6.600×10^{11}	6.600×10^{11}	1.062×10^{12}	0.000	0.000
388.62	0.0	0.0	0.0000	6.600×10^{11}	6.600×10^{11}	1.062×10^{12}	0.000	0.000
381	0.0	0.0	0.0000	5.740×10^{11}	5.740×10^{11}	1.062×10^{12}	0.000	0.000
198.12	0.0	0.0	0.0000	5.740×10^{11}	5.740×10^{11}	1.062×10^{12}	0.000	0.000
190.5	0.0	0.0	0.0000	1.481×10^{12}	1.200×10^{12}	1.062×10^{12}	0.000	0.000
0	0.0	0.0	0.0000	1.481×10^{12}	1.200×10^{12}	1.062×10^{12}	0.000	0.000

Table 24: XV-15 Blade Aerodynamic center location

Radial	Twist	Chord	Sweep	Anhedral
r/R	deg	r/R	r/R	r/R
0.09	0	0.1333	0	0
0.17	28.107	0.1333	−0.018 877 168	0.010 079 008
0.24	22.892	0.1333	−0.007 761 291	0.003 169 836
0.3	19.590	0.1341	$-6.141\,72 \times 10^{-5}$	−0.000 309 264
0.35	17.25	0.1347	0.002 785 122	−0.001 449 424
0.4	14.687	0.1347	0.003 559 017	−0.001 852 172
0.45	12.375	0.1347	0.004 332 913	−0.002 254 92
0.5	10.125	0.1347	0.005 106 808	−0.002 657 668
0.55	7.9375	0.1347	0.005 880 704	−0.003 060 416
0.6	5.875	0.1347	0.006 654 599	−0.003 463 164
0.65	3.844	0.1342	0.007 328 72	−0.003 859 21
0.7	1.8785	0.1285	0.007 902 552	−0.004 258 83
0.75	0.0555	0.1166	0.008 276 286	−0.004 670 731
0.8	−1.678	0.1056	0.008 650 653	−0.005 106 356
0.84	−3.062	0.0997	0.008 671 726	−0.005 487 147
0.88	−4.493	0.094	0.008 695 484	−0.005 898 687
0.92	−5.949	0.0865	0.008 524 23	−0.006 361 973
0.95	−7.062	0.0763	0.008 201 461	−0.006 761 769
0.97	−7.812	0.0621	0.008 319 484	−0.007 000 643
0.98	−8.175	0.0525	0.008 576 852	−0.007 092 643
0.99	−8.525	0.0429	0.008 537 389	−0.007 226 998
1	−8.875	0.0333	0.008 300 838	−0.007 394 59

C

XV-15 AIRFRAME DATA

Table 25: Wing main parameters

Parameter	Value	Unit
X Origin	7.89	m
Y Origin	0	m
Z Origin	2.2606	m
Span	4.906	m
Chord	1.600	m
Dihedral	2	deg
Sweep	−6.5	deg
θ_a	3	deg
θ_s	0	deg

Table 26: XV-15 wing inertial data in E50 format

STA	WEIGHT	XCG	YCG	ROTAPI	JX	JY	JP	EA
m	$kg\,m^{-1}$	m	m	deg	kg m	kg m	kg m	N
0	131.137 658 3	0.000	0.000	0.000	0.000	0.000	0.000	1.00×10^9
0.740 481 361	131.137 658 3	0.000	0.000	0.000	0.000	0.000	0.000	1.00×10^9
0.740 481 361	130.999 763 8	0.000	0.000	0.000	0.000	0.000	0.000	1.00×10^9
1.234 135 602	130.999 763 8	0.000	0.000	0.000	0.000	0.000	0.000	1.00×10^9
1.234 219 263	122.174 516 5	0.000	0.000	0.000	0.000	0.000	0.000	1.00×10^9
2.468 271 204	122.174 516 5	0.000	0.000	0.000	0.000	0.000	0.000	1.00×10^9
2.468 271 204	79.220 383 46	0.000	0.000	0.000	0.000	0.000	0.000	1.00×10^9
3.702 406 805	79.220 383 46	0.000	0.000	0.000	0.000	0.000	0.000	1.00×10^9
3.702 406 805	19.236 281 1	0.000	0.000	0.000	0.000	0.000	0.000	1.00×10^9
4.936 542 407	19.236 281 1	0.000	0.000	0.000	0.000	0.000	0.000	1.00×10^9

Table 27: XV-15 wing stiffness data in E50 format

STA	XNA	YNA	ROTAN	EJX (FLAP)	EJY (LAG)	GJ	XCT	YCT
m	m	m	deg	$N\,m^2$	$N\,m^2$	$N\,m^2$	m	m
0	0.000	0.000	0.000	1.06×10^7	2.87×10^{10}	2.87×10^{10}	0.000	0.000
0.740 481 361	0.000	0.000	0.000	1.06×10^7	2.87×10^{10}	2.87×10^{10}	0.000	0.000
0.740 481 361	0.000	0.000	0.000	1.06×10^7	3.22×10^7	8.04×10^6	0.000	0.000
1.234 135 602	0.000	0.000	0.000	1.06×10^7	3.22×10^7	8.04×10^6	0.000	0.000
1.234 219 263	0.000	0.000	0.000	1.06×10^7	3.22×10^7	8.04×10^6	0.000	0.000
2.468 271 204	0.000	0.000	0.000	1.06×10^7	3.22×10^7	8.04×10^6	0.000	0.000
2.468 271 204	0.000	0.000	0.000	1.06×10^7	3.22×10^7	8.04×10^6	0.000	0.000
3.702 406 805	0.000	0.000	0.000	1.06×10^7	3.22×10^7	8.04×10^6	0.000	0.000
3.702 406 805	0.000	0.000	0.000	1.06×10^7	3.22×10^7	8.04×10^6	0.000	0.000
4.936 542 407	0.000	0.000	0.000	1.06×10^7	3.22×10^7	8.04×10^6	0.000	0.000

Table 28: XV-15 Rigid body inertial data in global reference frame

Parameter	MASS	XCG	YCG	ZCG	IXX	IYY	IZZ	IZX	IZY
	kg	m	m	m	$kg\,m^2$	$kg\,m^2$	$kg\,m^2$	$kg\,m^2$	$kg\,m^2$
Fuselage	2822.44	7.015	0.000	−0.864	6941.26	59 505.3	59 120.78	4209.44	0
Nacelle	718.49	7.298	4.902	−0.057	489.88	489.88	0.93	0.00	0
Rotor	253.23	6.131	4.902	0.171	0.00	0.00	0.00	0.00	0
Pilot	84.00	5.467	0.419	1.283	0.00	0.00	0.00	0.00	0

Table 29: Optimal control parameters used for the coupled model

Parameter	Value	Unit
Mass m	5714	kg
Inertia moment arount pitch axis I_{yy}	62 718.8	$kg\,m^2$
Distance between center of mass and pole d_{cg}	−0.8132	m
Distance between tail A.C. and pole d_t	6.0326	m
Distance between wing A.C. and pole d_w	0.000	m
Tail surface S_t	3.2624	m^2
Wing surface S_w	15.68	m^2
Tail lift coefficient $C^t_{L/\alpha}$	4.58	rad^{-1}
Wing lift coefficient $C^w_{L/\alpha}$	4.63	rad^{-1}
Tail incidence α_{i^t}	0	deg
Wing incidence α_{i^w}	3	deg
Tail chord c_t	0.8365	m
Wing chord c_w	1.6002	m
Tail derivative η	0.7658	-
Tail moment coefficient C_{m^t}	0	-
Wing moment coefficient C_{m^w}	0	-

D

PILOT'S MODEL DATA

D.1 SCALING INTERPOLATION FUNCTIONS

UPPER LIMB MASSES:

$$\begin{aligned} M_{L_{hr}} &= 0.022 M_b + 0.485 \\ M_{R_{hr}} &= 0.016 M_b + 0.809 \\ M_{L_{fr}} &= 0.013 M_b + 0.246 \\ M_{R_{fr}} &= 0.020 M_b - 0.219 \\ M_{L_{hd}} &= 0.005 M_b + 0.485 \\ M_{R_{hd}} &= 0.007 M_b - 0.485 \end{aligned} \tag{259}$$

where $M_{L_{hr}}$ is the mass of the left humerus, $M_{L_{fr}}$ of the left forearm and $M_{R_{hd}}$ of the left hand.

INTRAVERTEBRAL STIFFNESS COEFFICIENTS:

$$\begin{aligned} K_a &= -0.0091\,\mathrm{BMI}^2 + 0.4873\,\mathrm{BMI} - 5.58 \\ K_{b_x} &= -0.8499\,\mathrm{BMI}^2 + 45.6931\,\mathrm{BMI} - 523.51 \\ K_{b_y} &= -0.2125\,\mathrm{BMI}^2 + 11.4233\,\mathrm{BMI} - 130.88 \\ K_{b_z} &= -0.0091\,\mathrm{BMI}^2 + 0.4873\,\mathrm{BMI} - 5.5830 \end{aligned} \tag{260}$$

INTRAVERTEBRAL DAMPING COEFFICIENTS:

$$\begin{aligned} C_a &= 0.1415\,\mathrm{BMI}^2 - 7.1640\,\mathrm{BMI} + 95.28 \\ C_{b_x} &= 17.0531\,\mathrm{BMI}^2 - 863.5008\,\mathrm{BMI} + 11484 \\ C_{b_y} &= 13.6424\,\mathrm{BMI}^2 - 690.8006\,\mathrm{BMI} + 9187.5 \\ C_{b_z} &= 2.3580\,\mathrm{BMI}^2 - 119.4000\,\mathrm{BMI} + 1588 \end{aligned} \tag{261}$$

D.2 COCKPIT DATA

Table 30: Cockpit configuration parameters. Moment of inertia of the control sticks are referred to the pivot axis. Pivot positions are referred to the pilot seat center, with the X axis pointing forward, the Y axis pointing laterally towards the left and the Z axis pointing upward

Parameter	Value	Units
Collective stick length	800	mm
Collective stick mass	3.000	mm
Collective stick moment of inertia	1.50	$kg\,m^{-2}$
Collective stick pivot X position	−400	mm
Collective stick pivot Y position	300	mm
Collective stick pivot Z position	−445	mm
Cyclic stick length	700	mm
Cyclic stick mass	1.860	kg
Cyclic stick moment of inertia, X-axis	0.69	$kg\,m^{-2}$
Cyclic stick moment of inertia, Y-axis	0.58	$kg\,m^{-2}$
Cyclic stick pivot X position	200	mm
Cyclic stick pivot Y position	0	mm
Cyclic stick pivot Z position	−457	mm

Table 31: Trim and balancing system's parameters, long and heavy (LH) configuration.

Parameter	Value	Units
Collective		
Balancing spring stiffness	17.5	$N\,m\,rad^{-1}$
Balancing system damping	2.4	$N\,m\,s\,rad^{-1}$
Trim spring stiffness 1	525	$N\,m\,rad^{-1}$
Trim spring stiffness 2	40	$N\,m\,rad^{-1}$
Trim spring stiffness 3	85	$N\,m\,rad^{-1}$
Trim system damping	13.2	$N\,m\,s\,rad^{-1}$
Longitudinal cyclic		
Balancing spring stiffness	5.5	$N\,m\,rad^{-1}$
Balancing system damping	2.3	$N\,m\,s\,rad^{-1}$
Trim spring stiffness 1	660	$N\,m\,rad^{-1}$
Trim spring stiffness 2	90	$N\,m\,rad^{-1}$
Trim spring stiffness 3	90	$N\,m\,rad^{-1}$
Trim system damping	26.5	$N\,m\,s\,rad^{-1}$
Lateral cyclic		
Balancing spring stiffness	5.5	$N\,m\,rad^{-1}$
Balancing system damping	2.6	$N\,m\,s\,rad^{-1}$
Trim spring stiffness 1	580	$N\,m\,rad^{-1}$
Trim spring stiffness 2	33	$N\,m\,rad^{-1}$
Trim spring stiffness 3	55	$N\,m\,rad^{-1}$
Trim system damping	25.5	$N\,m\,s\,rad^{-1}$

Table 32: Spine intervertebral deformable elements stiffness and damping coefficients of the reference subject: 34 years old, stature of 1.783 m, weight of 84 kg (resulting BMI 26.5kg m^{-2}) .

Label	K_a	K_{b_x}	K_{b_y}	K_{b_z}	C_a	C_{b_x}	C_{b_y}	C_{b_z}
	N m^{-1}	N m rad^{-1}	N m rad^{-1}	N m rad^{-1}	N s m^{-1}	N m s rad^{-1}	N m s rad^{-1}	N m s rad^{-1}
H-C1	665 540.8	10 646.1	44 978.0	58.4	22 260.4	482.5	579.0	141.8
C1-C2	363 022.3	10 646.1	101 200.6	58.4	8392.3	310.2	372.2	91.7
C2-C3	847 052.0	10 379.9	8995.6	105.1	12 745.7	94.8	110.3	27.4
C3-C4	919 656.4	11 444.5	11 244.5	140.1	13 360.5	103.4	124.1	31.0
C4-C5	960 798.9	12 775.3	13 493.4	163.4	13 160.3	112.0	131.0	32.2
C5-C6	1 170 141.8	15 436.8	17 991.2	210.1	14 568.6	129.2	151.6	36.9
C6-C7	1 227 015.3	16 235.2	24 737.9	233.5	15 033.2	146.5	179.2	44.1
C7-T1	1 614 239.0	22 356.7	41 604.7	338.5	21 481.2	241.3	289.5	71.5
T1-T2	847 052.0	11 737.3	15 742.3	163.4	19 872.8	189.6	227.5	56.0
T2-T3	1 452 089.1	19 375.8	31 484.6	245.1	23 976.0	249.9	296.4	72.7
T3-T4	1 815 111.3	22 729.3	47 226.9	326.8	27 485.9	310.2	372.2	91.7
T4-T5	2 541 155.9	33 721.4	78 711.6	490.3	31 724.9	396.3	468.7	115.6
T5-T6	2 299 141.0	31 299.4	78 711.6	490.3	31 503.3	413.6	489.4	120.3
T6-T7	2 178 133.6	30 181.6	78 711.6	490.3	30 667.0	413.6	489.4	120.3
T7-T8	1 815 111.3	29 622.6	78 711.6	490.3	23 654.3	344.7	413.6	101.3
T8-T9	1 815 111.3	32 789.8	86 582.7	572.0	22 553.4	344.7	413.6	101.3
T9-T10	1 815 111.3	35 957.0	86 582.7	572.0	23 304.0	361.9	427.4	104.8
T10-T11	1 815 111.3	45 458.6	94 453.9	653.7	24 111.8	387.7	461.8	113.2
T11-T12	1 815 111.3	43 595.6	78 711.6	817.1	11 430.4	163.7	199.9	48.8
T12-L1	2 178 133.6	42 291.4	70 840.4	980.5	12 745.7	163.7	193.0	47.7
L1-L2	2 577 458.1	40 987.3	70 840.4	980.5	12 874.4	146.5	179.2	44.1
L2-L3	2 420 148.4	40 242.1	70 840.4	980.5	13 918.1	163.7	199.9	48.8
L3-L4	2 420 148.4	40 801.0	70 840.4	980.5	15 833.9	189.6	227.5	56.0
L4-L5	2 262 838.8	33 721.4	62 969.3	898.8	17 227.8	198.2	241.3	59.6
L5-S1	1 778 809.1	29 250.0	7871.2	735.4	50 611.2	232.6	282.6	69.1

Table 33: Mass, inertia moments and Cartesian coordinates of vertebral and visceral nodes of the reference subject: 34 years old, stature of 1.783 m, weight of 84 kg (resulting BMI 26.5kg m^{-2})

Label	M	Ixx	Ixy	Ixz	Jyy	Jyz	Jzz	x	y	z
	kg	kg cm^2	kg cm^2	kg cm^2	kg cm^2	kg cm^2	kg cm^2	m	m	m
Head	5.418	222.449	0.000	0.000	212.254	0.000	259.524	0.0265	0.00	0.7449
C1	0.981	6.664	0.000	0.000	5.570	0.000	12.049	0.0101	0.00	0.6871
C2	0.981	6.664	0.000	0.000	5.570	0.000	12.049	0.0148	0.00	0.6723
C3	0.981	6.664	0.000	0.000	5.570	0.000	12.049	0.0178	0.00	0.6584
C4	0.981	6.664	0.000	0.000	5.570	0.000	12.049	0.0193	0.00	0.6476
C5	0.981	6.664	0.000	0.000	5.570	0.000	12.049	0.0202	0.00	0.6357
C6	1.084	9.769	0.000	0.000	6.080	0.000	15.850	0.0203	0.00	0.6223
C7	1.445	32.607	0.000	0.000	7.183	0.000	40.690	0.0196	0.00	0.6089
T1	1.638	47.993	0.000	0.000	6.896	0.000	158.921	0.0177	0.00	0.5925
T2	1.293	95.347	0.000	0.000	19.233	0.000	113.959	0.0145	0.00	0.5747
T3	1.398	10.872	0.000	0.000	26.657	0.000	131.736	0.0093	0.00	0.5538
T4	1.281	9.816	0.000	0.000	29.057	0.000	125.461	0.0019	0.00	0.5304
T5	1.416	106.637	0.000	0.000	35.546	0.000	138.011	−0.0068	0.00	0.5054
T6	1.438	107.684	0.000	0.000	40.977	0.000	143.239	−0.0134	0.00	0.4783
T7	1.576	112.911	0.000	0.000	49.764	0.000	153.694	−0.0172	0.00	0.4494
T8	1.598	111.864	0.000	0.000	51.330	0.000	154.732	−0.0178	0.00	0.4183
T9	1.707	113.959	0.000	0.000	57.077	0.000	158.921	−0.0154	0.00	0.3888
T10	1.629	104.551	0.000	0.000	55.825	0.000	152.646	−0.0091	0.00	0.3541
T11	0.383	2.706	0.000	0.000	11.892	0.000	3.096	0.0016	0.00	0.3162
T12	0.401	2.957	0.000	0.000	11.771	0.000	3.235	0.0151	0.00	0.2790
L1	0.342	4.653	0.000	0.000	9.602	0.000	5.496	0.0267	0.00	0.2399
L2	0.412	6.229	0.000	0.000	11.614	0.000	7.369	0.0327	0.00	0.2033
L3	0.521	6.164	0.000	0.000	13.736	0.000	7.980	0.0329	0.00	0.1696
L4	0.677	8.917	0.000	0.000	13.226	0.000	11.382	0.0273	0.00	0.1383
L5	0.561	4.977	0.000	0.000	9.204	0.000	7.656	0.0150	0.00	0.1089
S1	20.324	862.917	0.000	0.000	1309.669	0.000	1064.141	0.0034	0.00	0.0914
T10V	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.0475	0.00	0.3541
T11V	1.516	111.317	0.000	0.000	47.548	0.000	157.939	0.0449	0.00	0.3162
T12V	1.586	119.288	0.000	0.000	47.076	0.000	154.880	0.0458	0.00	0.2790
L1V	1.982	140.792	0.000	0.000	54.407	0.000	177.589	0.0407	0.00	0.2399
L2V	1.975	150.895	0.000	0.000	56.715	0.000	178.979	0.0396	0.00	0.2033
L3V	2.034	149.782	0.000	0.000	54.936	0.000	170.266	0.0406	0.00	0.1696
L4V	1.922	149.041	0.000	0.000	38.243	0.000	174.901	0.0434	0.00	0.1383
L5V	2.098	161.368	0.000	0.000	35.045	0.000	198.350	0.0438	0.00	0.1089
S1V	2.020	9.269	0.000	0.000	9.269	0.000	9.269	0.0438	0.00	0.0914
L-Limb	4.167	0.000	0.000	0.000	0.000	0.000	0.000	0.0093	0.2295	0.5446
R-Limb	4.173	0.000	0.000	0.000	0.000	0.000	0.000	0.0093	−0.2295	0.5446

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DECLARATION

I declare that this thesis was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification.

Milan, November 2022

Alessandro Cocco

COLOPHON

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